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French et al.

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(54) **SAFETY FOR FIREARM
ELECTROMAGNETIC FIRING SYSTEM**

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(73) Assignee: **Sturm, Ruger & Company, Inc.**

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 561 days.

(21) Appl. No.: **17/705,571**

(22) Filed: **Mar. 28, 2022**

(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation-in-part of application No. 17/158,139,
filed on Jan. 26, 2021, now Pat. No. 11,300,378,
(Continued)

(51) **Int. Cl.**
F41A 19/17 (2006.01)
F41A 19/10 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **F41A 19/17** (2013.01); **F41A 19/10**
(2013.01); **F41A 19/16** (2013.01); **F41A 19/58**
(2013.01); **F41A 19/59** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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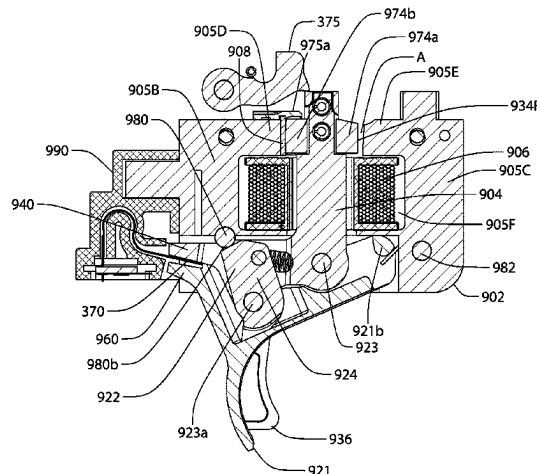
Primary Examiner — Reginald S Tillman, Jr.

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(57) **ABSTRACT**

An electronic trigger system with safety mechanism for
firearms includes a trigger unit comprising an electromag-
netic actuator operably coupled to the firing mechanism and
a programmable trigger mechanism microcontroller. The
actuator is changeable between a non-powered unactuated
position and powered actuated firing position via pulling
both an outer trigger and inner safety trigger of the trigger
unit. The actuator is operably interfaced with a movable
firing component of the firing mechanism operable to dis-
charge the firearm. In a blocking position of the safety
trigger, the safety trigger is configured to block outer trigger
movement which prevents energizing and actuating the
electromagnetic actuator thereby preventing the firearm
from discharging. Conversely in an unblocking position, the
safety trigger allows outer trigger movement sufficient to

(Continued)



discharge the firearm by either energizing the actuator when a pre-selected trigger pull force threshold is applied or applying sufficient trigger force to manually trip the actuator.

21 Claims, 94 Drawing Sheets

Related U.S. Application Data

which is a continuation-in-part of application No. 16/909,577, filed on Jun. 23, 2020, now Pat. No. 10,900,732, which is a continuation-in-part of application No. 16/530,545, filed on Aug. 2, 2019, now Pat. No. 10,690,430, which is a continuation of application No. 16/283,338, filed on Feb. 22, 2019, now Pat. No. 10,458,736, which is a continuation-in-part of application No. 15/908,883, filed on Mar. 1, 2018, now Pat. No. 10,228,208.

- (60) Provisional application No. 62/635,598, filed on Feb. 27, 2018, provisional application No. 62/468,632, filed on Mar. 8, 2017.

(51) Int. Cl.

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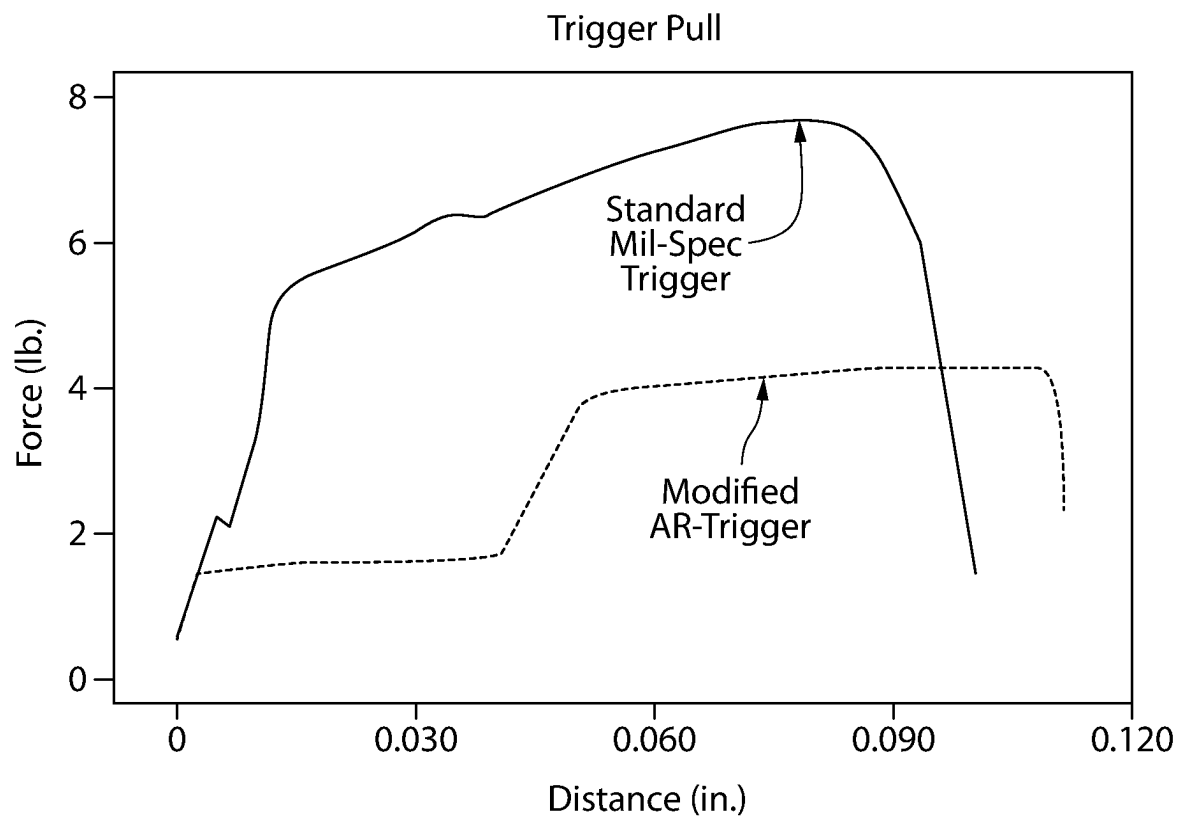
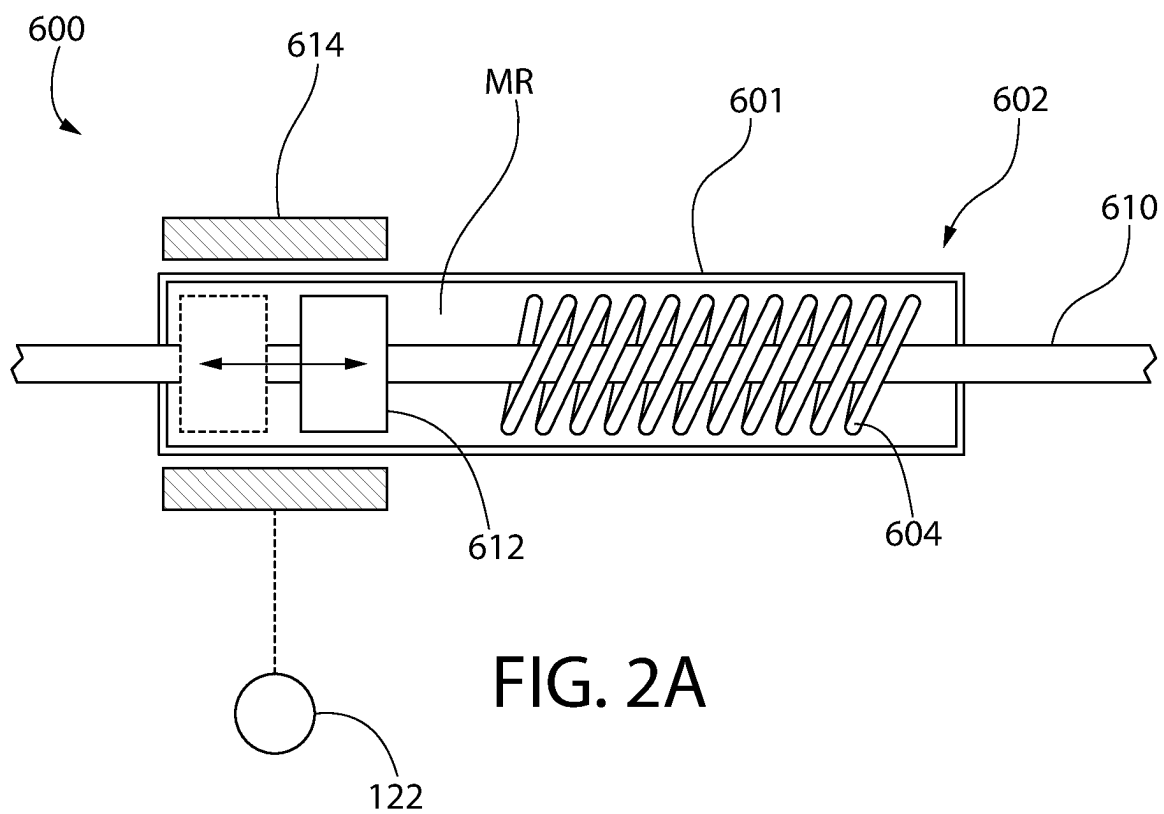
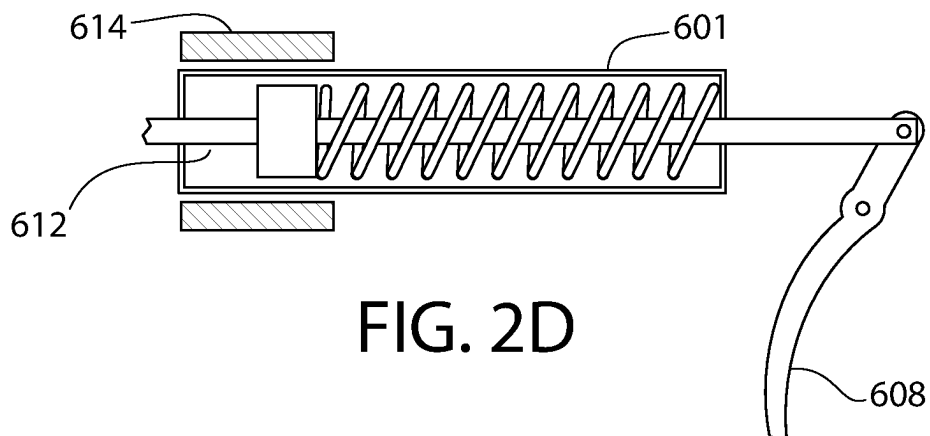
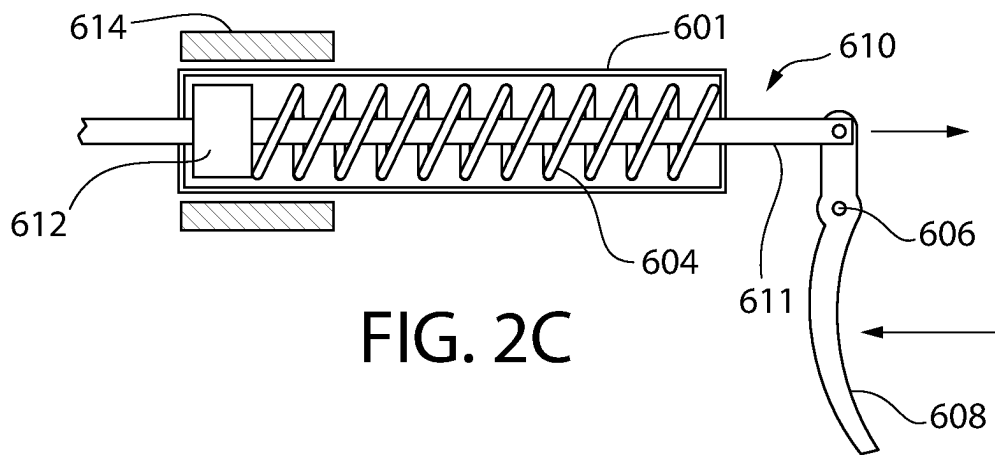
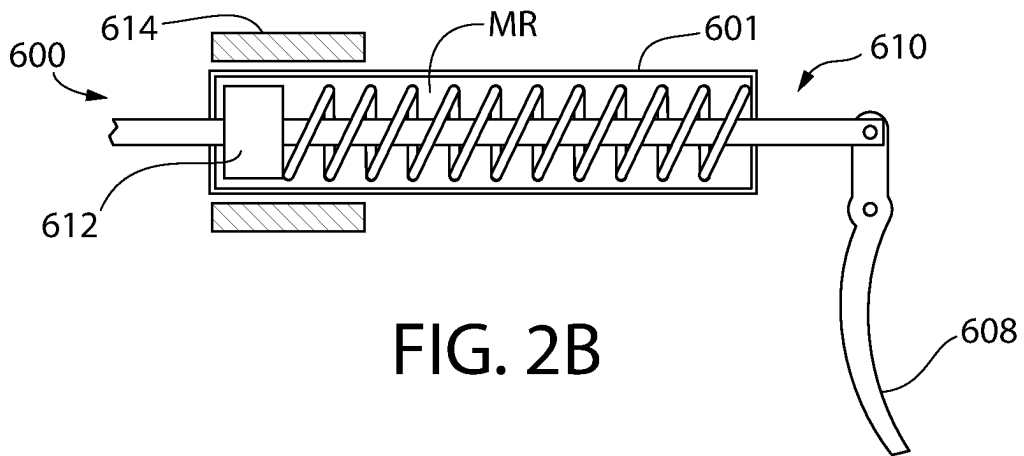


FIG. 1





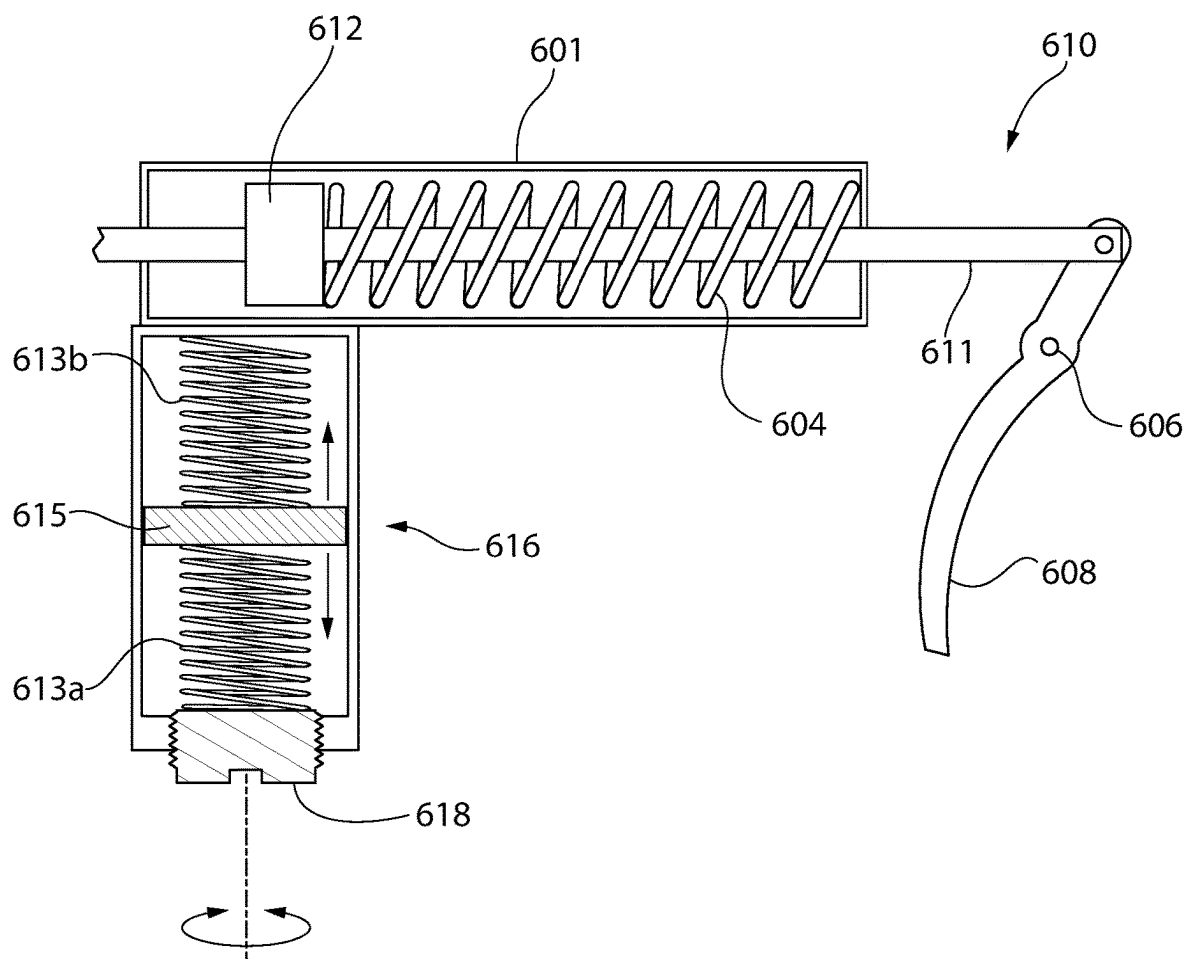


FIG. 3

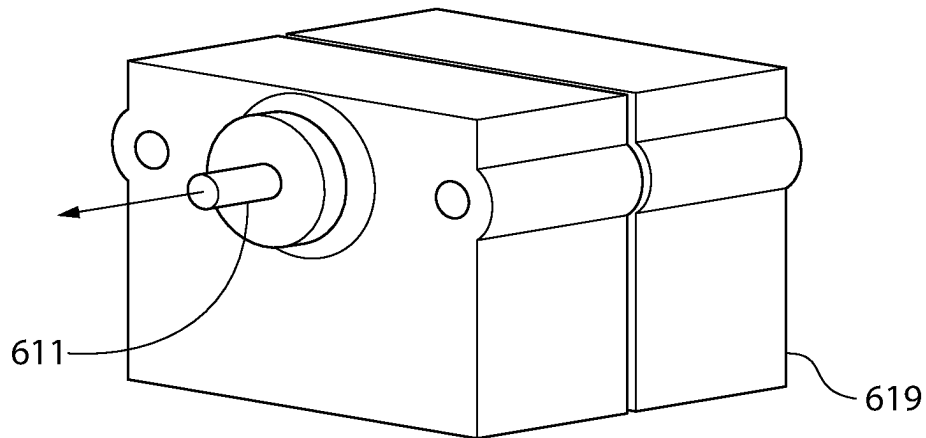


FIG. 4A

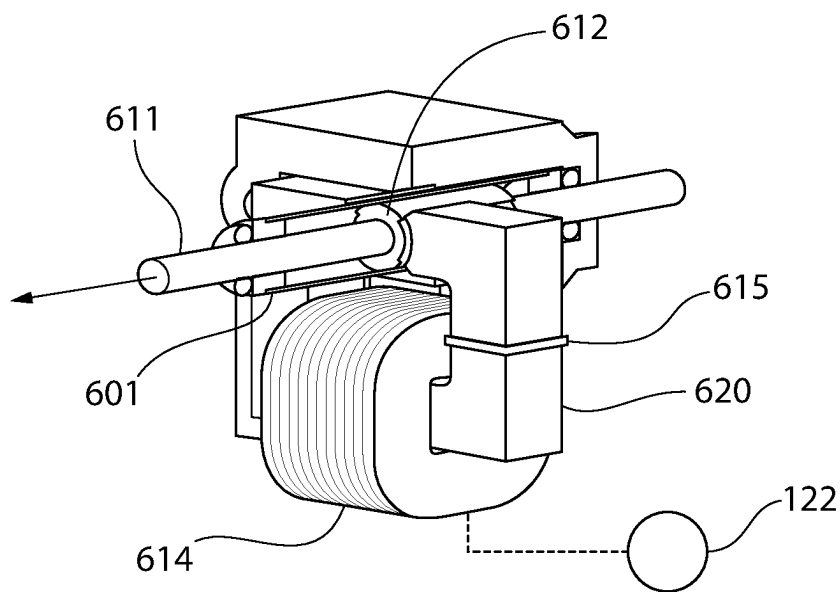


FIG. 4B

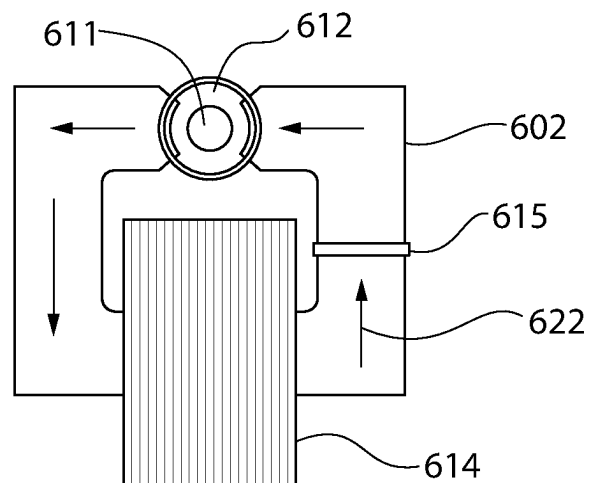


FIG. 4C

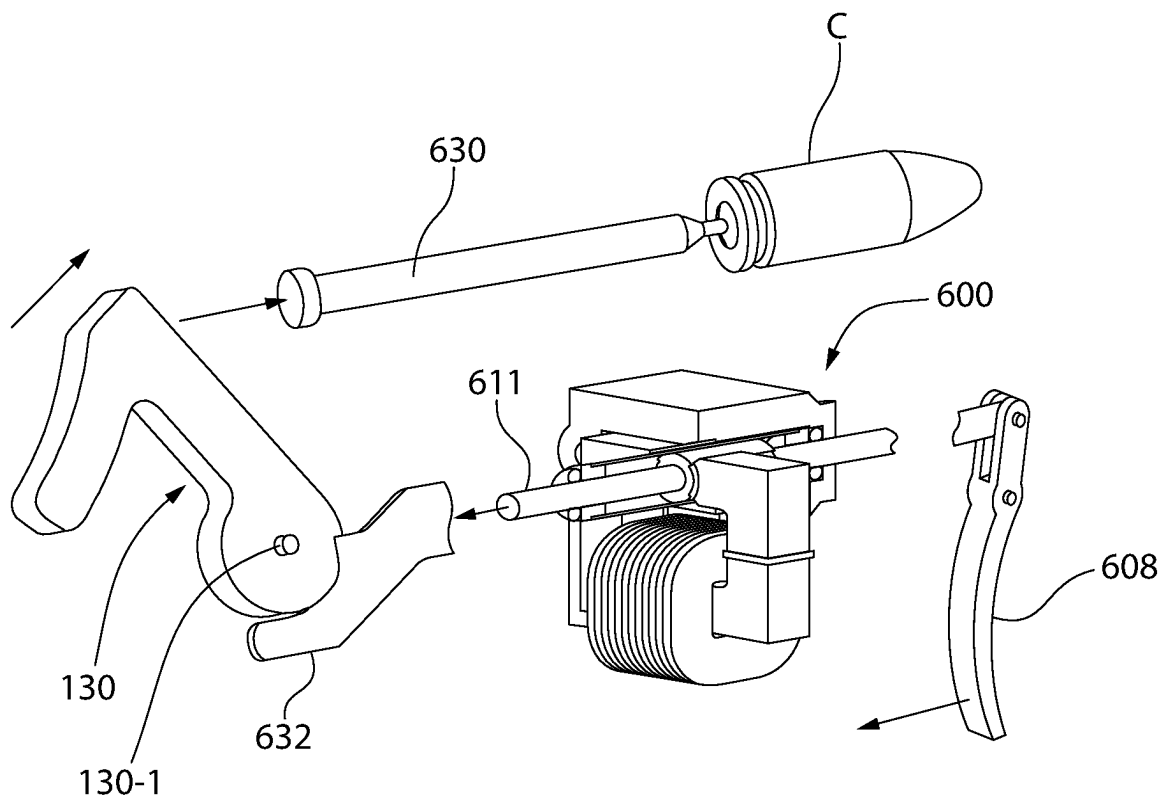


FIG. 5

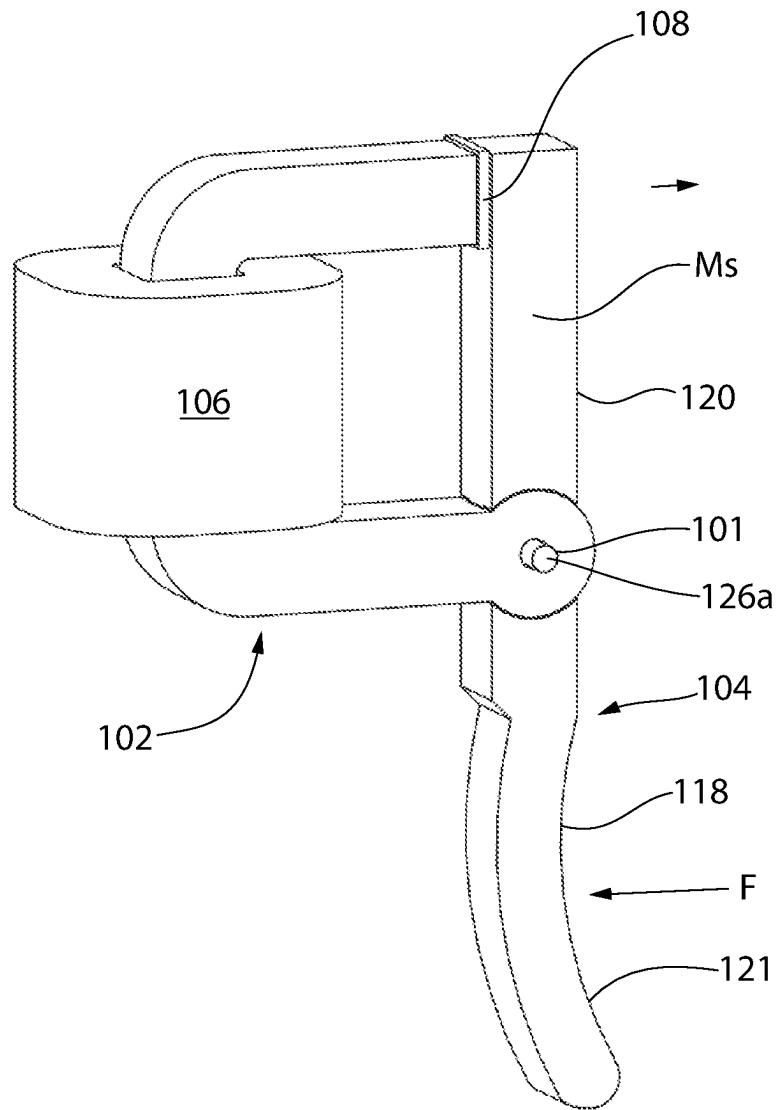


FIG. 6

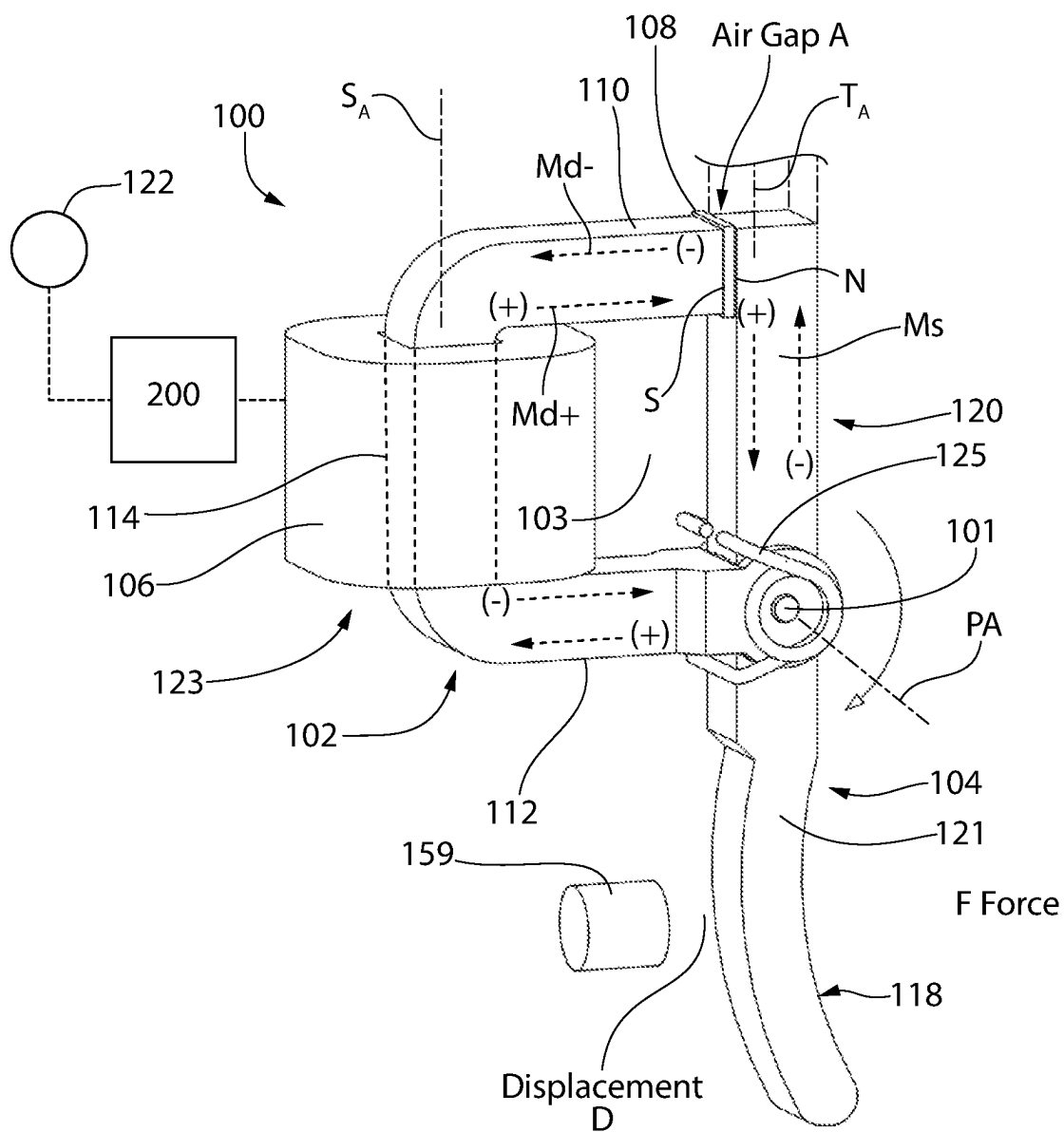


FIG. 7

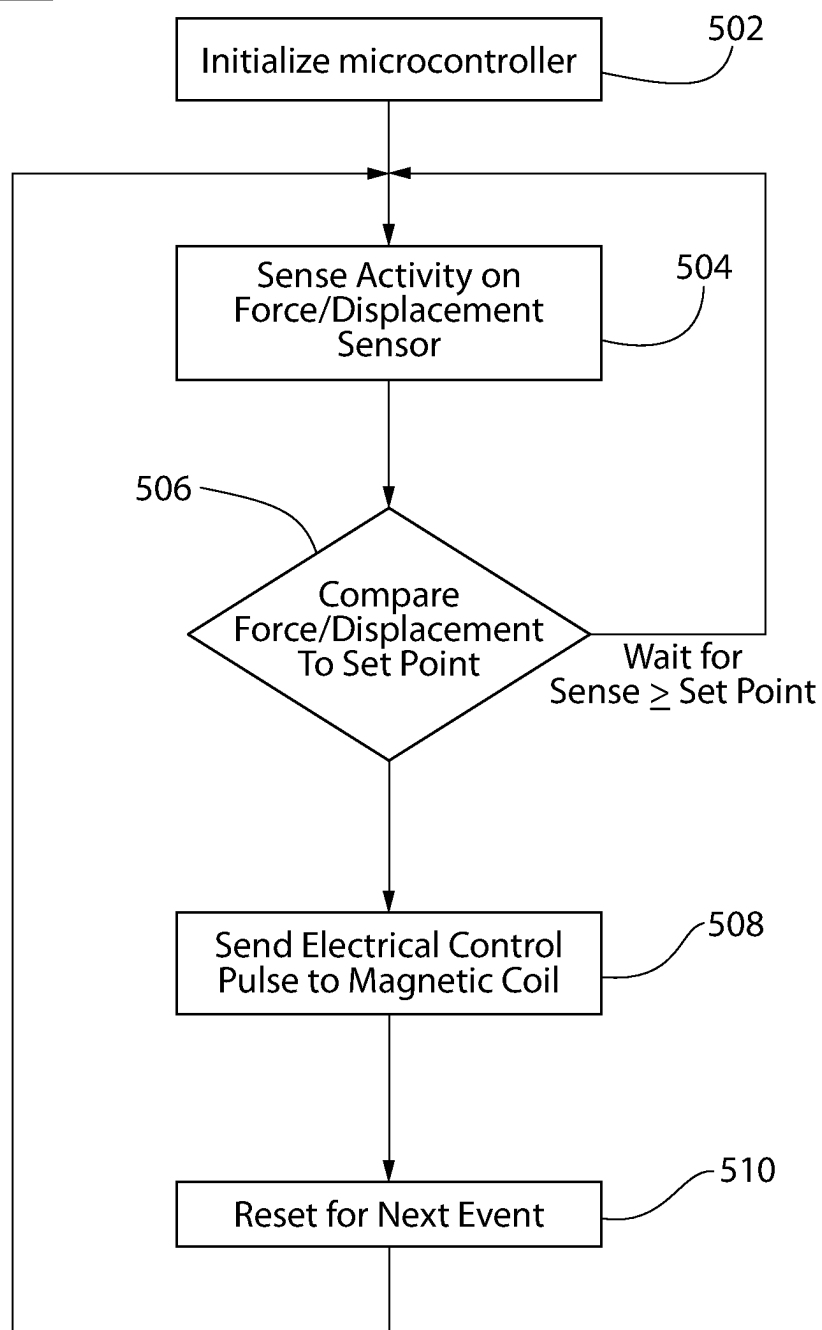
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FIG. 8

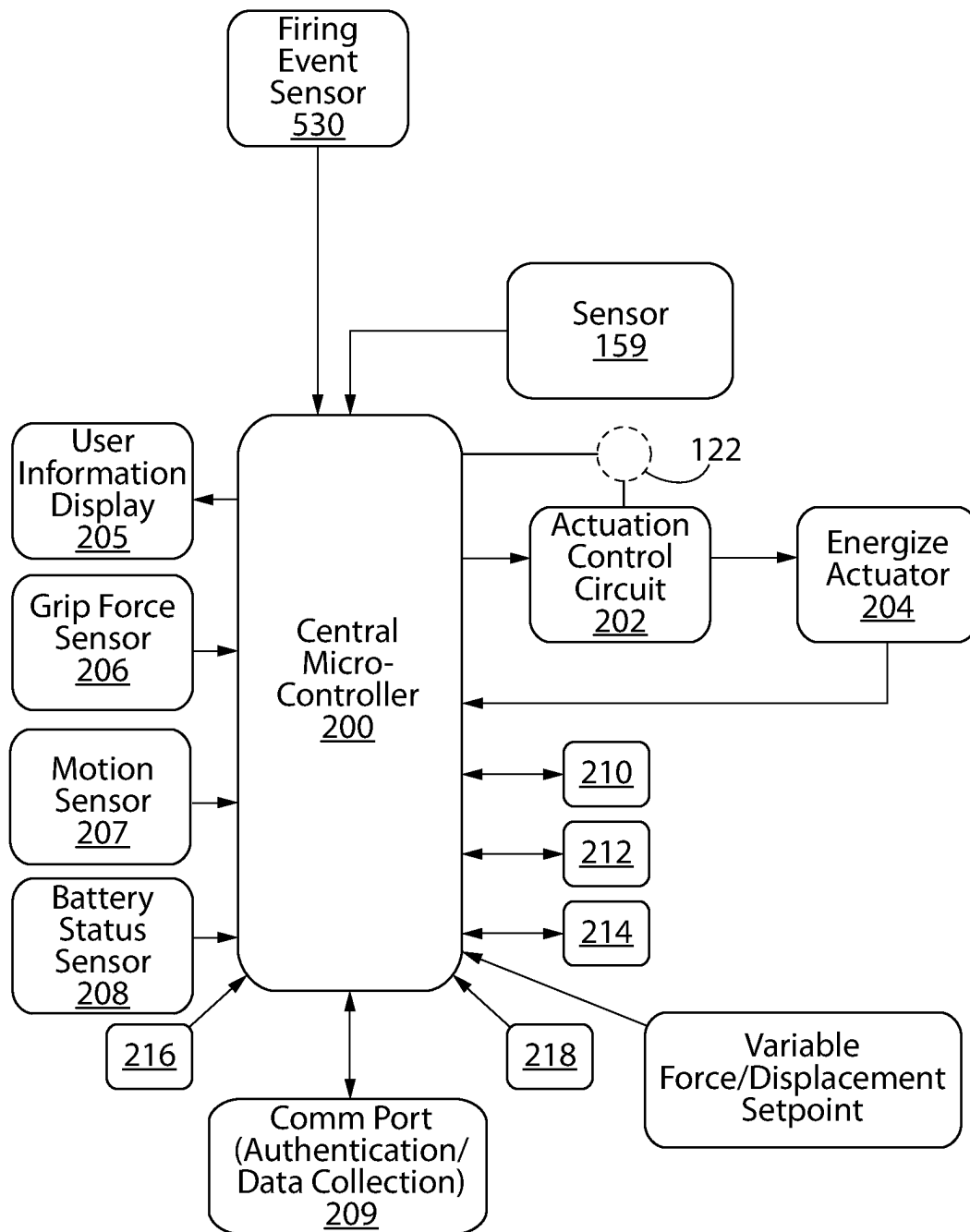


FIG. 9

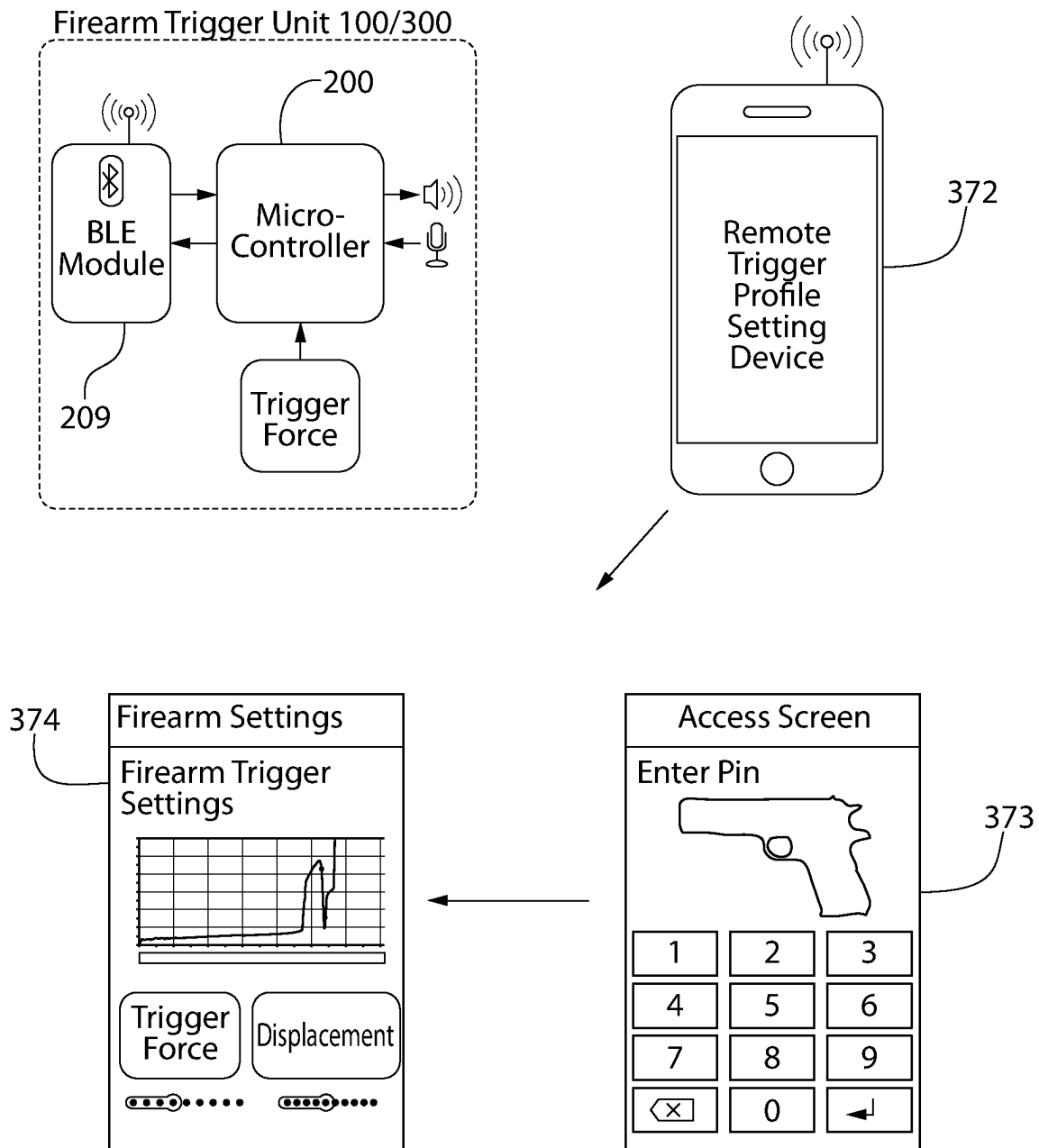


FIG. 10A

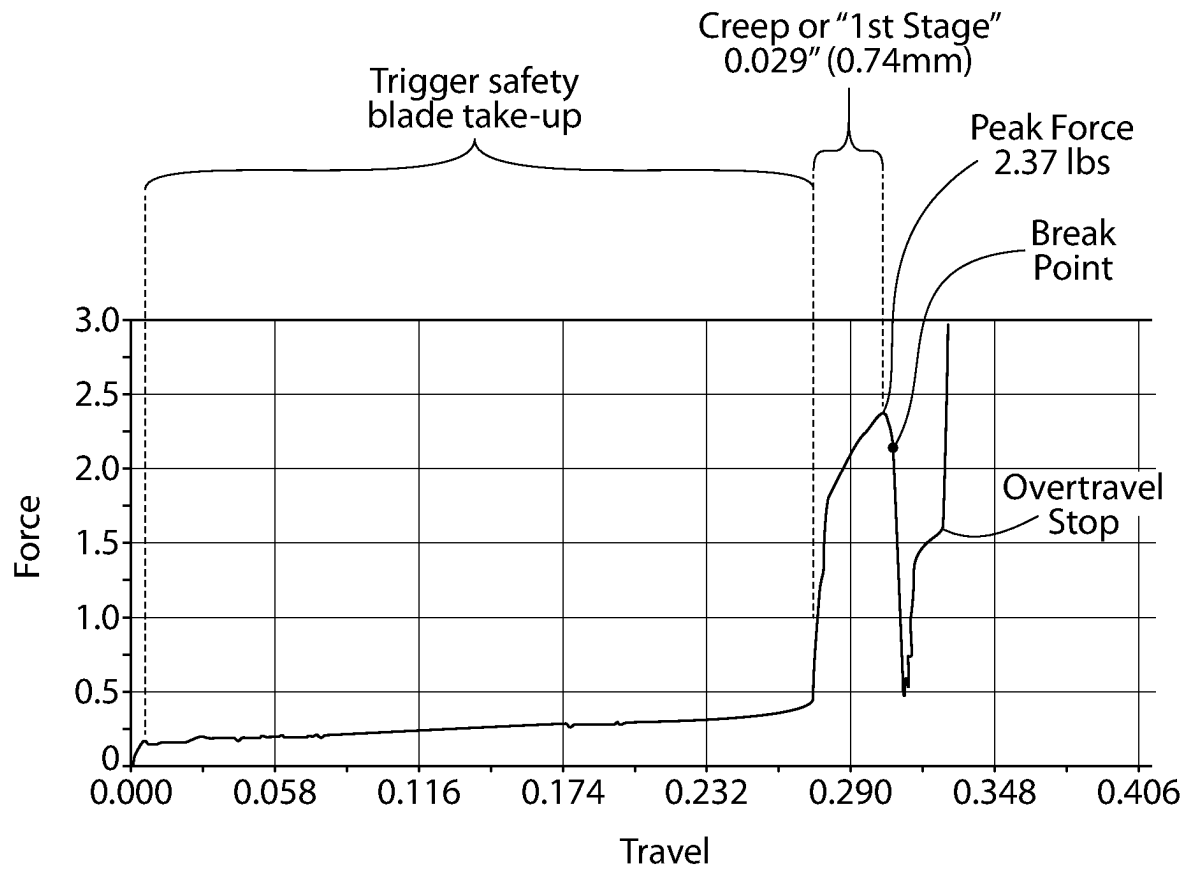


FIG. 10B

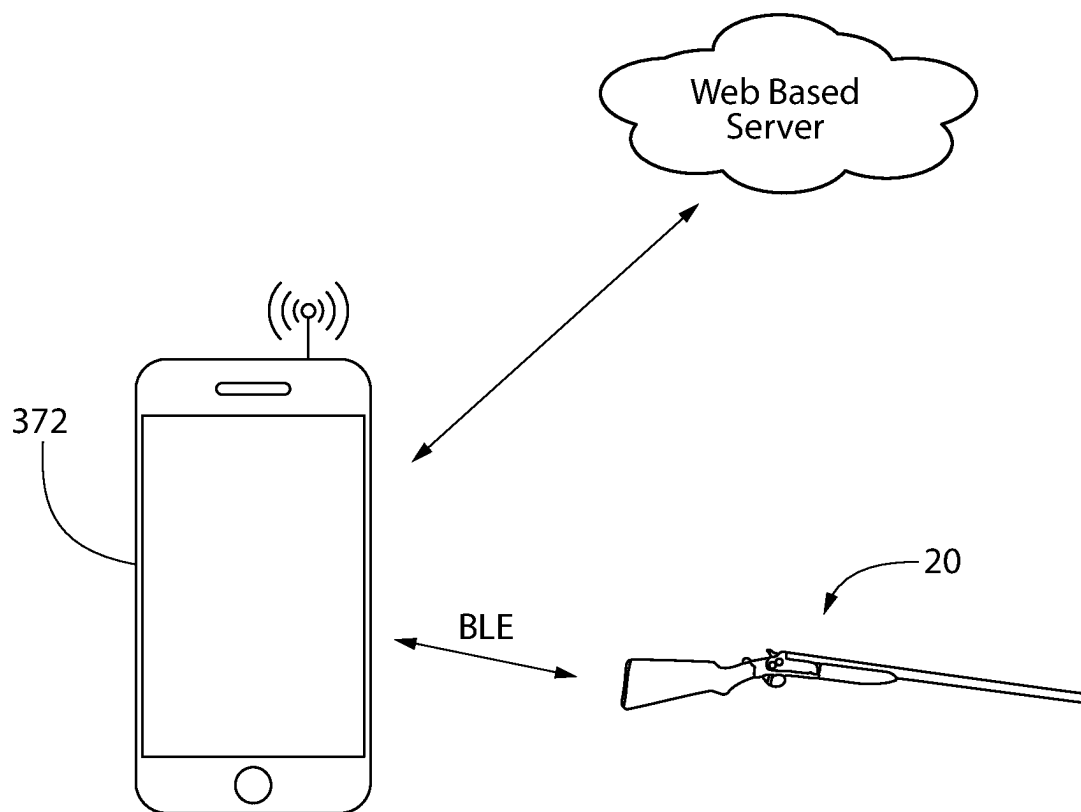


FIG. 11

Non-linear Force-Displacement
Curve
(segmented design)

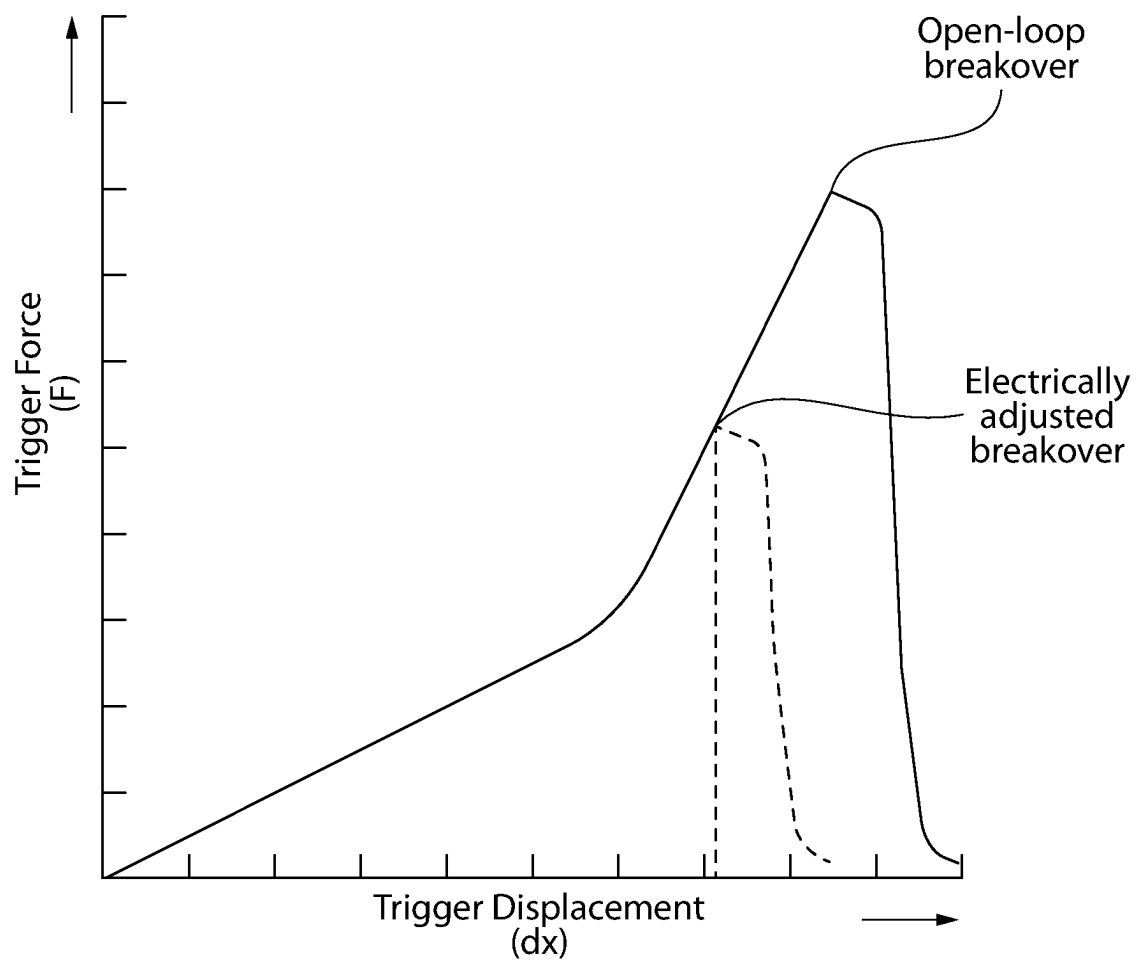


FIG. 12

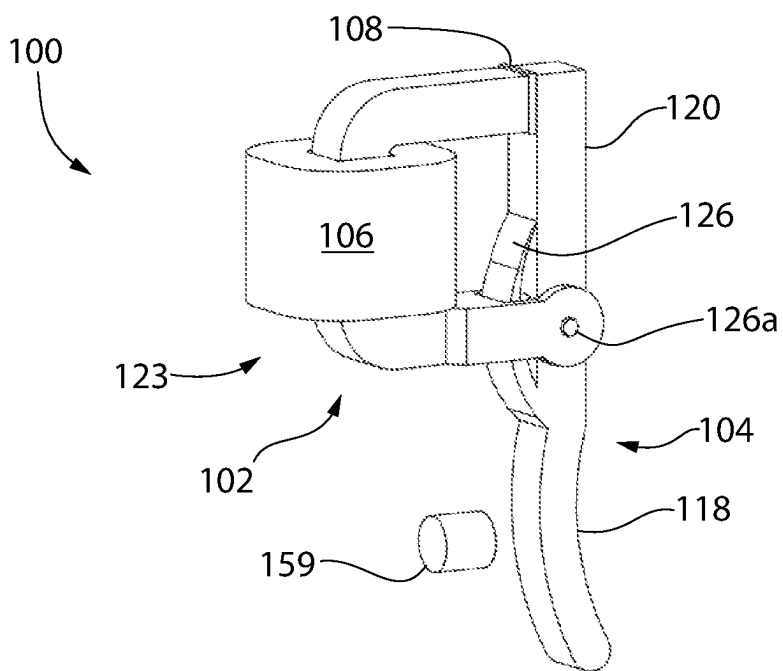


FIG. 13A

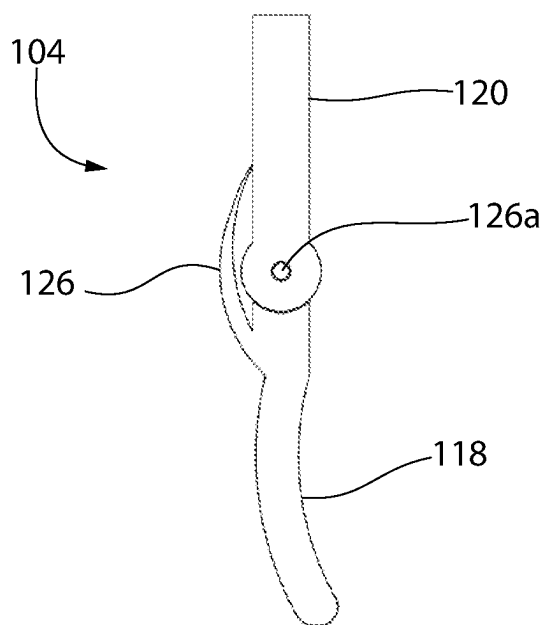


FIG. 13B

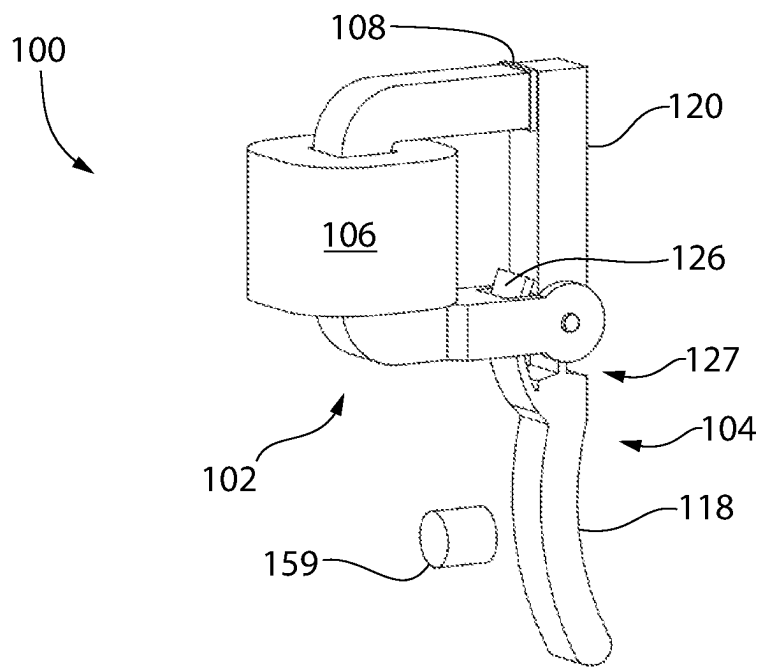


FIG. 14A

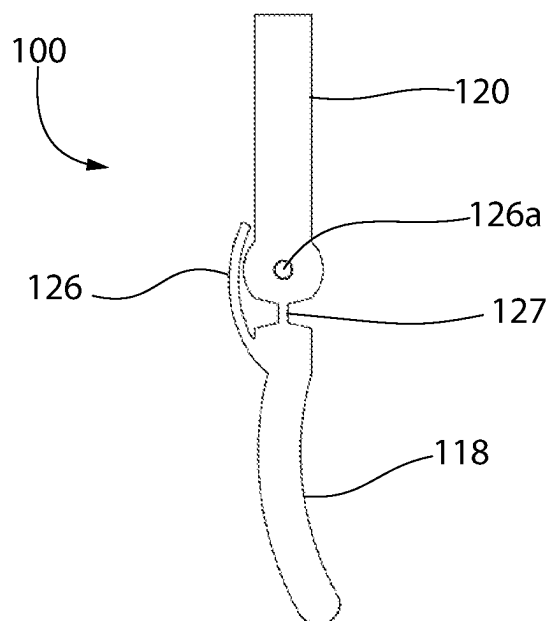


FIG. 14B

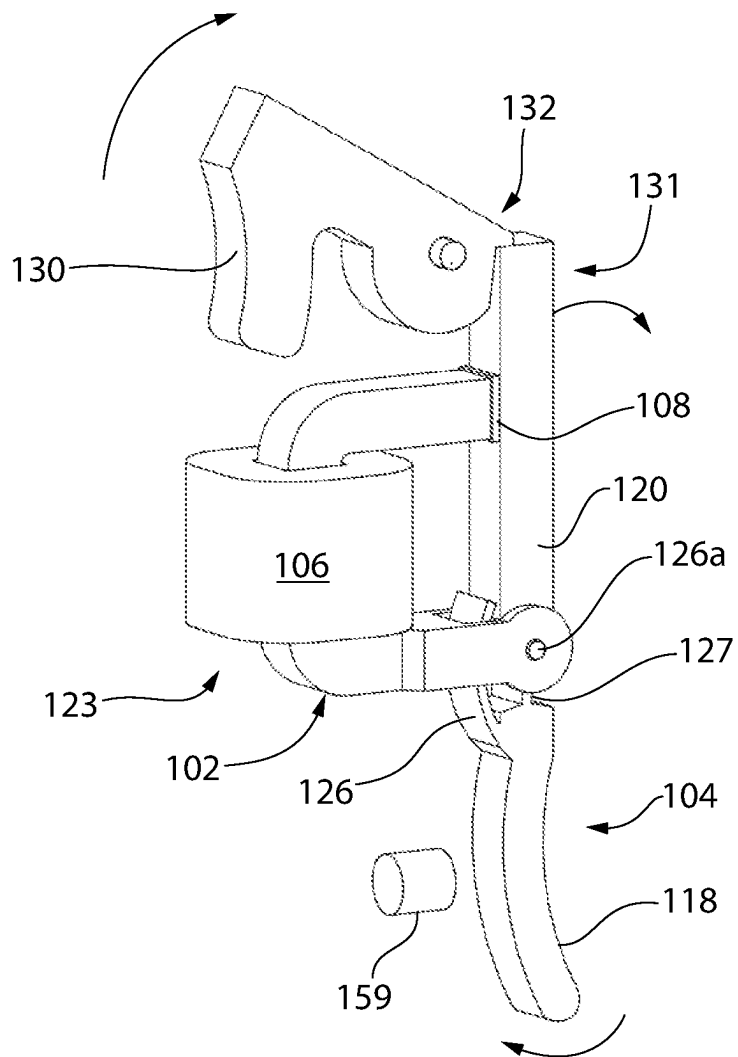


FIG. 15

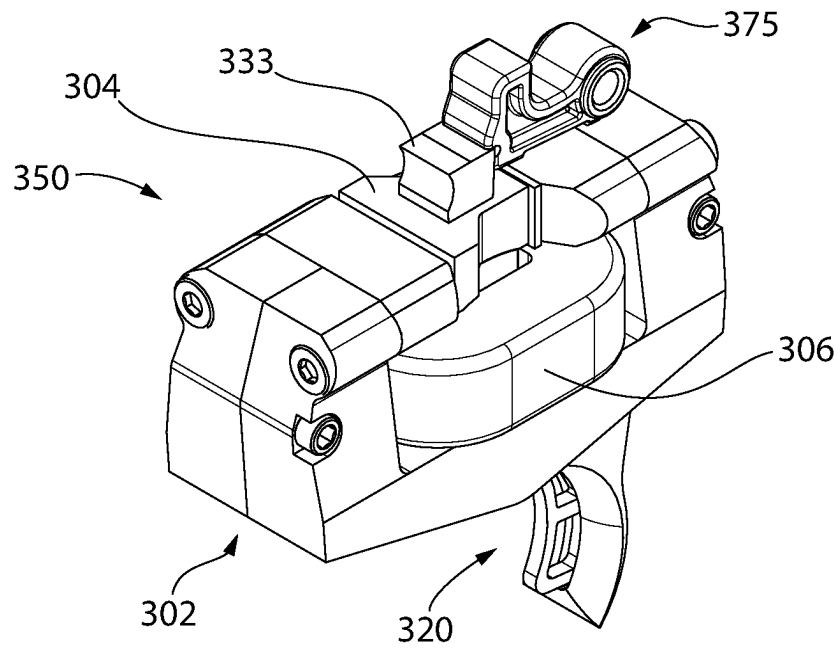


FIG. 16

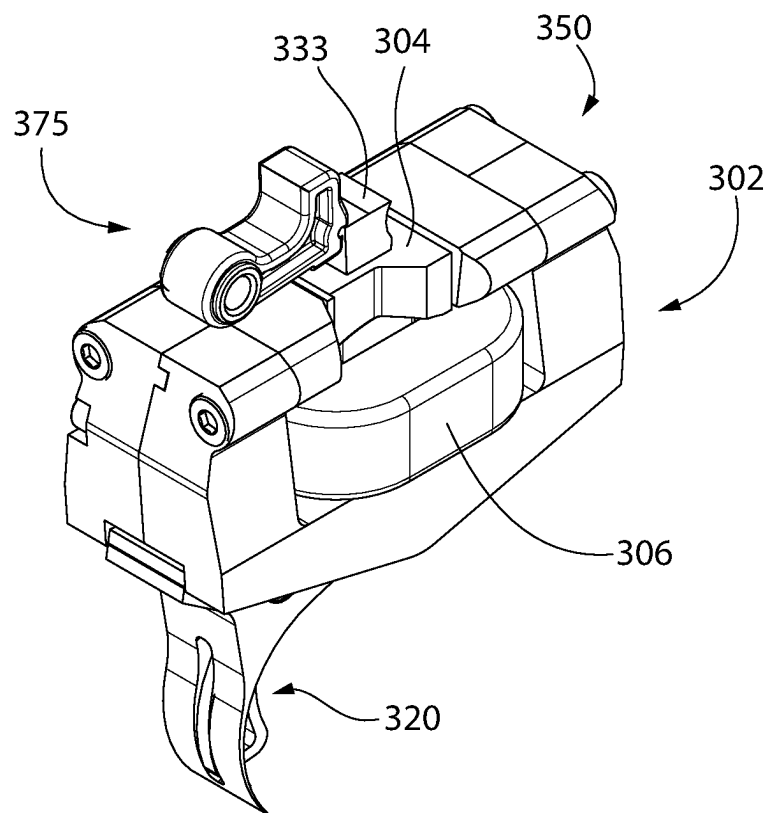


FIG. 17

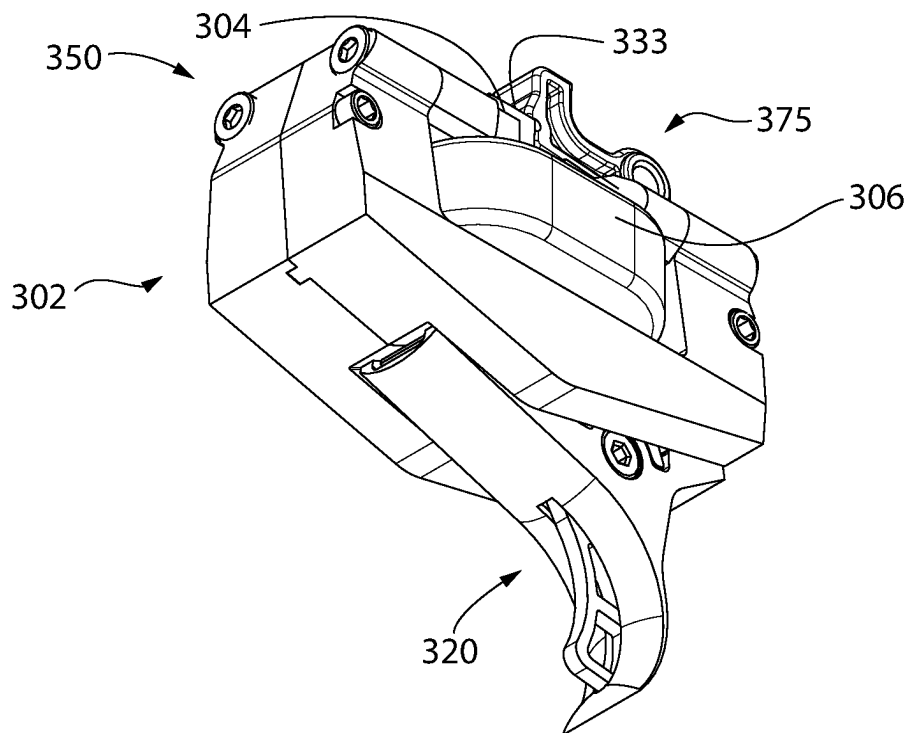


FIG. 18

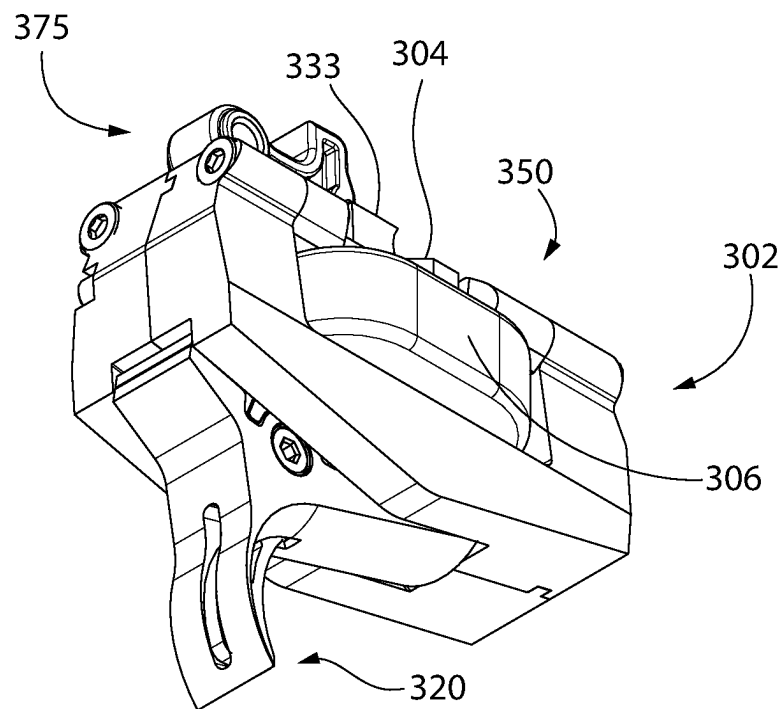


FIG. 19

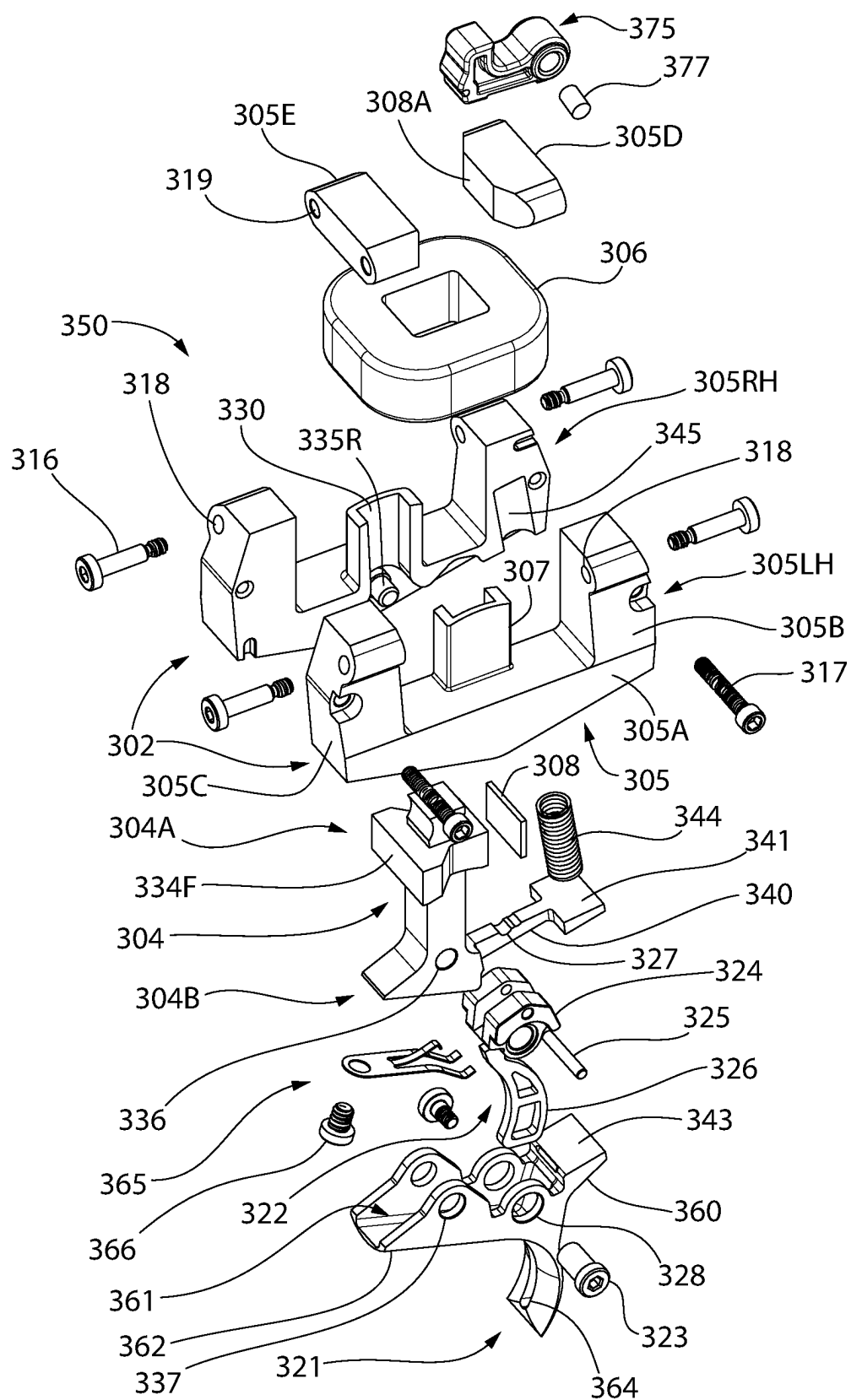


FIG. 20

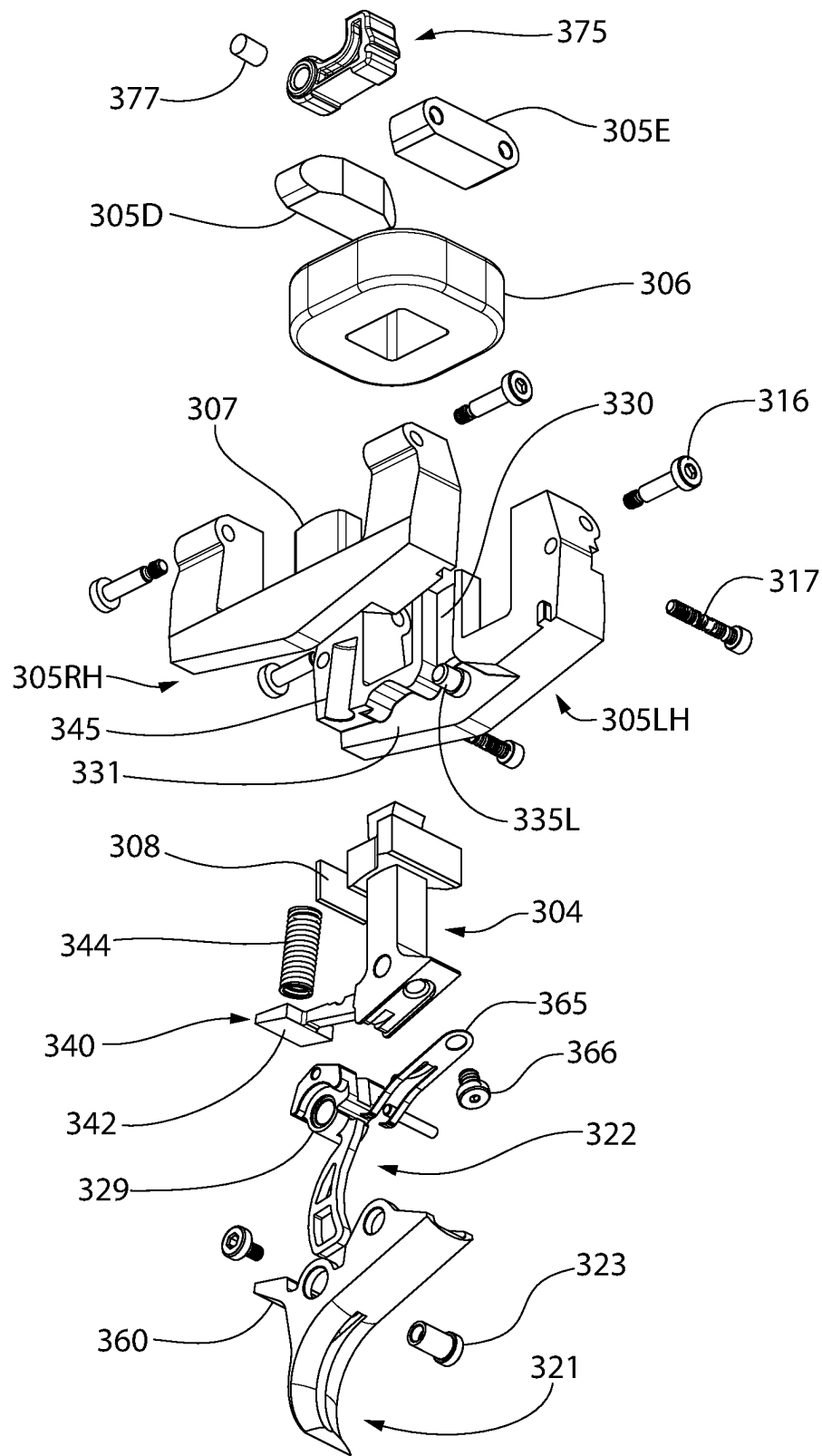


FIG. 21

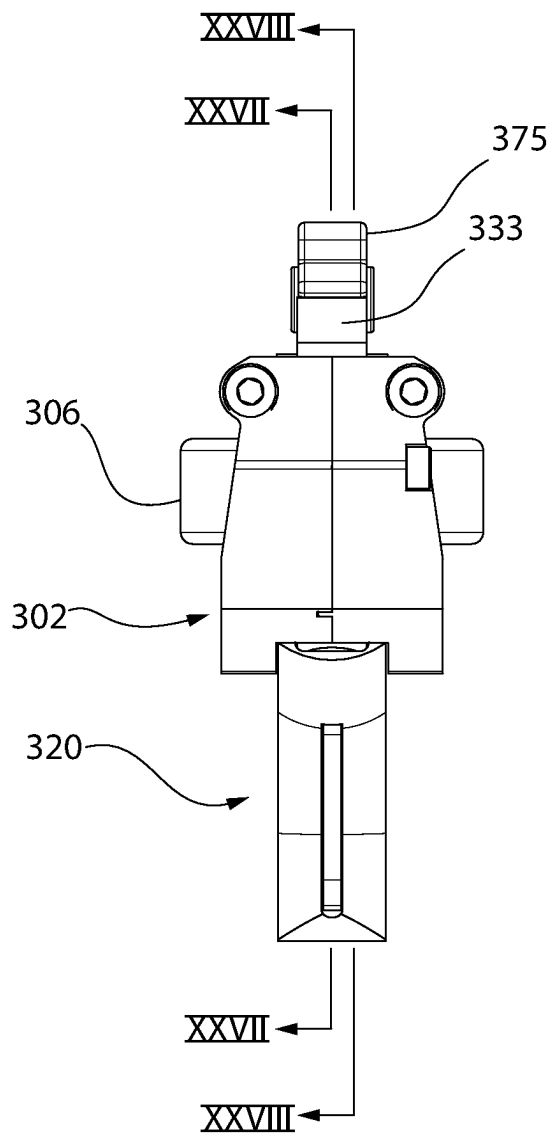


FIG. 22

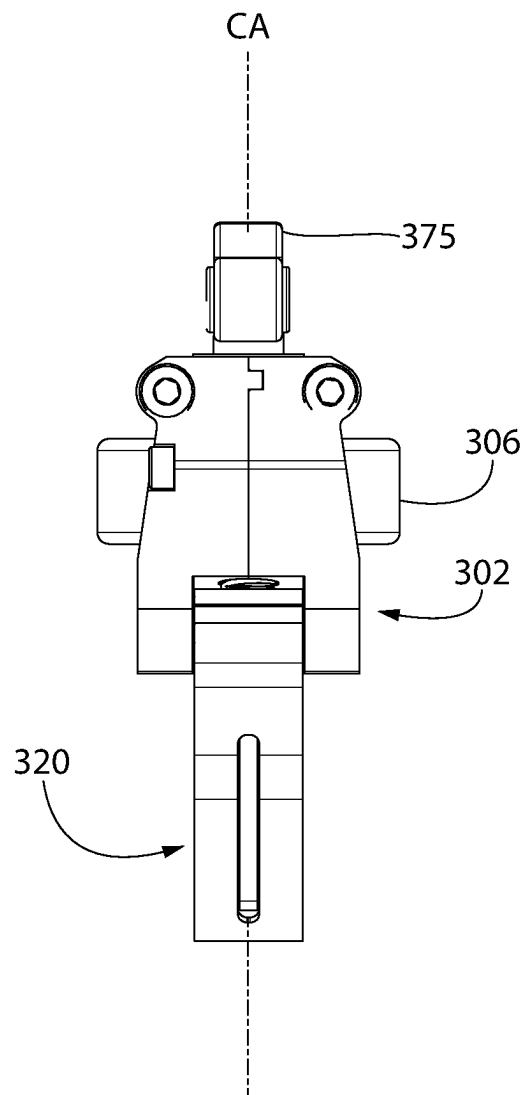


FIG. 23

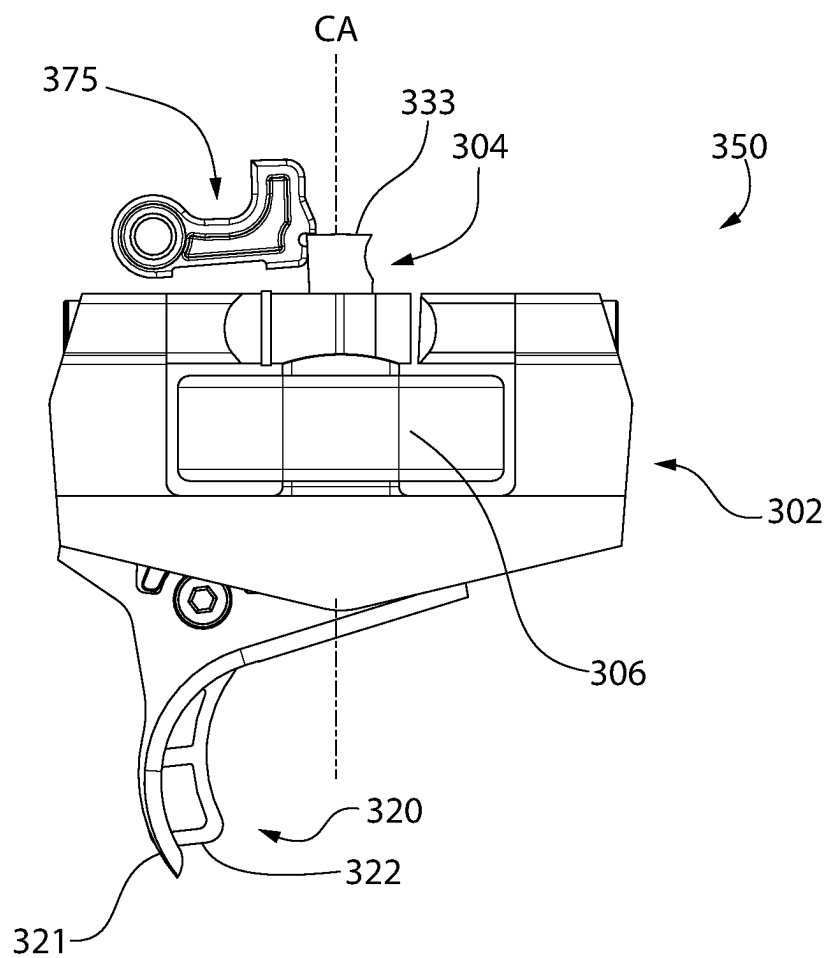
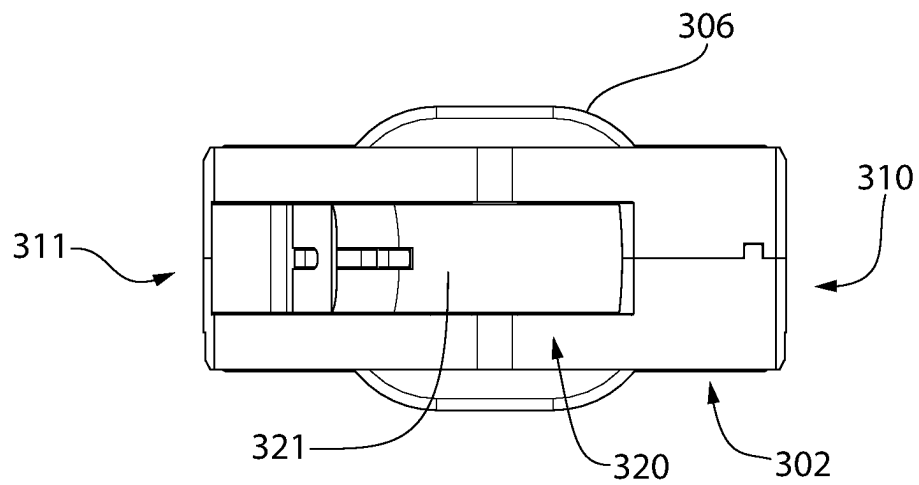
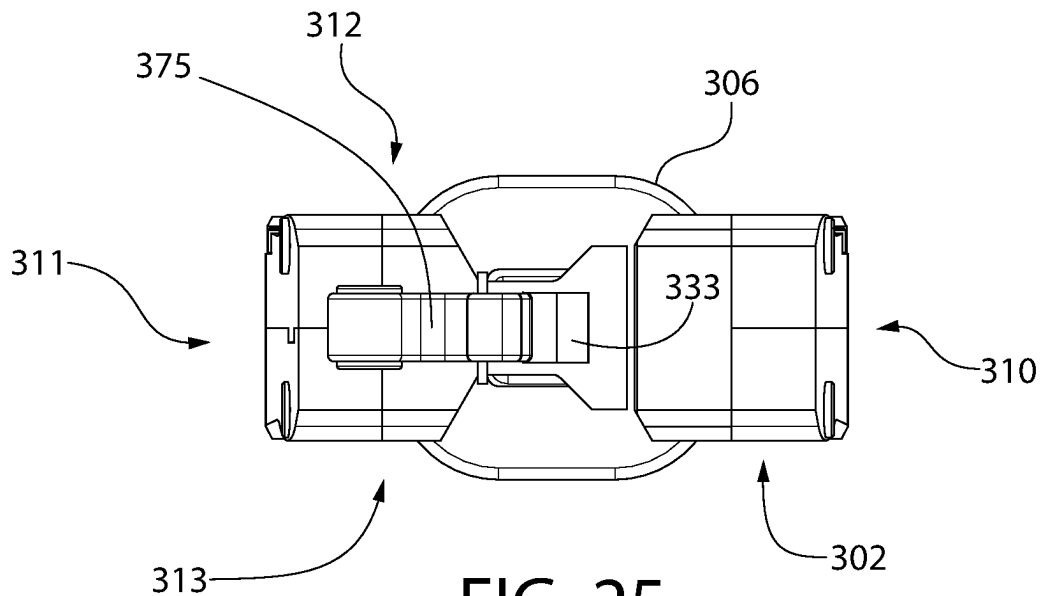


FIG. 24



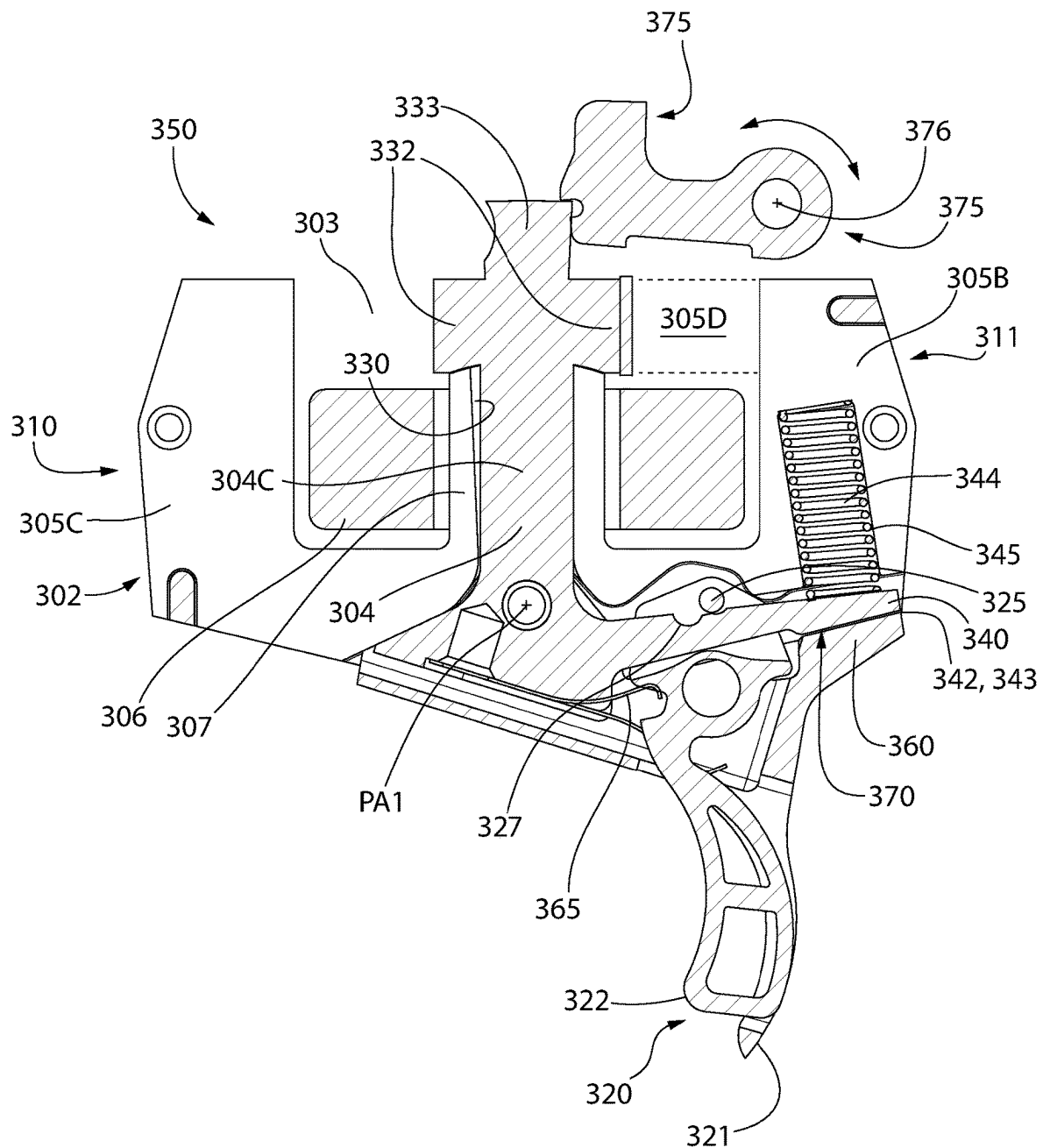


FIG. 27

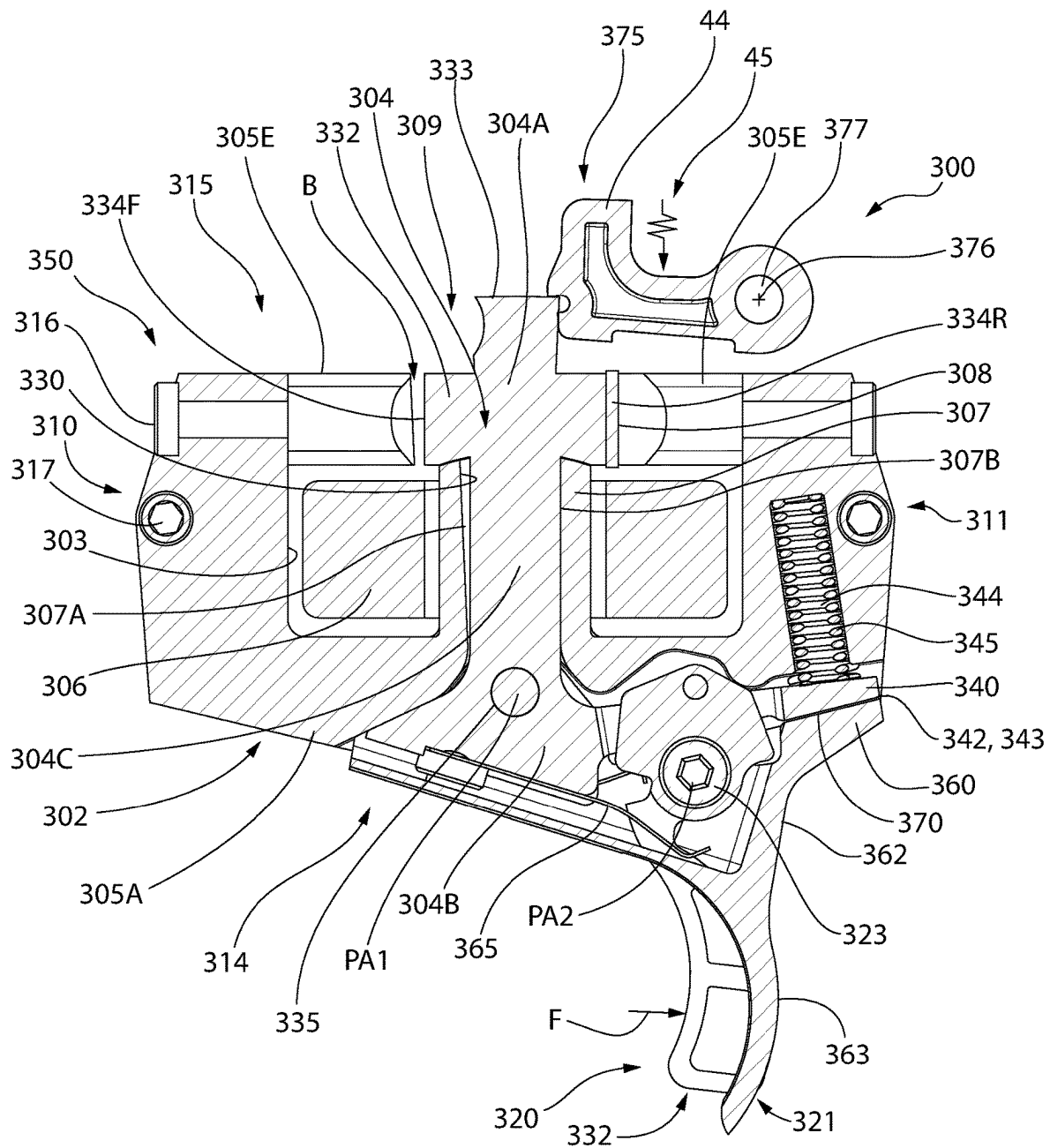


FIG. 28

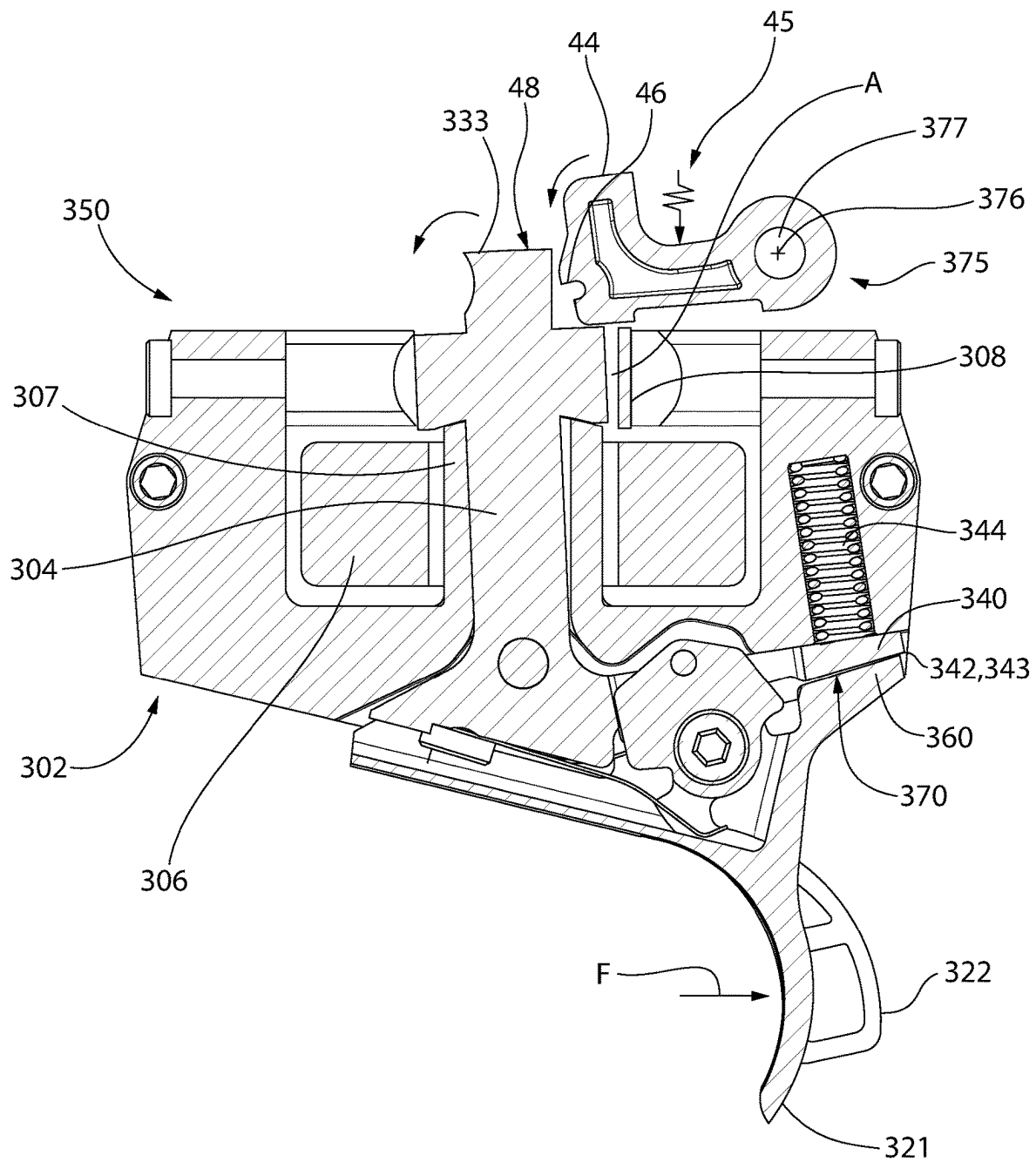


FIG. 29

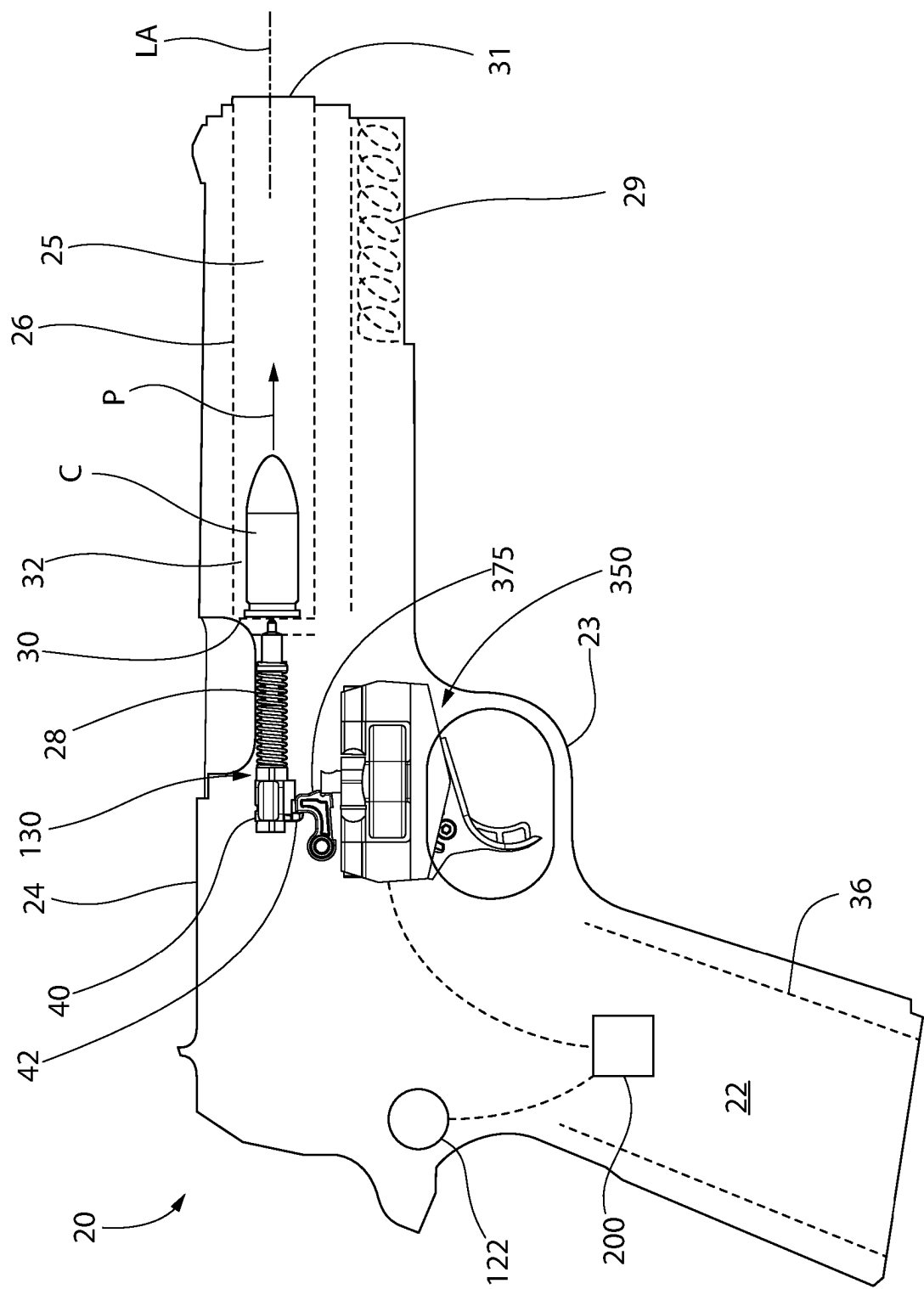


FIG. 30

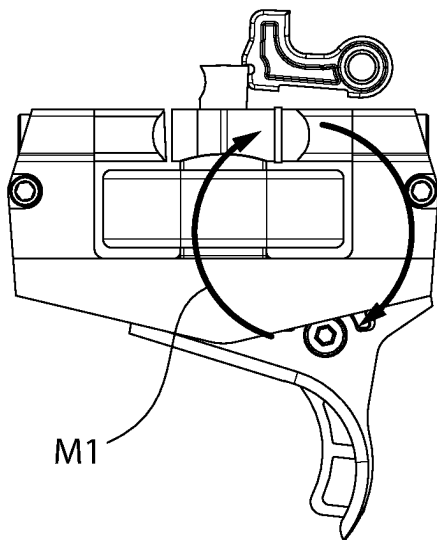


FIG. 31

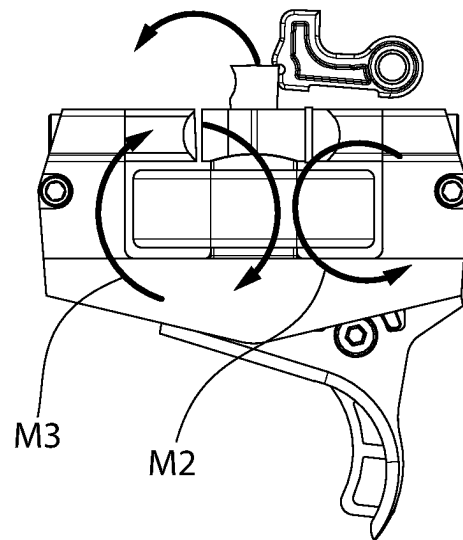
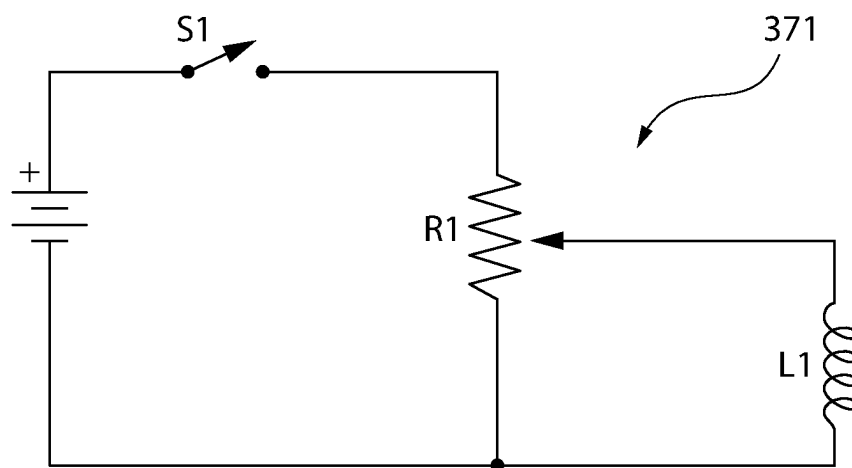


FIG. 32



Where:

S1 = trigger activation switch

R1 = force setpoint selection potentiometer

L1 = magnetic coil

FIG. 33

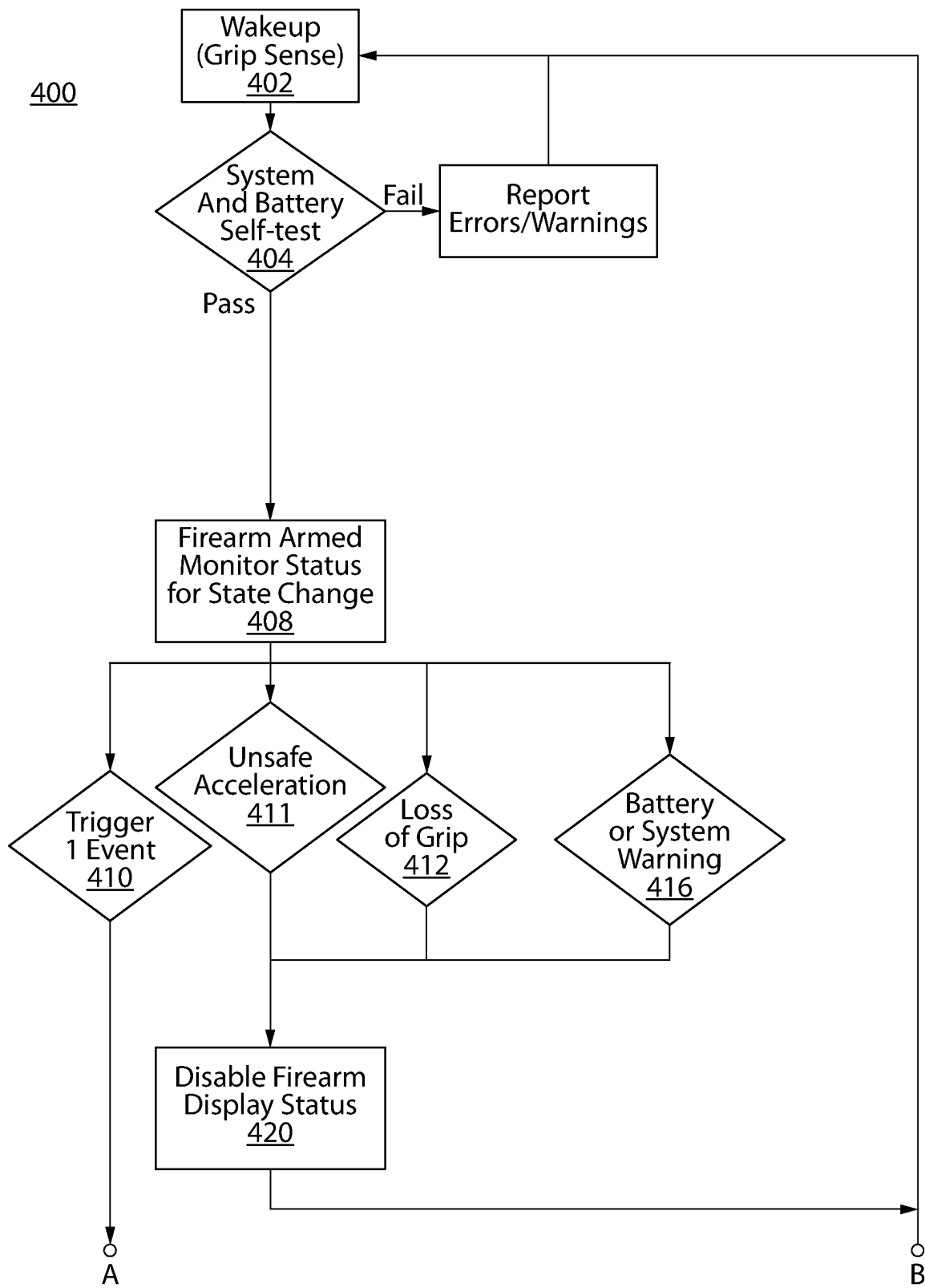


FIG. 34A

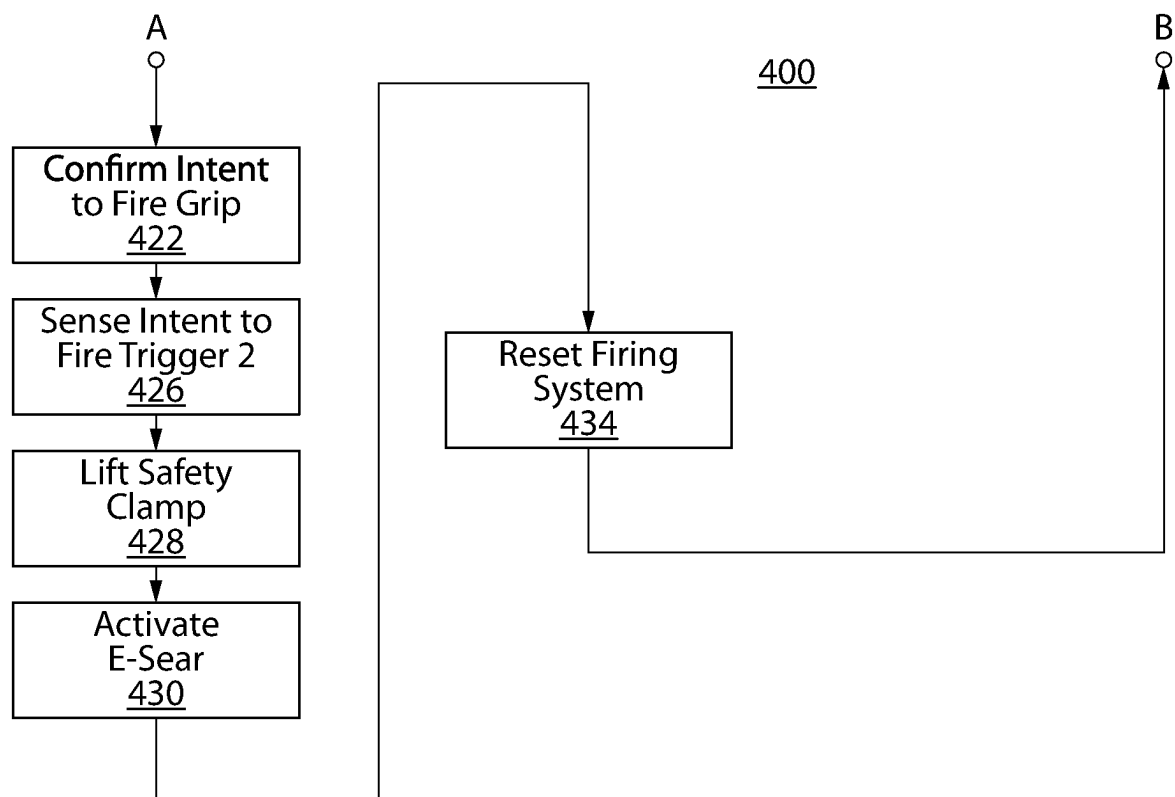


FIG. 34B

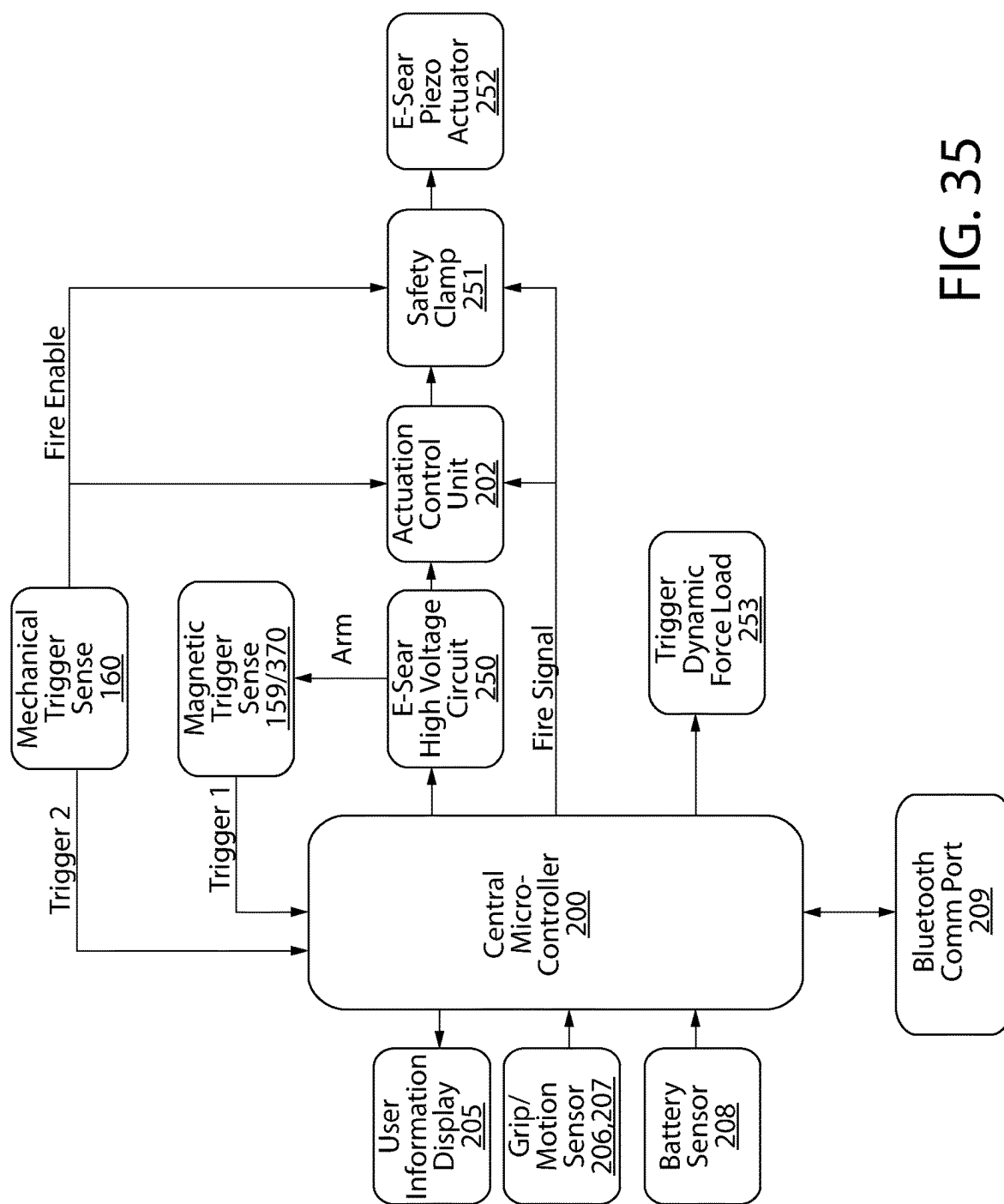


FIG. 35

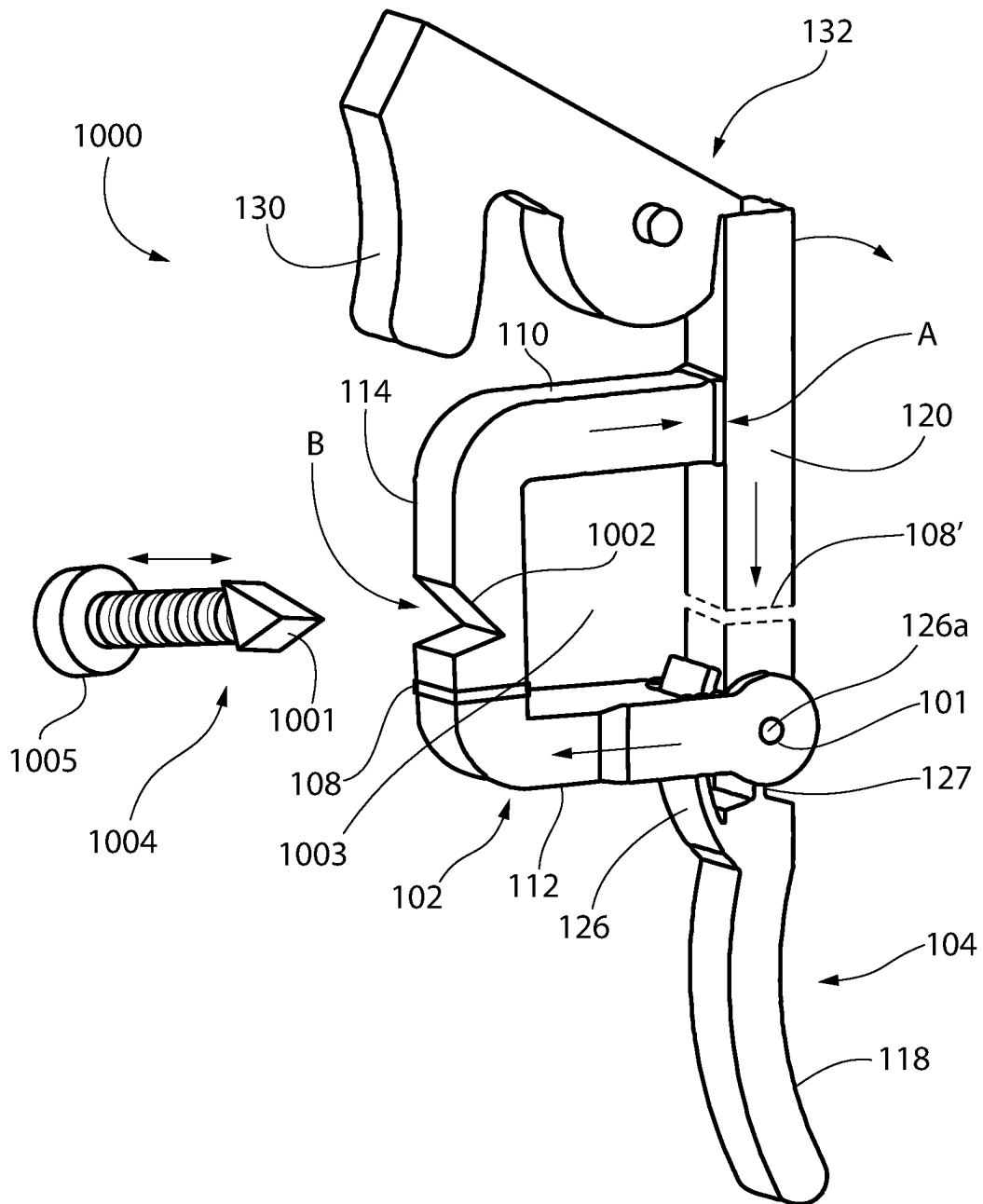


FIG. 36

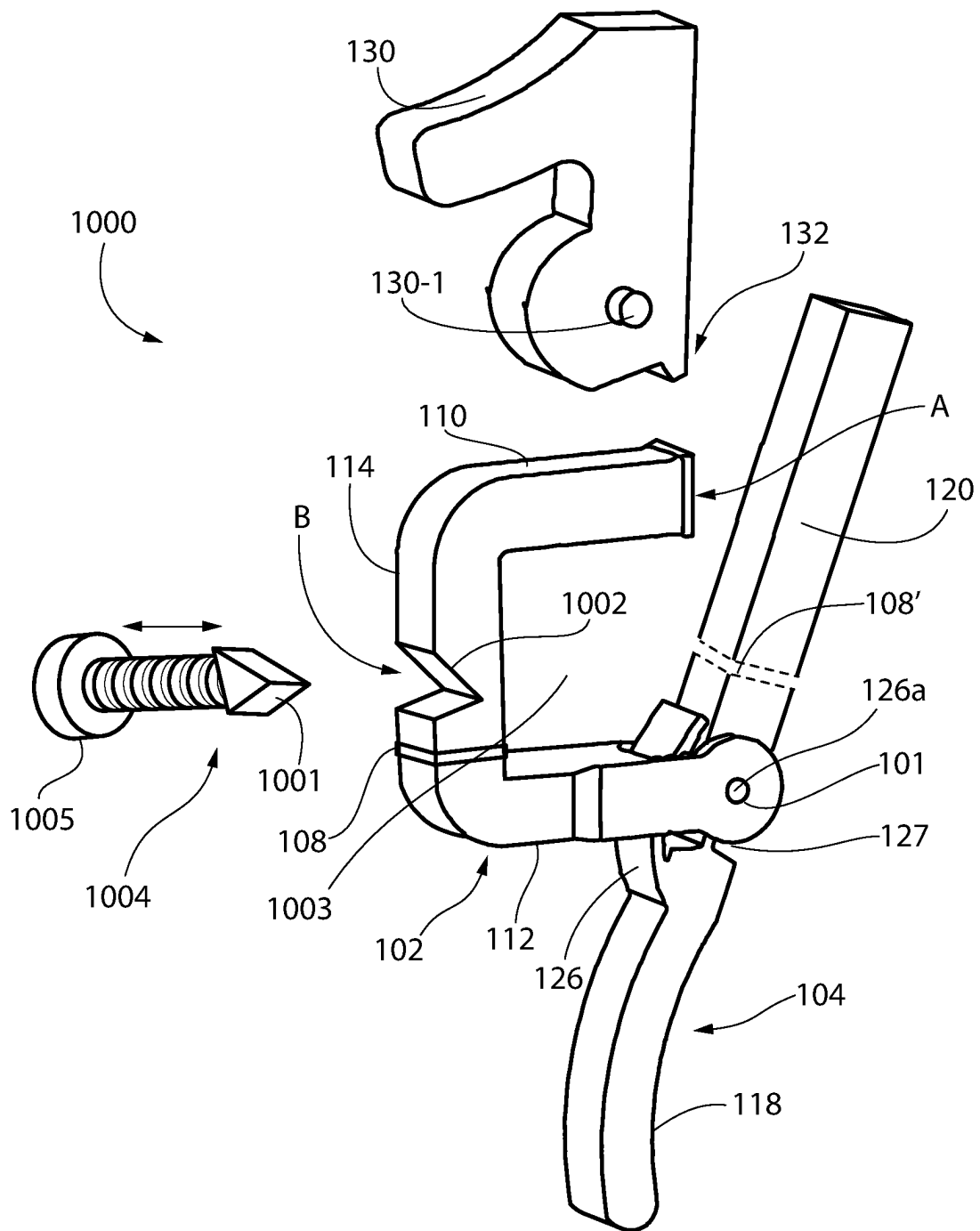


FIG. 37

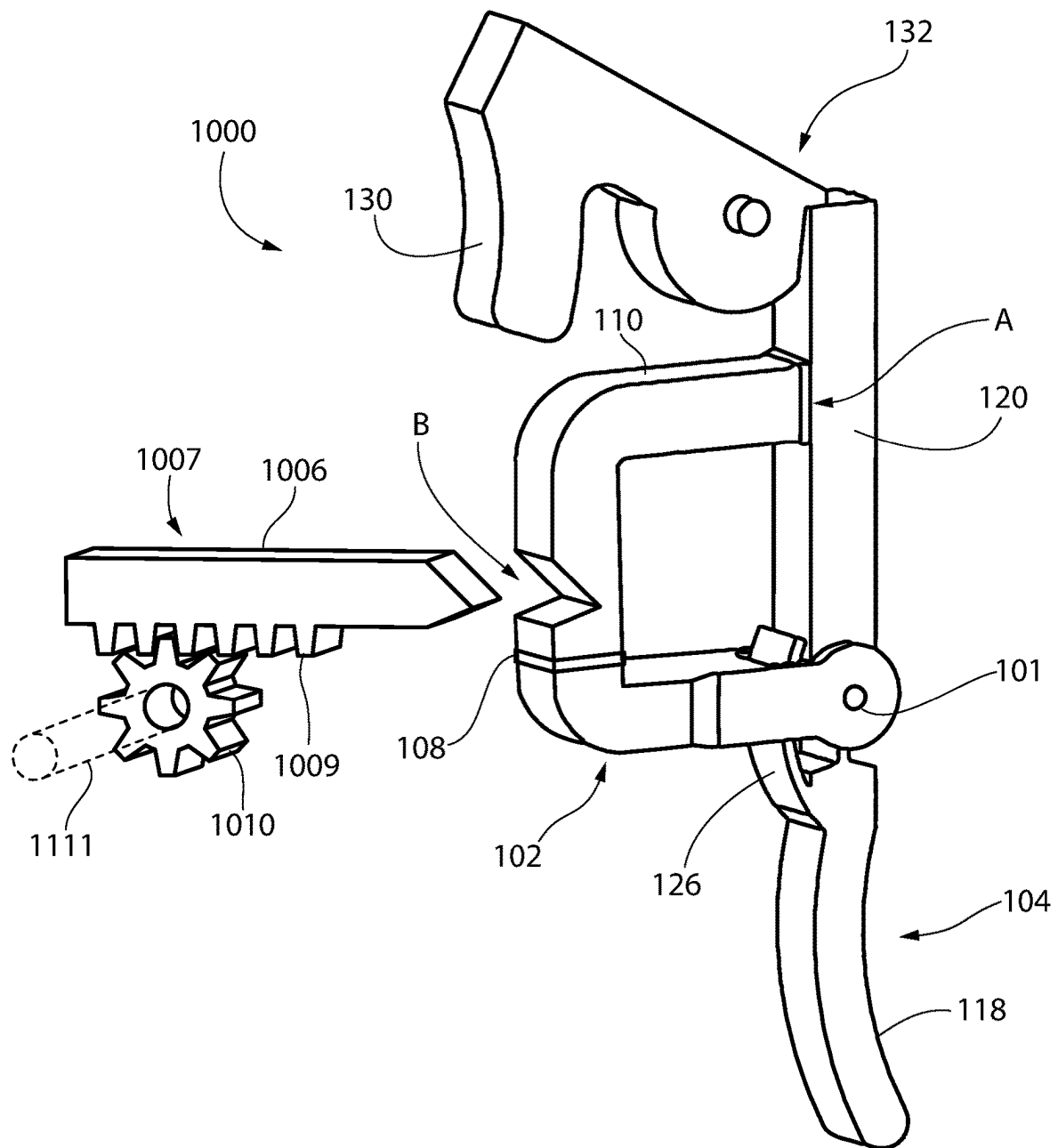
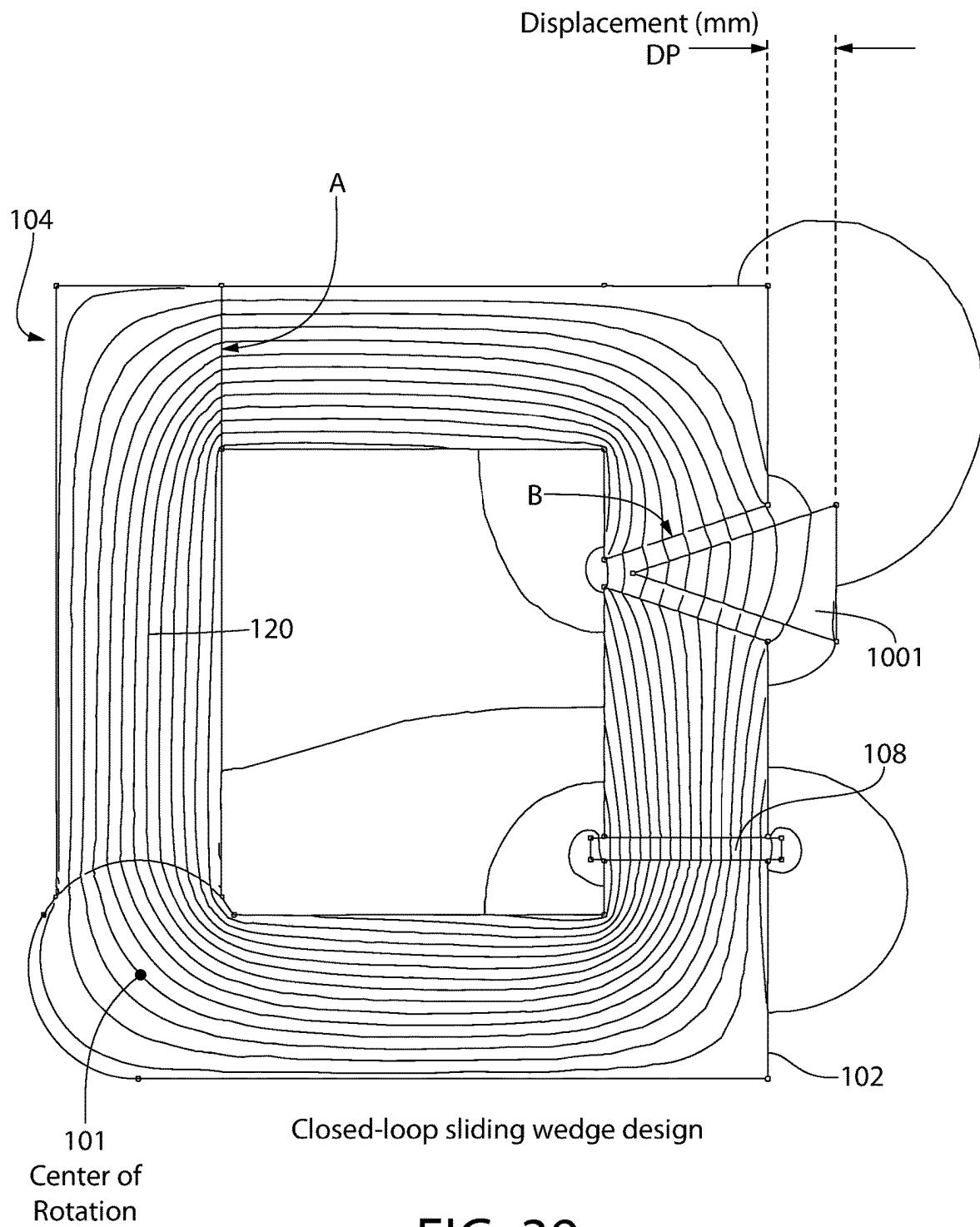


FIG. 38



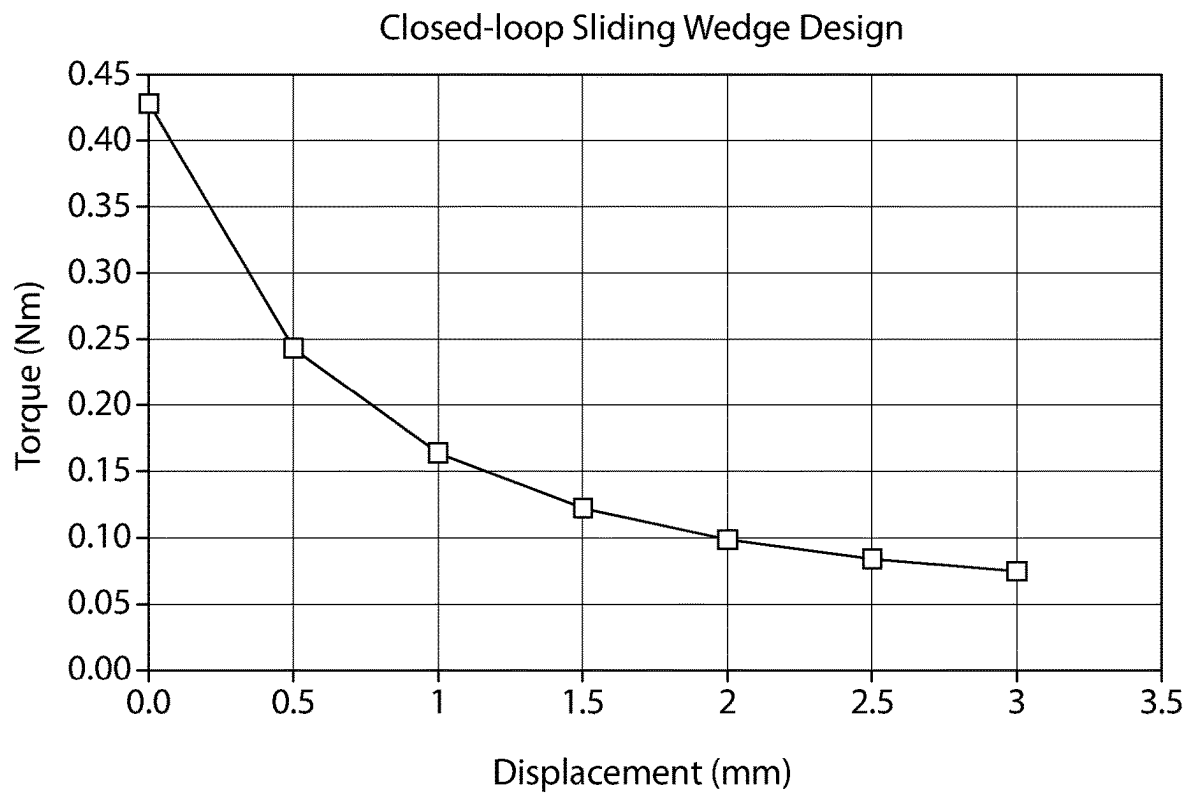


FIG. 40

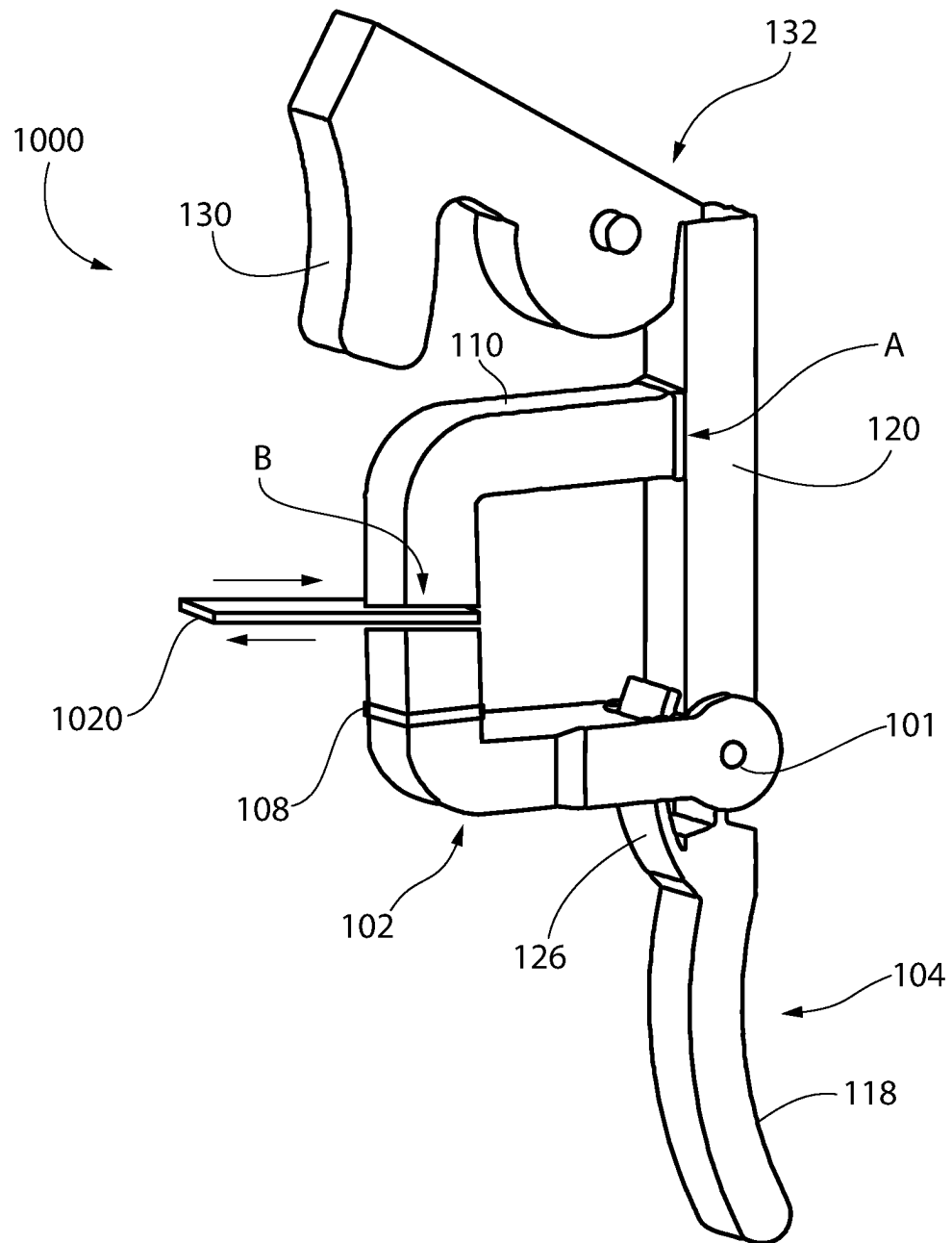
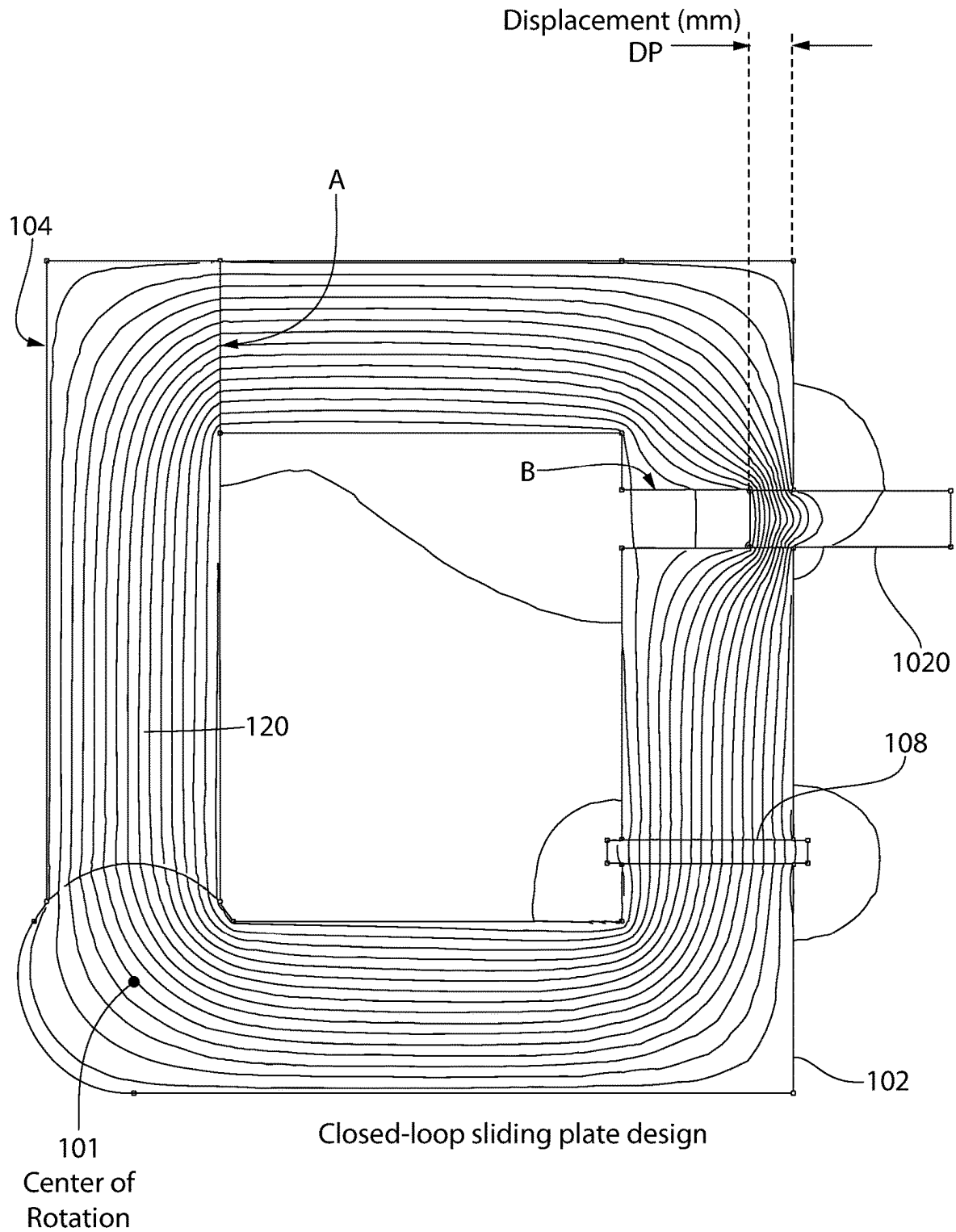


FIG. 41



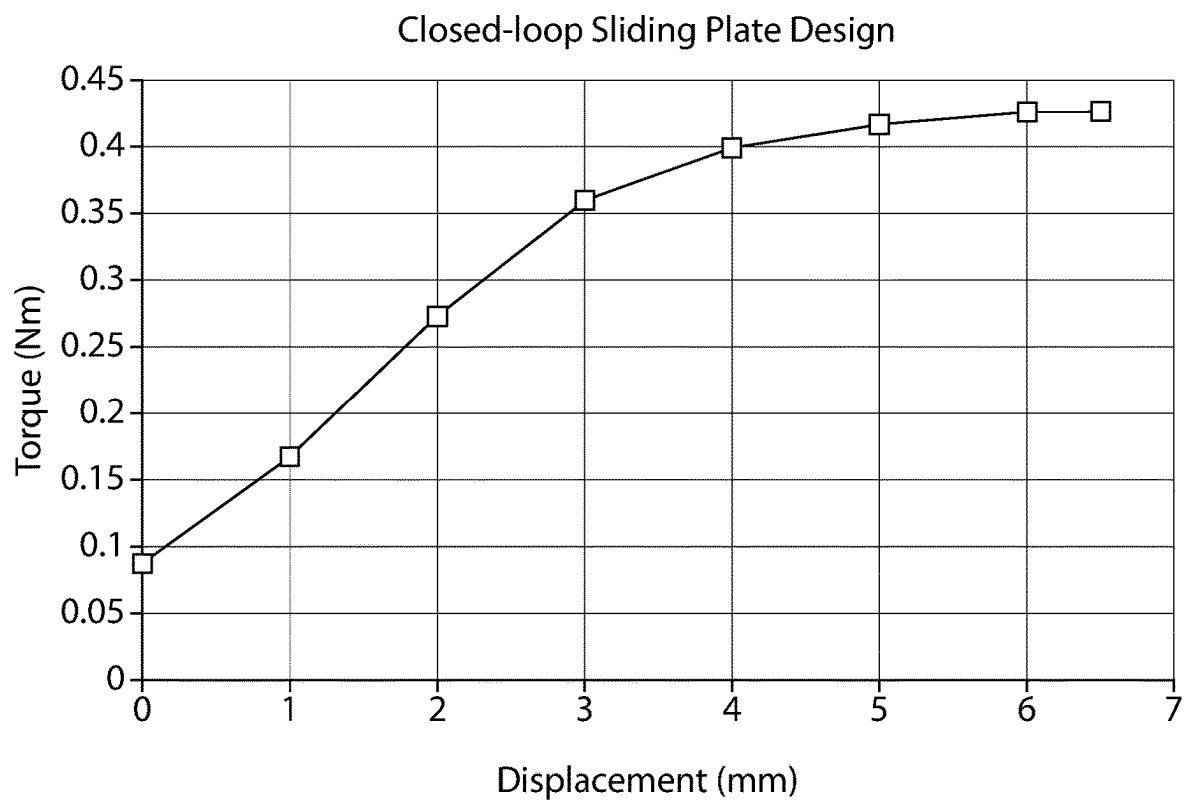


FIG. 43

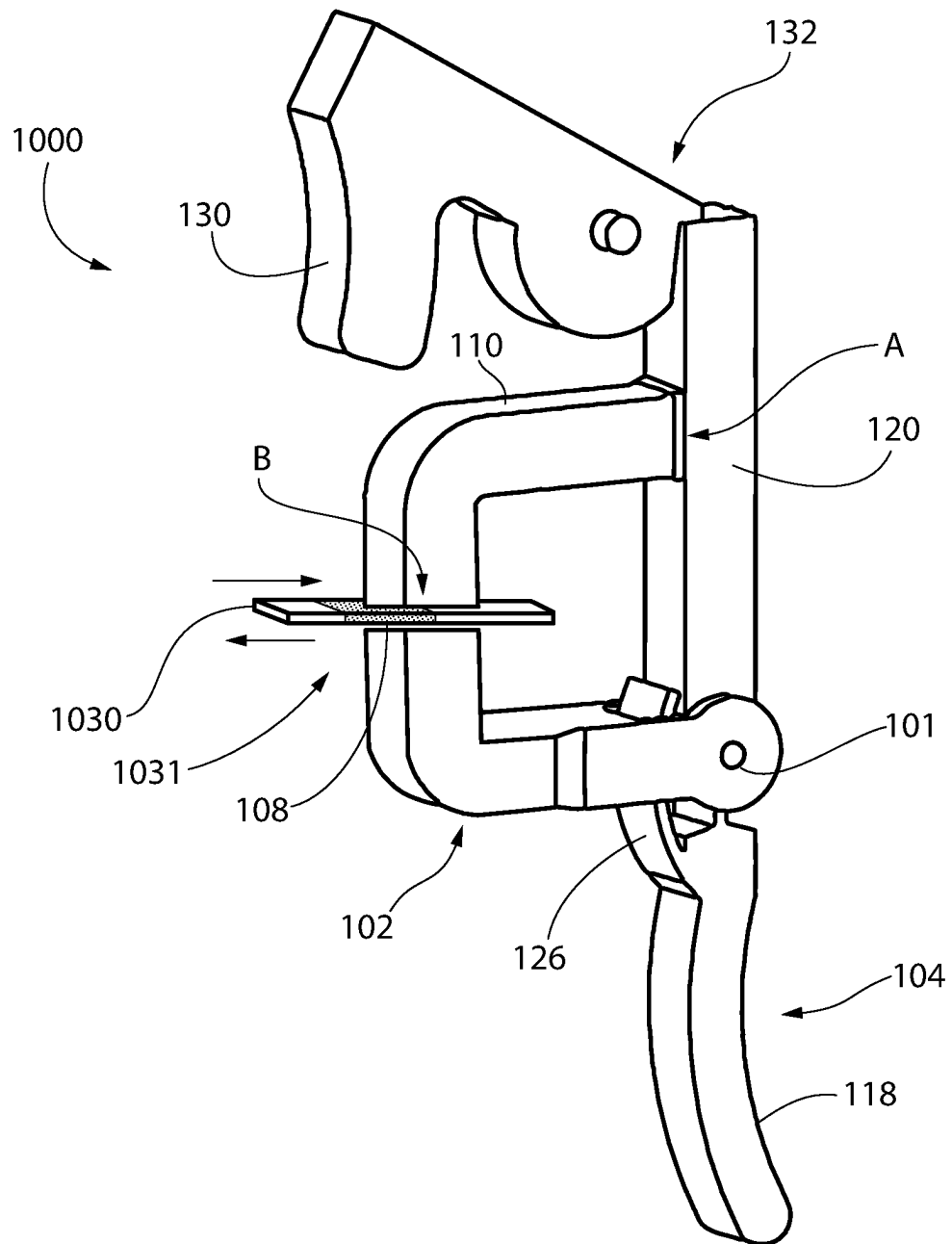


FIG. 44

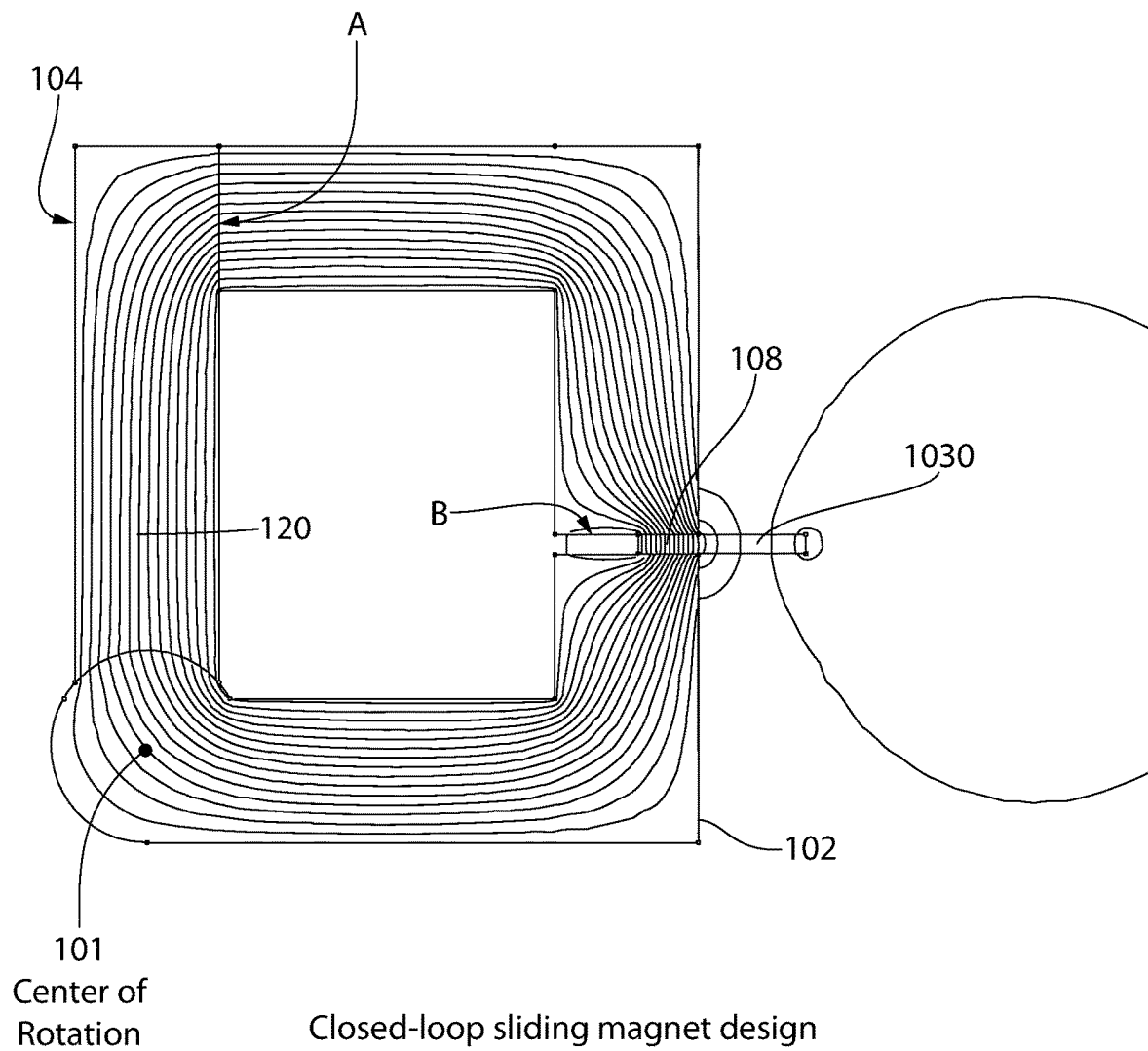


FIG. 45

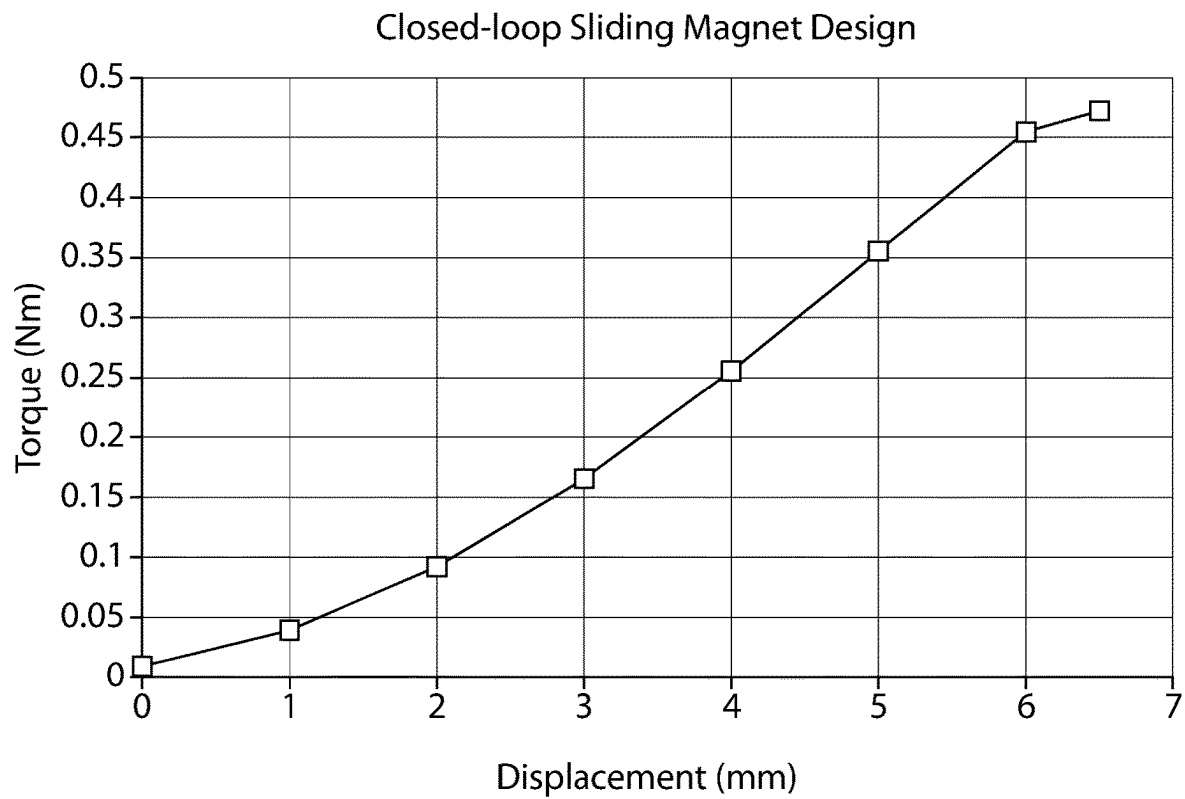


FIG. 46

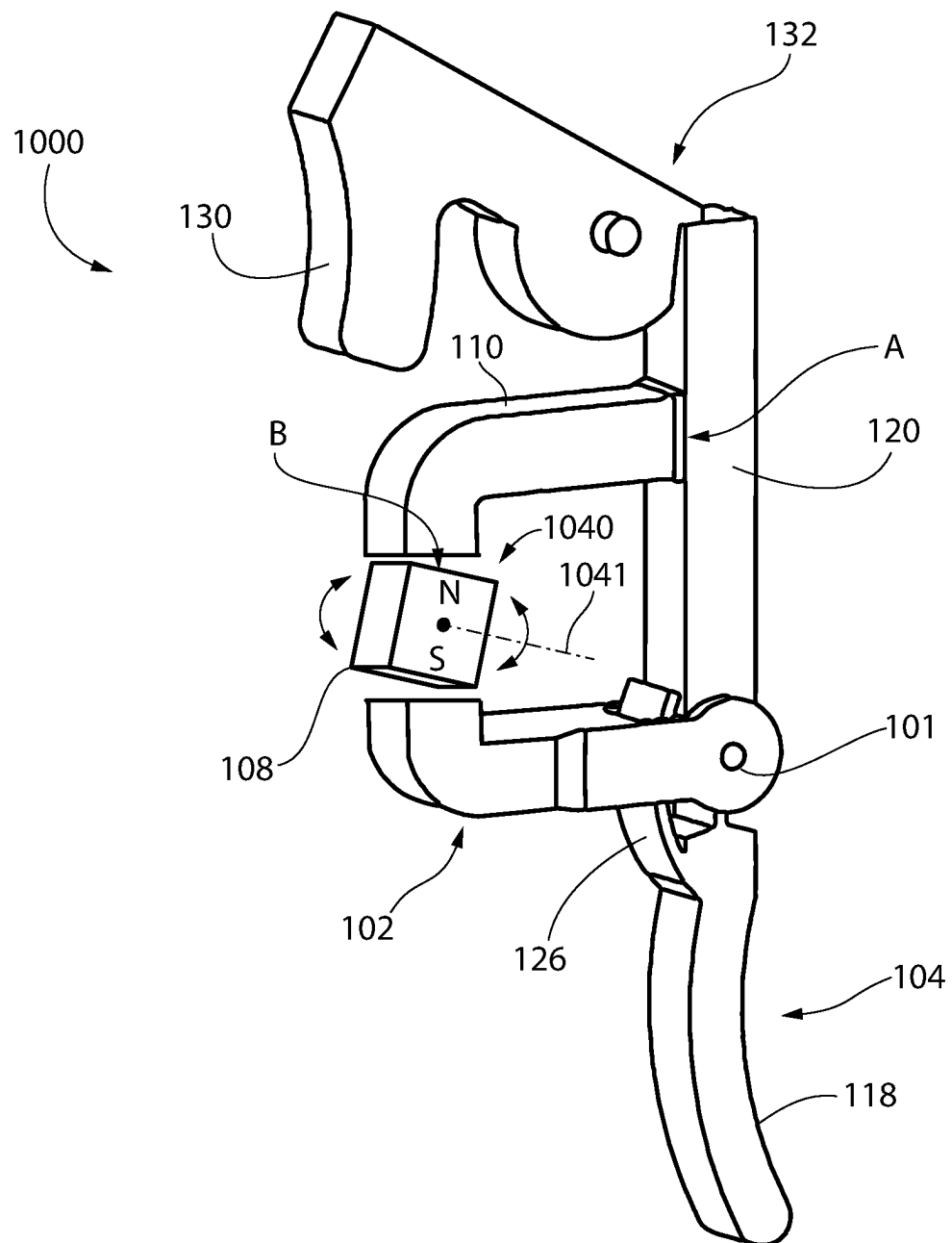


FIG. 47

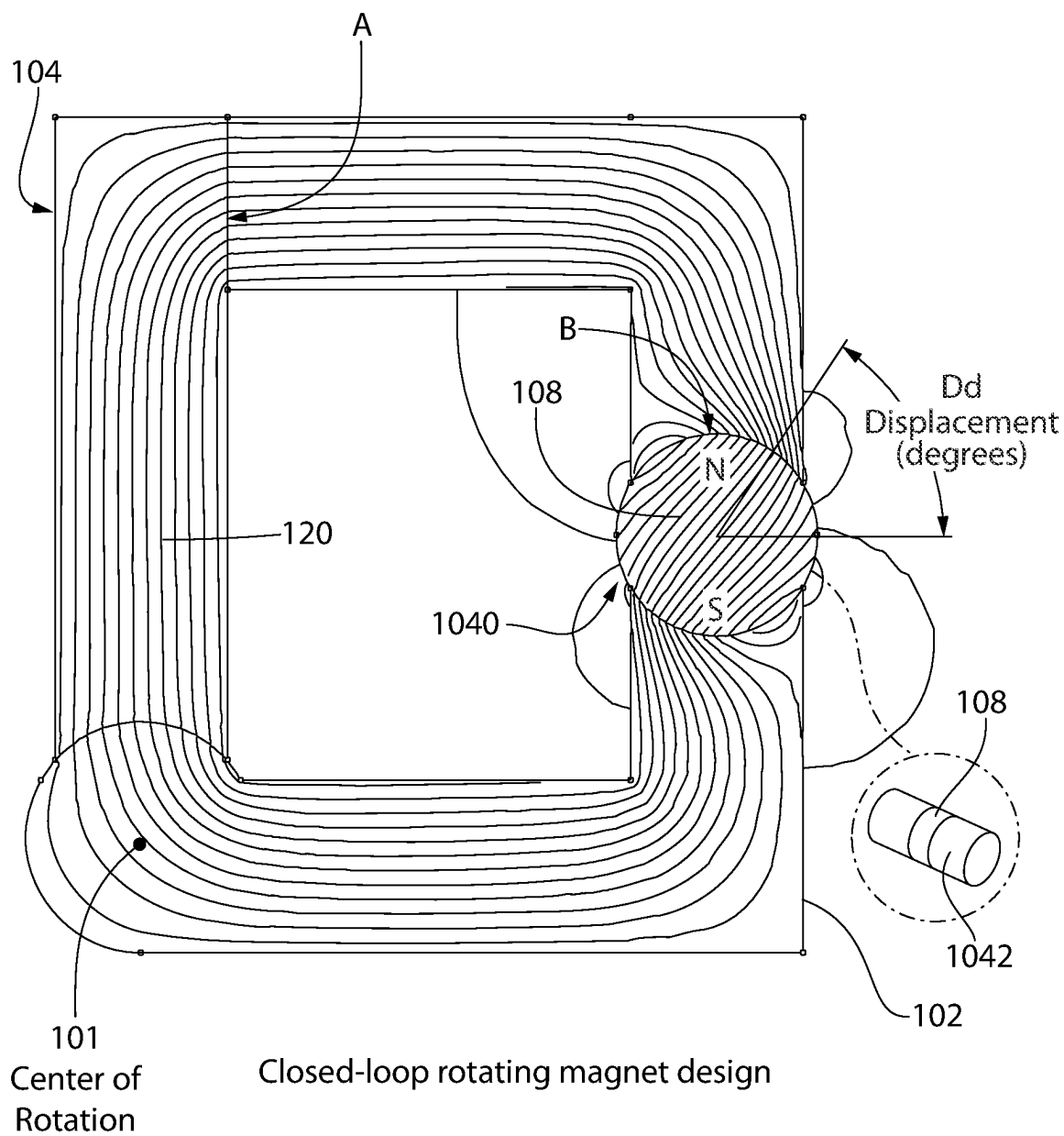


FIG. 48

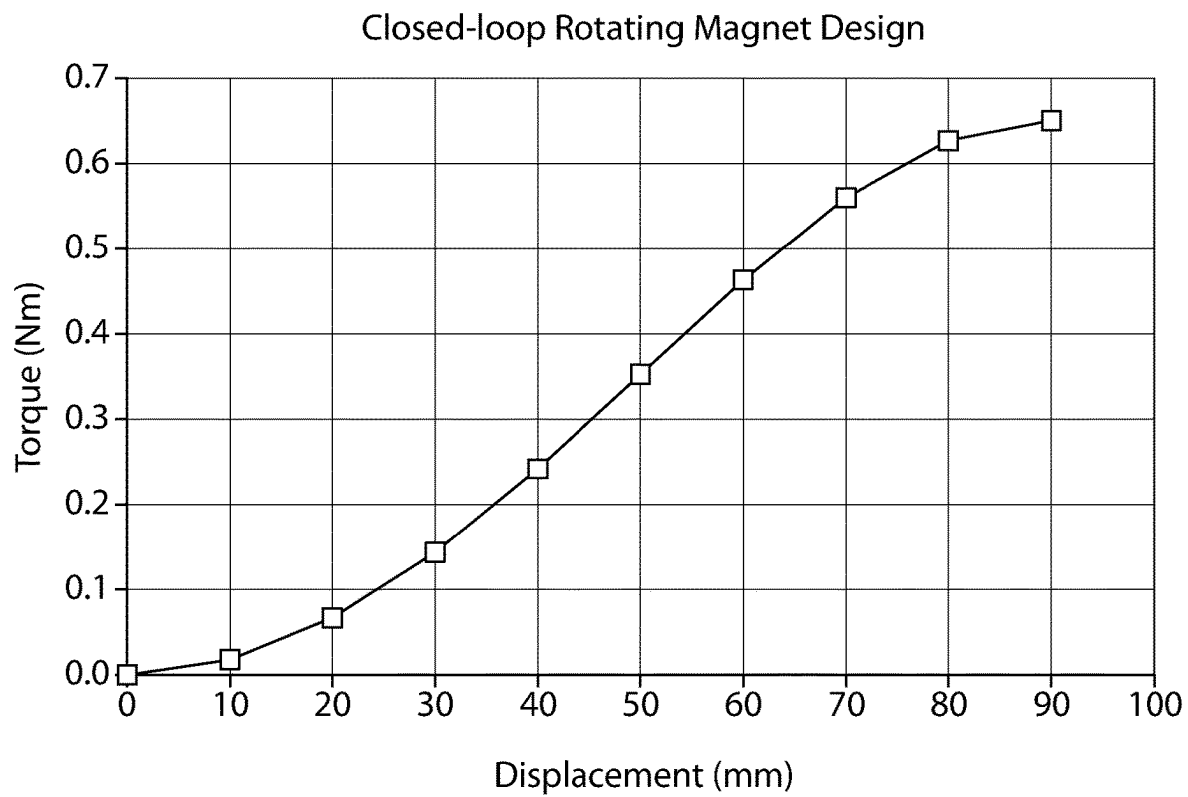


FIG. 49

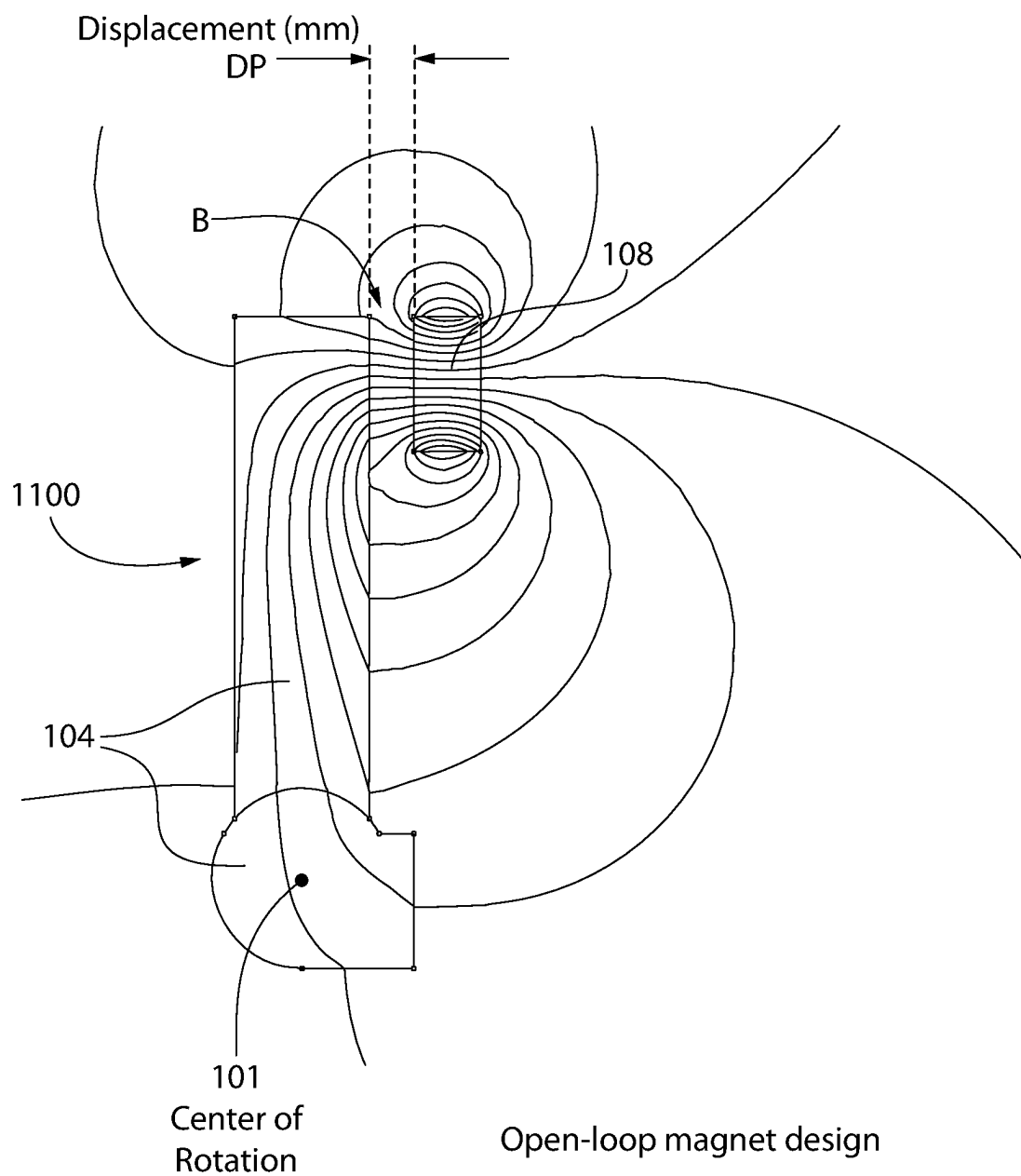


FIG. 50

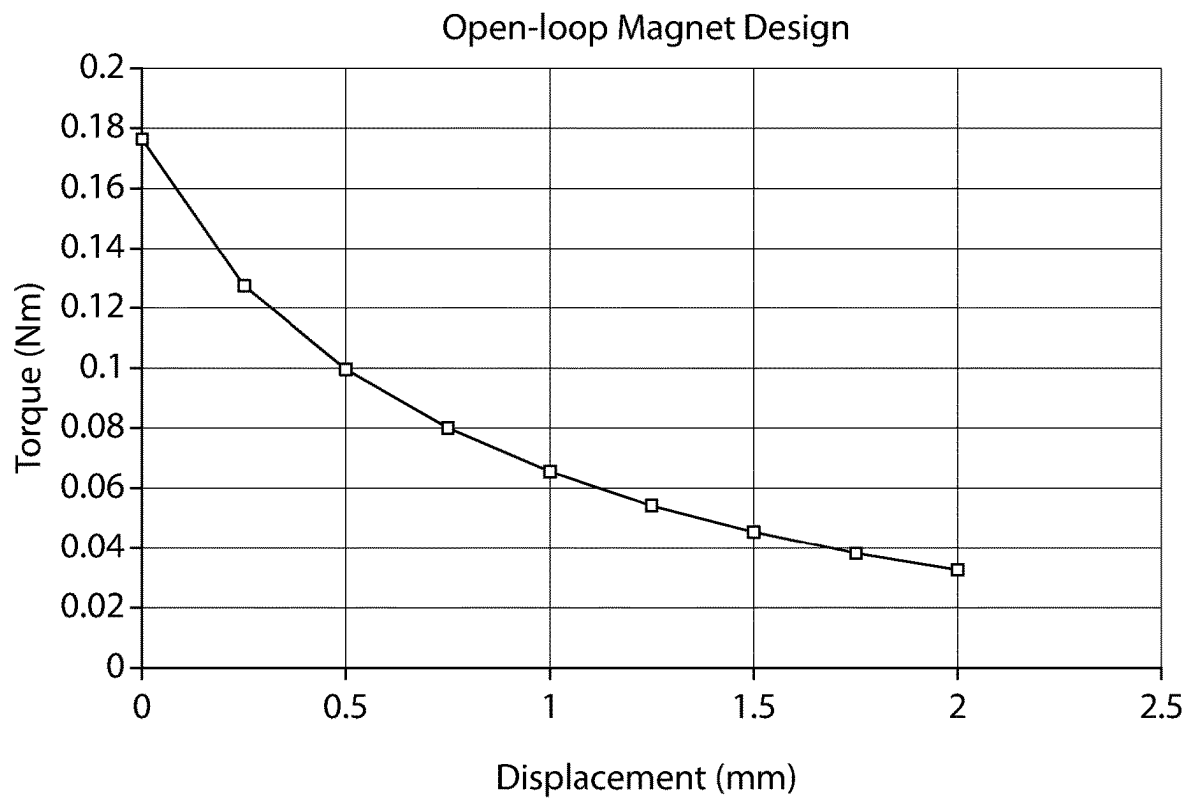


FIG. 51

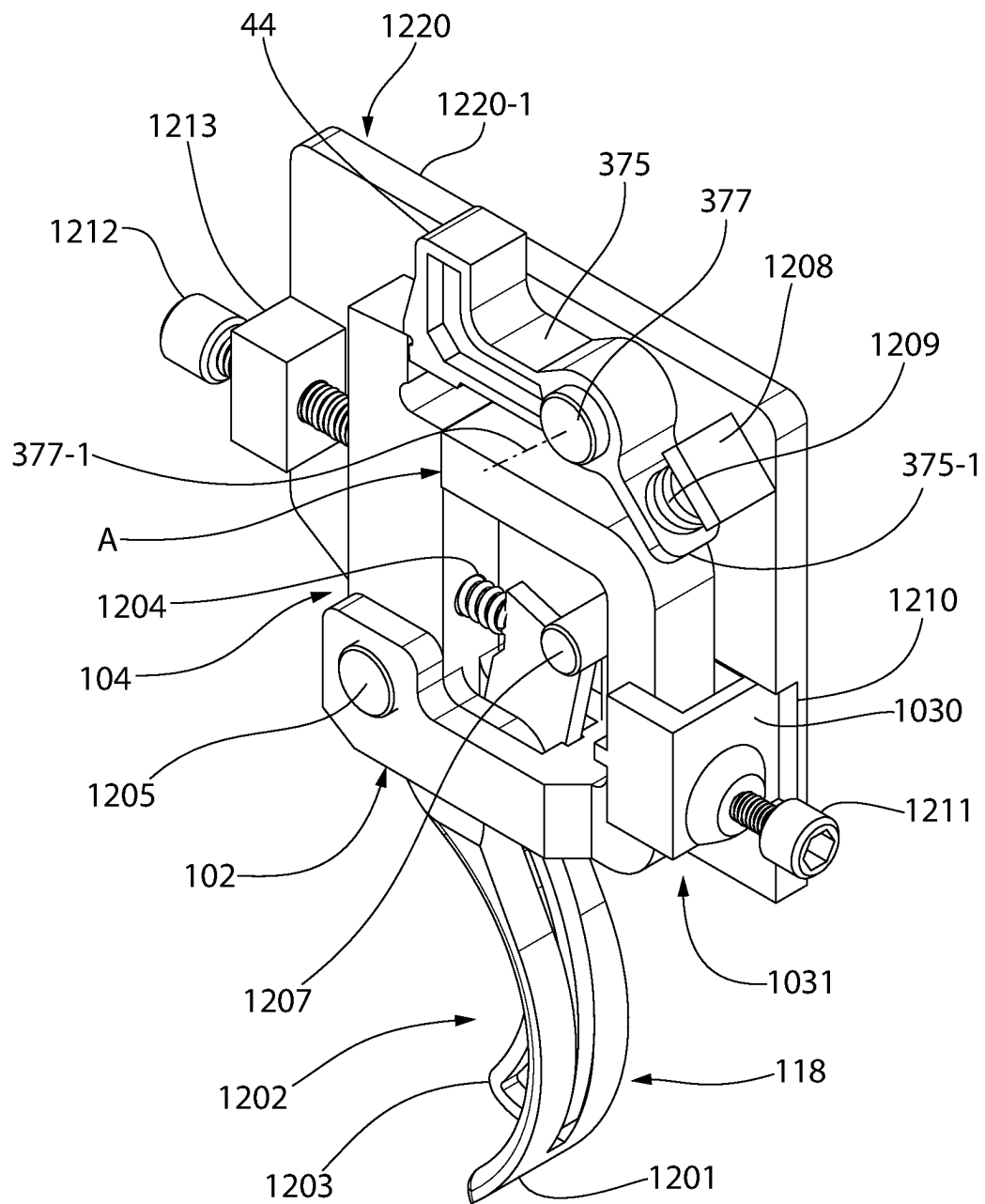


FIG. 52

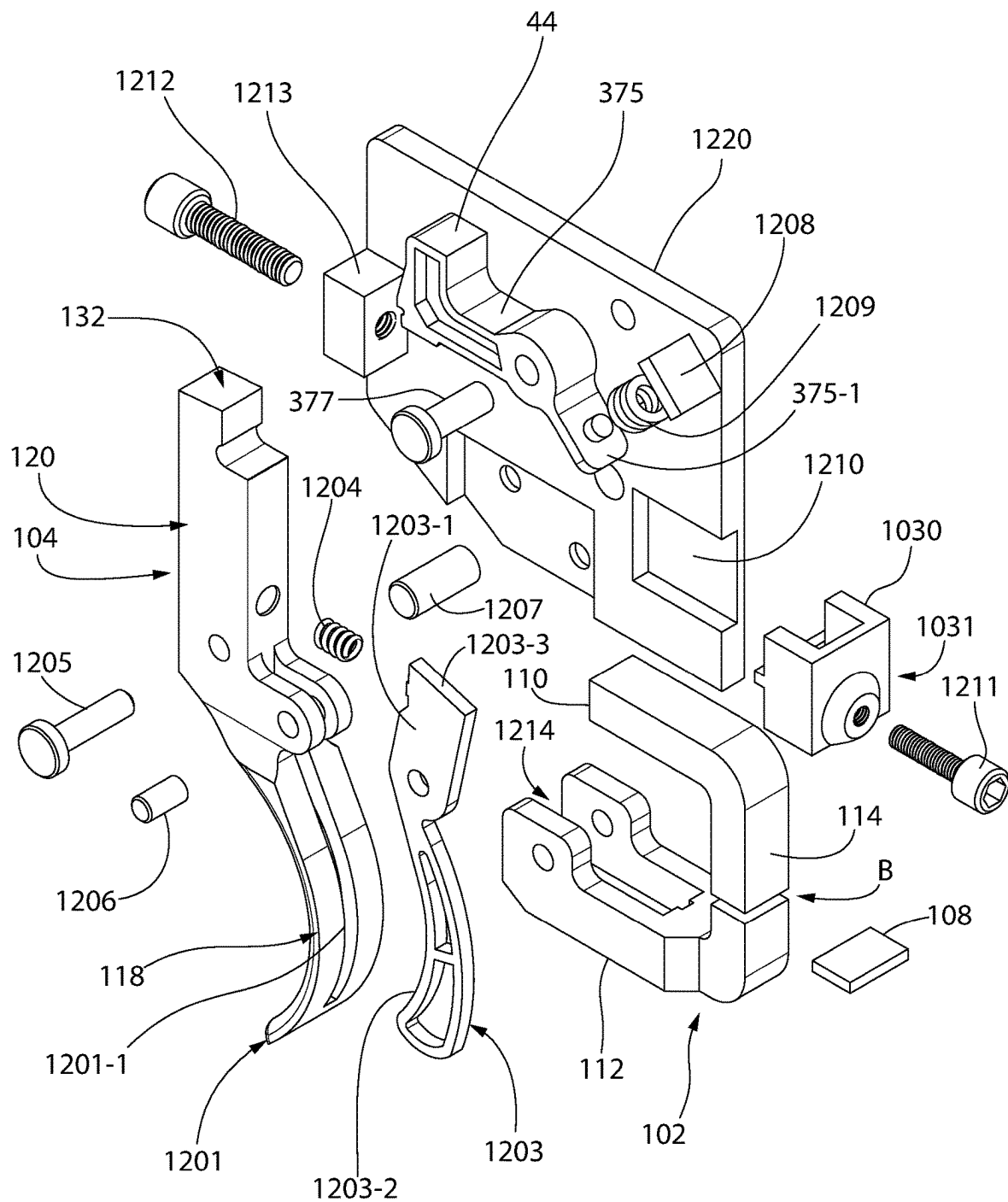


FIG. 53

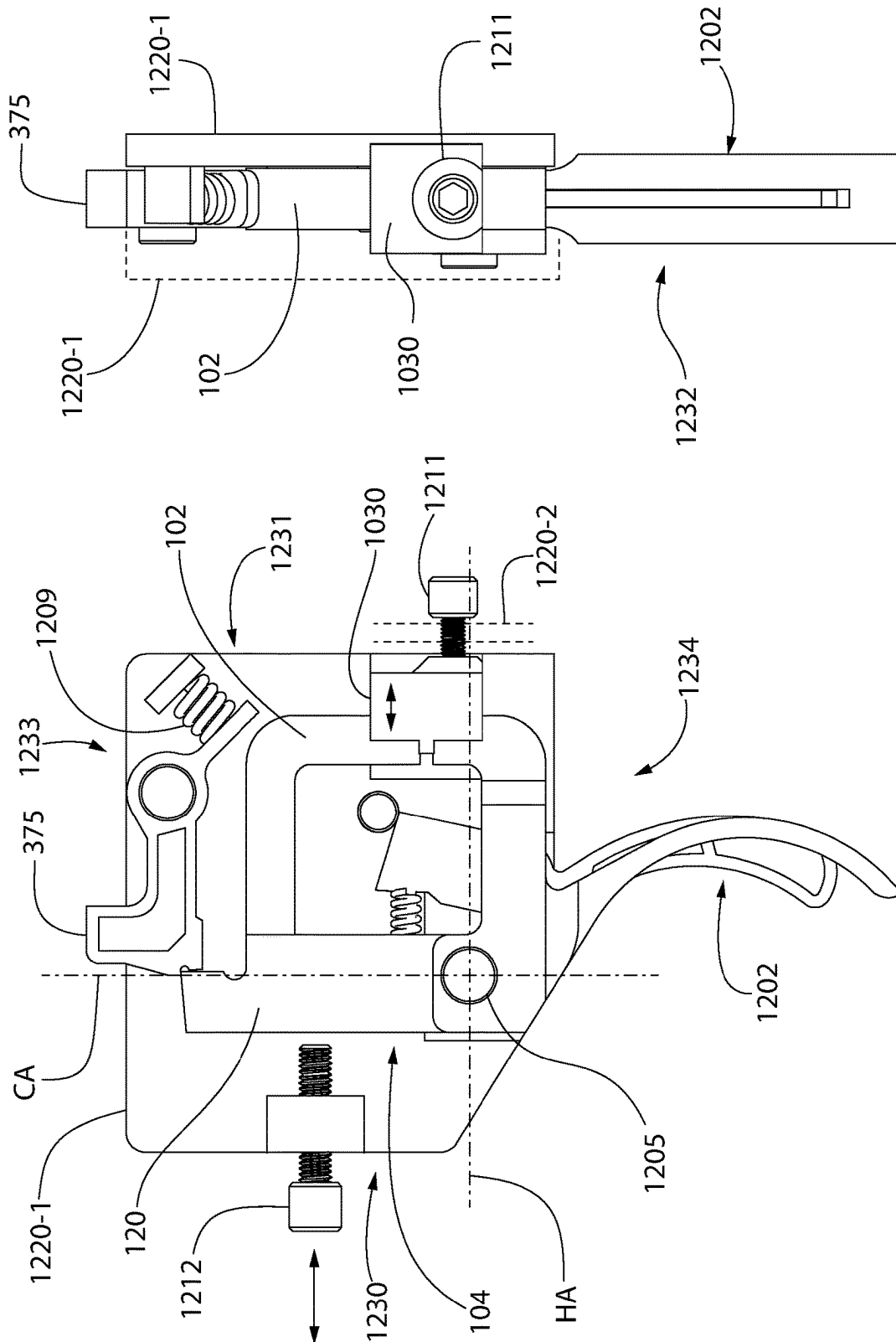


FIG. 54

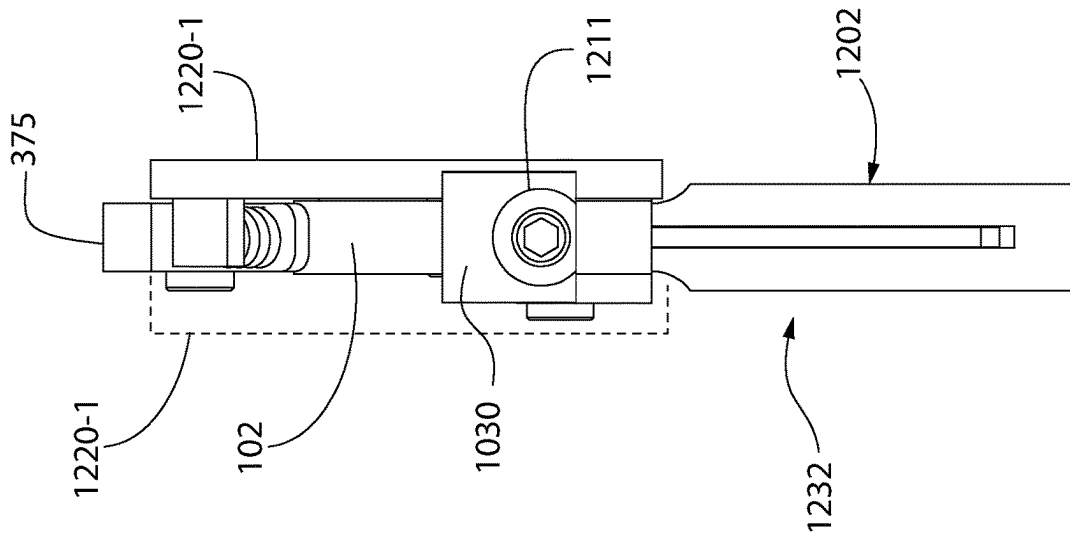


FIG. 55

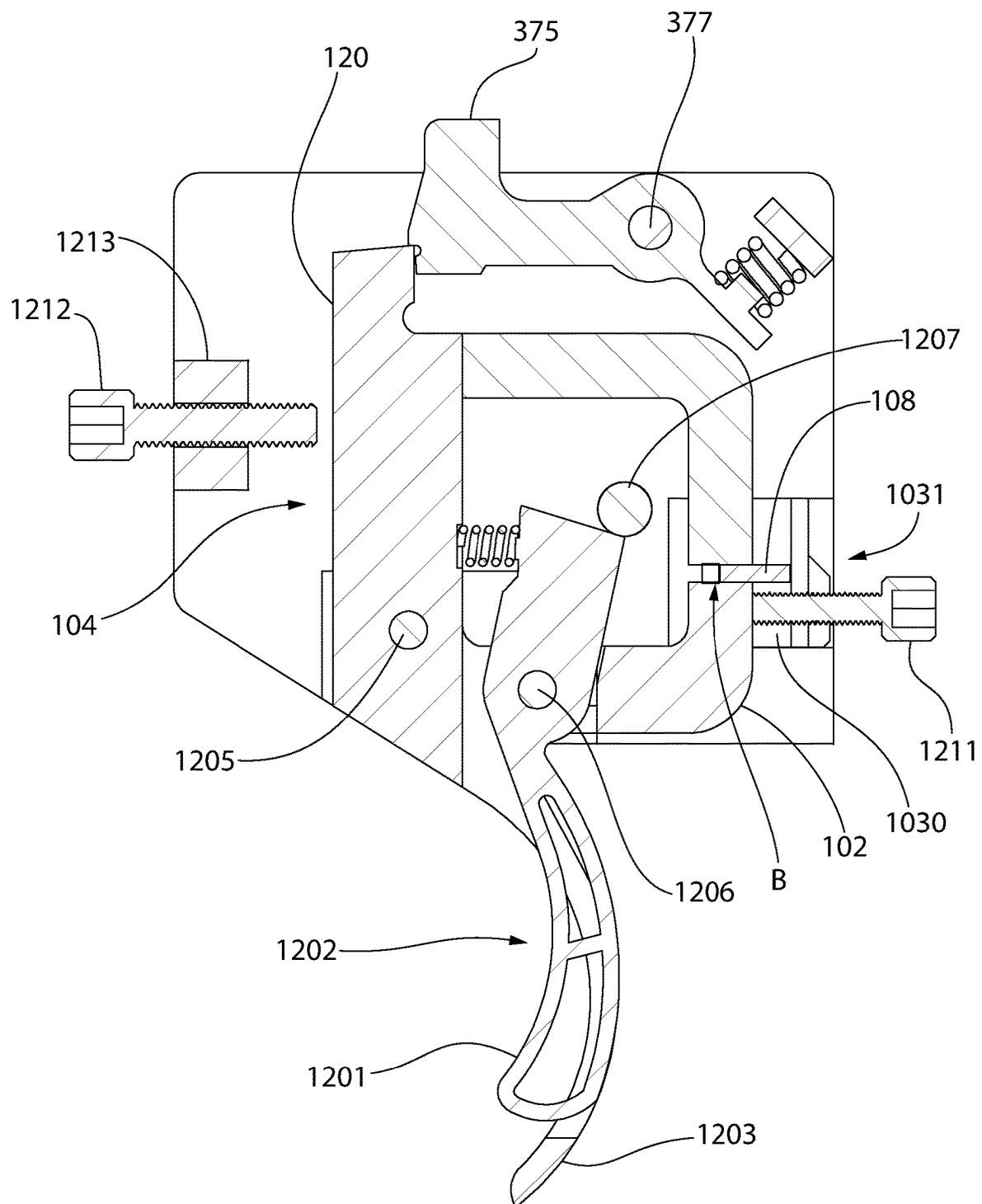


FIG. 56

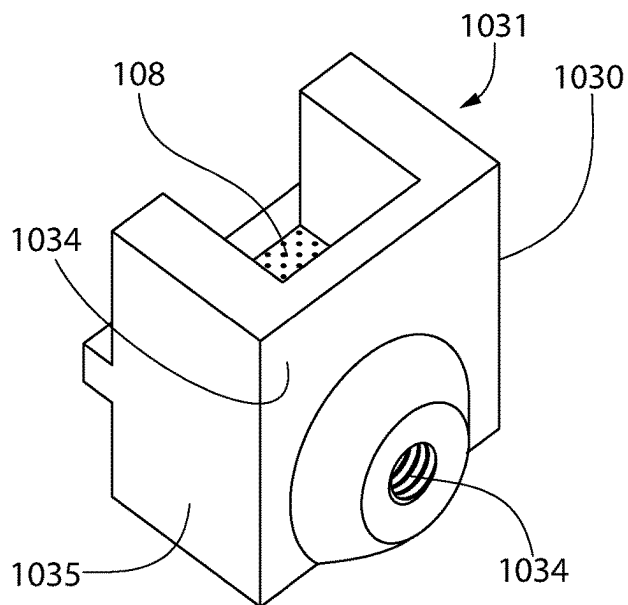


FIG. 57

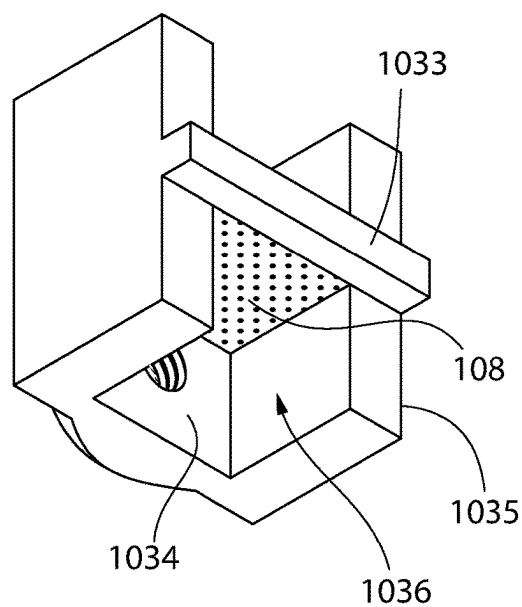


FIG. 58

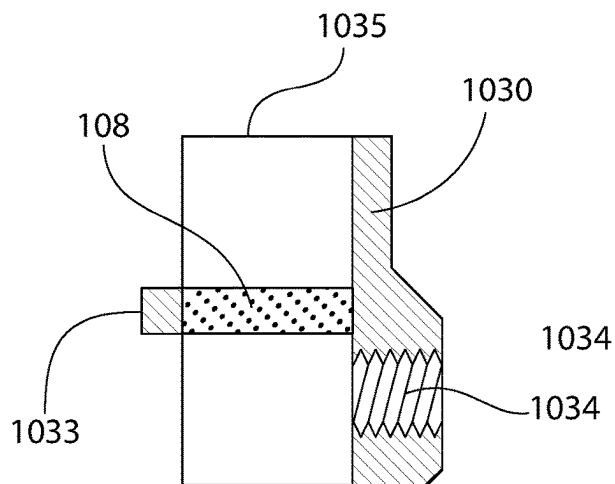


FIG. 59

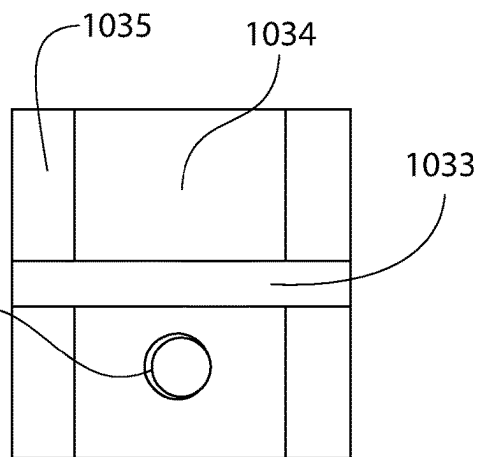


FIG. 60

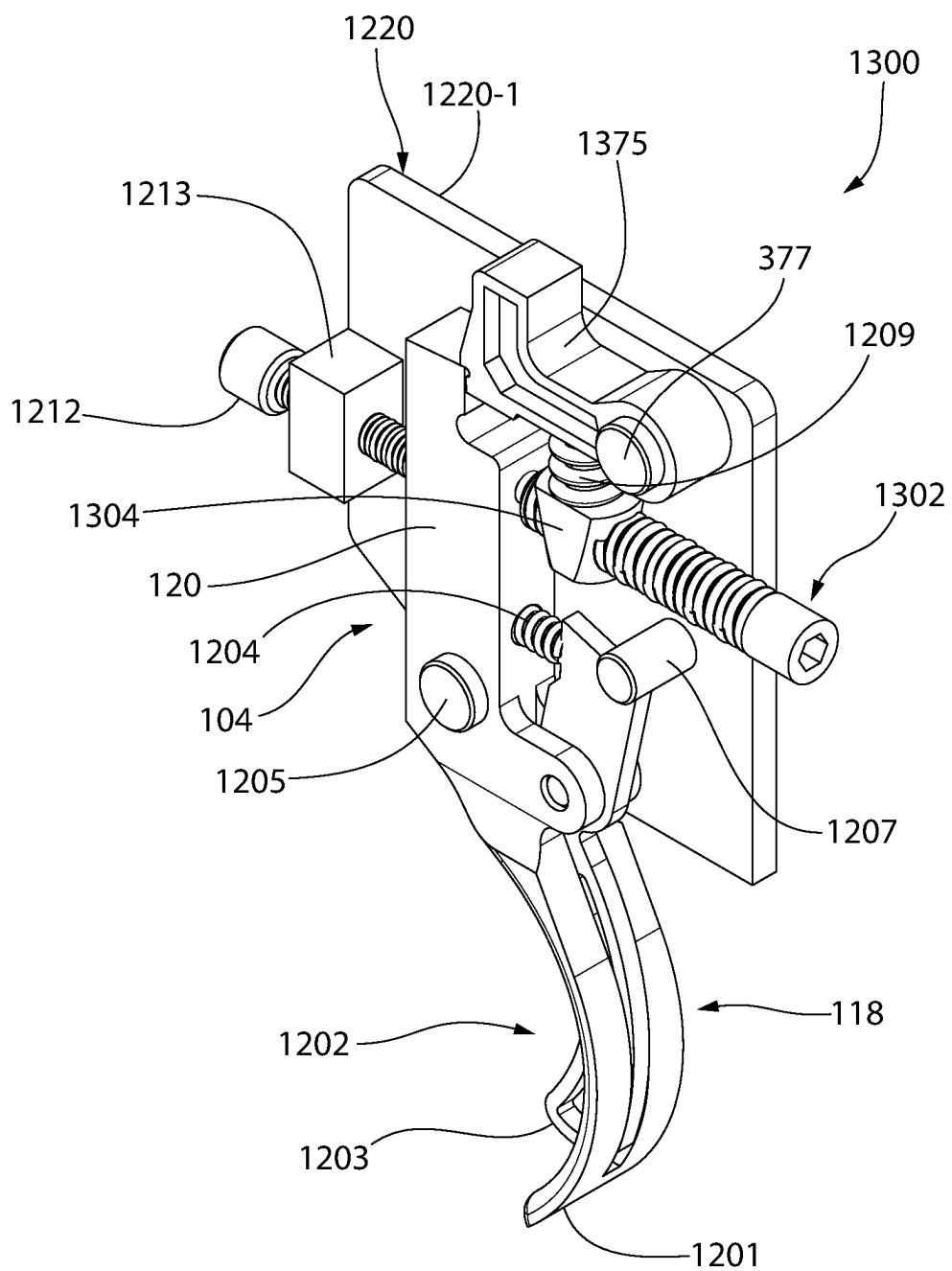


FIG. 61

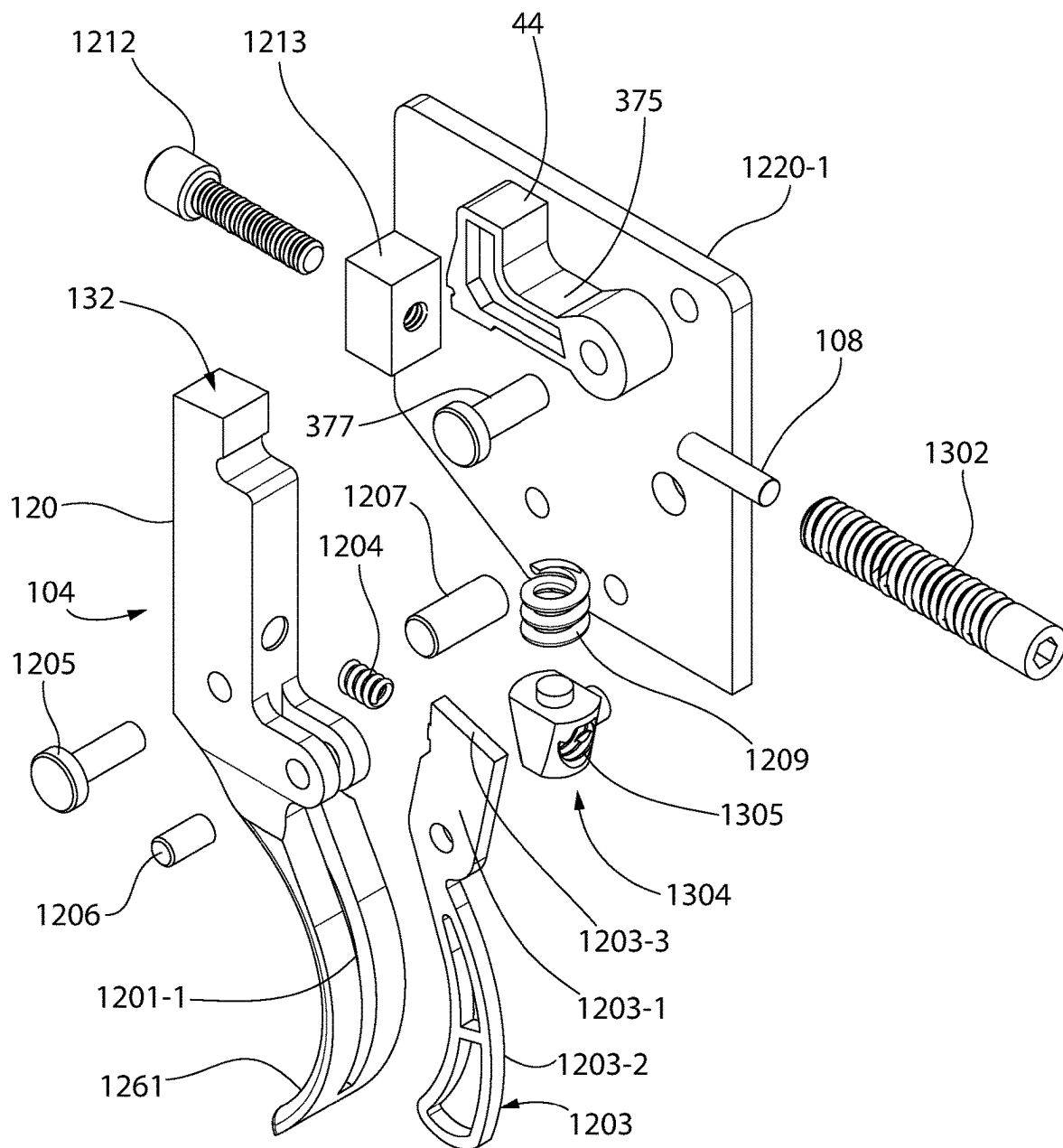


FIG. 62

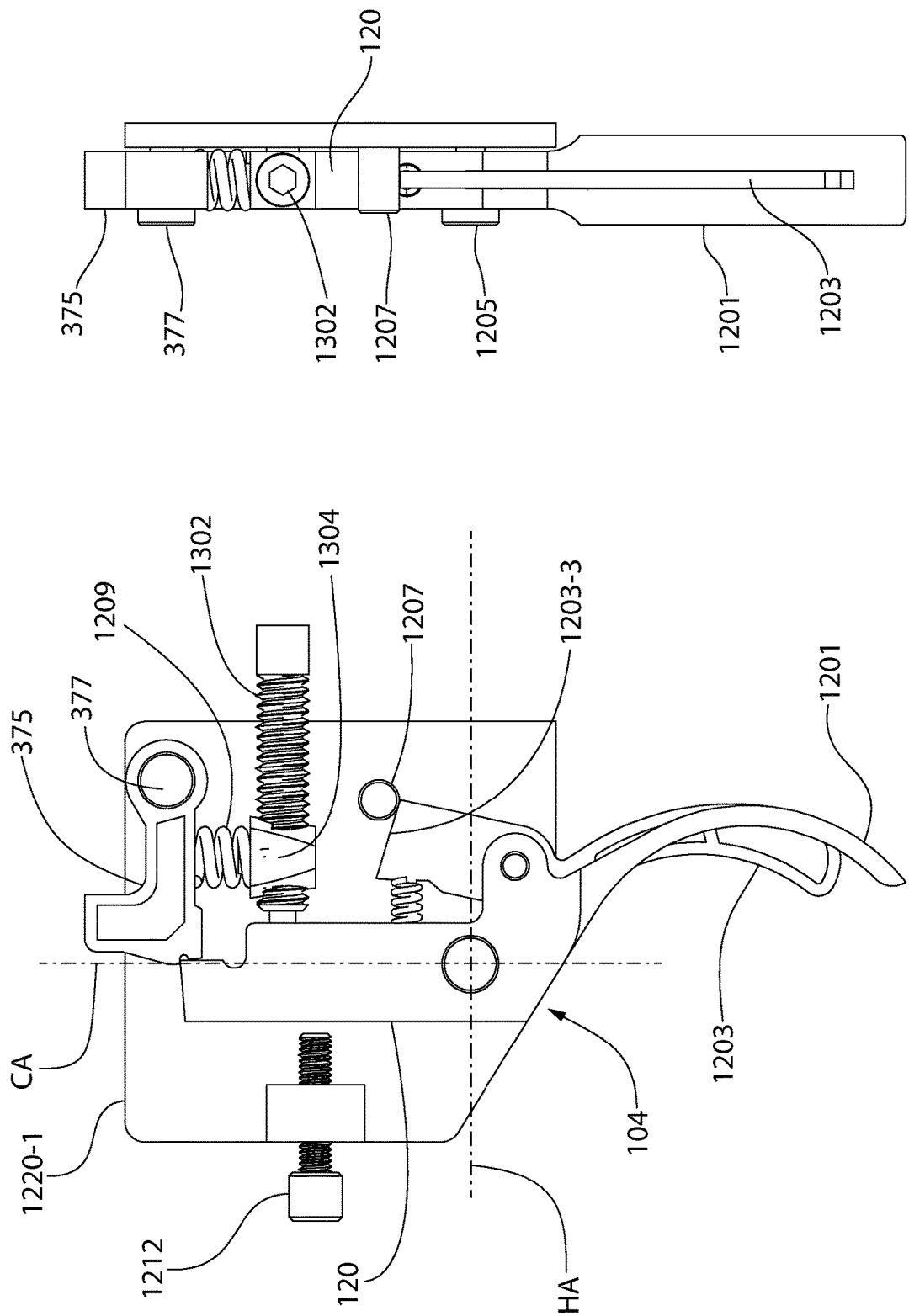


FIG. 64

FIG. 63

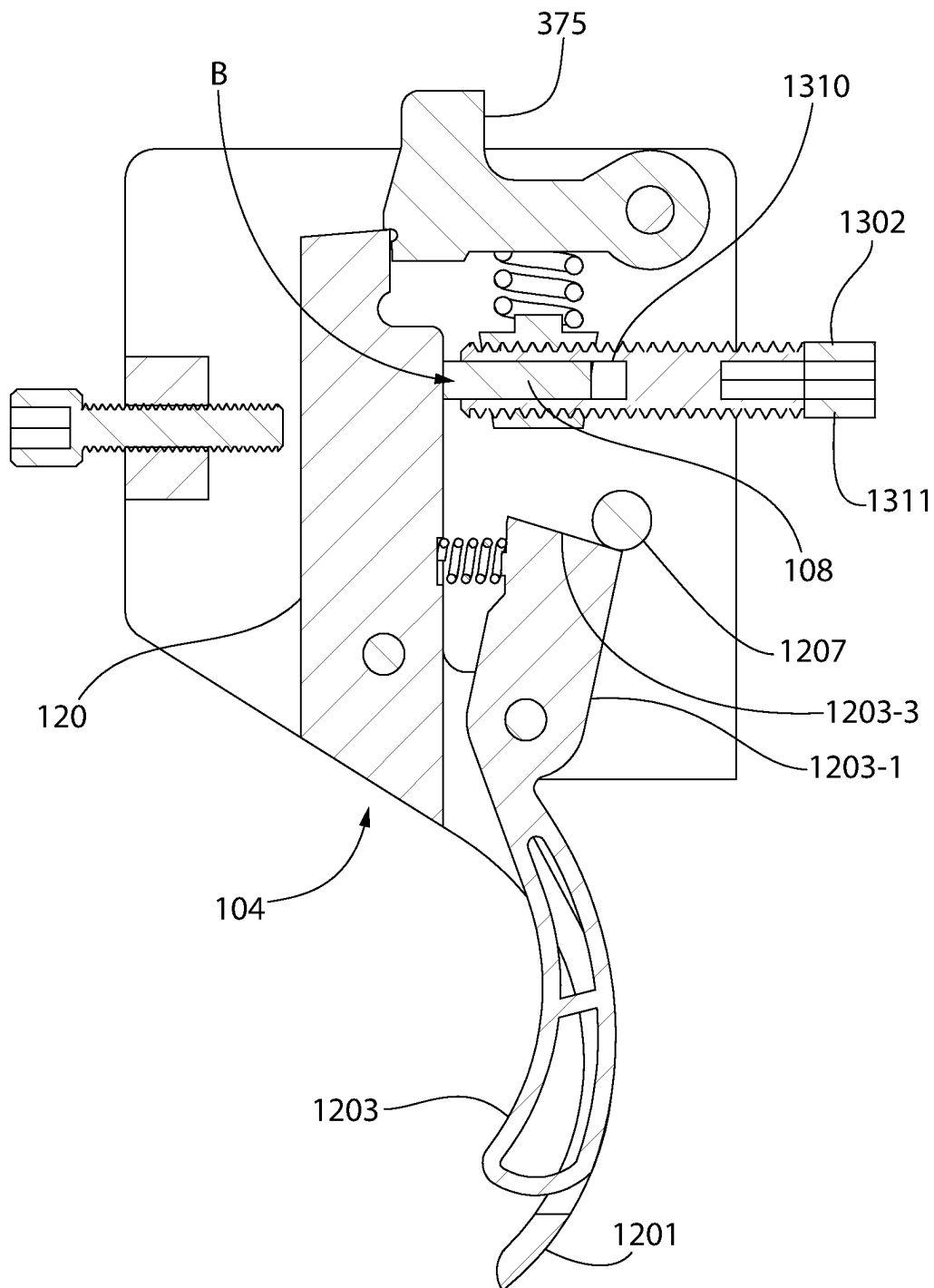


FIG. 65

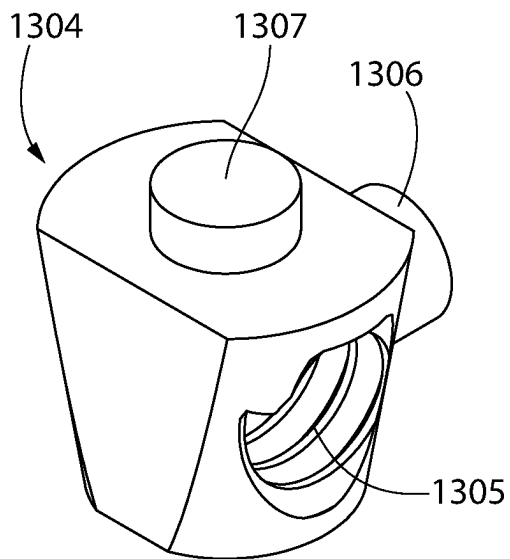


FIG. 66

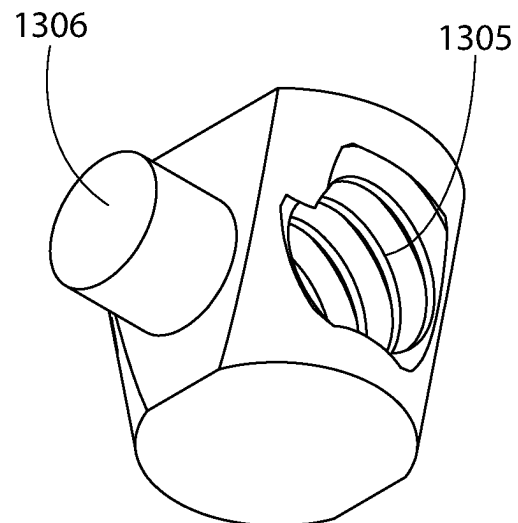


FIG. 67

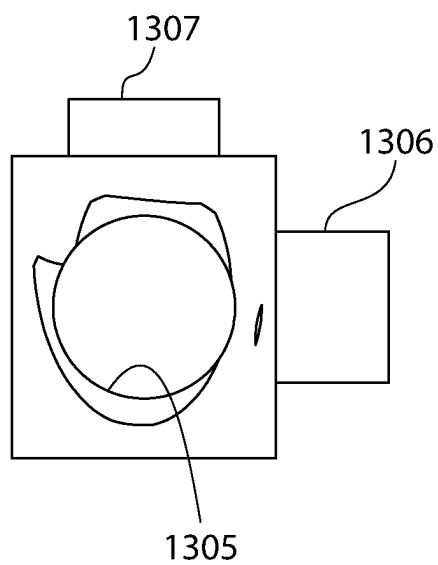


FIG. 68

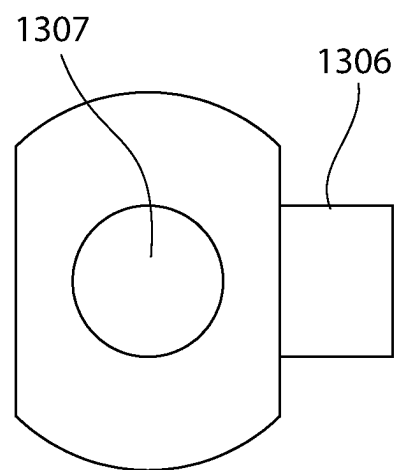


FIG. 69

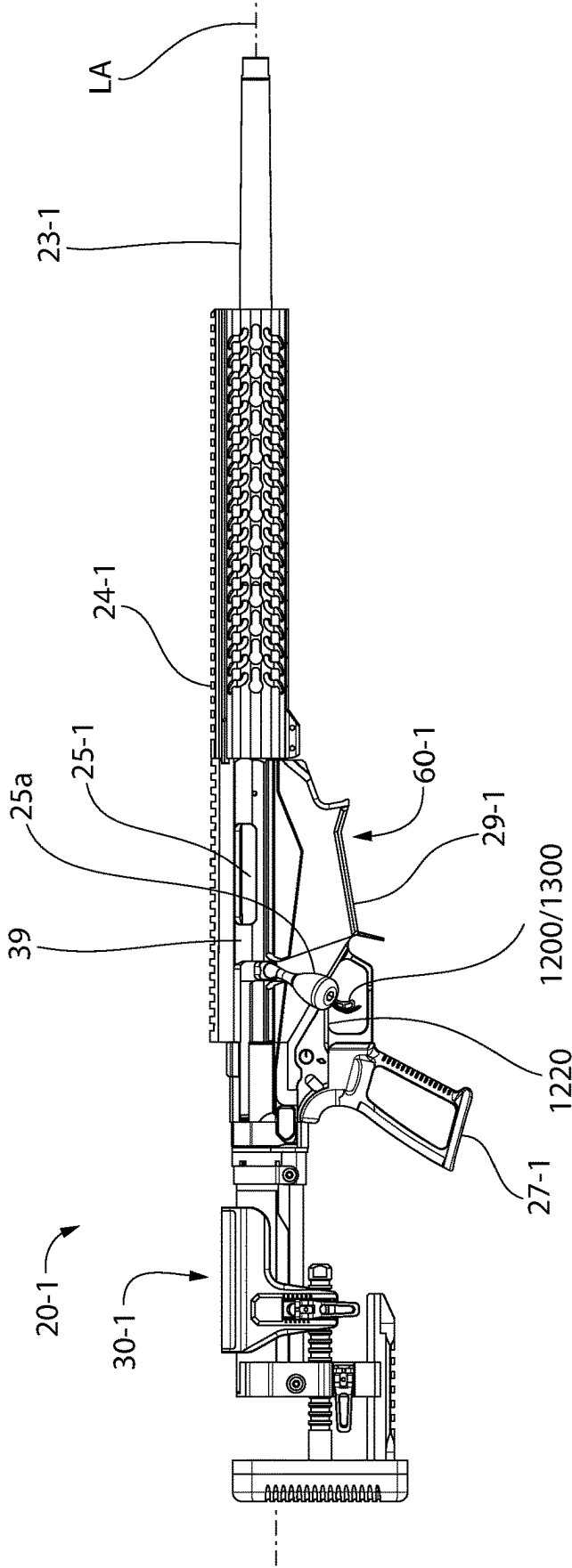


FIG. 70

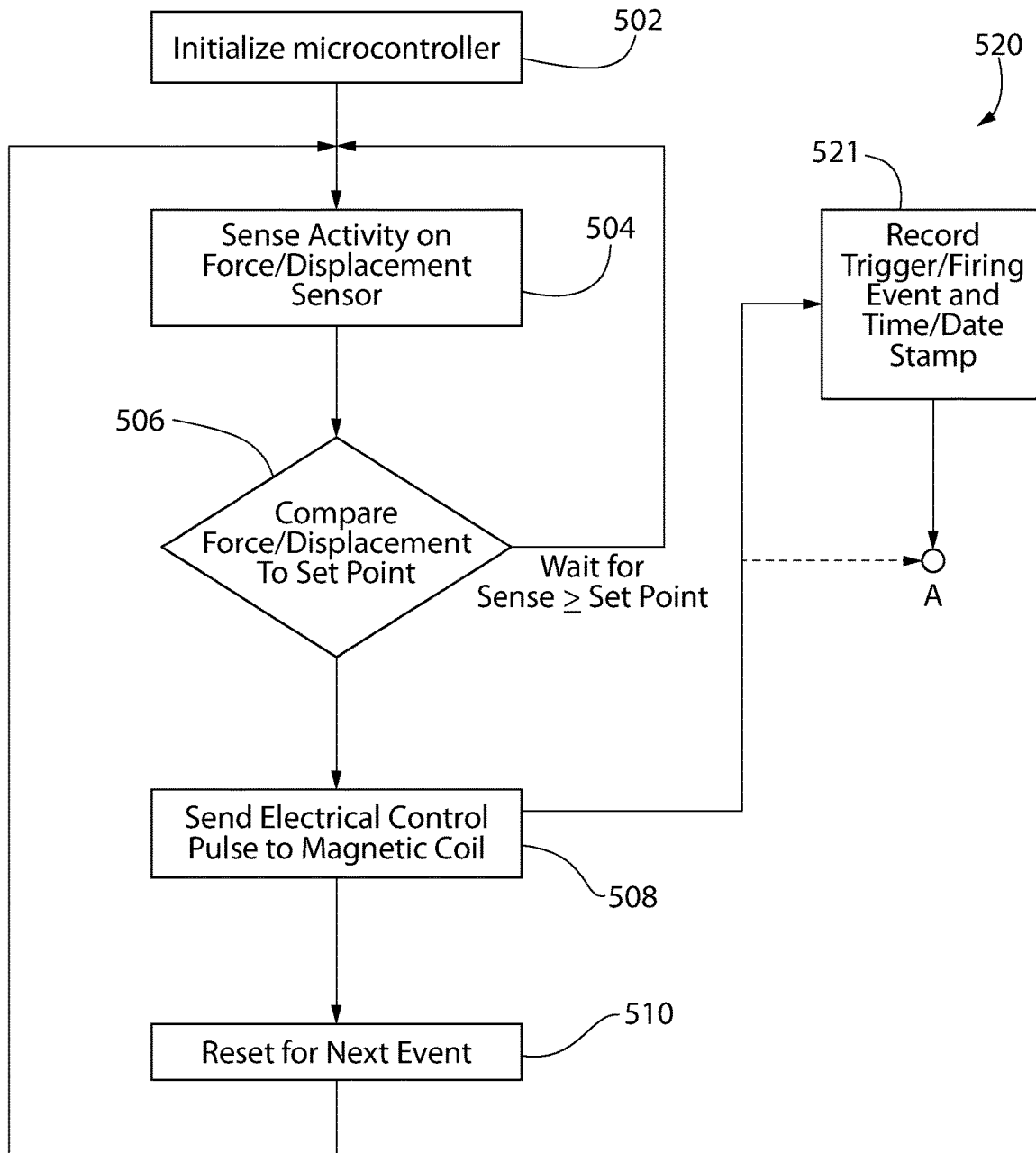
500

FIG. 71A

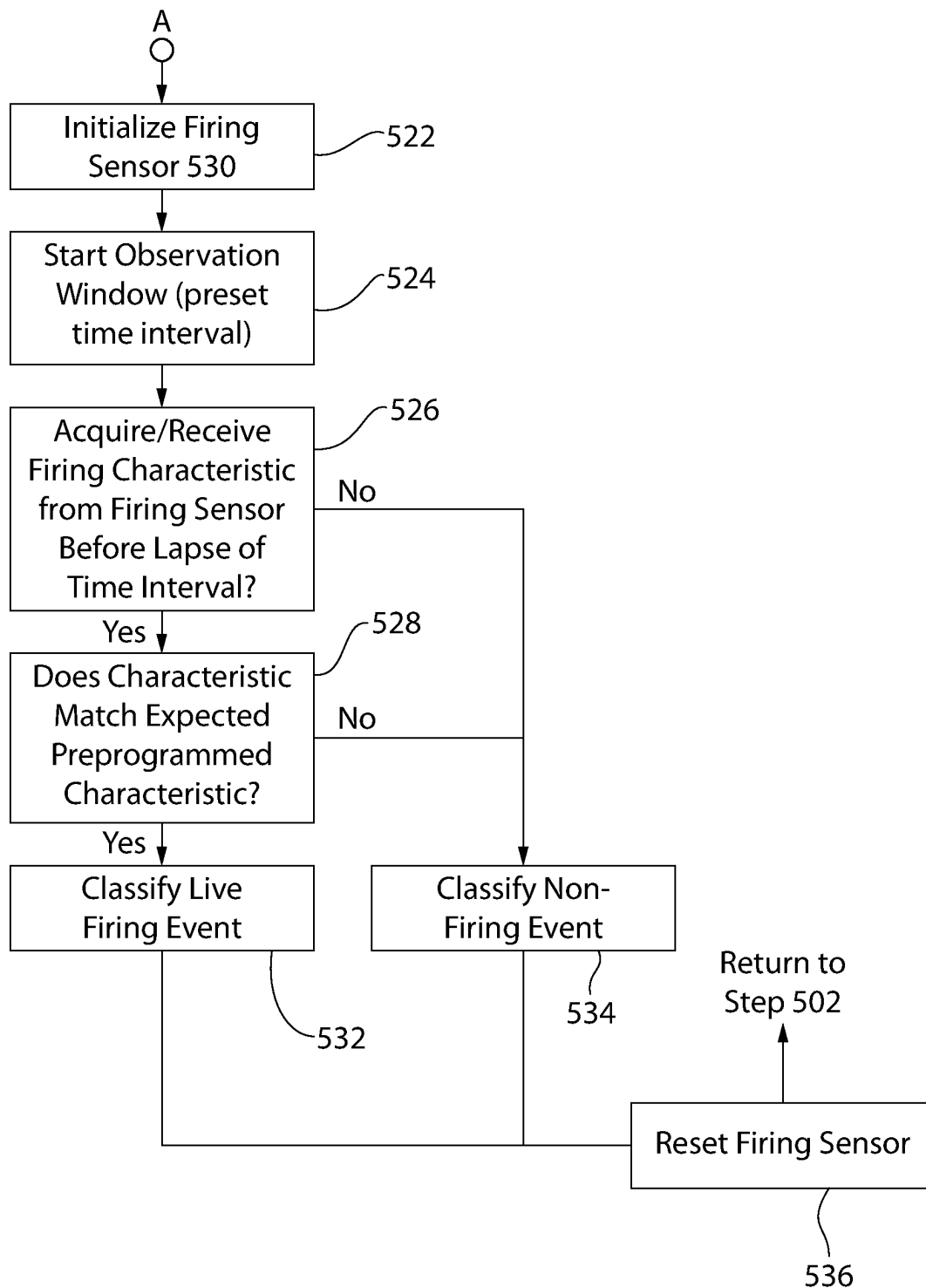


FIG. 71B

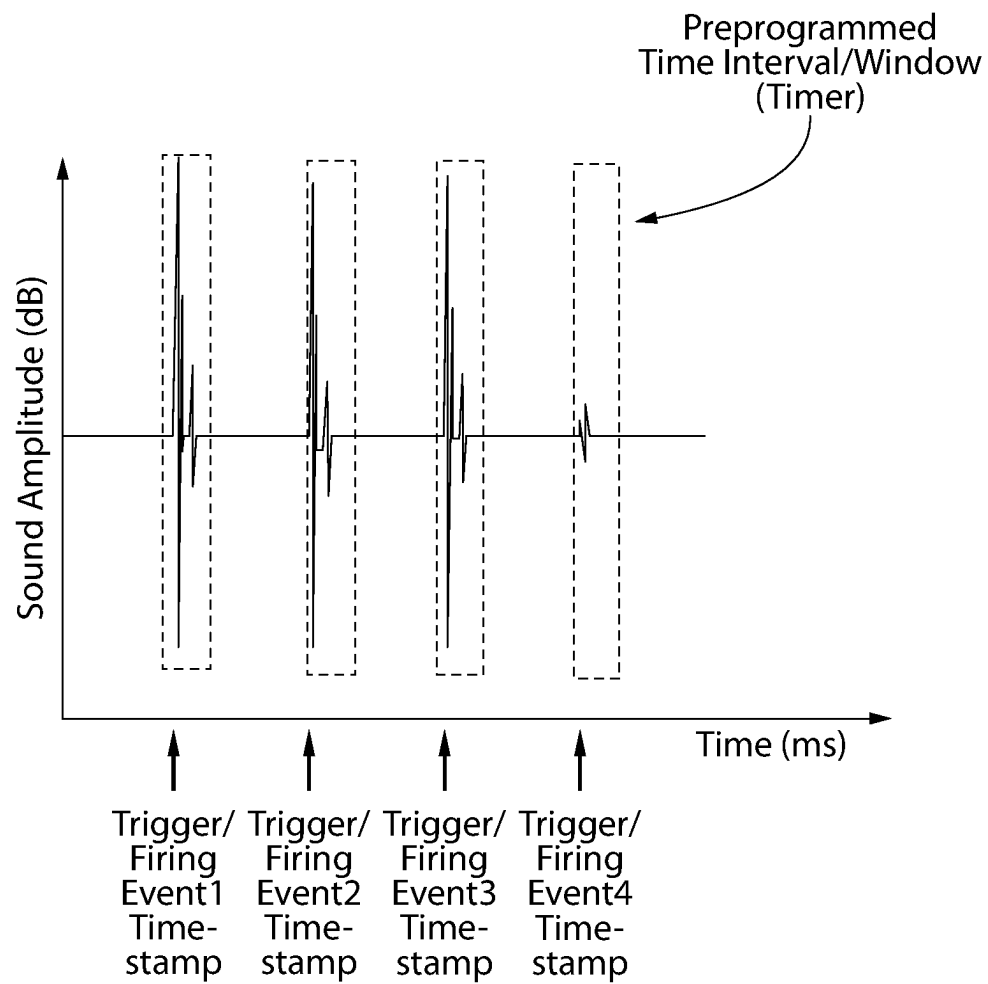


FIG. 72

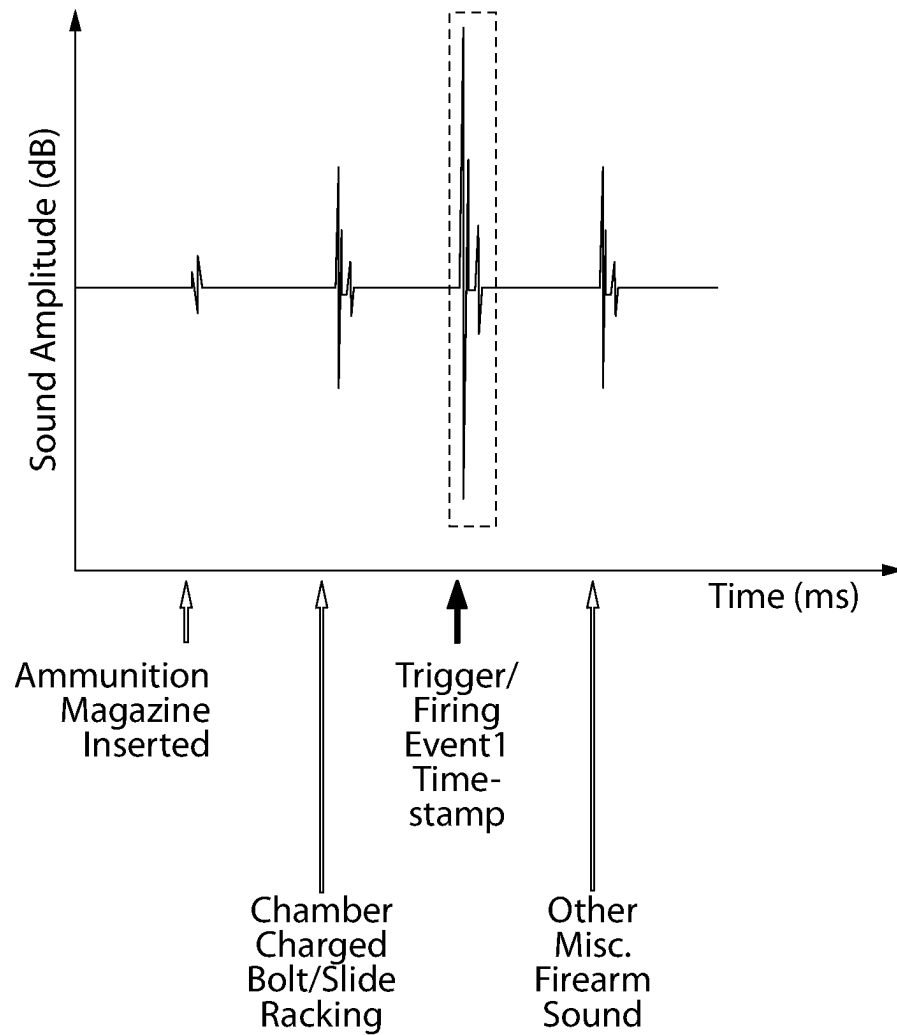


FIG. 73

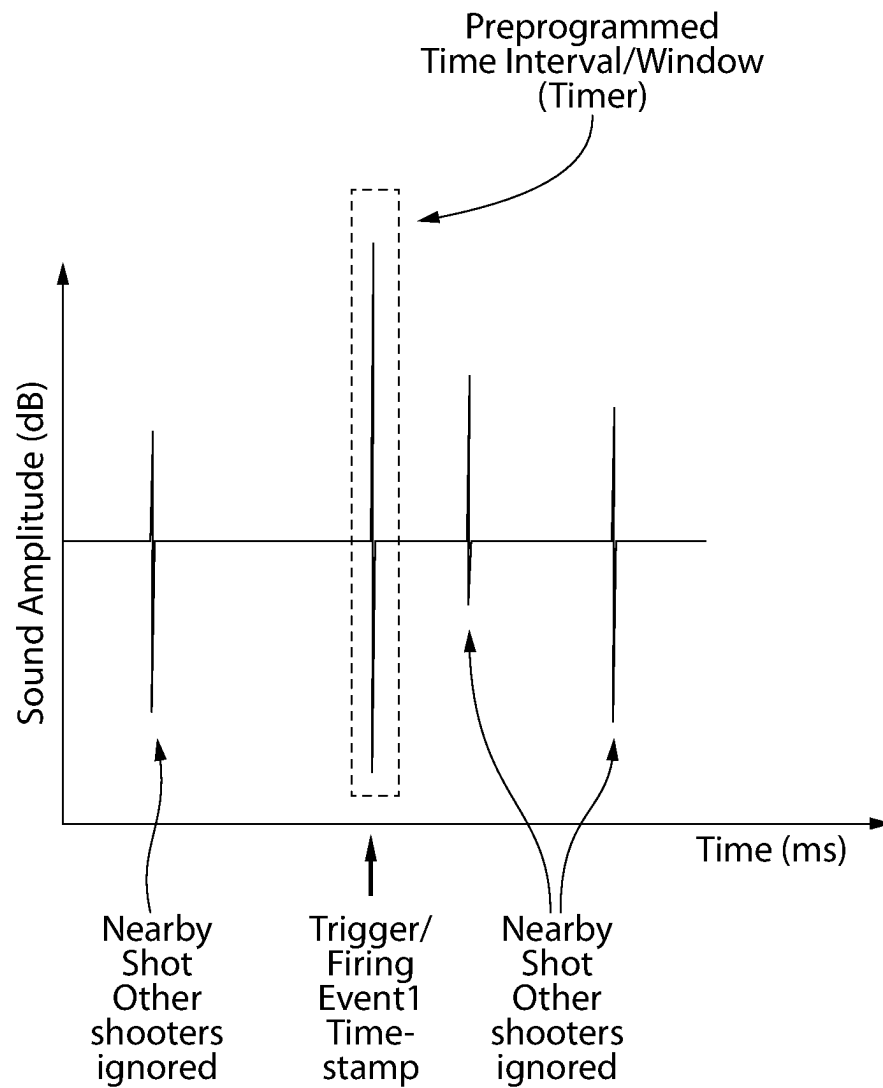


FIG. 74

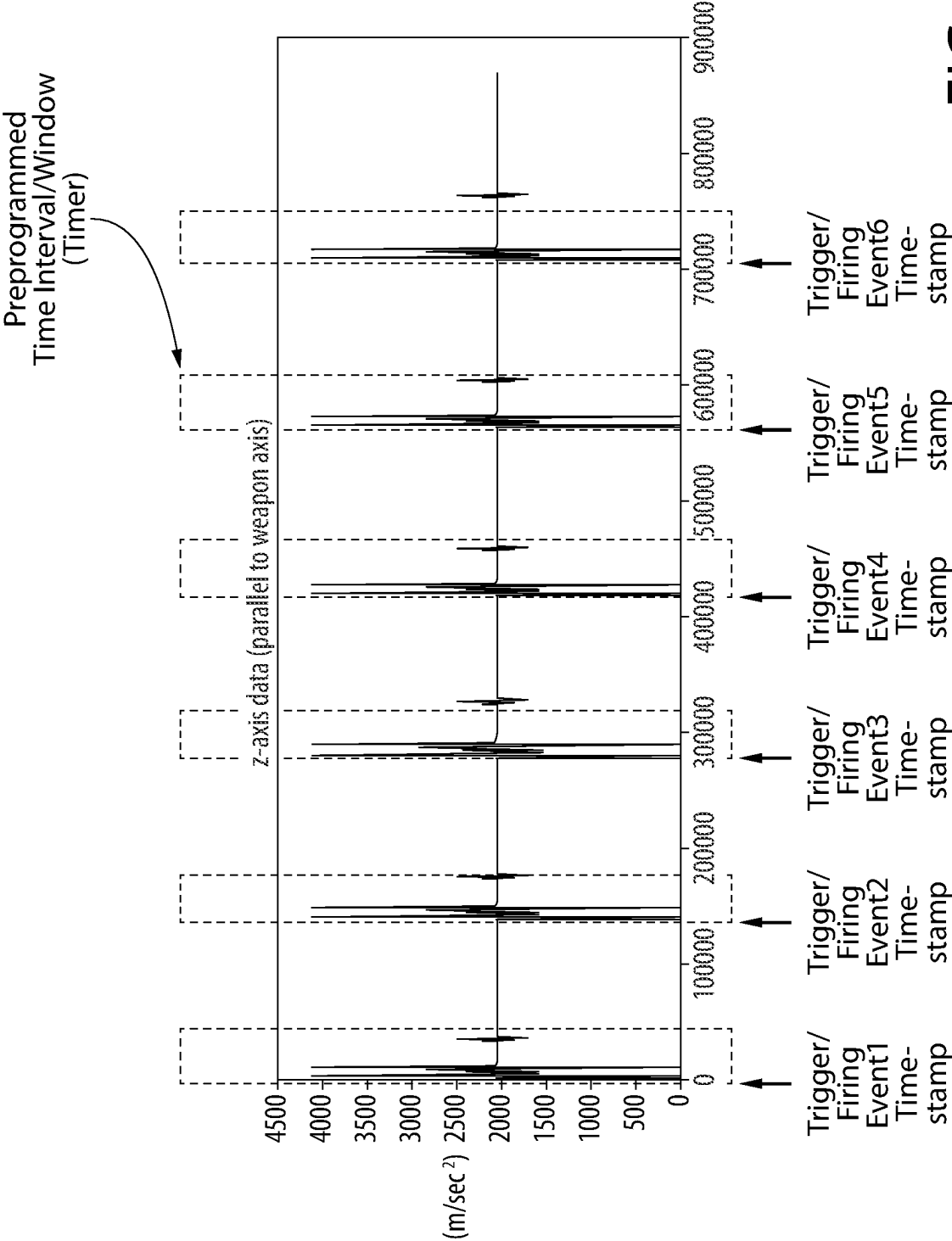
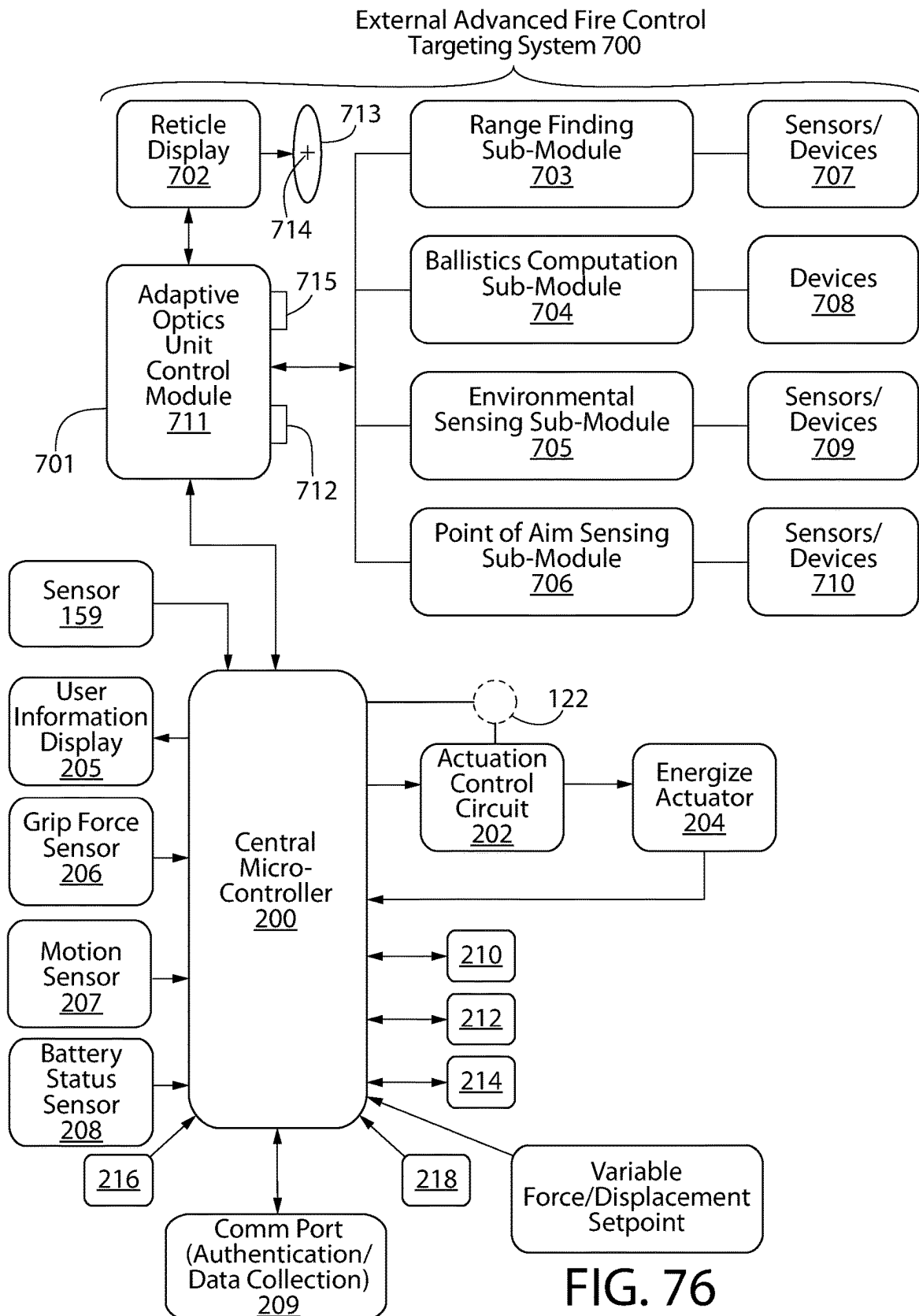


FIG. 75



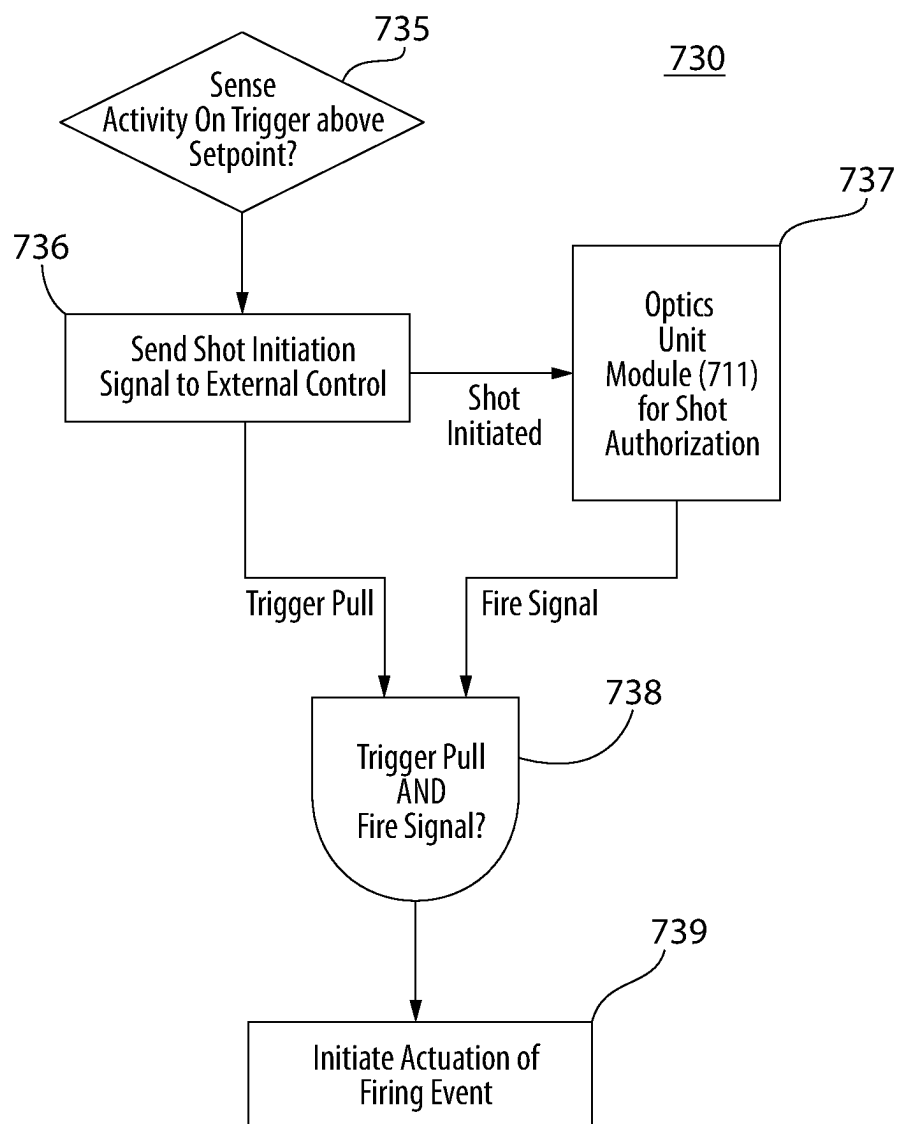


FIG. 77

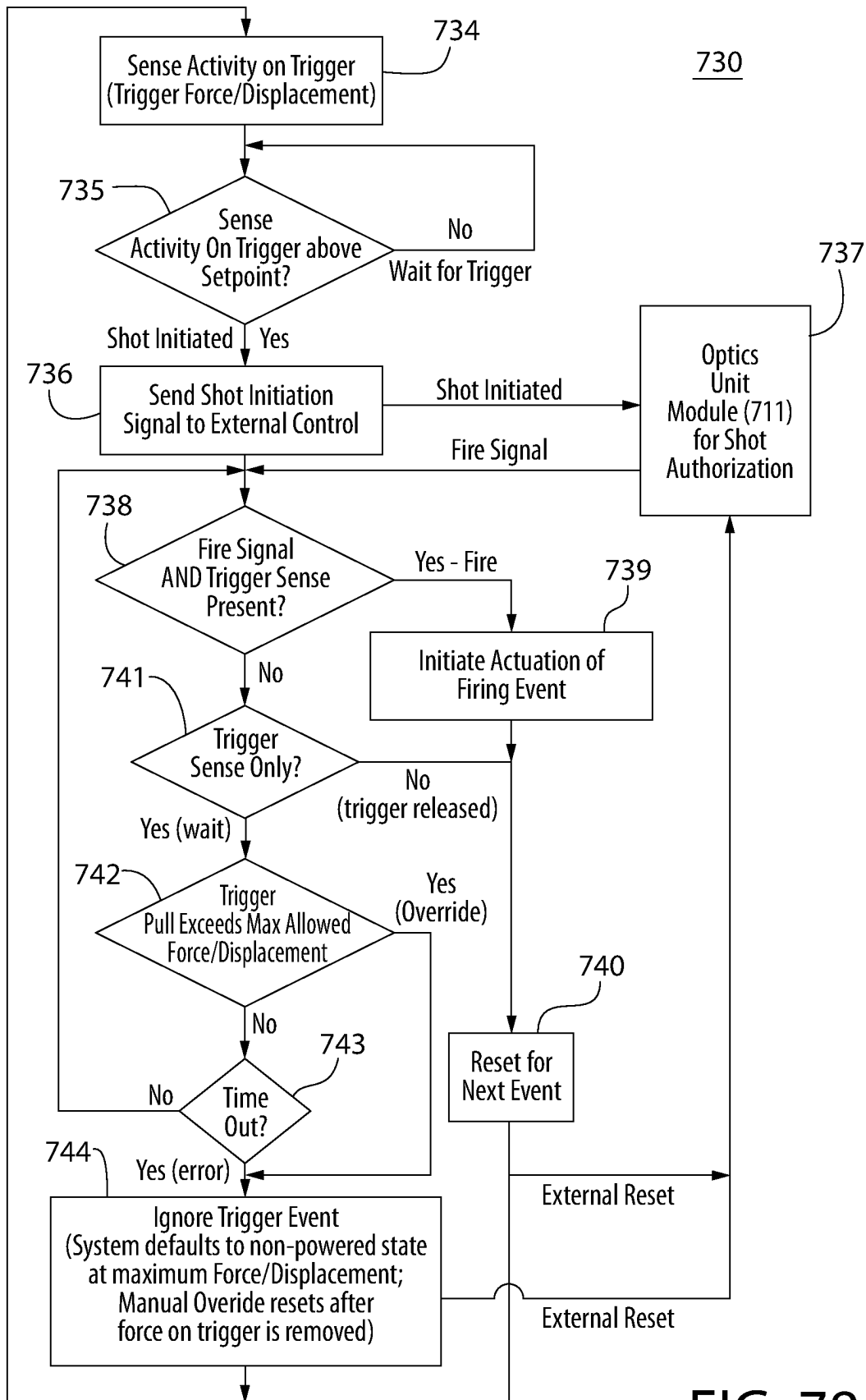


FIG. 78

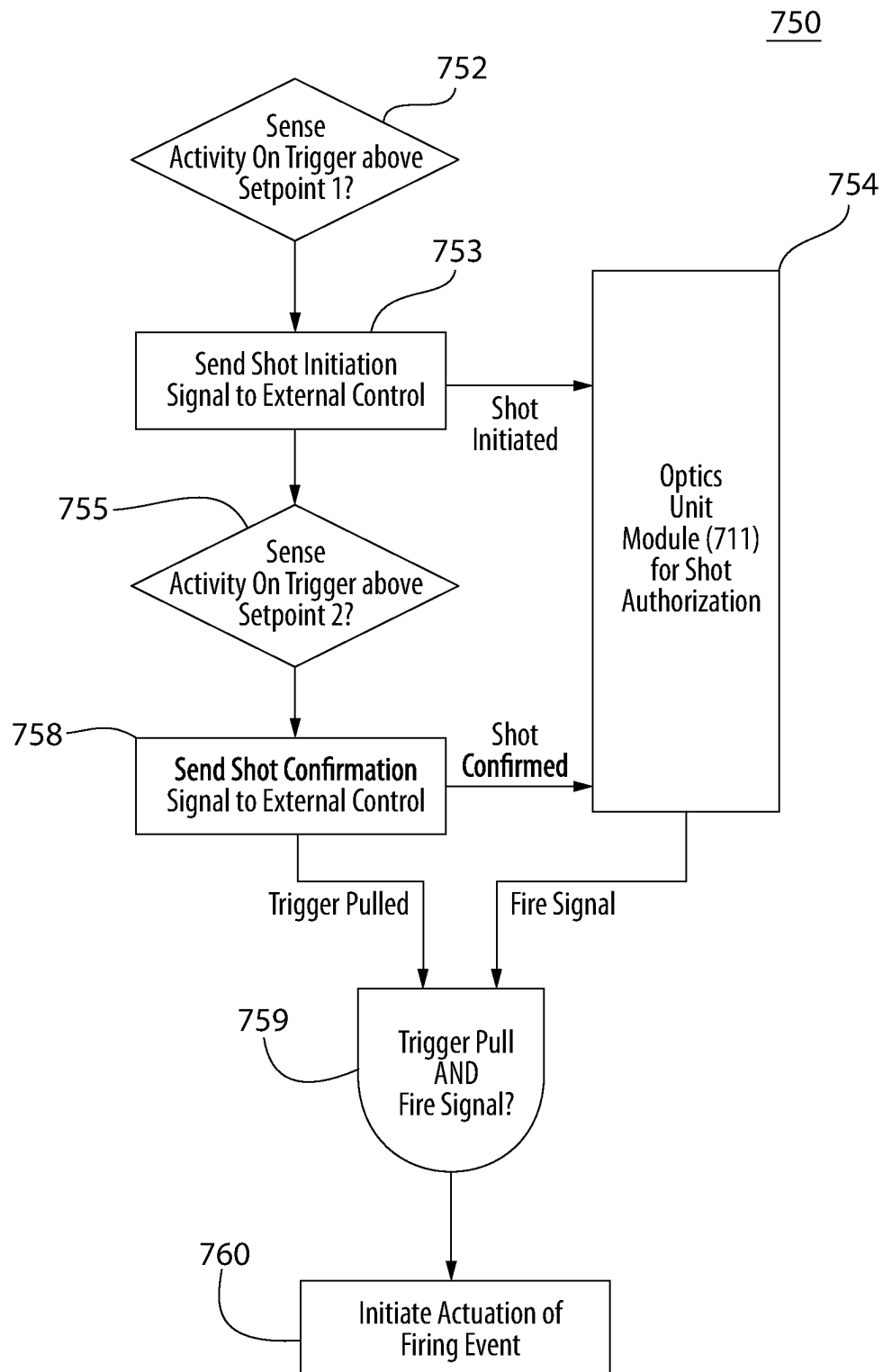
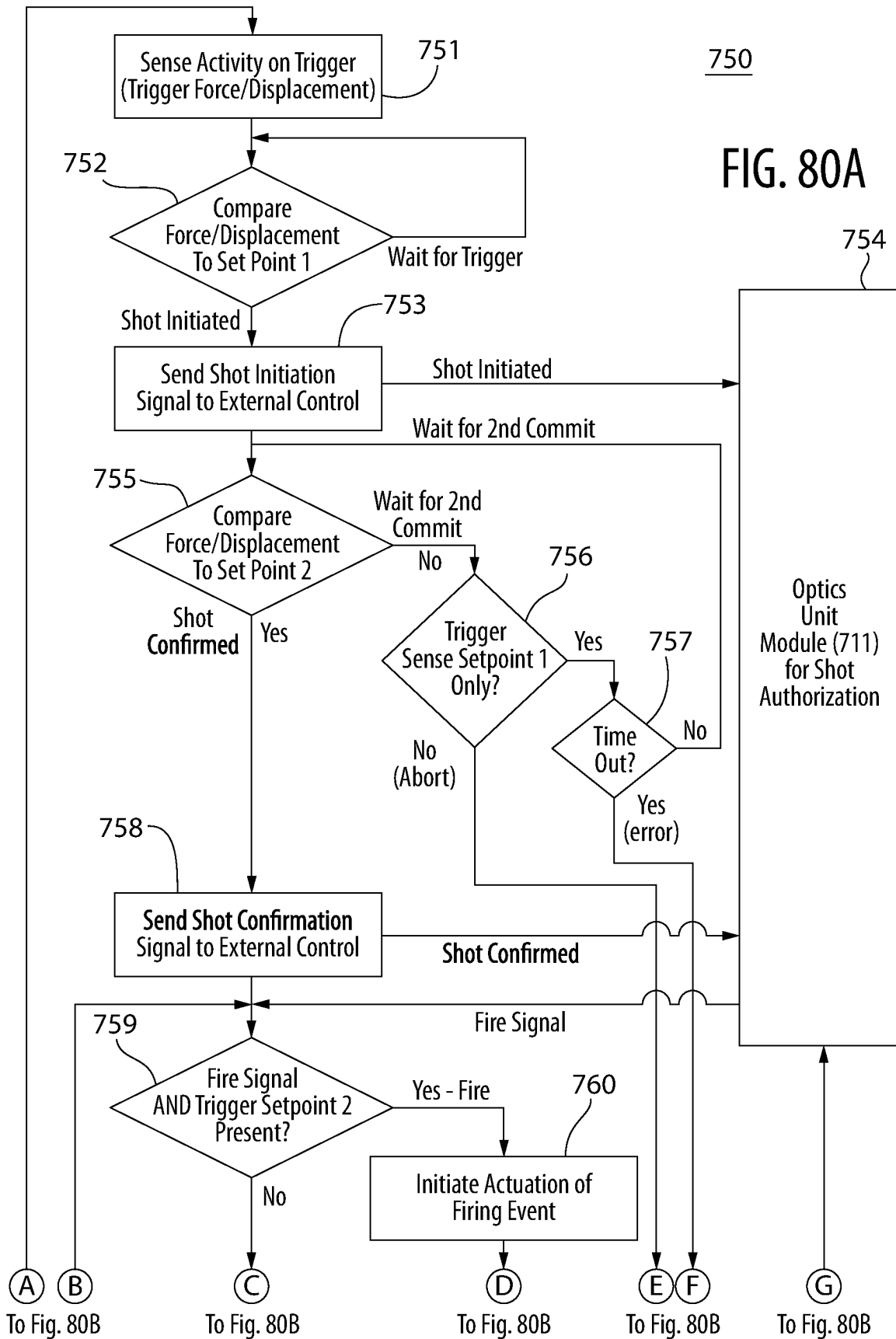


FIG. 79



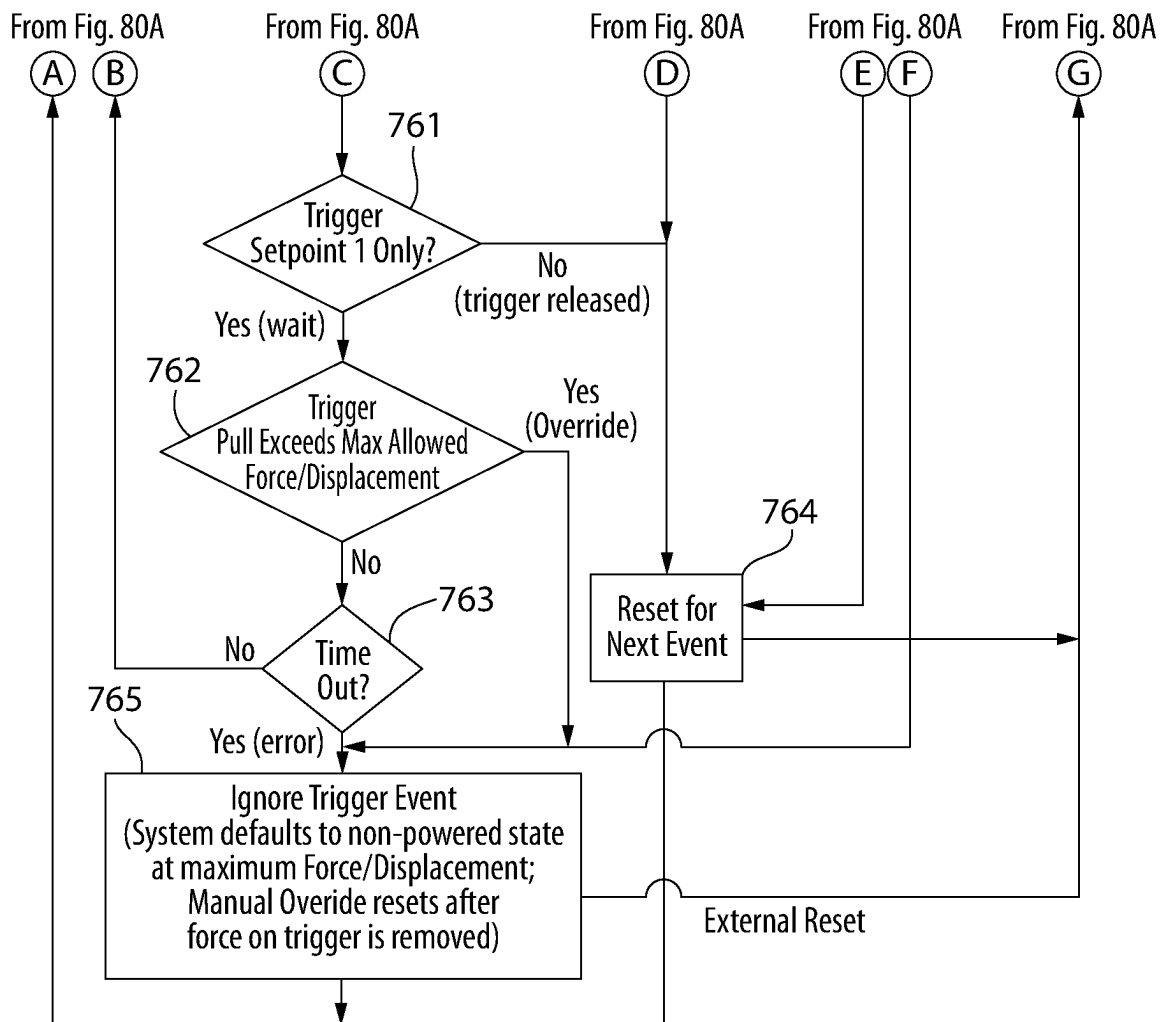


FIG. 80B

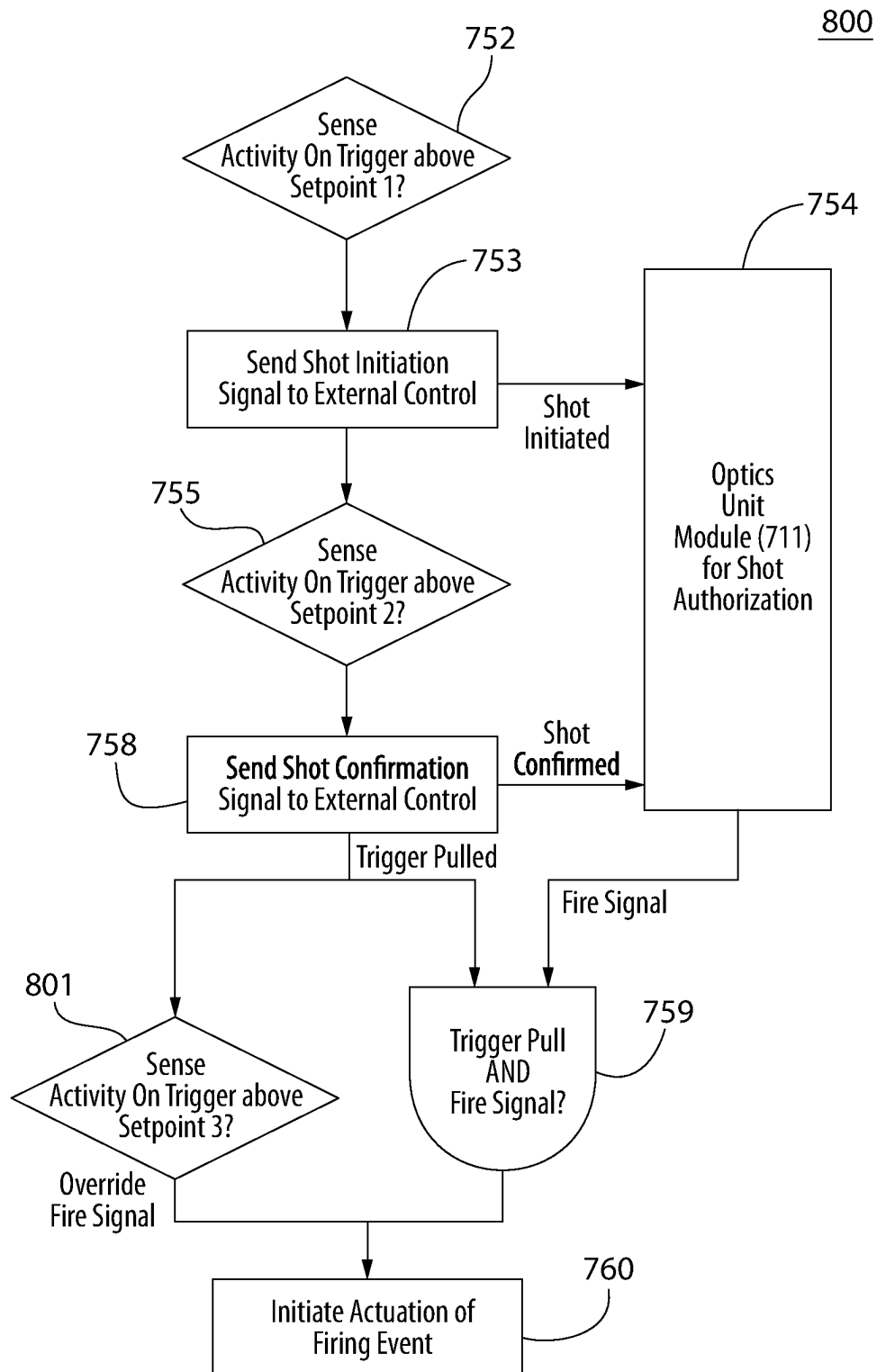
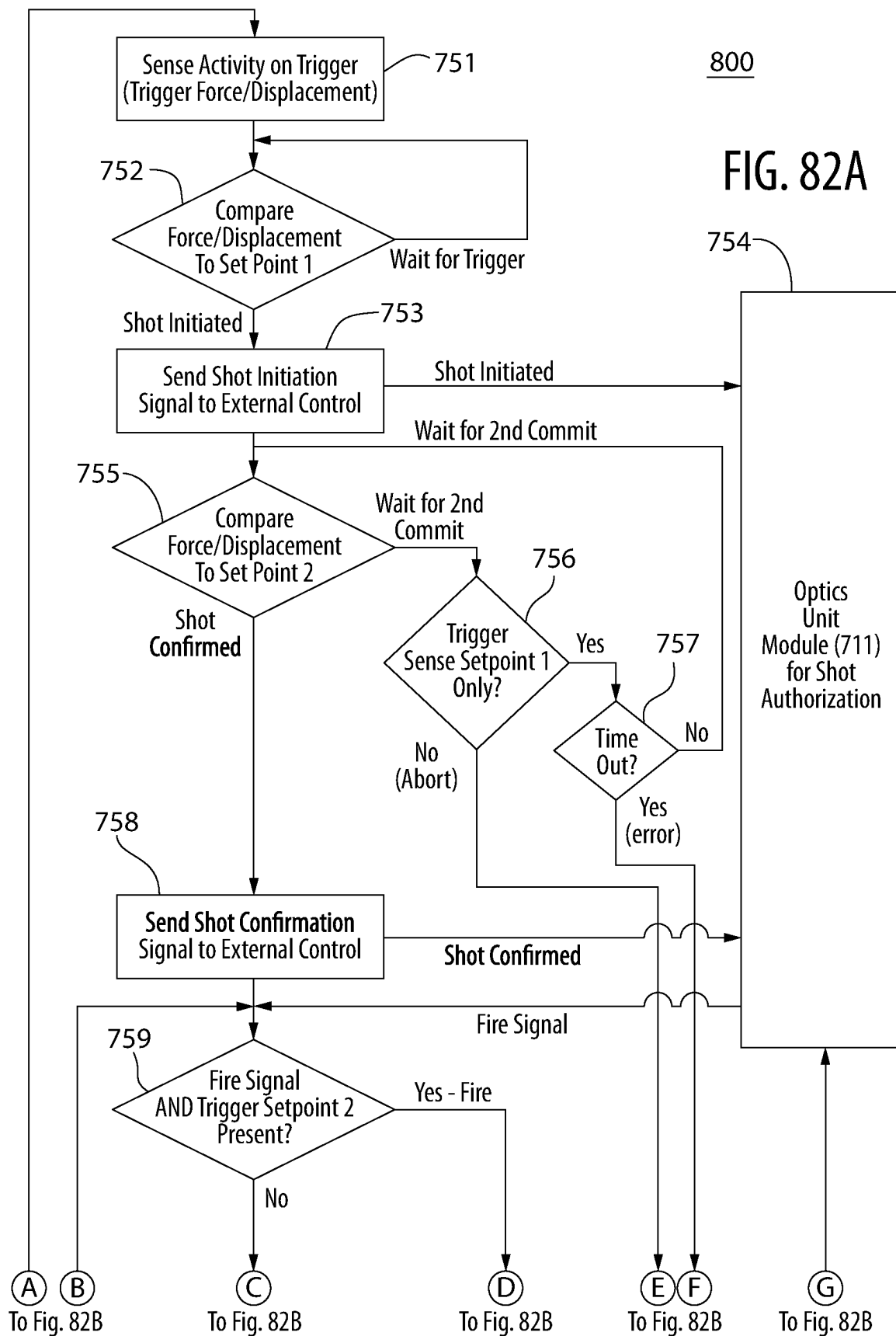


FIG. 81



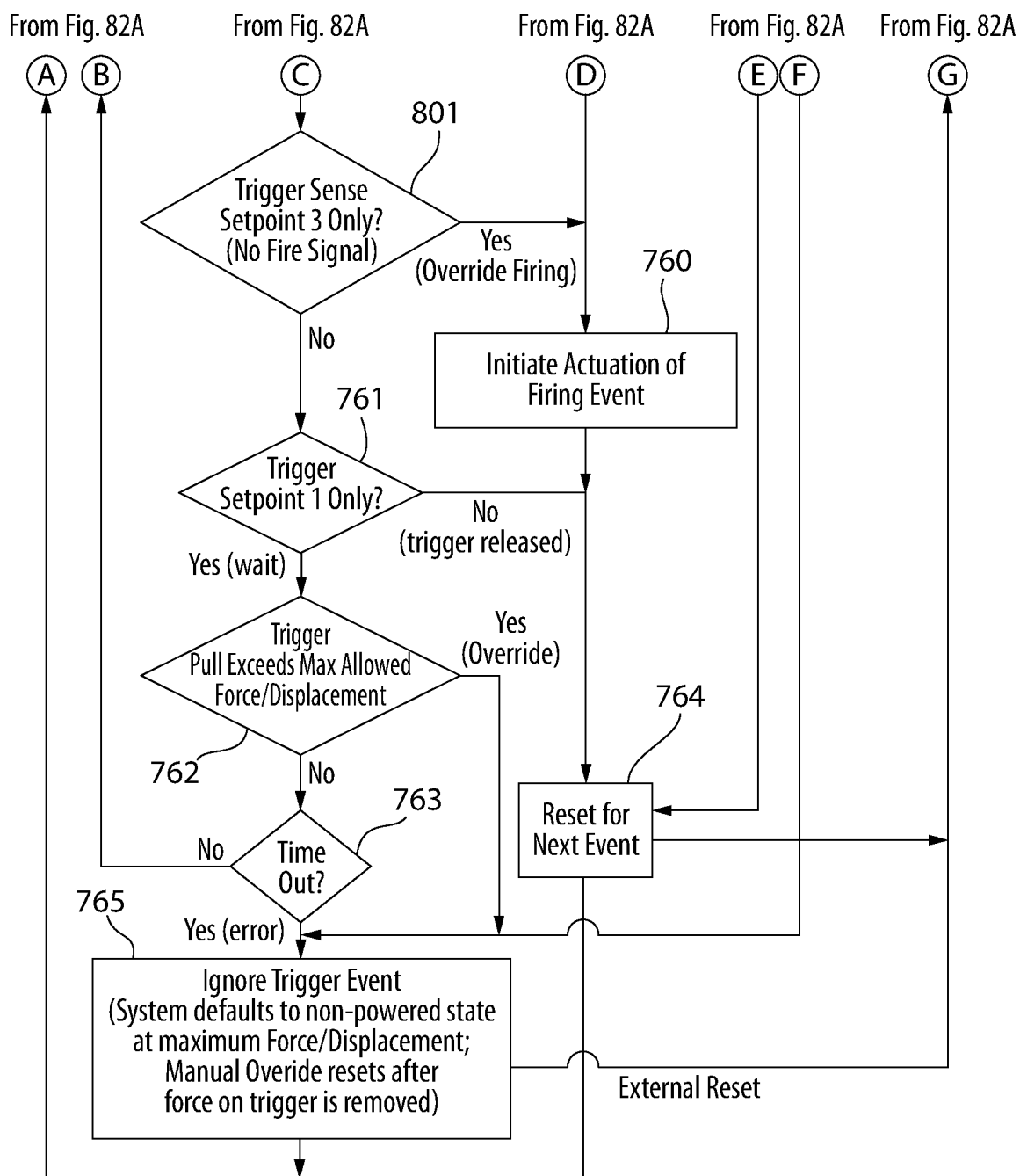


FIG. 82B

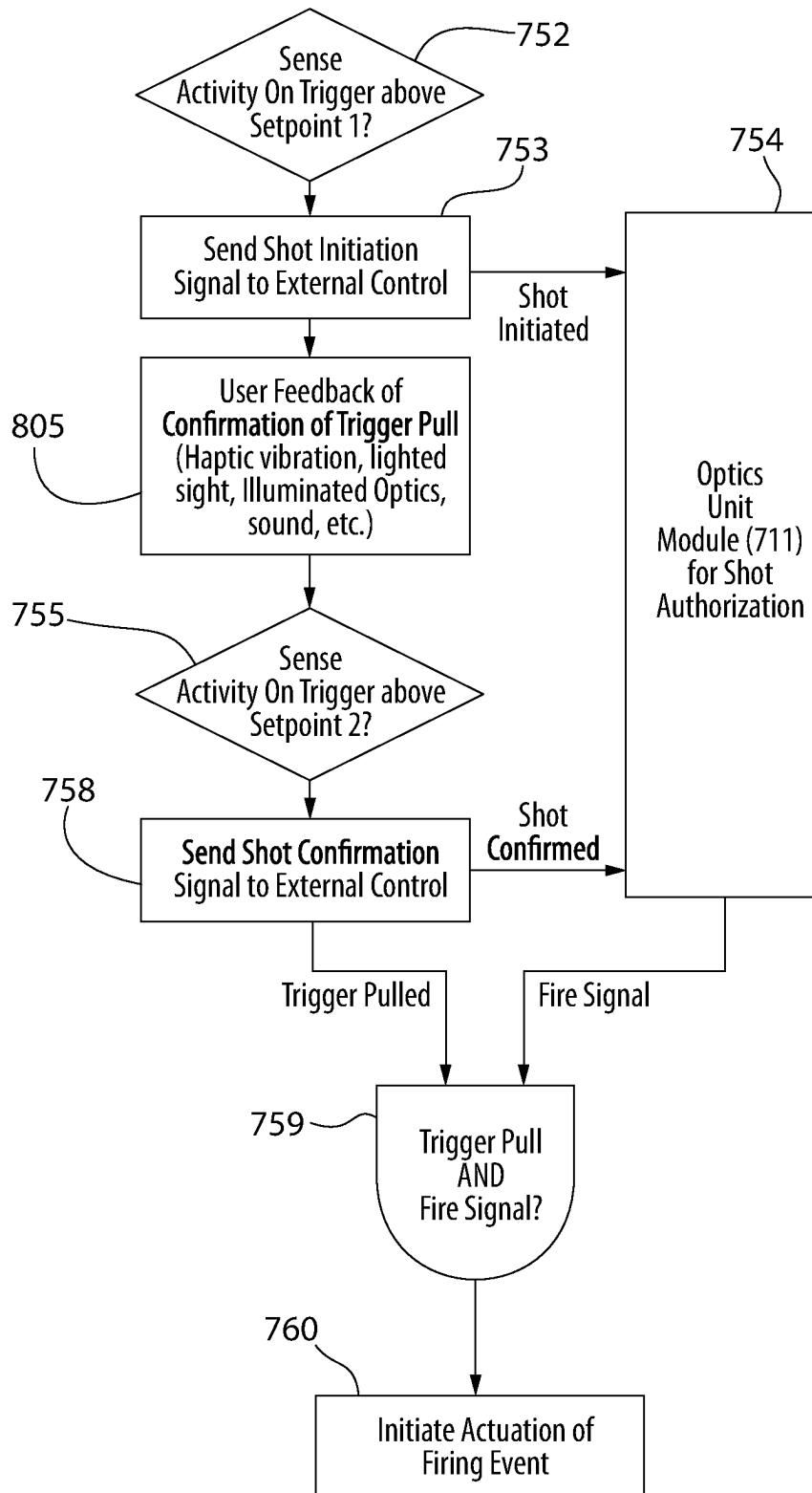


FIG. 83

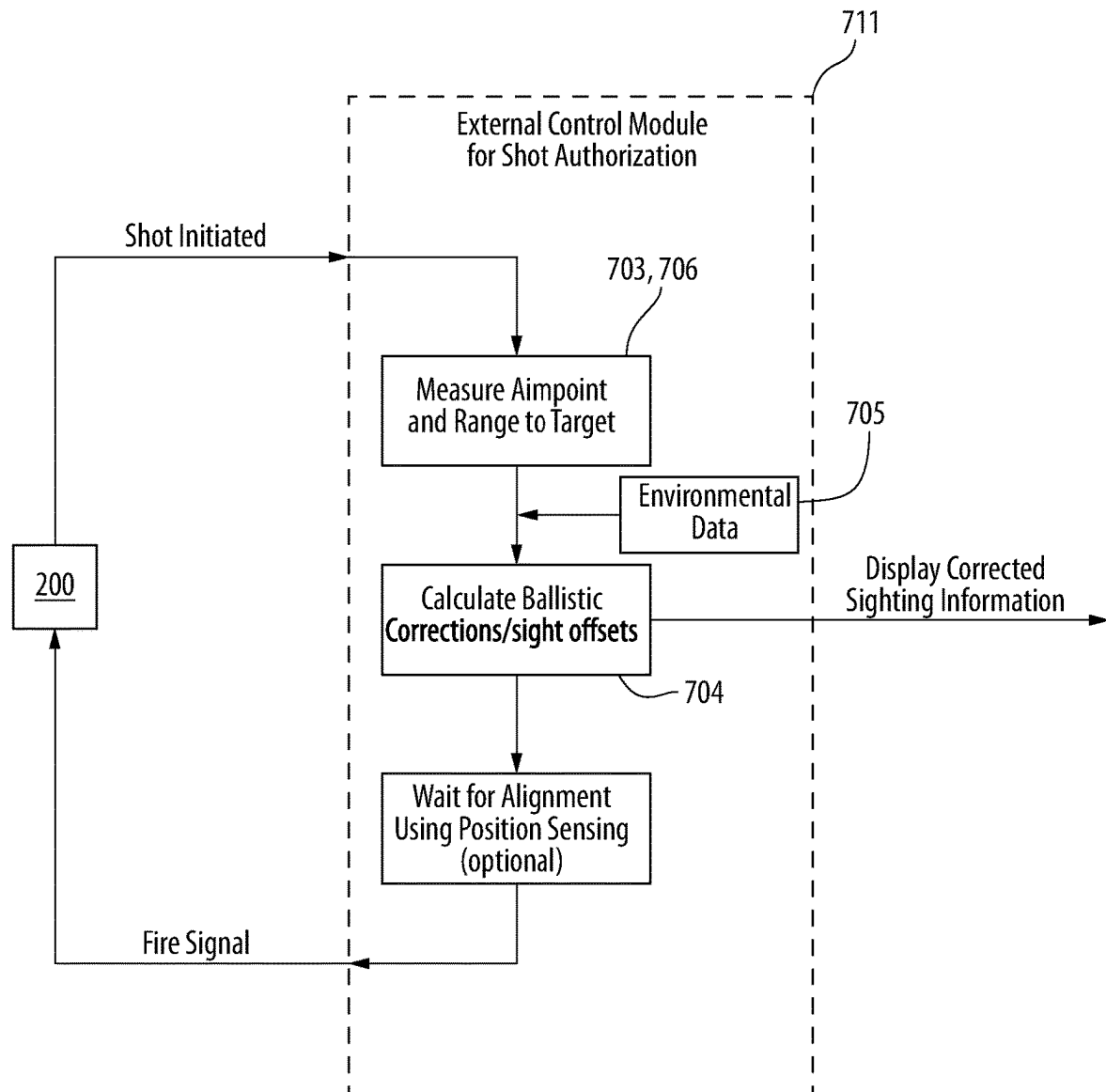


FIG. 84

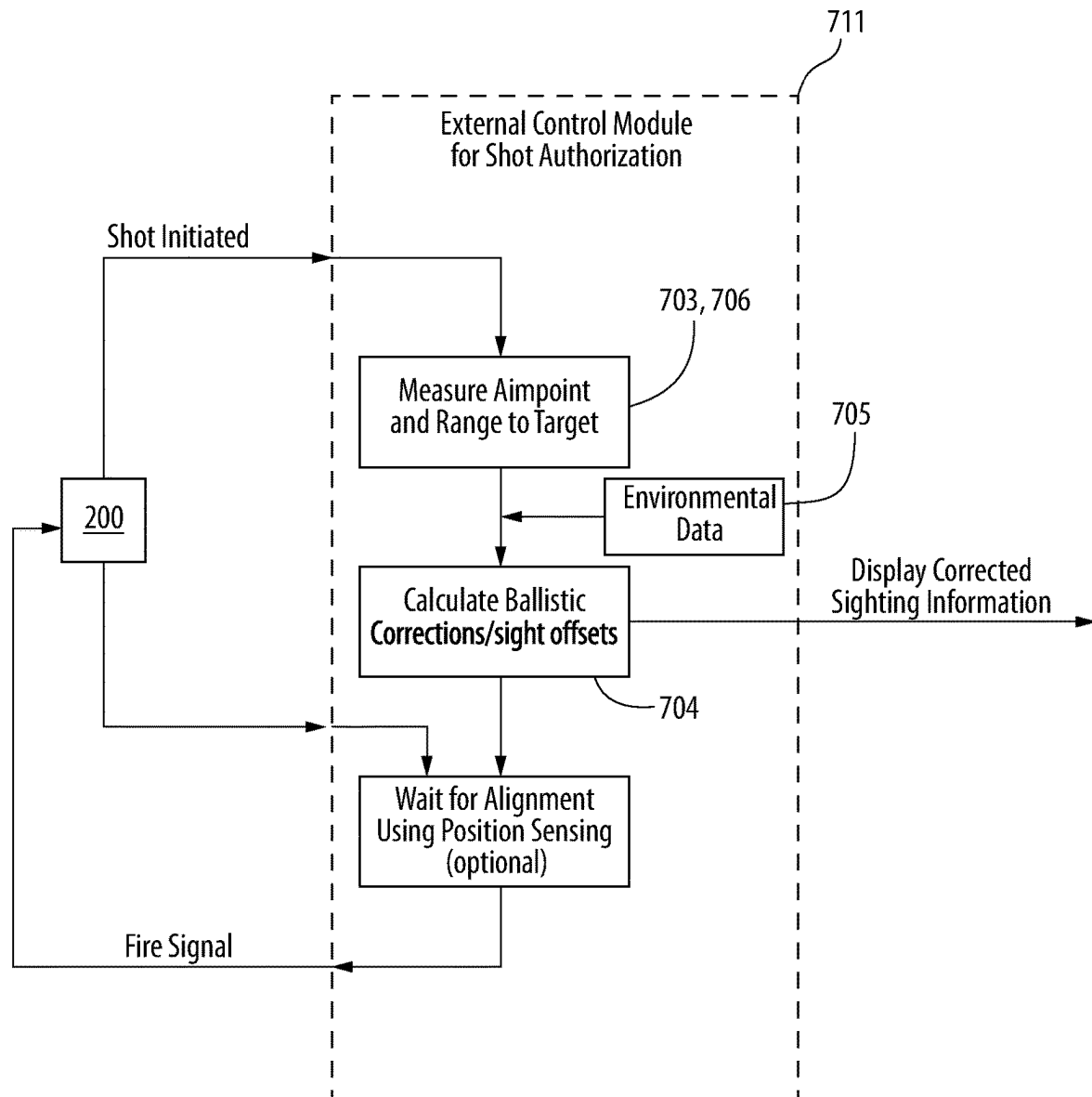


FIG. 85

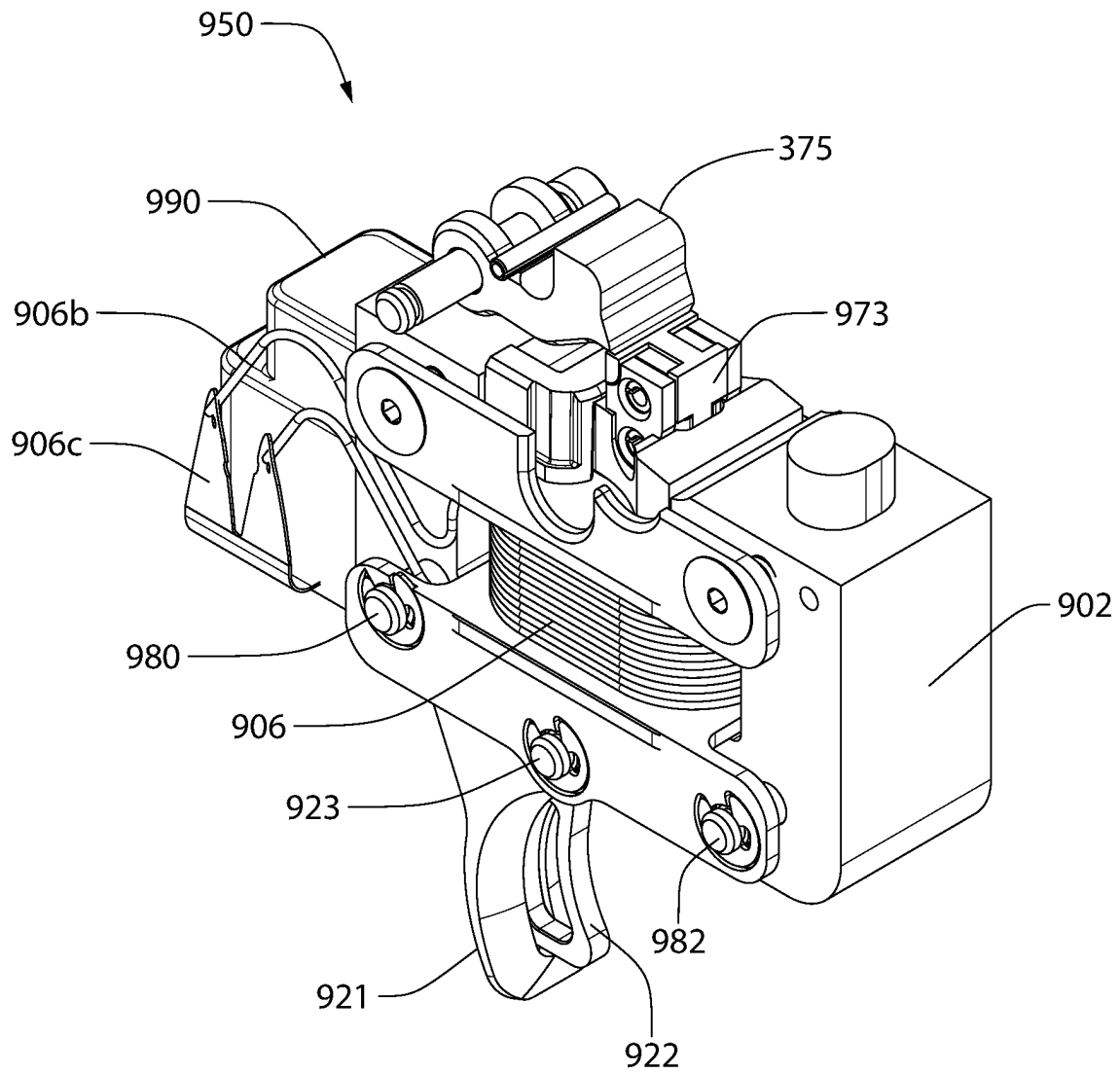


FIG. 86

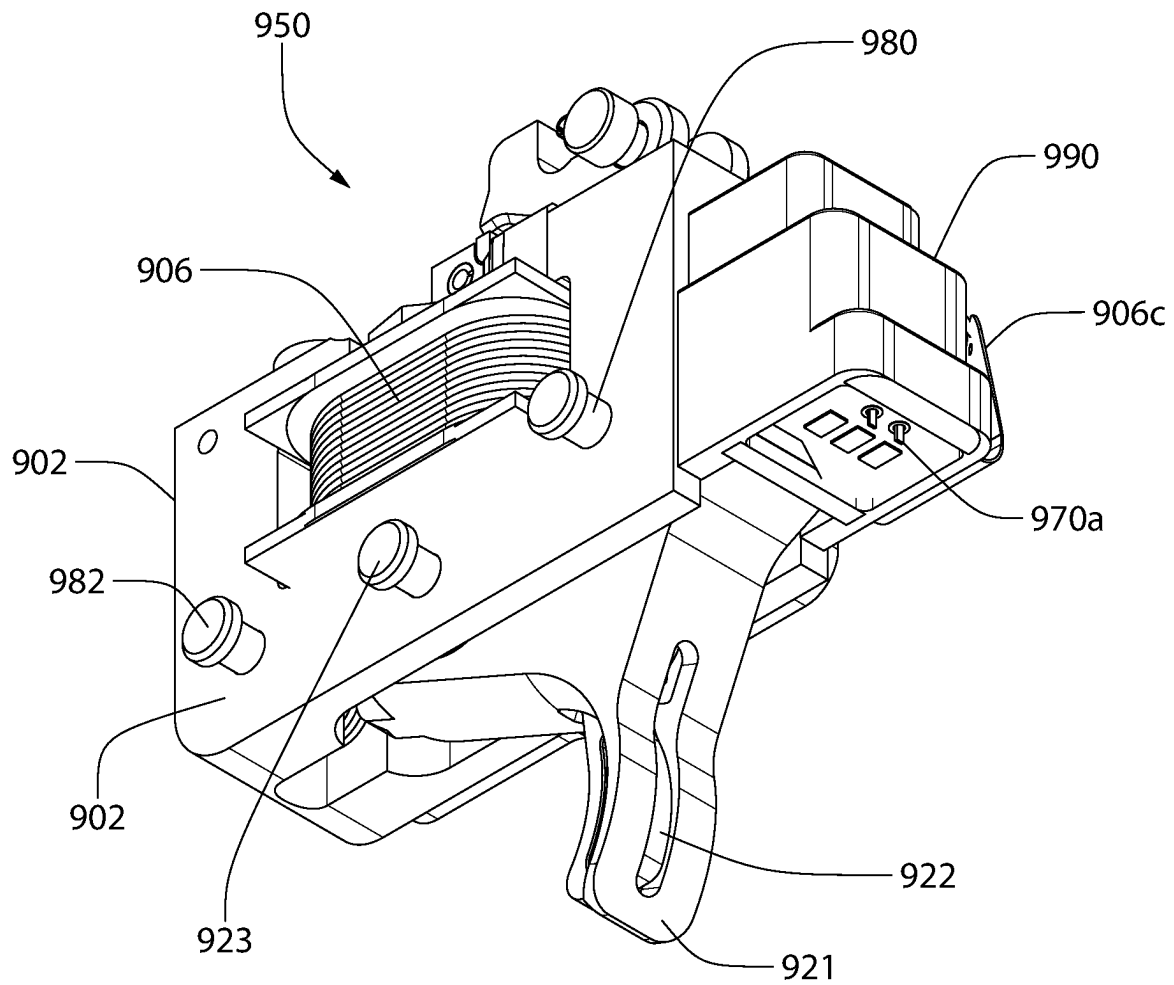


FIG. 87

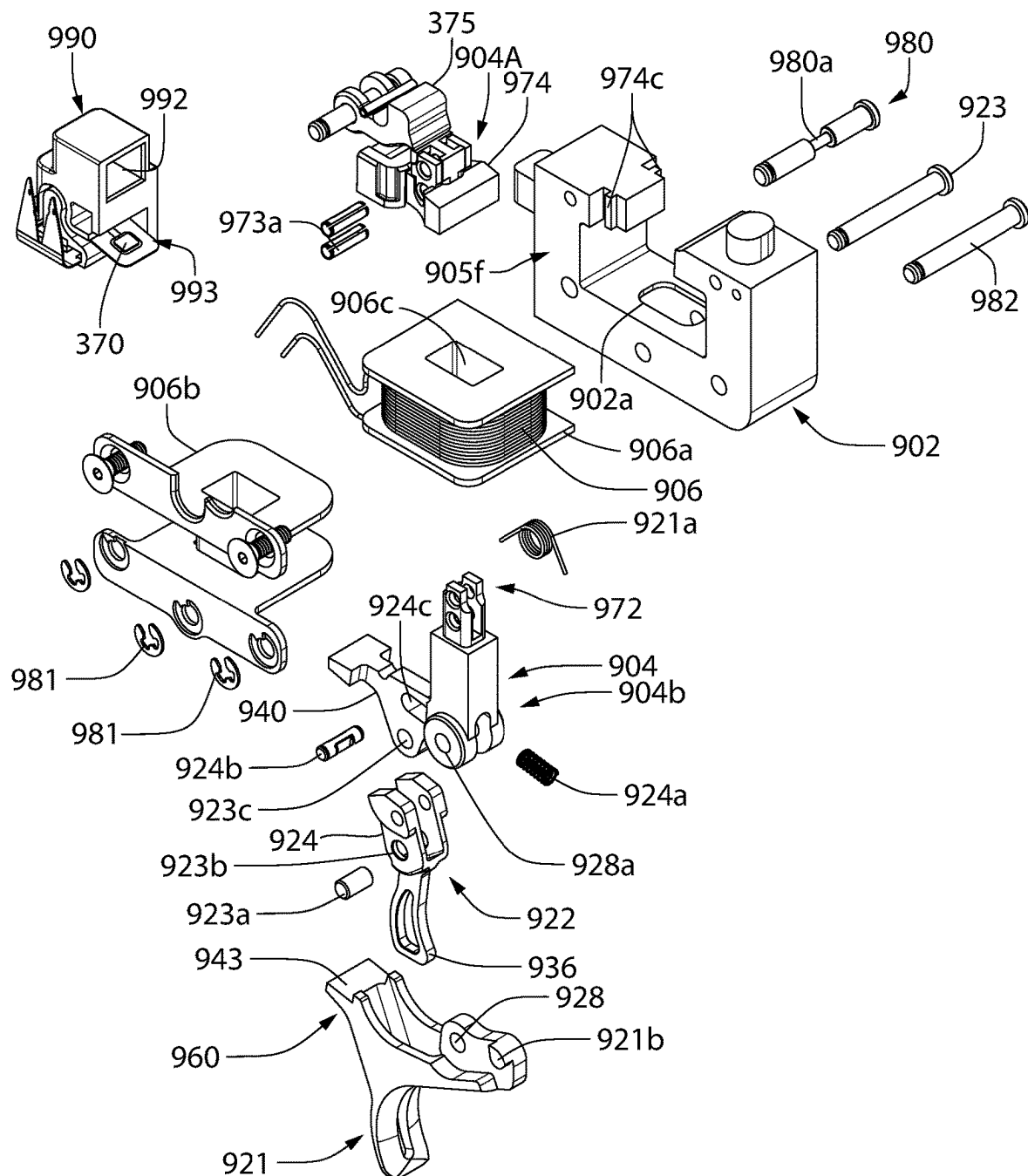


FIG. 88

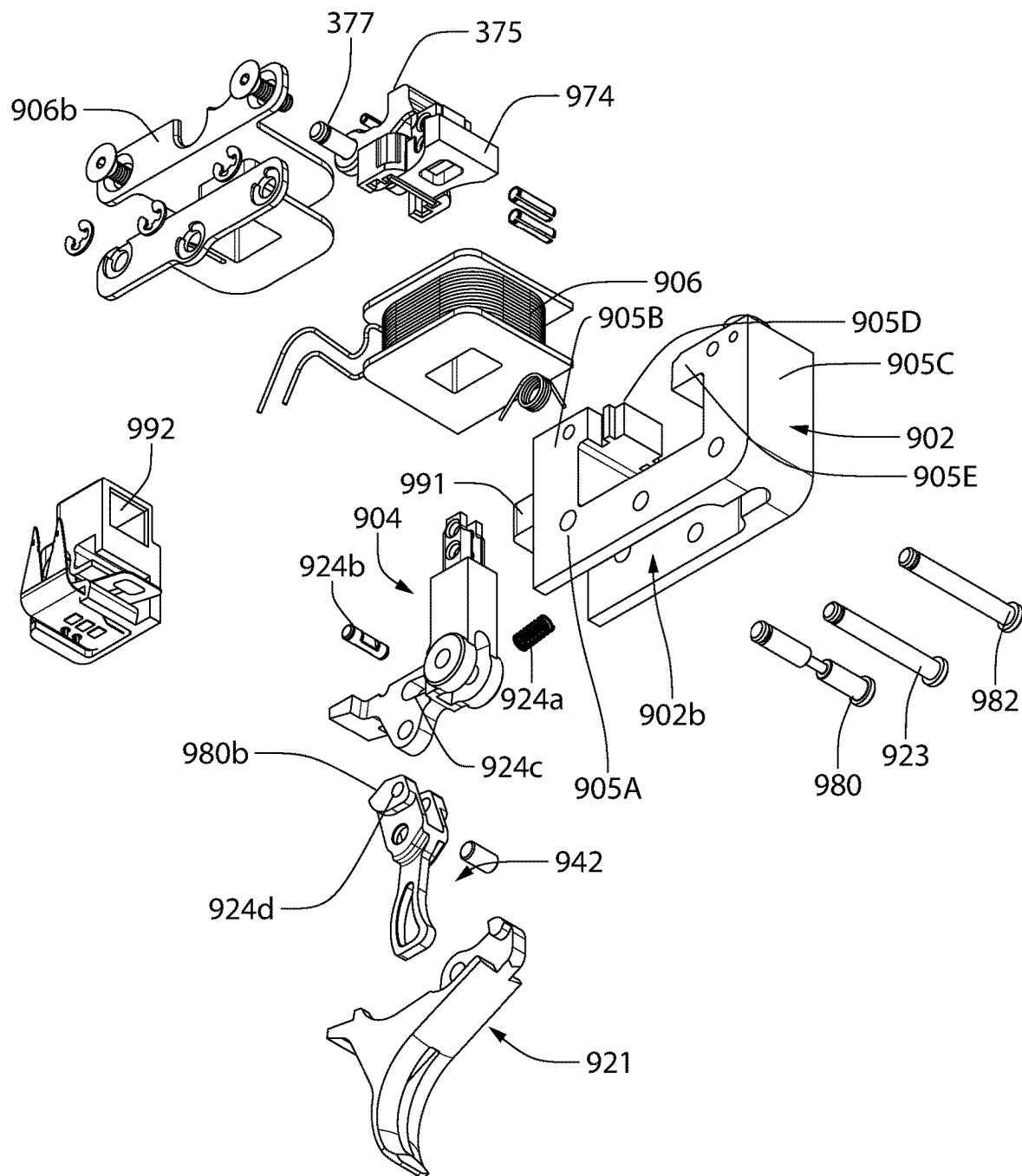


FIG. 89

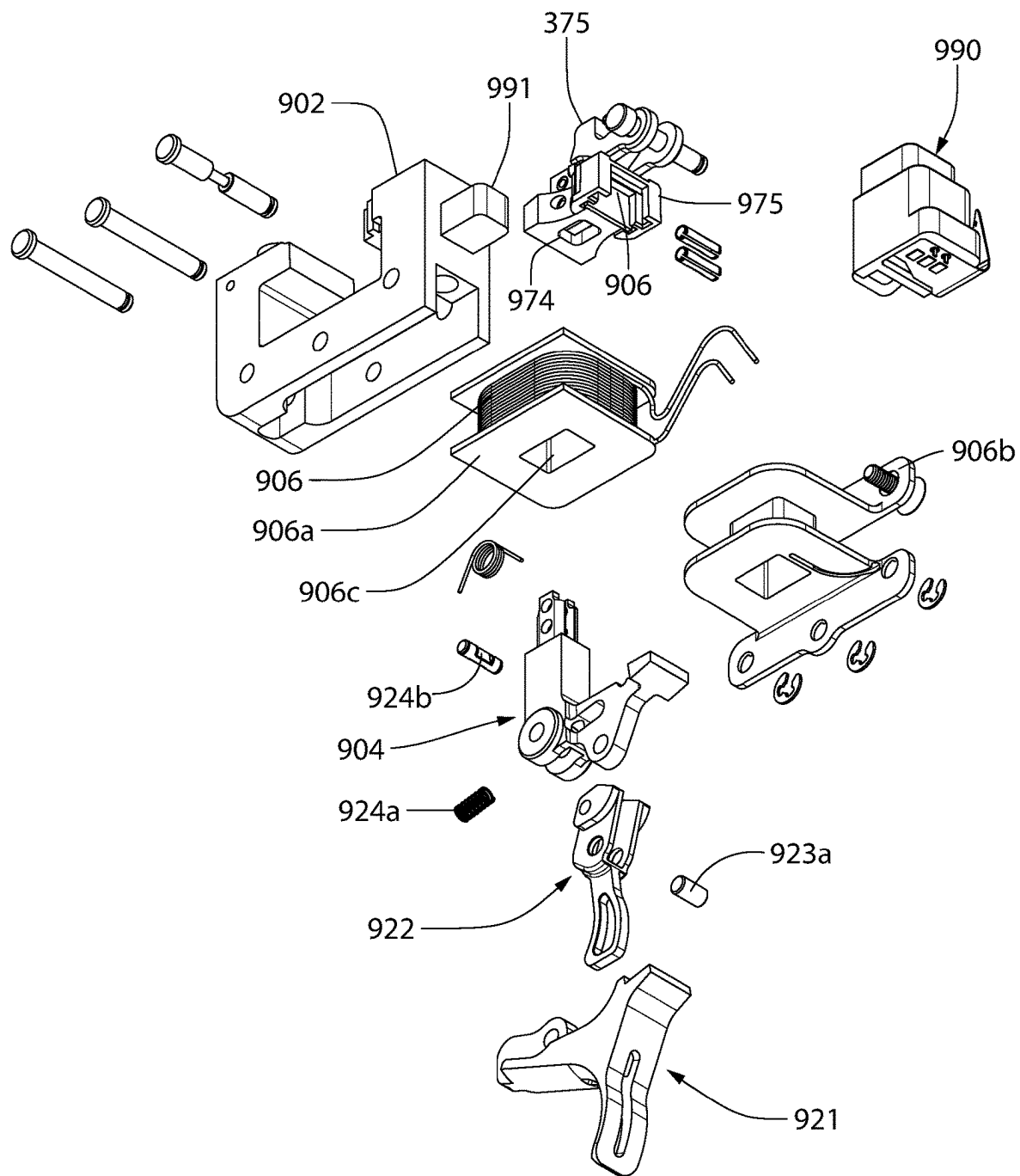


FIG. 90

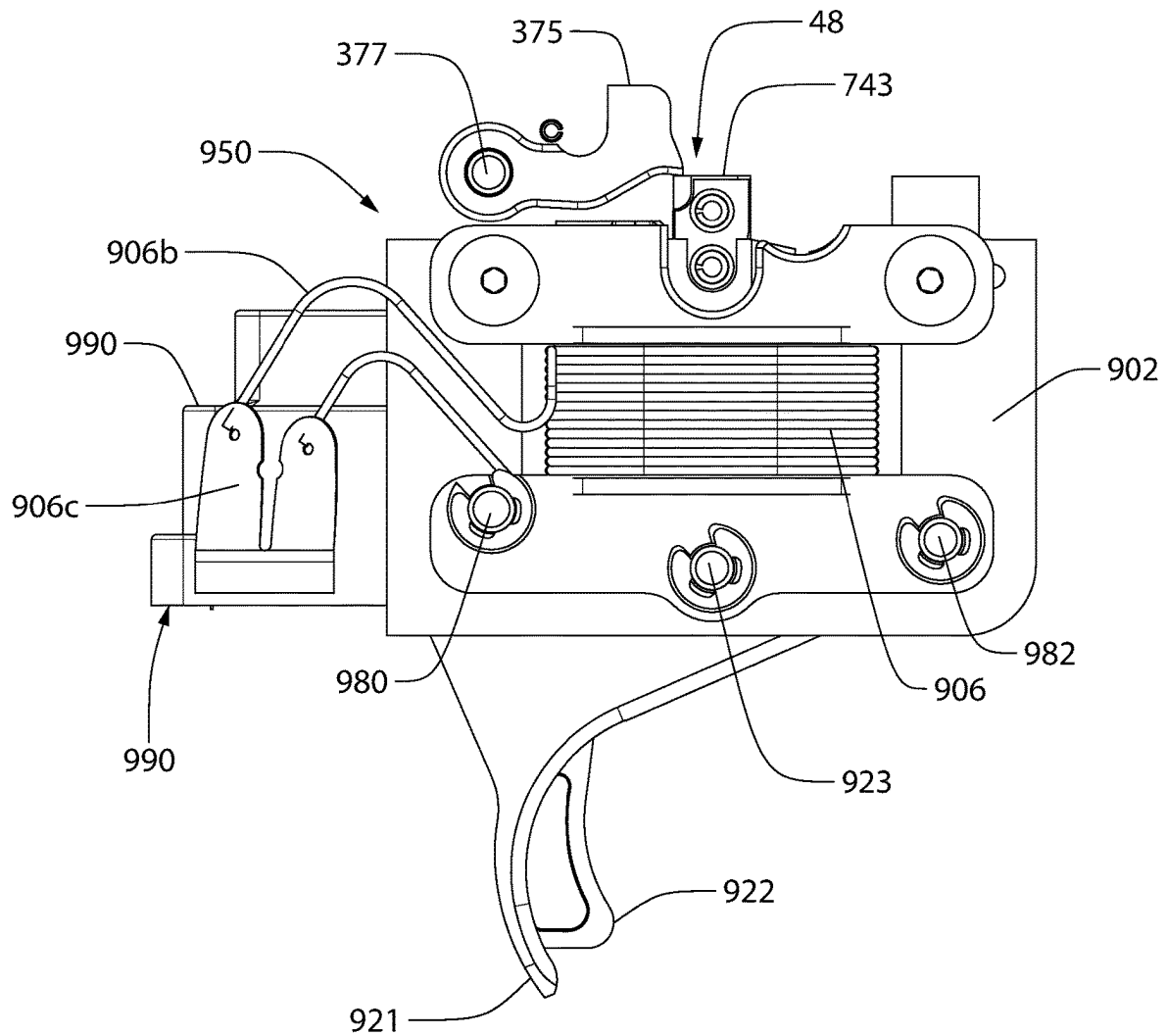


FIG. 91

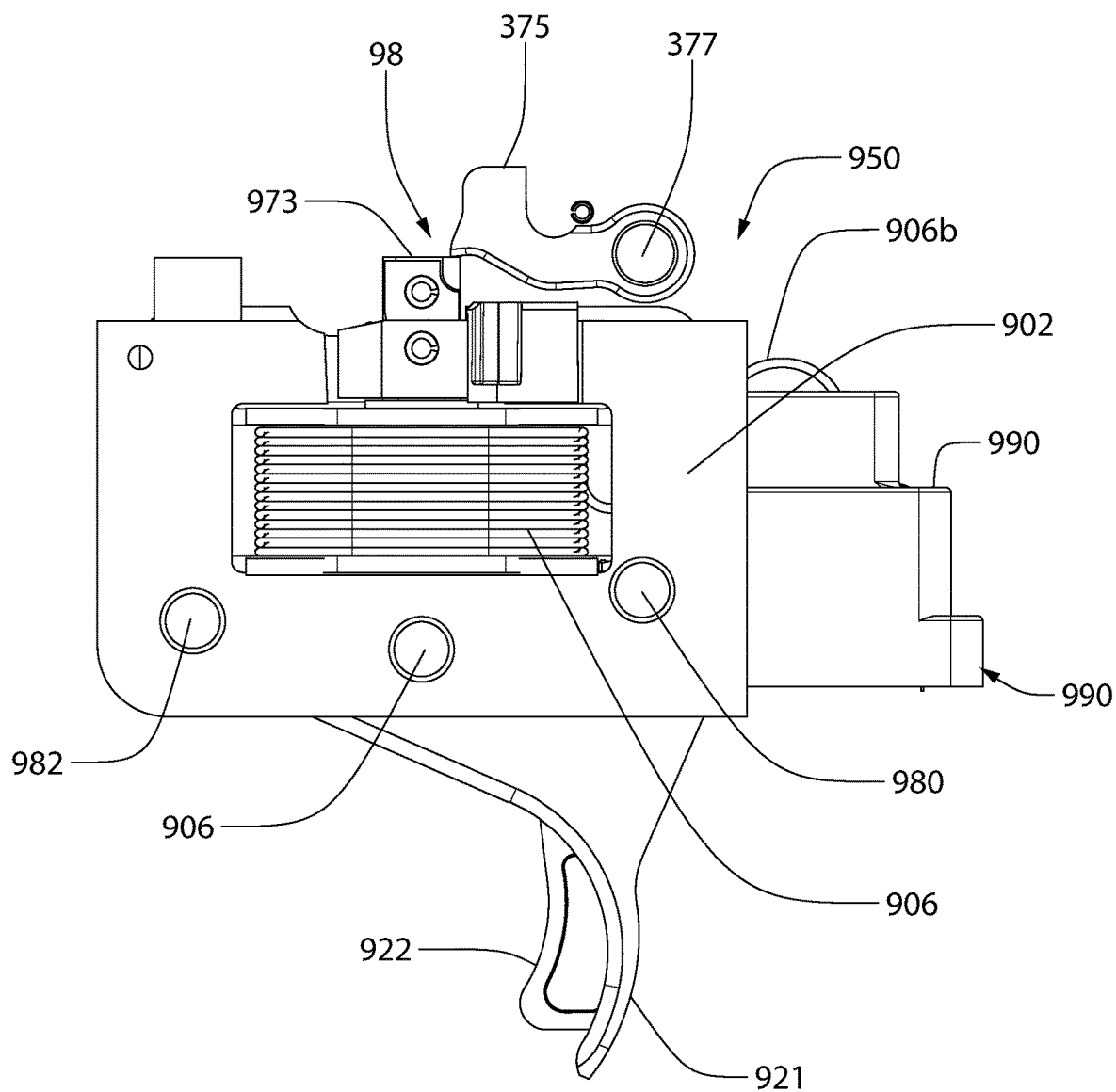


FIG. 92

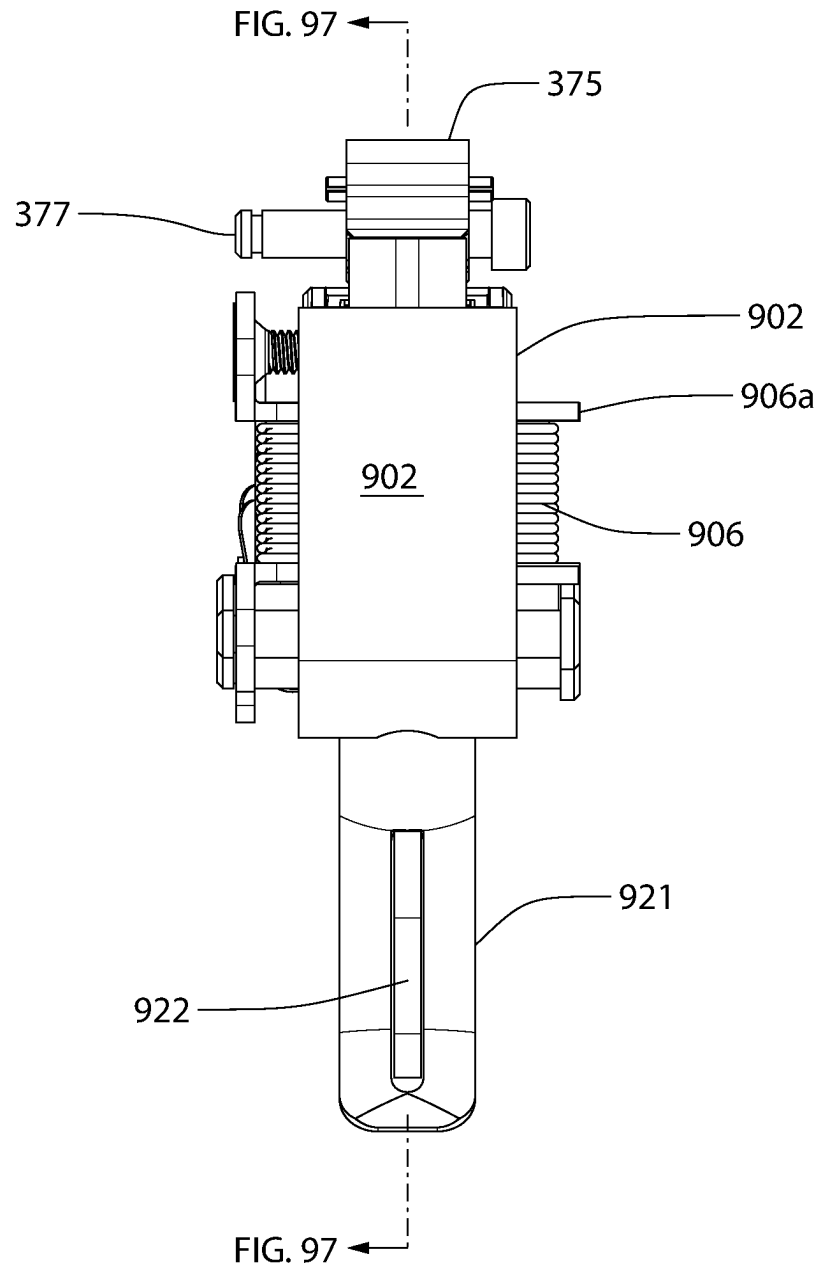


FIG. 93

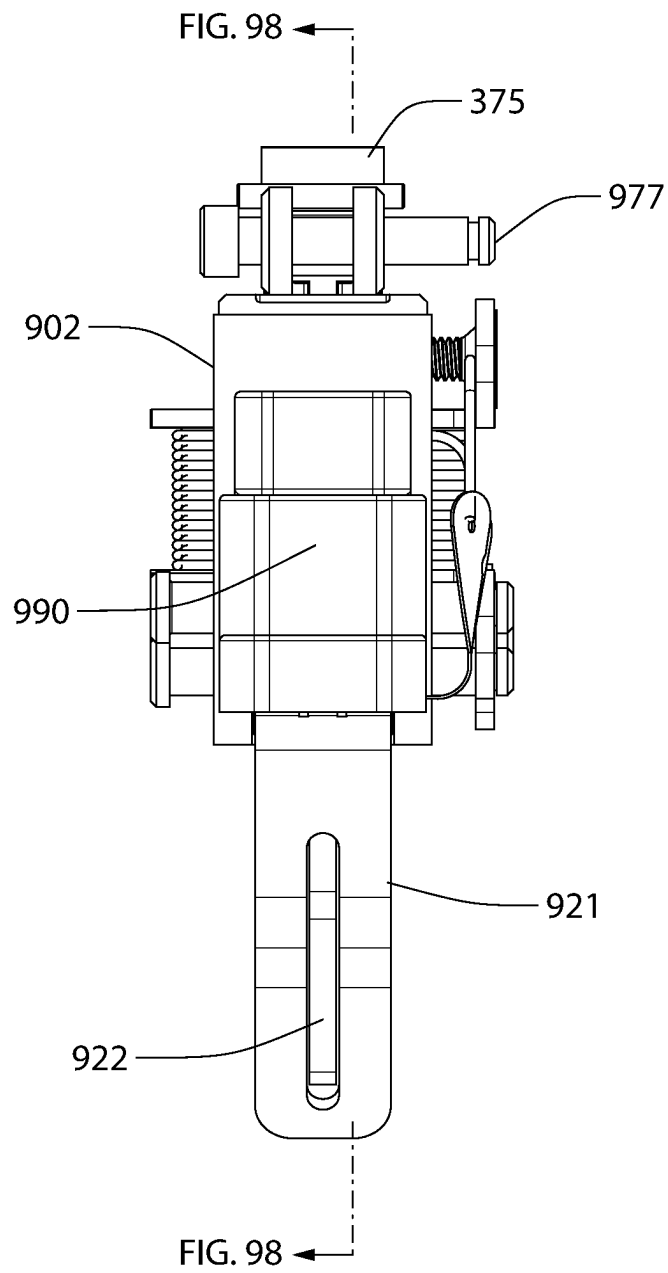


FIG. 94

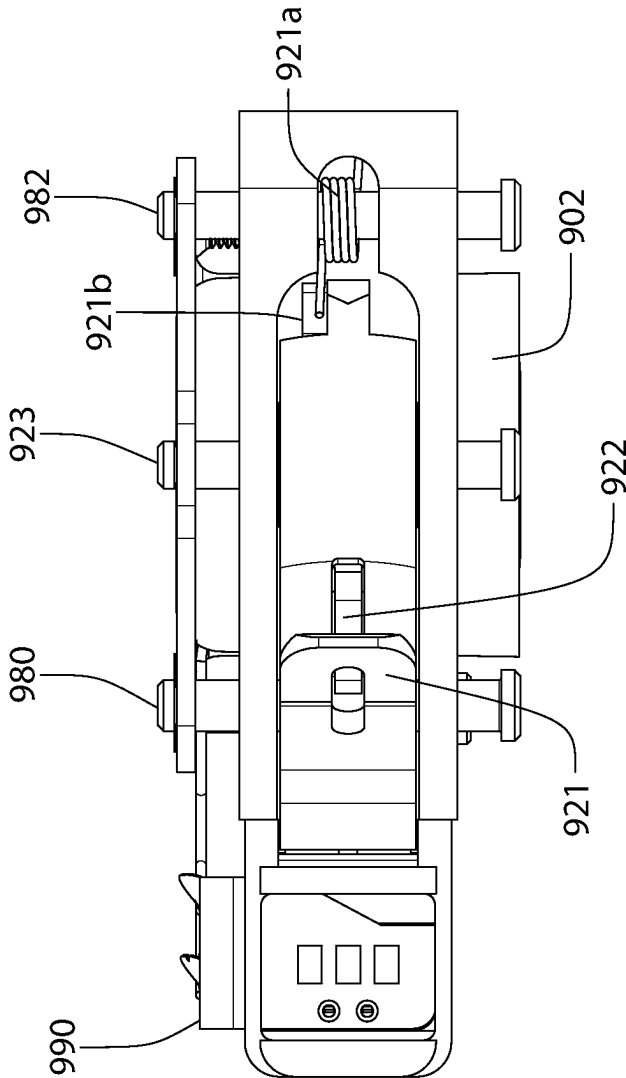


FIG. 95

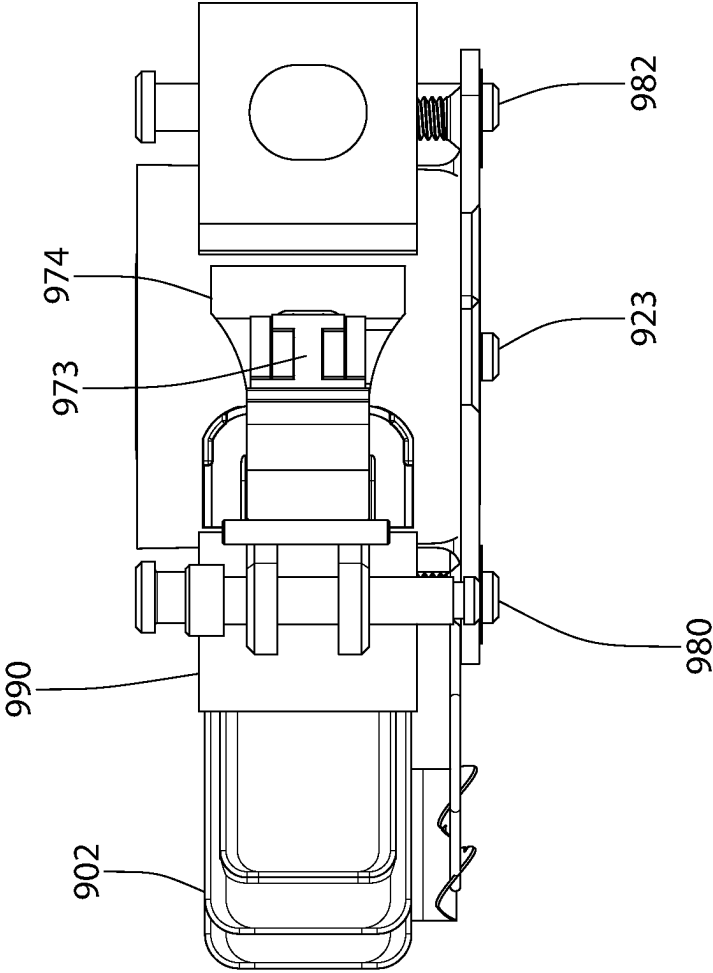


FIG. 96

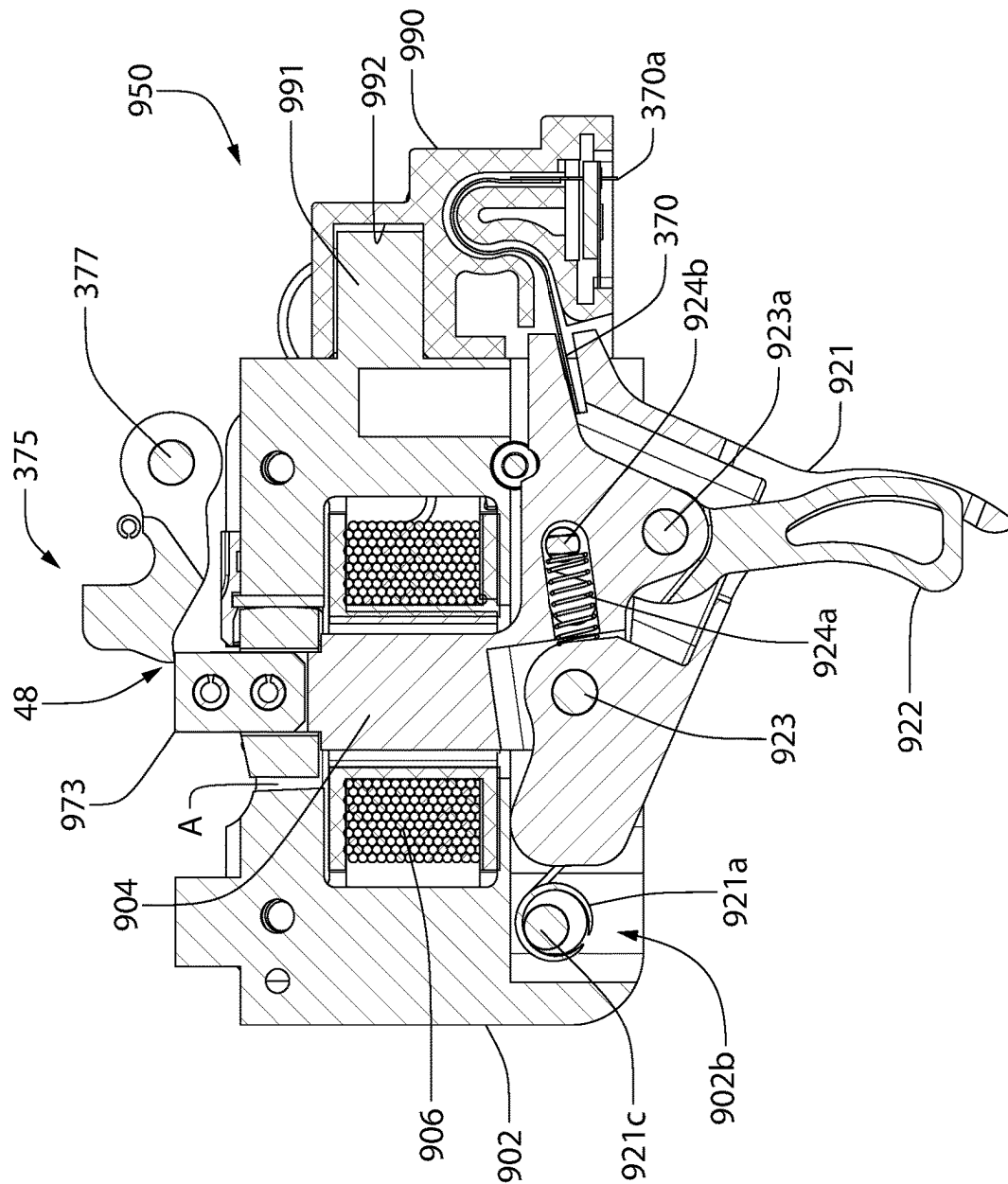
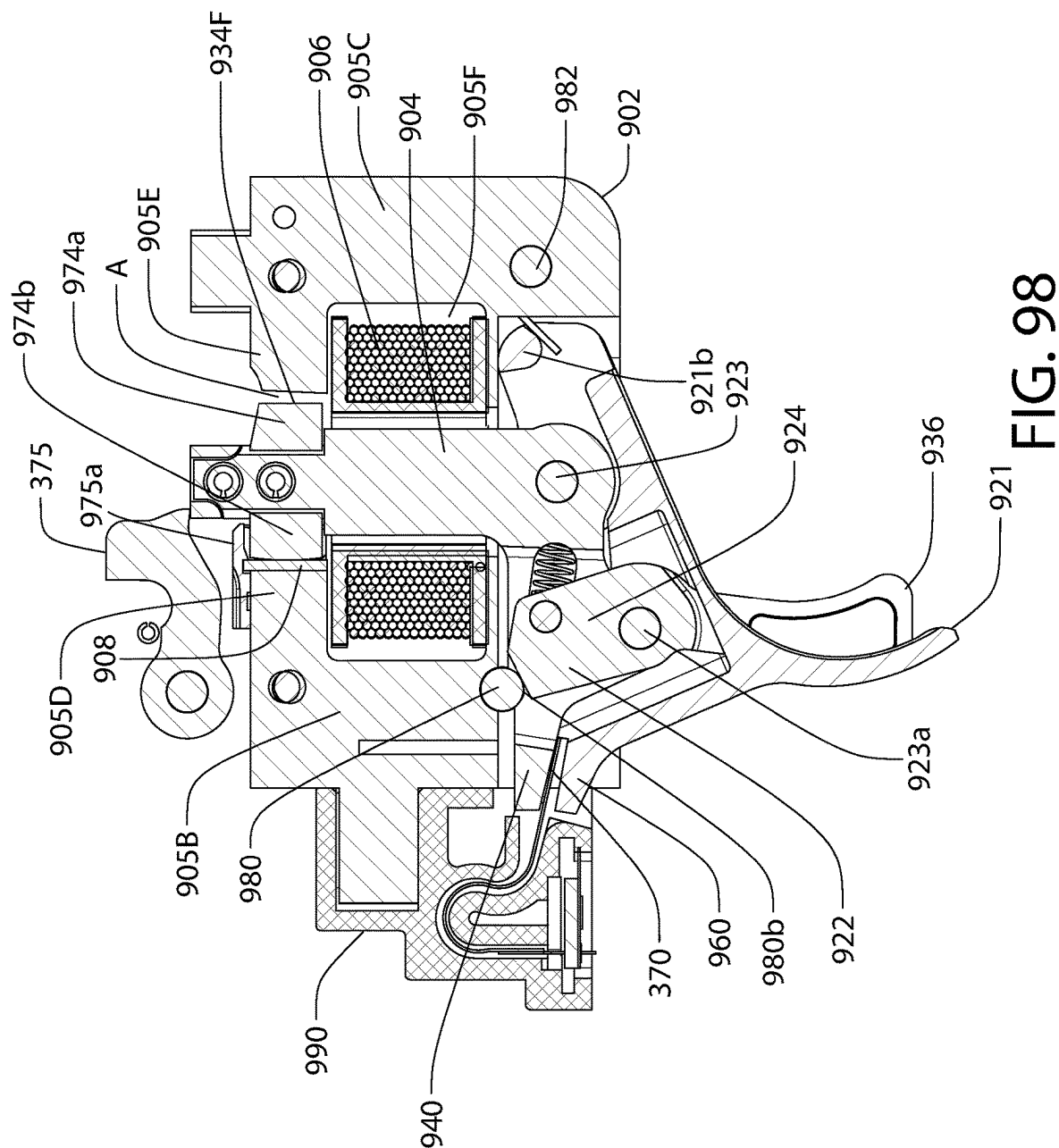


FIG. 97



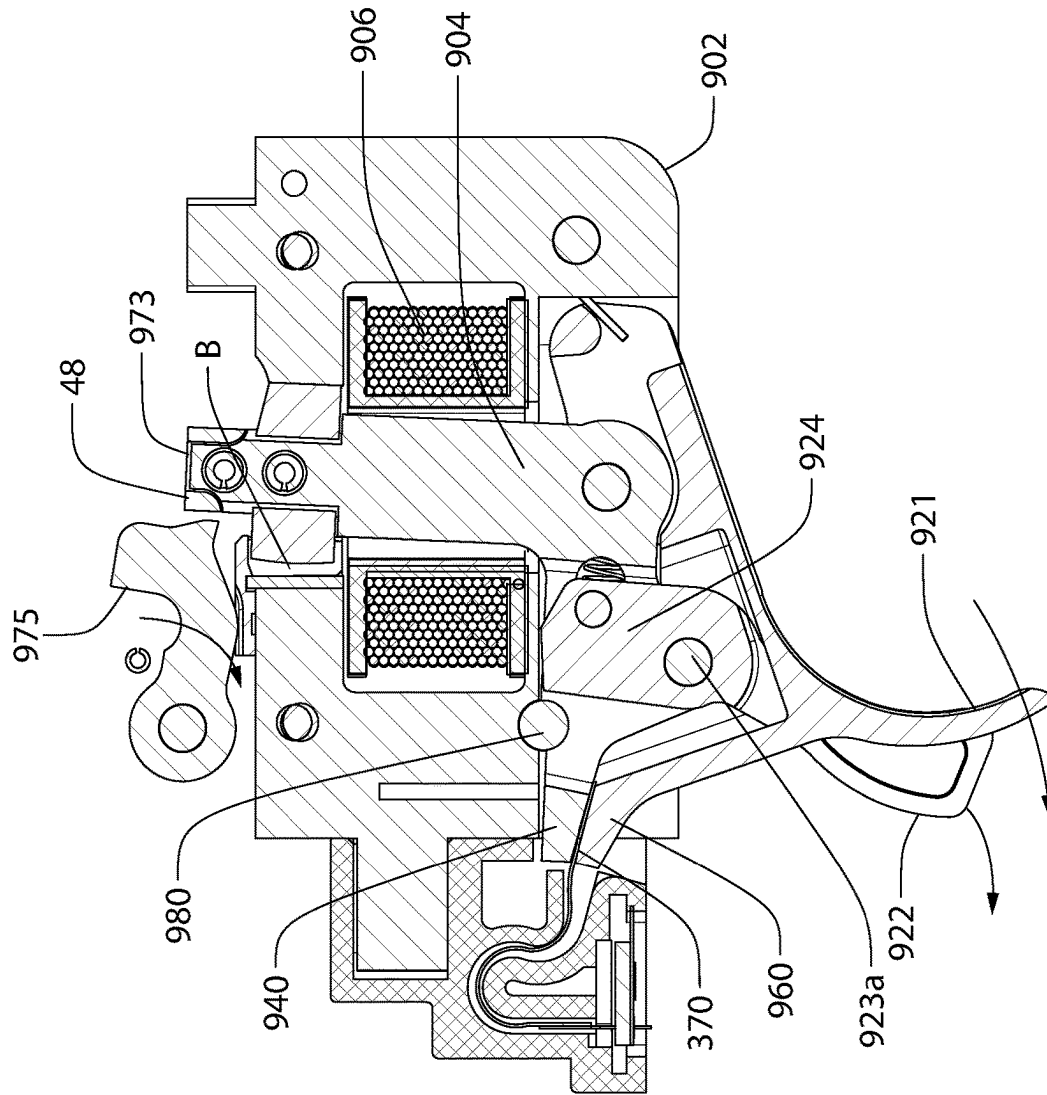


FIG. 99

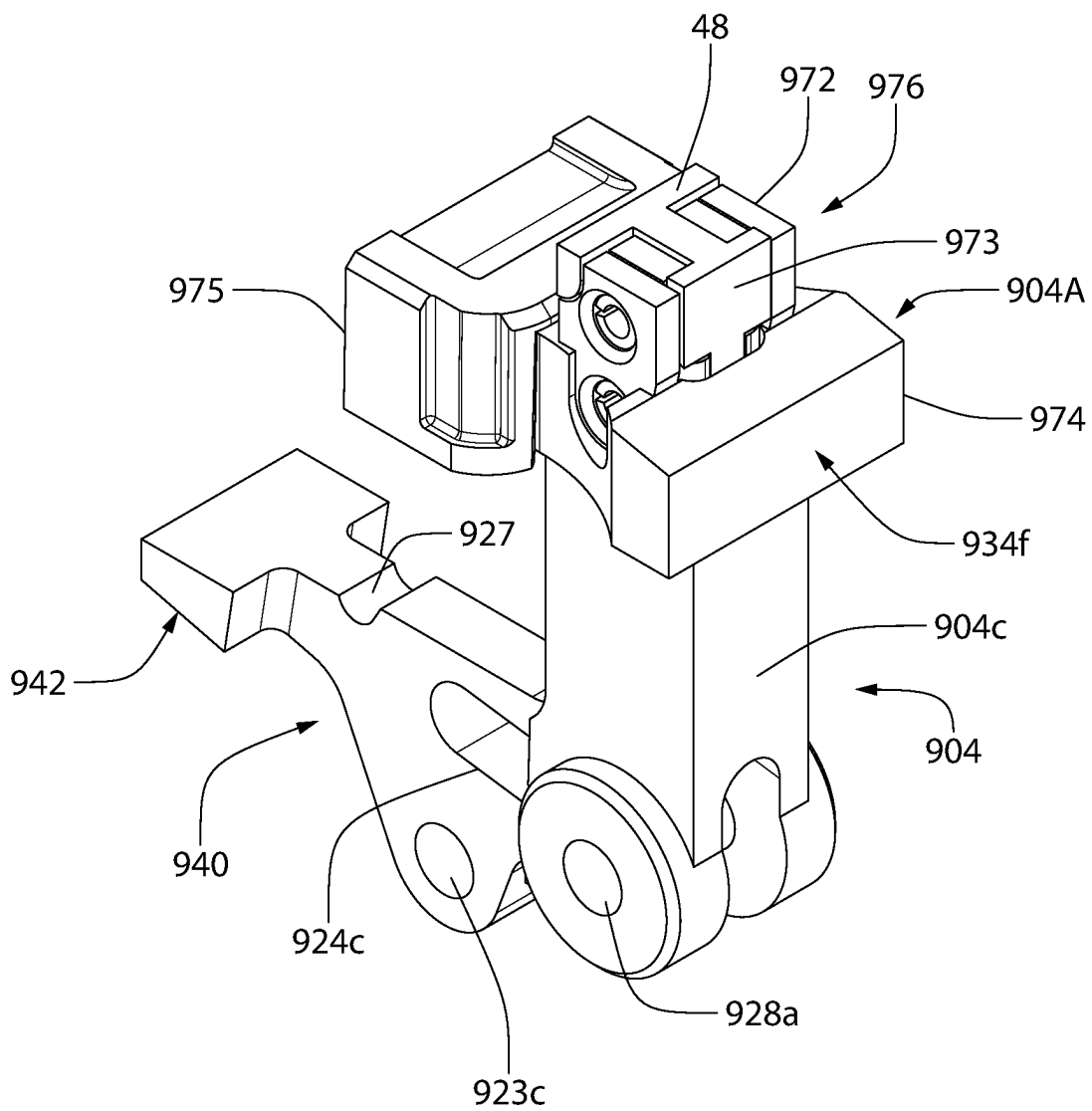


FIG. 100

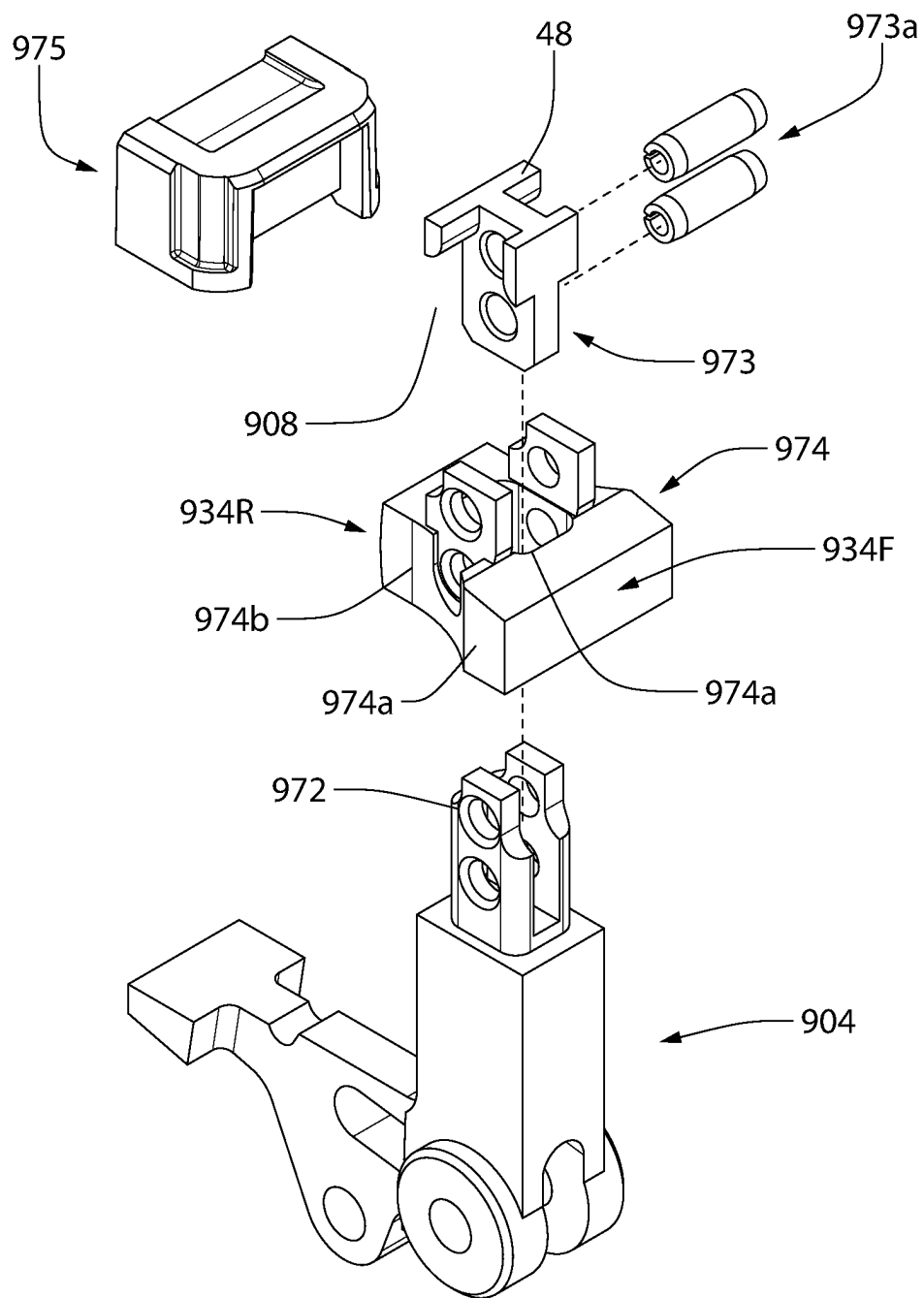


FIG. 101

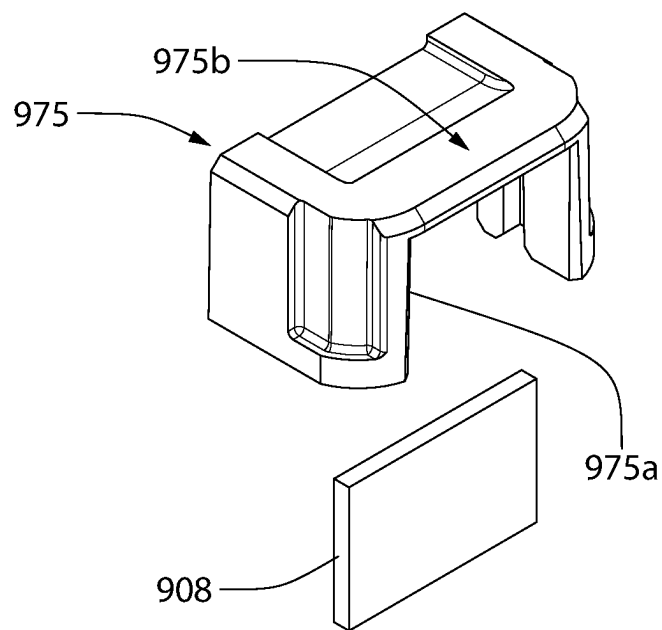


FIG. 102

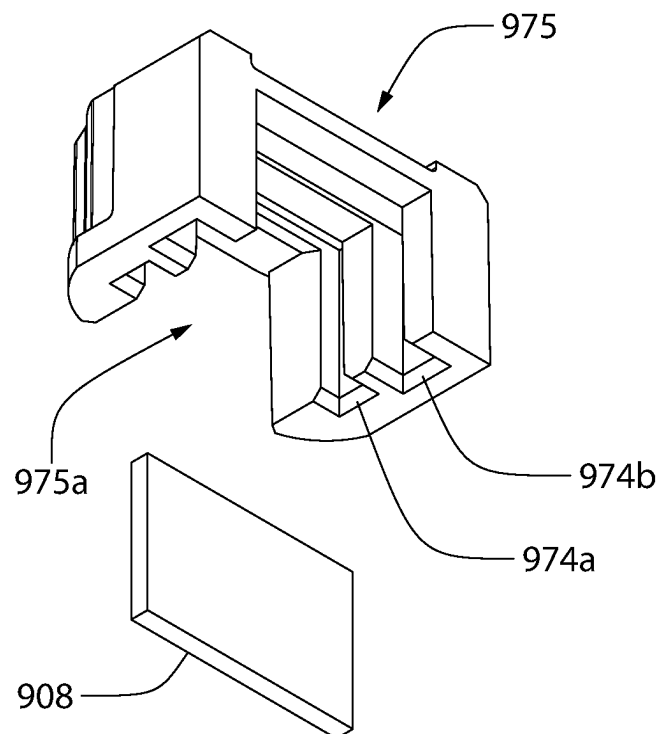


FIG. 103

SAFETY FOR FIREARM ELECTROMAGNETIC FIRING SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation-in-part of U.S. patent application Ser. No. 17/158,139 filed Jan. 26, 2021, which is a continuation-in-part of U.S. patent application Ser. No. 16/909,577 filed Jun. 23, 2020 (now U.S. Pat. No. 10,900,732); which is a continuation-in-part of U.S. patent application Ser. No. 16/530,545 filed Aug. 2, 2019 (now U.S. Pat. No. 10,690,430), which is a continuation of U.S. patent application Ser. No. 16/283,338 filed Feb. 22, 2019 (now U.S. Pat. No. 10,458,736), which: (1) claims priority to U.S. Provisional Application No. 62/635,598 filed Feb. 27, 2018; and (2) is a continuation-in-part of U.S. patent application Ser. No. 15/908,883 filed Mar. 1, 2018 (now U.S. Pat. No. 10,228,208), which claims the benefit of priority to U.S. Provisional Application No. 62/468,632 filed Mar. 8, 2017. The foregoing applications/patents are incorporated herein by reference in their entireties.

BACKGROUND OF THE DISCLOSURE

The present invention relates to firearms, and more particularly to an energizable electromagnetic trigger mechanism for the firing system of a firearm which provides a dynamically adjustable force and displacement profile for a trigger customizable by a user.

Traditional triggers for firearms provide a decisive intent-to-fire signal through mechanical motion that utilizes a displacement and force profile developed by using mechanical linkages, springs and the release of energy stored in a spring-biased hammer, striker, or sear. The trigger force and displacement curve or profile is normally fixed by these mechanical linkages and springs. A number of designs exist that provide adjustable characteristics for the force and displacement of the trigger using set screws, additional springs, or part changes to customize the force-displacement profile of firearm triggers mechanically.

An improved variable force trigger is desired which allows the trigger force-displacement profile to be more quickly and easily altered in a dynamically changeable manner without resort to strictly adjusting the position of mechanical components or physically exchanging such mechanical components and/or other hardware of the trigger mechanism.

SUMMARY OF THE DISCLOSURE

An electromagnetically variable firing system for a firearm according to the present disclosure includes a trigger assembly or mechanism having an electromagnetically-operated control device which allows the user to preselect and adjust the trigger pull force-displacement profile electronically in an expeditious non-mechanical manner in one embodiment. The preselected trigger force may be implemented automatically and dynamically during the course of a trigger pull event based on sensing an applied force to the trigger by the user to initiate the firing sequence.

The electromagnetic control device is an integral part of the trigger mechanism, which in turn operably interfaces with other components of the firing system for discharging the firearm. The electromagnetically variable firing system may include a movable energy storage device such as a spring-biased cockable striking member such as a pivotable

hammer or linearly-movable striker for striking a chambered ammunition cartridge or round, a movable sear operable to hold and release the hammer or striker from the cocked position, and other associated firing mechanism components which collectively operate together to discharge the firearm when actuated via a manual trigger pull. In some embodiments, the sear may be formed as an integral unitary structural part of the trigger mechanism instead of being a separate component.

In certain implementations, the trigger pull force and displacement profile is electrically/electronically adjustable via the trigger control device by changing or altering a magnetic field acting on a portion of the trigger mechanism, thereby increasing or decreasing resistance of the trigger to movement. The trigger pull force required may vary with displacement distance or travel of the trigger when actuated by the operator or user such that the initial trigger pull force may have an initial value or magnitude during the first stage or phase of the trigger pull (e.g. hard or easy) which is then followed by either a constant or varying different second values or magnitudes of trigger pull force during the subsequent and final phases of the trigger pull until the firearm is discharged.

To power, monitor, and control operation of the trigger control device and trigger mechanism including adjustment of the trigger pull force and displacement profile, the firearm may include a control system including a suitable power source (e.g. battery) mounted to a frame of the firearm or module attached thereto, and a programmable electronic processor such as a microprocessor or microcontroller including circuitry, memory, data storage devices, sensors, sensor and drive circuits, communication devices and interfaces (e.g. wired or wireless protocols), and other electronic devices, components, and circuits necessary for a fully functional microprocessor based control system. The microcontroller may preferably be disposed onboard the firearm. The microcontroller is operably coupled to the power source to control via an actuation control circuit to energize or de-energize the trigger control device.

In one embodiment, the electromagnetically-operated trigger control device may comprise a magnetorheological fluid device or operator which is selectably alterable electrically/electronically via the microcontroller to vary the trigger pull force and displacement profile characteristics.

In another embodiment, the electromagnetically-operated trigger control device may comprise a magnetic device or operator such as an electromagnetic snap actuator of a non-bistable design which is selectably alterable electrically/electronically via the microcontroller to vary the trigger pull force and displacement profile characteristics by altering the magnet field force of the trigger mechanism. The electromagnetic actuator forms an integral part of the trigger mechanism, and in some embodiments may constitute substantially the entirety of the trigger mechanism with minimal appurtenances for operational simplicity and reliability. The electromagnetic actuator may generally include a stationary yoke attached to the firearm frame, a rotatable member pivotably movable relative to the yoke, and an electromagnet coil electrically connected to the on-firearm electric power source. In some implementations, the trigger mechanism may be configured to establish a closed single or double flux loop that limits susceptibility to external magnetic fields which might inadvertently change the trigger pull force or displacement of the trigger mechanism. This completely contained flux loop around the permanent mag-

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net optimizes the magnetic coupling force between the yoke and rotating member making this design inherently resistant to external magnetic fields.

Certain implementations of the control device may also employ mechanical components to assist with adjusting the trigger pull force and displacement profile. The trigger control device may be used as an on/off safety in some embodiments, and/or to vary trigger pull force which may be adjusted by the user to meet personal preferences.

Embodiments of the present electromagnetic trigger mechanisms may be employed with any type of trigger-operated small arms including without limitation as some examples pistols, revolvers, long guns (e.g. rifles, carbines, shotguns), grenade launchers, etc. Accordingly, the present invention is expressly not limited in its applicability and breadth of use.

Accordingly, embodiments of the present invention provide a trigger mechanism or assembly for use in a firearm that provides a changeable and variable force of resistance (i.e. trigger pull force) as the trigger moves and is displaced in distance.

The foregoing or other embodiments of the present invention may control the change in resistance force dynamically during the actual displacement of the trigger linkage by the operator or user at the time of operation.

The foregoing or other embodiments of the present invention provide that the trigger force can be controlled by varying the viscosity of a magnetorheological fluid incorporated into the trigger mechanism.

The foregoing or other embodiments of the present invention provide that the trigger force can be controlled by varying the magnetic field of an electromagnetic snap actuator incorporated into and configured as a trigger mechanism or assembly for discharging the firearm.

The foregoing or other embodiments of the present invention provide that the trigger force can be programmed remotely from an external smartphone, tablet, personal wearable device, or other remote device using a wireless communications standard such as Bluetooth, BLE (Bluetooth Low Energy), NFC (Near-Field Communication), LoRa (Long Range wireless), WiFi, or a proprietary wireless protocol or other protocol.

The foregoing or other embodiments of the present invention may be configured to capture cycle count and direct sensing of the trigger mechanism for the implementation of data collection on the performance and operation of the device. Shot counting, shot timing, pre-fire trigger analysis, and post firing performance analysis can be tied to internal sensing of the trigger event and electrically interfaced to the user through external electronic devices, such as without limitation cellphones, tablets, pads, wearables, or web applications.

In one aspect, an electromagnetically variable trigger force firing system comprises: a frame; a striking member supported by the frame for movement between a rearward cocked position and forward firing position for discharging the firearm; an electromagnetic actuator trigger unit affixed to the frame and comprising: a stationary yoke comprising an electromagnet coil; a rotating member movable about a pivot axis relative to the stationary yoke and operable for releasing the striking member from the cocked position to the firing position; a trigger operably engaged with the rotating member, the trigger manually movable by a user from a first position to a second position which rotates the rotating member for discharging the firearm; and a permanent magnet generating a static magnetic field in the stationary yoke and rotating member, the static magnetic field

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creating a primary resistance force opposing movement of the trigger when pulled by the user; an electric power source operably coupled to the coil; the electromagnet coil when energized generating a user-adjustable secondary magnetic field interacting with the static magnetic field, the secondary magnetic field operating to change the primary resistance force dynamically during a trigger pull event initiated by the user.

In another aspect, an electromagnetic firing system for a firearm comprises: a frame; a striking member supported by the frame and movable between a rearward cocked position and forward firing position for discharging the firearm; an electromagnetically adjustable trigger mechanism operably coupled to the striking member for discharging the firearm, the trigger mechanism comprising an electromagnetic actuator including: a stationary yoke comprising an electromagnet coil operably coupled to an electric power source, the coil having an energized state and a de-energized state; a rotating member pivotably coupled to the stationary yoke for movement between an unactuated and actuated positions, the rotating member operably coupled to the striking member for moving the striking member from the cocked position to the firing position; a trigger movably coupled to the stationary yoke and interacting with the rotating member, the trigger manually movable by a user from a first actuation position to a second actuation position which rotates the rotating member for discharging the firearm; and a permanent magnet generating a static magnetic flux in the yoke and rotating member, the static magnetic flux creating a primary resistance force opposing movement of the trigger when pulled by the user; a programmable microcontroller operably coupled to the electromagnetic actuator of the trigger mechanism and pre-programmed with a trigger force setpoint, the microcontroller configured to: receive an actual trigger force applied to the trigger by a user and measured by a trigger sensor communicably coupled to the microcontroller; compare the actual trigger force to the preprogrammed trigger force setpoint; and selectively energize the electromagnetic actuator based on the comparison of the actual trigger force to the trigger force setpoint; wherein the electromagnet coil when energized generates a user-adjustable secondary magnetic flux interacting with the static magnetic field, the secondary magnetic field operating to increase or decrease the primary resistance force when the trigger is pulled by the user.

In another aspect, an electromagnetic firing system for a firearm comprises: a frame; a striking member supported by the frame and movable between a rearward cocked position and forward firing position for discharging the firearm; a pivotable sear configured to selectively hold the striking member in the cocked position; an electromagnetic actuator trigger mechanism supported by the frame, the trigger mechanism configured to create a dual loop magnetic flux circuit and comprising: a stationary yoke comprising an electromagnet coil operably coupled to an electric power source, the coil having an energized state and a de-energized state; a rotating member pivotably coupled to the stationary yoke about a pivot axis, the rotating member movable between an unactuated position engaging with the sear and an actuated position disengaging the sear; a trigger operably engaged with the rotating member and manually movable by a user for applying an actual trigger force on the rotating member; and a permanent magnet generating a static magnetic flux holding the rotating member in the unactuated position, the permanent magnet generating a static magnetic flux creating a primary resistance force opposing movement of the trigger when pulled by the user; a programmable

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microcontroller operably coupled to the power source and communicably coupled to a trigger sensor configured to sense the applied trigger force, the microcontroller when detecting the applied trigger force being configured to transmit an electric pulse to the electromagnet coil of the trigger mechanism; the electromagnet coil when energized generating a secondary magnetic flux interacting with the static magnetic field, the secondary magnetic field being configurable by the user via the microcontroller to increase or decrease the primary resistance force when the trigger is pulled by the user.

In another aspect, an electromagnetically variable trigger system comprises: a frame; an electromagnetic actuator trigger unit affixed to the frame and comprising: a stationary yoke comprising an electromagnet coil; a rotating member movable about a pivot axis relative to the stationary yoke; a trigger operably engaged with the rotating member, the trigger manually movable by a user from a first position to a second position which rotates the rotating member; and a permanent magnet generating a static magnetic field in the stationary yoke and rotating member, the static magnetic field creating a primary resistance force opposing movement of the trigger when pulled by the user; an electric power source operably coupled to the coil; the electromagnet coil when energized generating a user-adjustable secondary magnetic field interacting with the static magnetic field, the secondary magnetic field operating to change the primary resistance force dynamically during a trigger pull event initiated by the user. The trigger system may further comprise an electronic actuation control circuit operably coupled between to the power source and coil, the actuation control circuit configurable by the user to selectively energize the coil upon detection of a trigger pull and de-energize the coil in an absence of the trigger pull, and a trigger sensor communicably coupled to the actuation control circuit and operable to detect movement of the trigger initiated by the user.

The present application further discloses non-electric magnetic only trigger mechanisms of the closed and open magnetic loop designs.

According to one aspect, a closed loop magnetically variable trigger force trigger mechanism for a firearm comprises: a stationary yoke configured for mounting to the firearm; a rotatable trigger member pivotably coupled to the stationary yoke about a pivot axis, the trigger member and stationary yoke collectively configured to form a closed magnetic loop; an openable and closeable first air gap formed between the trigger member and the stationary yoke; a permanent magnet arranged to generate a static magnetic field in the closed magnetic loop, the static magnetic field creating a primary resistance force opposing movement of the trigger member when pulled by the user; a control insert selectively movable relative to a second control air gap formed in the yoke which attenuates the static magnetic field, the control insert constructed and operable to change the static magnetic field; wherein the static magnetic field is changeable via varying position of the control insert relative to the control air gap to adjust a trigger pull force of the trigger mechanism.

In another aspect, a closed loop magnetically variable trigger force trigger mechanism for a firearm comprises: a stationary yoke configured for mounting to the firearm; a rotatable trigger member pivotably movable about a pivot axis relative to the stationary yoke, the trigger member and stationary yoke collectively configured to form a closed magnetic loop; an openable and closeable first air gap formed between the trigger member and the stationary yoke;

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a control insert selectively movable into and out of a second control air gap formed in the yoke which attenuates the static magnetic field, the control insert operable to change the static magnetic field; the control insert comprising a non-magnetic carrier and a permanent magnet operable to generate a static magnetic field in the closed magnetic loop, the static magnetic field creating a primary resistance force opposing movement of the trigger member when pulled by the user; wherein the static magnetic field is changeable via varying position of the permanent magnet in the control insert relative to the second control air gap to adjust a trigger pull force of the trigger mechanism.

In another aspect, a closed loop magnetically variable trigger force trigger mechanism for a firearm comprises: a stationary yoke configured for mounting to the firearm; a rotatable trigger member pivotably movable about a pivot axis relative to the stationary yoke, the trigger member and stationary yoke collectively configured to form a closed magnetic loop; an openable and closeable first air gap formed between the trigger member and the stationary yoke; a control insert comprising a permanent magnet rotatably disposed in a second control air gap formed in the yoke which attenuates the static magnetic field, the permanent magnet operable to generate a static magnetic field in the closed magnetic loop, the static magnetic field creating a primary resistance force opposing movement of the trigger member when pulled by the user; wherein the static magnetic field is changeable via rotating the permanent magnet of the control insert relative to the second control air gap to adjust a trigger pull force of the trigger mechanism.

In another aspect, a method for adjusting the trigger pull force of a closed loop magnetically variable trigger force trigger mechanism for a firearm comprises: providing a stationary yoke configured for mounting in the firearm, a rotating trigger member pivotably movable about a pivot axis relative to the stationary yoke, the trigger member and stationary yoke collectively configured to form a closed magnetic loop, and an openable and closeable first air gap being formed between the trigger member and the stationary yoke; providing a control insert comprising a non-magnetic carrier and a permanent magnet operable to generate a static magnetic field in the closed magnetic loop, the static magnetic field creating a primary resistance force opposing movement of the trigger member when pulled by the user; rotating an actuator operably coupled to the control insert in a first direction to advance the permanent magnet into a second control air gap formed in the stationary yoke, the magnet creating a first static magnetic field strength in the closed magnetic loop which resists movement of the trigger member relative to the stationary yoke at the first air gap; rotating the actuator in an opposite second direction to withdraw the magnet from the second control air gap, the magnet creating a second static magnetic field strength in the closed magnetic loop less than the first magnetic field strength; wherein the strength of the static magnetic field is changeable via varying position of the permanent magnet in the control insert relative to the second control air gap in order to adjust a trigger pull force of trigger mechanism.

The present disclosure further discloses a microcontroller-operated firing event (shot) tracking system.

In one aspect, an electromagnetic firing system for a firearm with firing event tracking comprises: an electromagnetic actuator trigger unit comprising: a stationary yoke configured for mounting to the firearm; a rotating member movable about a pivot axis relative to the stationary yoke and operably coupled to a firing mechanism of the firearm; a trigger operably coupled to the rotating member, the

trigger manually movable by a user from a first position to a second position which rotates the rotating member for discharging the firearm; and a permanent magnet generating a static magnetic field in the stationary yoke and rotating member, the static magnetic field creating a primary resistance force opposing movement of the trigger when pulled by the user; a magnetic coil operably coupled to an electric power source and the yoke or rotating member; the magnetic coil when energized generating a user-adjustable secondary magnetic field interacting with the primary resistance force which changes a trigger pull force required to be exerted by a user to overcome the primary resistance force and discharge the firearm in response to a trigger pull event; a programmable microcontroller configured to detect the trigger pull event and selectively energize the coil via the power source in accordance with a user-selected trigger force or displacement setpoint preprogrammed into the microcontroller thereby defining a firing event; the microcontroller further configured to record and store each firing event and an associated time/date stamp.

In another aspect, an electromagnetic firing system for a firearm with firing event tracking comprises: a trigger unit mounted in the firearm, the trigger unit comprising: an electromagnetic actuator including a stationary yoke, a rotating member movable about a pivot axis relative to the stationary yoke and operably coupled to a firing mechanism of the firearm, a trigger operable when pulled by a user to move the rotating member between an unactuated position and an actuated position for discharging the firearm, and a magnetic coil when energized generating a user-adjustable magnetic field which changes a trigger pull force required to be exerted by a user on the trigger to discharge the firearm; a programmable microcontroller operably coupled to the electromagnetic actuator and configured to selectively energize the coil for discharging the firearm in response to detecting a trigger pull event; the microcontroller further configured to count each energization of the coil as indicative of a firing event and record the firing event.

In another aspect, a method for tracking firing events in a firearm with an electromagnetic firing system comprises: mounting a trigger unit in the firearm, the trigger unit comprising a trigger and an electromagnetic actuator operably coupled to the trigger and a firing mechanism of the firearm, the actuator including a magnetic coil which when energized moves the actuator from an unactuated position to an actuated position which discharges the firearm; providing a programmable microcontroller operably coupled to the actuator, the microcontroller configured to detect a trigger pull event and selectively energize the coil for discharging the firearm in response thereto; the microcontroller: detecting the trigger pull event; energizing the coil of the actuator via a power source; counting energizing the coil as indicative of a firing event; and recording the firing event in memory.

The present disclosure further discloses an interruptible electronic trigger system with microcontroller-operated electromagnetic actuator trigger unit operably interfaced with an external advanced fire control targeting system. The microcontroller-operated targeting system is configured to interact with the trigger unit and perform ballistics computations to assist the firearm user in accurately aiming the firearm and acquiring the target. The targeting system may be embodied in an adaptive optics unit mountable to the firearm for use in sighting the target by the user.

In one aspect, an interruptible electronic trigger system for a firearm comprises: an electromagnetic actuator trigger unit configured for mounting to the firearm, the trigger unit

comprising: a stationary yoke; a rotating member movable about a pivot axis relative to the stationary yoke and operably coupled to a firing mechanism component operable to discharge the firearm; a trigger operably coupled to the rotating member, the trigger manually movable by a user from a first position to a second position for discharging the firearm; a permanent magnet generating a static magnetic field in the stationary yoke and rotating member, the static magnetic field creating a primary resistance force opposing movement of the trigger when pulled by the user; a coil operably coupled to an electric power source and the yoke or rotating member, the coil when energized operable to rotate the rotating member and discharge the firearm; a programmable trigger unit microcontroller operably coupled to the trigger unit, the trigger unit microcontroller configured to: detect a trigger pull event; send a shot initiation signal to a fire control targeting system operably coupled to the trigger unit microcontroller; and receive a shot authorization signal returned from the fire control targeting system in response to receiving the shot initiation signal. The electronic trigger system is operable to revert to manual firing mode thereby allowing the user to fire the firearm mechanically when the system senses that an applied trigger force or displacement of the trigger exceeds a preprogrammed maximum allowable trigger pull force or displacement limit.

According to another aspect, a firearm with interruptible electromagnetic trigger system comprises: an electronic trigger unit mounted to the firearm and operable to discharge the firearm, the trigger unit including an electromagnetic actuator comprising: a rotating member operably coupled with a firing mechanism component movable to discharge the firearm, the rotating member rotatable about a pivot axis to actuate the firing mechanism component for discharging the firearm; a trigger operably coupled to the rotating member, the trigger manually movable by a user between first and second positions; a permanent magnet generating a static magnetic field in the rotating member, the static magnetic field creating a primary resistance force opposing movement of the rotating member and trigger when pulled by the user; the coil when energized generating a secondary magnetic field in the rotating member which overcomes the primary resistance force and rotates the rotating member to discharge the firearm; a programmable trigger unit microcontroller operably coupled to the electromagnetic actuator of the trigger unit, the trigger unit microcontroller further operably coupled to an external fire control targeting system and configured to: detect user activity on the trigger sensed by a trigger sensor communicably coupled to the trigger unit microcontroller, the user trigger activity comprising an applied trigger force or trigger displacement; compare the user activity on the trigger to a preprogrammed first trigger setpoint; and transmit a shot initiation signal to the fire control targeting system when the user activity on the trigger exceeds the first trigger setpoint. The electronic trigger system is operable to revert to manual firing mode thereby allowing the user to fire the firearm mechanically when the system senses that an applied trigger force or displacement of the trigger exceeds a preprogrammed maximum allowable trigger pull force or displacement limit.

According to another aspect, a method for discharging a firearm with an interruptible firing system comprises: providing an electronic trigger unit operably coupled to a power source and mounted to the firearm, the trigger unit comprising a programmable trigger unit microcontroller, a trigger, and an electromagnetic actuator operably coupled to the trigger and a firing mechanism of the firearm, the actuator

including a magnetic coil which when energized moves the actuator from a ready-to-fire unactuated position to an actuated firing position which discharges the firearm; providing a fire control targeting system comprising an electronic adaptive optics unit mounted to the firearm for sighting a target, the adaptive optics unit including a programmable targeting microcontroller operably coupled to the trigger unit microcontroller; the trigger unit microcontroller detecting trigger activity initiated by a user, the trigger activity comprising a trigger pull force or displacement; the trigger unit microcontroller sending a shot initiation signal to the targeting microcontroller when the trigger activity exceeds a preprogrammed first trigger setpoint; the targeting microcontroller sending a shot authorization control signal to the trigger unit microcontroller in response to receiving the shot initiation signal; the trigger unit microcontroller energizing the actuator in response to receiving the shot authorization signal which changes the actuator from the unactuated position to the firing position which discharges the firearm.

The present application further discloses a trigger safety mechanism which adds an extra level of protection and permits an electronic firing system with electromagnetic actuator trigger unit to pass gun drop safety tests.

In one aspect, a firearm with electronic firing system comprises: a chassis or frame; a trigger mechanism coupled to the chassis or frame, the trigger mechanism comprising an electromagnetic actuator including a moving member configured and operable to discharge the firearm when actuated; the trigger mechanism further comprising: a movable outer trigger; an inner safety trigger pivotably movable with respect to the outer trigger; and a coil operably coupled to an electric power source and the electromagnetic actuator, the coil when energized operable to generate a magnetic field in the electromagnetic actuator which actuates the moving member; wherein the inner safety trigger is movable between a first position which prevents actuation of the electromagnetic actuator, and a second position which enables the electromagnetic actuator to discharge the firearm when the outer trigger is pulled. The electromagnetic actuator may comprise: a stationary yoke; the moving member being disposed at least partially within the yoke, the moving member rotatably movable about a pivot axis relative to the yoke and operably coupled to a firing mechanism component operable to discharge the firearm; the coil being disposed at least partially within the yoke; the outer trigger operably coupled to the moving member and manually movable by a user from a first unpulled position to a second pulled position for discharging the firearm.

In another aspect, a method for discharging a firearm with an electronic firing system comprises: providing an electronic trigger unit operably coupled to a power source and mounted to the firearm, the trigger unit comprising a programmable trigger unit microcontroller, a trigger assembly including an outer trigger and inner safety trigger movable between blocking and unblocking positions, and an electromagnetic actuator operably coupled to the trigger assembly and a firing mechanism of the firearm, the actuator including a magnetic coil which when energized via pulling the outer trigger changes the actuator from a ready-to-fire unactuated position to a firing actuated position which discharges the firearm; blocking movement of the outer trigger assembly by positioning the inner safety trigger in the blocking position which prevents energizing the actuator and changing the actuator from the unactuated to actuated position; moving the inner safety trigger of the trigger assembly from the blocking position to the unblocking position which allows

movement of the outer trigger; and pulling the outer trigger which energizes the actuator and changes the actuator from the unactuated position to actuated position to discharge the firearm.

These and other features and advantages of the present invention will become more apparent in the light of the following detailed description and as illustrated in the accompanying drawings.

BRIEF DESCRIPTION OF DRAWINGS

The features of the exemplary embodiments will be described with reference to the following drawings where like elements are labeled similarly, and in which:

FIG. 1 is a graph depicting variation in trigger pull force versus displacement (distance) for two different trigger actions or mechanisms;

FIG. 2A is a side cross-sectional view of a control device comprising an electromagnetic magnetorheological fluid piston assembly for a trigger mechanism of a firearm;

FIGS. 2B-D show sequential views of the piston assembly thereof embodied in a variable force trigger mechanism during different stages in the process of pulling the trigger;

FIG. 3 is a side cross-sectional view thereof including an alternative embodiment of a user-adjustable magnetic control device for altering the trigger pull force comprised of a permanent magnet control linkage that provides the magnetic field in lieu of an electromagnetic shown in FIGS. 2A-D;

FIG. 4A is a perspective view of a housing incorporating the foregoing magnetorheological fluid piston assembly and a user-adjustable electromagnetic control device for altering the trigger pull force;

FIG. 4B is a partial cutaway view thereof showing the coiled electromagnetic device which includes a permanent magnet in greater detail;

FIG. 4C is an end view thereof showing a closed loop magnetic flux path or circuit formed by the electromagnetic device incorporated with the magnetorheological fluid piston assembly;

FIG. 5 is a perspective view showing the magnetorheological fluid piston assembly and electromagnetic control device incorporated in a firing mechanism or system of a firearm;

FIG. 6 is a perspective view of an electrically variable and adjustable electromagnetic trigger mechanism comprising an electromagnetic control device in the form of an electromagnetic actuator designed with a single magnetic flux loop;

FIG. 7 is a perspective view of a second embodiment thereof adding spring assist and control feedback from a trigger displacement sensor;

FIG. 8 is a control logic diagram of a process implemented by a programmable microprocessor-based microcontroller for controlling operation of the electromagnetic trigger mechanism;

FIG. 9 is a system block diagram of the programmable microcontroller based control system for monitoring and operating the electromagnetic trigger mechanism;

FIG. 10A is a diagram showing a wireless communication and control system interfacing with the microcontroller for use with the electromagnetic trigger mechanism which is programmable via an external/remote electronic device;

FIG. 10B is a graph of an example trigger pull force versus displacement (travel) curve showing various stages trigger force during a trigger pull sequence and an illustrating a breakpoint in the trigger release profile;

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FIG. 11 is a diagram showing a variable force trigger wireless data collection and communication smart application;

FIG. 12 is a graph of trigger pull force versus displacement (travel or distance) of a non-linear force displacement curve for a segmented trigger design;

FIG. 13A is a perspective view of an electrically variable and adjustable electromagnetic trigger mechanism comprising an electromagnetic control device and including a non-linear leaf spring;

FIG. 13B is a side view of the trigger member thereof in isolation;

FIG. 14A is a perspective view thereof including a secondary spring flexing member joining an upper rotating member of the trigger mechanism with a lower trigger member;

FIG. 14B is a side view of the trigger member thereof in isolation;

FIG. 15 is a perspective view thereof with the upper rotating member of the electromagnetic trigger mechanism configured as a sear for interacting with a firing system component for discharging the firearm;

FIGS. 16 and 17 are front and rear top perspective views respectively of a second embodiment of an electromagnetic trigger mechanism comprising an electromagnetic actuator designed with a dual closed magnetic flux loop;

FIGS. 18 and 19 are front and rear bottom perspective views respectively thereof;

FIGS. 20 and 21 are exploded top and bottom perspective views respectively thereof;

FIGS. 22 and 23 are front and rear end views respectively thereof;

FIG. 24 is a right side view thereof;

FIGS. 25 and 26 are top and bottom views respectively thereof;

FIG. 27 is a first left side cross-sectional view thereof showing the electromagnetic actuator trigger mechanism in an unactuated ready-to-fire position or state;

FIG. 28 is a second left side cross-sectional view thereof showing the same;

FIG. 29 is a view thereof showing the electromagnetic actuator trigger mechanism in an actuated fire position or state;

FIG. 30 is a right side view of a firearm in the form of a pistol incorporating the electromagnetic actuator trigger mechanism;

FIGS. 31 and 32 show magnetic flux paths in the electromagnetic actuator trigger mechanism in a de-energized state (FIG. 31) and energized state (FIG. 32);

FIG. 33 is a schematic diagram of a manually adjustable potentiometer which may be used to control operation of the electromagnetic actuator;

FIGS. 34A and 34B are first and second parts of a control logic diagram of a fire-by-wire electric firing system for a firearm implemented by the microcontroller;

FIG. 35 is a system block diagram of the programmable microcontroller based control system for monitoring and operating the fire-by-wire firing system;

FIG. 36 is a side view of a first non-electric embodiment of a closed magnetic loop trigger mechanism comprising a sliding soft magnetic material wedge with trigger mechanism in a ready-to-fire position;

FIG. 37 is a side view thereof showing the trigger mechanism in the pulled firing position;

FIG. 38 is a side view a second non-electric embodiment of a closed magnetic loop trigger mechanism comprising a

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sliding soft magnetic material wedge but with an alternative actuator mechanism for translating the sliding wedge;

FIG. 39 shows computer-modeled magnetic flux lines generated by the trigger mechanism of FIGS. 36 and 38;

FIG. 40 shows the results of finite element analysis (FEA) of trigger mechanism of FIGS. 36 and 38 in a trigger pull force (Torque) versus displacement (Dp) profile graph;

FIG. 41 is a side view of a third non-electric embodiment of a closed magnetic loop trigger mechanism comprising a sliding soft magnetic material plate;

FIG. 42 shows computer-modeled magnetic flux lines generated by the trigger mechanism of FIG. 41;

FIG. 43 shows the results of finite element analysis (FEA) of trigger mechanism of FIG. 41 in a trigger pull force (Torque) versus displacement (Dp) profile graph;

FIG. 44 is a side view of a fourth non-electric embodiment of a closed magnetic loop trigger mechanism comprising a sliding magnet;

FIG. 45 shows computer-modeled magnetic flux lines generated by the trigger mechanism of FIG. 44;

FIG. 46 shows the results of finite element analysis (FEA) of trigger mechanism of FIG. 44 in a trigger pull force (Torque) versus displacement (Dp) profile graph;

FIG. 47 is a side view of a fifth non-electric embodiment of a closed magnetic loop trigger mechanism comprising a rotating magnet;

FIG. 48 shows computer-modeled magnetic flux lines generated by the trigger mechanism of FIG. 47;

FIG. 49 shows the results of finite element analysis (FEA) of trigger mechanism of FIG. 47 in a trigger pull force (Torque) versus displacement (Dp) profile graph;

FIG. 50 is a side view of a non-electric embodiment of an open magnetic loop trigger mechanism comprising a moving magnet and showing the computer-modeled magnetic flux lines generated;

FIG. 51 shows the results of finite element analysis (FEA) of trigger mechanism of FIG. 50 in a trigger pull force (Torque) versus displacement (Dp) profile graph;

FIG. 52 is a side perspective view of a preferred embodiment of a non-electric closed magnetic loop trigger mechanism of the sliding magnet design;

FIG. 53 is an exploded view thereof;

FIG. 54 is a side view thereof;

FIG. 55 is a rear view thereof;

FIG. 56 is a side cross-sectional view thereof;

FIG. 57 is a top rear perspective view of the non-magnetic magnet carrier of the trigger mechanism of FIG. 52;

FIG. 58 is a bottom front perspective view thereof;

FIG. 59 is a side cross-sectional view thereof;

FIG. 60 is a front view thereof;

FIG. 61 is a side perspective view of a preferred embodiment of a non-electric open magnetic loop trigger mechanism of the movable magnet design;

FIG. 62 is an exploded view thereof;

FIG. 63 is a rear view thereof;

FIG. 64 is a side view thereof;

FIG. 65 is a side cross-sectional view thereof;

FIG. 66 is a top rear perspective view of the magnet holder mounting block of the trigger mechanism of FIG. 61;

FIG. 67 is a bottom side perspective view thereof;

FIG. 68 is a rear view thereof;

FIG. 69 is a top view thereof;

FIG. 70 is a right side view of a long gun in the form of a rifle incorporating a trigger housing including the trigger mechanisms of FIG. 52 or 61.

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FIGS. 71A and 71B are first and second parts of a control logic diagram of a firing event tracking system implemented by the microcontroller;

FIG. 72 is a graph showing the acoustic signatures produced by discharging a firearm in sound amplitude (decibels/dB) versus time (milliseconds) for a series of different trigger/firing events;

FIG. 73 is a graph showing a comparison of acoustic signatures produced by a trigger/firing event resulting in discharge of the firearm to other non-fire events not resulting in discharge measured in sound amplitude (decibels/dB) versus time (milliseconds);

FIG. 74 is a graph showing acoustic signatures produced by discharging a firearm for a trigger/firing event initiated by the shooter of interest using a firearm equipped with the present firing event tracking system in comparison to those produced by other nearby shooters, in sound amplitude (decibels/dB) versus time (milliseconds); and

FIG. 75 is a graph showing motion/acceleration signatures produced by discharging a firearm in acceleration (meters per second²) versus time for a series of different trigger/firing events.

FIG. 76 is a system block diagram of the programmable microcontroller based electronic trigger control system for monitoring and operating the electromagnetic trigger mechanism in an interruptible firing mode in conjunction with an operably and communicably coupled external fire control targeting system;

FIG. 77 is a high level control logic flow diagram of a single stage interruptible electronic trigger system control scheme;

FIG. 78 is a lower level control logic flow diagram thereof showing additional sub-steps of the basic process in FIG. 77;

FIG. 79 is a high level control logic flow diagram of a two stage interruptible electronic trigger system control scheme;

FIGS. 80A and 80B are first and second parts of a lower level control logic flow diagram thereof showing additional sub-steps of the basic process in FIG. 79;

FIG. 81 is a high level control logic flow diagram of a three stage interruptible electronic trigger system control scheme;

FIGS. 82A and 82B are first and second parts of a lower level control logic flow diagram thereof showing additional sub-steps of the basic process in FIG. 81;

FIG. 83 is a high level control logic flow diagram showing the addition of tactile, visual, and/or audible confirmation to the user during the firing sequence of the single stage interruptible electronic trigger system control scheme;

FIG. 84 is a high level control logic flow diagram showing an example of the general control process which may be implemented by the fire control targeting system and interaction with the interruptible electronic trigger system for the single stage trigger control scheme; and

FIG. 85 is a high level control logic flow diagram showing an example of the general control process which may be implemented by the fire control targeting system and interaction with the interruptible electronic trigger system for the two stage trigger control scheme.

FIGS. 86 and 87 are front and rear top perspective views respectively of another embodiment of an electromagnetic trigger mechanism comprising an electromagnetic actuator designed with a dual closed magnetic flux loop and trigger safety mechanism;

FIG. 88 is a top front exploded perspective view thereof;

FIG. 89 is a bottom front exploded perspective view thereof;

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FIG. 90 is a bottom rear exploded perspective view thereof;

FIGS. 91 and 92 are right and left side views respectively thereof;

FIGS. 93 and 94 are front and rear end views respectively thereof;

FIGS. 95 and 96 are bottom and top views respectively thereof;

FIG. 97 is a left side transverse cross sectional view thereof;

FIG. 98 is a right side transverse cross sectional view thereof showing the electromagnetic trigger mechanism in an unactuated position and inner safety trigger of the trigger assembly in a blocking position;

FIG. 99 is a right side transverse cross sectional view thereof showing the electromagnetic trigger mechanism in an actuated position via a trigger pull and inner safety trigger of the trigger assembly in an unblocking position;

FIG. 100 is a perspective view of the moving or rotating member of the electromagnetic trigger mechanism and associated components;

FIG. 101 is an exploded view thereof;

FIG. 102 is a top perspective view of the magnet holder of the electromagnetic actuator detachably mountable on the actuator yoke; and

FIG. 103 is a bottom perspective view thereof.

All drawings are schematic and not necessarily to scale. Any reference herein to a whole figure number (e.g. FIG. 2) which may include several subpart figures (e.g. FIGS. 2A, 2B, 2C, etc.) shall be construed as a reference to all subpart figures unless explicitly noted otherwise. Numbered parts appearing in some figures which appear un-numbered in other figures are the same parts unless explicitly noted otherwise.

DETAILED DESCRIPTION

The features and benefits of the invention are illustrated and described herein by reference to example ("exemplary") embodiments. This description of exemplary embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as "lower," "upper," "horizontal," "vertical," "above," "below," "up," "down," "top" and "bottom" as well as derivative thereof (e.g., "horizontally," "downwardly," "upwardly," etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation. Terms such as "attached," "affixed," "connected," and "interconnected," refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. Accordingly, the disclosure expressly should not be limited to such exemplary embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features.

As used throughout, any ranges disclosed herein are used as shorthand for describing each and every value that is within the range. Any value within the range can be selected as the terminus of the range.

The dynamics of the trigger feel is one of the most important aspects of the shooter's experience, impacting accuracy, repeatability, and safety of the firearm. A conventional trigger pull consists of three stages: take-up or pre-travel, the break-over point of release of stored energy in the hammer, striker, or sear, and finally over-travel. In a conventional trigger mechanism, these stages are fixed by the springs, linkages, and mechanical components that make up the trigger system. An adjustable trigger allows adjustments to the travel distance, force, and feel of the trigger pull during one or more of these stages or phases.

The desired trigger pull force and displacement characteristic is dependent upon the type of firearm, application, safety, reliability, and individual preferences. For example, a shooter may wish for a medium to heavy trigger pull weight for hunting and a significantly lighter and different feel for competition shooting. FIG. 1 shows a comparison of a conventional military spec trigger pull force profile versus a modified version of an AR type rifle trigger exhibiting a lower pull force profile over the range from the initial trigger pull through release of the hammer or striker of the firearm.

The current state of the art for making changes in the trigger pull force requirement and shape of the force profile (e.g. between a heavy and light trigger pull) is to physically adjust spring or linkage tensions within the trigger mechanism or directly replace existing and install alternate parts to attain the desired trigger force and displacement characteristics. These approaches both limit the shape of the possible trigger force verses displacement curve and the timing of how it can be adjusted. Additionally, the adjustment is usually only possible over a narrow range of trigger pull forces unfortunately due to physical limitations of the physical trigger mechanism components.

The present invention includes a novel trigger mechanism which allows the trigger pull force and displacement to be controlled by a magnetic field. By actively adjusting the magnetic field, dynamic real-time variability of the trigger pull force over a wide range of displacement can advantageously be achieved. In addition, the "feel" of the trigger may be improved by tailoring this force-displacement curve to provide a large range of variation that is not possible with conventional mechanical springs, linkages, and levers.

One method disclosed herein to control the force-displacement profile may be to use a rheological fluid. An electric or magnetic field can influence the viscosity of certain fluids. This characteristic can be exploited to design a variable force trigger for firearms, turn on or off a manual safety feature, or provide active damping of recoil.

Magnetorheological (MR) fluids have the unique property of changing from a free-flowing liquid to a semi-solid state in the presence of a magnetic field. This dynamically changeable viscosity property has significant potential for control applications in firearms. Currently, magnetorheological fluids, such as the commercially available MRF-132DG by LORD Corporation, provide a range of fast response time, dynamic yield strength, temperature resistance to meet the needs of an adjustable force trigger system in firearms. Other materials such as ferro-fluids, electrorheological fluids, and devices based on the Giant Electrorheological effect may also provide a reliable alternative to the use of magneto-rheological fluids in this application. Embodiments of Dynamic Variable-Force Trigger Using MR Fluids

Magneto-rheological (MR) fluids can respond almost instantly to varying levels of a magnetic field precisely and proportionally for controlled force loading. By dynamically adjusting the viscosity of the MR fluid, it is possible to construct a dynamically variable trigger force apparatus. If the movement of a trigger transfer linkage is constrained by using an MR fluid-filled spring loaded piston as disclosed herein, the viscosity of the MR fluid using a magnetic field, we can then be dynamically changed. The resulting viscosity change results in a significant change in force loading necessary to move the trigger transfer linkage to the fire position, which translates into a user-variable trigger pull force resistance opposing movement of the trigger linkage.

FIGS. 2A-D and 4-5 depict one embodiment of an electromagnetic MR fluid actuator 600 comprising an MR fluid-filled piston assembly 602 comprising a disk-shaped piston 612 movably disposed inside an MR fluid-filled cylinder 601. An electromagnet coil 614 is wound around a portion of the cylinder 601 and operably coupled to an electric power source 122 onboard the firearm and further described herein. The piston 612 is spring loaded so that the trigger linkage 610 would have a low return spring force sufficient to reliably return the trigger to its original vertical ready-to-fire position with the MR fluid in its free-flowing most liquid state (i.e. lowest viscosity condition). Approximately 1.0 lbs. might be a good baseline in one example for spring force imparted by piston spring 604. By increasing a magnetic field via the electromagnet coil 614 operably coupled to a power source 122, applied in such a way as to change the viscosity of the MR fluid, the force necessary to move the trigger bar could be adjusted upward to as much as 10-15 lbs. force in some embodiments. The trigger linkage 610 may comprise an elongated rod 611 pivotally coupled to a trigger member 608 rotatable about a transverse pivot axis 606 formed by a pin. Trigger member 608 may be mounted to a frame of a firearm.

In a basic implementation of a simple non-electromagnetic MR fluid actuator shown in FIG. 3, the magnetic field may be created by a spatially adjustable permanent magnet 615 mounted in close proximity to the piston cylinder 601 via an adjustable mechanical linkage 616. The linkage 616 may comprise a permanent magnet 615 slideably disposed inside a guide tube 616 and acted upon by a pair of springs 613a and 613b. One spring is disposed on each side of the permanent magnet. By adjusting the linkage up or down using a rotary adjustment device 618 such as set-screw or other manual device, the position of the permanent magnet 615 relative to the piston cylinder 601 can be adjusted. In one embodiment, the guide tube 616 may be disposed perpendicularly to the piston cylinder 601. Other arrangements are possible. This allows the relationship of the magnetic field in respect to the MR fluid filled spring-loaded piston to be changed for increasing or decreasing the viscosity of the MR fluid (i.e. viscosity increasing with decreasing proximity to cylinder). This simple non-electromagnetic adjustment means can be used by the user to increase or decrease the trigger pull force required to actuate the firing mechanism of the firearm (e.g. trigger linkage 610). This would allow for a user selectable fixed trigger force profile.

By replacing the permanent magnet 615 with an electromagnet coil 614 as already described herein, one can dynamically change the MR fluid viscosity and hence resulting trigger pull force-displacement profile examples of which are shown in FIG. 1. This would allow a number of force profiles to be defined, selected, and implemented under electrical control. For example, one might want a very high trigger force when used in a self-defense, holstered, or

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concealed carry situation. Or one might choose a very light trigger force when target shooting, something in between when recreational shooting, or perhaps a different trigger force for the first round and lighter trigger profile for subsequent shots.

FIGS. 4A-C depicts an embodiment of a complete electromagnetic MR fluid actuator **600** assembly according to one embodiment. The actuator **600** may be mounted at least partially or fully inside a housing **619** which is configured for mounting to a frame of a firearm. Actuator **600** further comprises a stationary magnetic yoke **620** around which the electromagnet coil **614** (shown only schematically in FIGS. 2A-D) may be wound. Coil **614** is operably connected to the power source **122**, which may be a battery. In this embodiment, a permanent magnet **615** is mounted to the yoke **620** to create a static or fixed magnetic field which may be biased to automatically maintain the trigger in the upright ready-to-fire position shown in FIG. 2B when the trigger is not pulled by the user. The yoke **602** is configured to form a single closed flux loop with lines of flux represented by flux arrows **622**. When energized, the coil **614** creates a secondary electromagnetic field which interacts with the static magnetic field and dynamically changes the viscosity of the MR fluid and trigger pull force required to move the trigger **608**.

FIG. 5 shows the complete electromagnetic MR fluid actuator **600** embodied in a firing mechanism of a firearm. The firing mechanism may comprise a movable spring-biased striking member **130** which may be a rotatable hammer about hammer pin **130-1** as shown or alternatively a linear movable striker (not shown). The striking member **130** is arranged to strike the rear end of a firing pin **630** which in turn strikes a chambered ammunition cartridge **C** held in the barrel of the firearm. The striking member **130** is movable between a rearward cocked and forward firing position. A sear **632** is releasably engaged with the striking member **130** which is held in the cocked position by sear. The sear **632** is operably coupled to the trigger rod **611** at a rear end opposite the front end of the rod which is pivotably coupled to the trigger **608**. Pulling the trigger which has a trigger pull force-displacement profile created by energizing the coil **614** moves the sear, which releases the striking member **130** to strike the firing pin and discharge the firearm. Variations of the firing mechanism are possible for use with the electromagnetic MR fluid actuator **600**. The actuator **600** and its operation to energize and adjust the MR fluid viscosity and trigger pull force may be adjusted and control via a suitable programmed microcontroller **200**; an example of which is discussed elsewhere herein. In some embodiments, the electromagnetic MR fluid actuator **600** may be configured to be additive during one portion or phase of the trigger pull, and changed to subtractive over another portion or phase of the pull based on the trigger displacement distance via properly configuring the control logic executed by the microcontroller which controls the electric power supplied to the electromagnet coil **614**. For example, a higher initial trigger pull force may be desired for the initial portion or phase of the trigger pull and a lower pull force for the remaining portion or phase of the trigger pull as the trigger continues to move rearward. The timing of when each phase is initiated, its duration, and change in value or magnitude of the pull force required may be selected via appropriately programming and configuring the microcontroller **200**.

Using multiple magnetic force concentration points, or a piston plunger port configuration that extends through an adjustable magnetic field during the full travel of the trigger,

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it is possible to dynamically change the viscosity (trigger force) during a single trigger pull. Such a configuration allows dynamically changing force verses displacement curves of an unlimited nature that could allow custom trigger feel optimized for certain users and use profiles.

Another embodiment related to the variable force-displacement effect is the use of MR fluids as an ON/OFF Trigger Safety. Movement of a trigger transfer mechanism would move freely through a MR fluid reservoir when no magnetic field is applied. When a magnetic field is applied to the MR fluid, its yield stress increases inhibiting movement of the trigger transfer mechanism. Ideally the use of a permanent magnet could be used as a fail-safe always on trigger safety.

In its most basic form, this could be implemented by a permanent magnet mounted on a mechanical linkage that could be manually moved in and out of the critical proximity to the MR fluid like a manual safety lever. While functional this provides no advantage over a conventional mechanical safety.

To take full advantage of the magnetic on/off nature of the MR fluid, an electro-magnet may be included to control the on/off function. This would allow an electrical signal to control the on/off function of the trigger. The reversible and almost instantaneous changes from a free-flowing liquid to a semi-solid with high yield strength would allow the safety to be electrically controlled based on control logic.

Only when an electromagnet is actuated would the effects of the permanent magnet be nulled and allow the MR fluid become more liquid and allow free movement of the trigger mechanism (reference FIG. 5).

To minimize power consumption, an enhancement to the concept would place a fixed permanent magnet in place so that the trigger linkage is in the blocked state when at rest. To reverse the MR fluid back to a flowing liquid state, a secondary electro-magnet could be energized to balance out the permanent magnets field. In this configuration, the electromagnet could enable the trigger operation at almost the point that the operator fires while using no power at any other time. The default static unpowered state of the system would be in the no-fire or ready-to-fire condition.

While the use of a MR fluid could be used as a standalone ON/OFF trigger safety feature, the preferred embodiment would combine this active safety feature with a dynamic variable force trigger configuration that acts as both an adjustable trigger force and trigger on/off safety. By applying a fixed permanent magnet field in proximity to the MR fluid filled piston, sufficient to block movement when the firearm is not required to operate, we would have the features of a firearm safety. The magnet field could then be nulled out by the addition of a reverse magnetic field using an electro-magnet and thus enabling the dynamic variable force trigger features.

Embodiments of Dynamic Variable-Force Trigger Using Electromagnetic Actuators

Another embodiment for dynamically controlling the displacement force profile of a firearm trigger utilizes magnetic fields to directly constrain the movement of the trigger linkage until a preselected release force is reached. In one embodiment, a combination of a continuous primary static magnetic field and an intermittently acting dynamic electromagnetic field may be used. FIGS. 6 and 7 depict non-limiting examples of an electrically-variable electromagnetic trigger release mechanism or simply "electromagnetic trigger mechanism" is presented. FIG. 6 depicts a one-piece

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rotating trigger member whereas FIG. 7 depicts a trigger member in which an upper portion is pivotably movable relative to the lower portion.

The electromagnetic trigger mechanism 100 generally comprises an electromagnetic snap actuator 123 configured as a trigger assembly for discharging the firearm. The trigger mechanism 100 forms an integral part of the firing system or mechanism of the firearm itself, and does not merely act on the firing mechanism. Actuator 123 is configured as a release type actuator which directly or indirectly releases the energy in the energy storage device such as a spring-biased striking member (e.g. rotatable hammer or linearly movable striker) operable to strike a chambered cartridge positioned in the barrel of the firearm. If a sear which releases the striking member is built directly into the release actuator 123 as shown in FIG. 15, then the actuator is directly releasing the hammer or striker. If the sear is a separate secondary component as shown in FIGS. 16-29, then the release actuator can release the sear which in turn releases the hammer or striker. In either case, energy applied to the actuator directly results in the firing of the weapon.

Referring now again to FIGS. 6 and 7, trigger mechanism 100 includes a magnetic stationary yoke 102, a rotating trigger member 104, and an electromagnet coil 106 disposed and wound around a portion of the stationary yoke. The yoke 102 may be fixedly and rigidly but removably attached to the frame 22 of the firearm 20 (see, e.g. FIG. 30), receiver 39, or trigger housing 1220 (see, e.g. FIG. 70) by any suitable manner, including for example without limitation entrapment in an open trigger unit receptacle of the frame, fasteners, couplers, pins, interlocking features, etc. The mode of attachment is not limiting of the invention. The trigger mechanism 100 may have a generally annular shape in one embodiment which is collectively formed in part by the yoke 102 and in the remaining part by the rotating trigger member 104 to form the annulus. An open central space 103 is defined by the trigger mechanism 100. This space 103 provides room for receiving a portion of the coil 106 when wound around the trigger mechanism.

The stationary yoke 102 of the electromagnetic trigger mechanism 100 may be substantially C-shaped in one embodiment including a horizontal upper portion 110, horizontal lower portion 112 spaced apart and parallel to the upper portion, and a vertical intermediate portion 114 extending between the upper and lower portions. The intermediate portion 114 is integrated with captive ends of the upper and lower portions 110, 112 being a unitary structural part of the entire yoke 102 in one embodiment. The portions 110, 112, and 114 may have any suitable transverse cross-sectional shape including polygonal such as rectilinear as shown, non-polygonal (e.g. circular), or combinations thereof which lend themselves to winding the coil 106 thereto. Although the stationary yoke 102 is illustrated herein as have a C-shaped configuration, it will be appreciated that other configurations of the yoke are possible and may be used.

The rotating trigger member 104 may have a vertically elongated and substantially linear shaped body in one embodiment as shown. The rotating trigger member 104 may lie in the same vertical reference plane as the yoke 102 and is pivotably movable within that plane. The vertical reference plane may intersect the longitudinal axis of the firearm in one embodiment.

Rotating trigger member 104 is pivotably disposed in the frame of the firearm. In one embodiment, rotating trigger member 104 may be pivotably coupled to stationary yoke 102 via pivot 101 formed by cross pin 126a which defines

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a pivot axis PA of rotation oriented transversely to the longitudinal axis LA of the firearm (see, e.g. FIG. 30). As shown in FIGS. 6 and 7, rotating trigger member 104 may be pivotably coupled to the lower portion 112 of yoke 102 at a terminal end thereof. The rotating trigger member 104 and lower portion 112 are thus each configured to receive pivot 101 therethrough for forming the pivotable coupling. Any suitable type of pivot connection may be used for pivot 101, such as without limitation a pin or rod as some examples so long as the rotating trigger member 104 may be moved relative to the yoke 102. The rotating trigger member 104 defines an axis of tilt TA which is angularly movable with respect to a stationary axis SA defined by the vertical portion 114 of yoke 102 when the trigger mechanism is activated.

It will be appreciated that in alternative embodiments, for example, the rotating trigger member 104 may alternatively be pivotably mounted to the frame 22 of the firearm 20 instead of via the pivot 101 to achieve the same manner of movement relative to the yoke 102. Either arrangement may be used in various embodiments to best fit the design of the firearm in which the trigger mechanism 100 will be used.

With continuing reference to FIGS. 6 and 7, the rotating trigger member 104 includes a lower trigger segment or portion 118 below pivot 101 and an upper working segment or portion 120 above pivot 101. These portions may simply be referred to herein as lower and upper portions 118, 120 for brevity. In the case of FIG. 7, the lower portion 118 is pivotably movable relative to the upper portion. The lower portion 118 is configured to define a trigger 121 in one embodiment, and may include an arcuately curved shape typical of some forms of a firearm trigger for better engaging a user's finger. The upper portion 120 forms part of the magnetic flux circuit of the electromagnetic trigger mechanism 100 and is arranged to selectively and releasably engage the stationary yoke 102. In one embodiment, the rear surface of the upper portion 102 is engageable with the upper portion 110 of the yoke 102 as shown. The combination of the C-shaped yoke 102 and upper portion 120 of the rotating trigger member 104 including the pivot portion including the pivot 101 collectively define an openable and closeable annulus and magnetic flux loop via operation of the trigger (see magnetic flux path arrows). The lower portion 118 therefore may be considered to extend downwards from the annulus.

In one embodiment, as shown in FIG. 15, the upper portion 120 of the rotating trigger member 104 may be vertically elongated forming an extension that projects upwards beyond the upper portion 110 of yoke 102. This extension defines a sear 131 integrally formed with the trigger member. A sear surface 132 formed on the sear 131 is operably engageable with the striking member 130 (a pivotable hammer in the illustrated embodiment) to selectively hold or release the striking member 130 in/from the rearward cocked position for discharging the firearm. The sear surface 132 may be formed on the upward facing top surface on the top end of the sear 131 in one embodiment. In this example embodiment, the striking member 130 is a pivotable hammer. In other embodiments, the striking member 130 may be linearly movable and cockable striker well known in the art which operably interfaces with the sear 131. In yet other possible implementations, the sear surface 132 may operably interface with a separately rotatable sear disposed in the firearm frame which in turn interfaces with the striking member 130 similarly to that shown in FIG. 30. Numerous other variations and locations and configurations of sears and sear surfaces on the rotating trigger member 104

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may of course be used. It bears noting that the vertically elongated extension of the upper portion 120 of trigger member 104 to form sear 131 may of course be provided in any of the trigger mechanisms 100 shown in FIGS. 6, 7, 13, and 14.

The terminal end portion of upper portion 110 of yoke 102 and terminal end portion of the upper portion 120 of rotating trigger member 104 are movable together and apart via the pivoting action of the rotating trigger member 104 relative to the stationary yoke 102. Accordingly, an openable and closeable air space or gap A is formed at the interface between the yoke 102 and rotating trigger member 104. The rotating trigger member 104 is pivotably and manually movable between two actuation states or positions by a user. Rotating trigger member 104 is movable between a first unactuated or rest position physically engaged with the yoke 102 when the trigger is not pulled, and a second actuated or fire position disengaged from the yoke 102 when the trigger is pulled to discharge the firearm. In the actuated position, air gap A is opened whereas the gap is closed in the unactuated position. Also in the actuated position, the axis of tilt TA of the rotating trigger member 104 is obliquely oriented and angled to the stationary axis SA defined by yoke 102, whereas the axis of tilt TA is parallel to axis SA when the rotating trigger member is in the upright unactuated position.

With continuing reference to FIGS. 6 and 7, the electromagnet coil 103 of the trigger mechanism 100 is electrically coupled to and energized by an electric power source 122 (see, e.g. FIG. 1) of suitable voltage and current to control operation of the trigger mechanism for adjusting the trigger pull force and profile. The power source 122 is preferably mounted to the firearm and may comprise a single use or rechargeable replaceable battery in some embodiments. In one embodiment, an electric coil 106 wound primarily around and supported by the upright or vertical intermediate portion 114 of the stationary yoke 102 may be provided as shown which collectively forms an electromagnet. Operation of the trigger mechanism 100 such as for controlling the firing mechanism of a firearm or other applications is further described herein. In one embodiment, a protective casing such as an electrical resin encapsulate or potting compound may be provided to at least partially enclose and protect the coil 106.

The stationary yoke 102 and rotating trigger member 104 may be formed of any suitable soft magnetic metal capable of being magnetized, such as without limitation iron, low-carbon steel, nickel-iron, cobalt-iron, etc.

The trigger mechanism 100 in one embodiment includes a preferably strong permanent magnet 108 which creates a relatively high threshold static magnetic attractive or holding force between the yoke 102 and rotating trigger member 104 which acts to draw these two components into mutual engagement. This static and primary resistance force created by the magnetic field between yoke and trigger member acts to inhibit movement of the rotating trigger member 104 about its pivot axis PA between its two actuation positions when trigger 121 is pulled by a user. The magnetically-induced static resistance corresponds to a trigger pull force required to be exerted and surpassed by the user in order to rotate the trigger member sufficiently to discharge the firearm. The magnet 108 may have a flat rectilinear plate-like shape in one embodiment; however, other shapes may be used. Magnet 108 biases the rotating trigger member 104 into the first unactuated position engaged with the upper portion 110 of yoke 102 at magnet 108.

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Permanent magnet 108 may be disposed anywhere within the magnetic loop formed by the yoke 102 and the movable upper portion 120 of rotating trigger member 104. In one embodiment, the magnet 108 may be mounted on the front terminal end of the upper portion 110 of the yoke. Alternatively, the magnet 108 may be disposed on the rear surface of the rotating trigger member 104 and positioned to engage upper portion 110 of the yoke 102. The magnet 108 may therefore be interposed directly between the movable upper portion 120 of the rotating trigger member 104 and stationary yoke 102 to maximize the magnetic attraction of the rotating trigger member to the magnet 108. Other less preferred but still satisfactory locations for mounting the magnet 108 on yoke 102 may alternatively be used. Magnet 108 preferably may be dimensioned and has a cross-sectional area approximately commensurate with and similar to the dimensions and cross-sectional area of the yoke 102 or rotating trigger member in or on which the magnet is arranged.

The present invention further provides a user-selectable and dynamically variable secondary electromagnetic field generated when the electromagnetic actuator 123 is energized. This secondary electromagnetic field interacts with the primary static magnetic field produced by the permanent magnet 108. By electrically and preferentially biasing the magnet flux in the closed loop of the actuator 123 to add or detract from the static magnetic field using the actuator's electromagnet, a dynamically variable trigger pull force or resistance and profile is created which can be selected by the user to meet personal preferences. When coil 106 of the trigger mechanism snap actuator 123 is not energized, a trigger pull force sufficient to only overcome the primary fixed or static magnetic field force of the permanent magnet 108 on the rotating trigger member 104 would be needed to initiate and displace the trigger through a trigger pull event. This allows the trigger member to be actuated in the event power is lost to the actuator 123 (e.g. depleted battery charge).

Electrical energy supplied to the actuator coil 103 and its concomitant dynamically changeable electromagnetic field created when the coil is energized can be made additive or subtractive to the static magnetic field flux generated by the permanent magnet 108 such as by changing the polarity of the electric power. For example, if the user wishes to increase the pull force required over a portion of the travel or displacement of the trigger, the microcontroller 200 may be programmed to change polarity of power source 122 to make the electromagnetic field of the snap actuator additive. In such a setup, the electromagnetic lines of flux of the actuator when energized circulate and act in the same direction in the single closed flux loop as the static magnetic flux generated in the trigger mechanism 100 by the permanent magnet 108. The flux density increases at the air gap A. This increases the magnetic attraction between the yoke 102 and rotating trigger member 104, thereby concomitantly increasing the resistance to rotation of the trigger member by the user making it harder to further pull the trigger (i.e. heavier trigger pull).

Conversely, if the user wishes to decrease the pull force over the travel of the trigger, the microcontroller may be programmed to change polarity of power source 122 to make the electromagnetic field of the snap actuator subtractive. In such a setup, the electromagnetic lines of flux of the actuator when energized circulate and act in the opposite direction in the closed flux loop as the static magnetic flux generated in the trigger mechanism 100 by the permanent magnet 108. The flux density decreases at the air gap A. This

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decreases the magnetic attraction between the yoke **102** and rotating trigger member **104**, thereby concomitantly decreasing the resistance to rotation of the trigger member by the user making it easier to further pull the trigger (i.e. light trigger pull).

The magnitude of the peak trigger pull force required to fully actuate the electromagnetic trigger mechanism **100** may also be altered by the user. This may be achieved in one embodiment by configuring the actuation control circuit **202** associated with microcontroller **200** to increase or decrease the output voltage to the electromagnet coil **106** of snap actuator **123** from power source **122** which passes through and is controlled by the actuation control circuit **202** (reference FIG. 9). This results in either a decrease or increase in the peak trigger pull force required to be exerted on the rotating trigger member **104** by the user to pull and fully actuate the trigger mechanism **100**. This parameter may be configured in conjunction with preprogramming the actuator **123** to operate the secondary electromagnetic field in either the additive or subtractive mode described above, thereby advantageously creating a highly customized the trigger pull force-displacement profile or curve in accord with user preferences.

It bears noting that inclusion of the permanent magnet **108** also advantageously conserves energy by reducing power consumption. The static magnetic field of the permanent magnet **108** automatically maintains the rotating trigger member **104** of electromagnetic trigger mechanism in the unactuated state or position at rest. Accordingly, the magnetic field generated when the coil **106** of the trigger mechanism snap actuator **123** is energized is not required at all times such as when the trigger **121** is not pulled to simply hold the rotating trigger member **104** in the vertical unactuated state or position. To minimize power consumption, the trigger mechanism actuator therefore only needs to be energized once the trigger (i.e. rotating trigger member **104**) is pulled, which is sensed by trigger sensor **159** and the control system. After the trigger pull is completed and the firearm is discharged, the actuator coil may be de-energized until the next trigger pull cycle. This arrangement and mode of operation advantageously extends battery life of the power source **122**. Accordingly, the permanent magnet **108** provides energy conservation benefits in addition to creating the initial trigger pull force and primary resistance to movement of the electromagnetic trigger mechanism **100**.

As shown in FIG. 7, the stationary yoke **102** and rotating trigger member **104** of the snap actuator **123** are configured to create a magnetic circuit having a single closed flux loop or path. By orienting the north pole N and south pole S of permanent magnet **108** in any direction, a magnetic static holding force is created which draws the rotating member **104** to the stationary yoke **102**. As one non-limiting example, assuming the north pole N were facing towards the rotating trigger member **104** as illustrated, the static magnetic flux circulates or flows through the flux circuit between the north and south magnetic poles in the clockwise direction indicated by solid static magnetic flux field arrows Ms. This draws the rotating member **104** and yoke **102** together at permanent magnet **108** to hold the trigger mechanism in the unactuated ready-to-fire position shown. When the power source **122** is configured via microcontroller **200** to operate in the “additive” mode as previously described (based on the polarity of the electric pulse sent to the actuator), the dynamic or active magnetic flux circulates or flows through the flux circuit when energized in the same clockwise direction indicated by dashed dynamic magnetic flux arrows “Md+”. This intensifies and increases the mag-

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netic field and attraction between the yoke **102** and rotating member **104** which equates to a greater trigger pull force requirement to fully actuate the trigger mechanism. Conversely, when the power source **122** is configured by microcontroller **200** to operate in the “subtractive” mode as previously described (based on a reverse polarity of the electric pulse sent to the actuator), the dynamic or active magnetic flux circulates or flows through the flux circuit when energized in the opposite counterclockwise direction indicated by dashed dynamic magnetic flux arrows “Md-”. This lessens or decreases the magnetic field and attraction between the yoke **102** and rotating member **104**, which equates to a lesser trigger pull force (i.e. resistance) required by the user to fully actuate the trigger mechanism. In some embodiments, the active magnetic flux field can complete the trigger pull for the user upon detection of a trigger pull event. It bears noting that the actuator **123** would still operate in a similar manner if the north N and south S poles of permanent magnet **108** were reversed from the illustrated position which still creates a magnetic attractive force pulling the rotating member **104** to the yoke **102**.

FIG. 9 shows one non-limiting embodiment of a control system which enables user selectable, programmable, and precisely timed adjustment of the trigger pull force/displacement profile during a trigger pull event via application of electric control current to the electromagnetic actuator **123** of the trigger mechanism **100**. The control system includes programmable microcontroller **200** for monitoring and controlling operation of the electromagnetic trigger mechanism snap actuator and other aspect of the firearm operation in general. An actuation control circuit **202** operably coupled to power source **122** forms a control interface between the microcontroller **200** and electromagnetic actuator **123**. In some configurations, the microcontroller **200** may actually form an integral part of the actuation control circuit **202** which is mounted on the same circuit board as opposed to being a separate component electrically coupled to the control circuit. This creates a “smart” control circuit **202**.

Microcontroller **200** includes a programmable processor **210**, a volatile memory **212**, and non-volatile memory **214**. The non-volatile memory **214** may be any type of non-removable or removable semi-conductor non-transient computer readable memory or media. Both the volatile memory **212** and the non-volatile memory **214** may be used for saving sensor data received by the microcontroller **200**, for storing program instructions (e.g. control logic or software), and storing operating parameters (e.g. baseline parameters or setpoints) associated with operation of the actuator control system. The programmable microcontroller **200** may be communicably and operably coupled to a user display **205**, a geolocation module **216** (GPS), grip force sensor **206**, motion sensor **207**, battery status sensor **208**, audio module **218** to generate sound, and a communication module **209** configured for wired and/or wireless communications with other off-firearm external electronic devices configured to interface with the microcontroller. The geolocation module **216** generates a geolocation signal, which identifies the geolocation of the firearm (to which the programmable controller is attached), and communicates the geolocation signal to the programmable microcontroller **200**, which in turn may communicate its location to a remote access device. The audio module **218** may be configured to generate suitable audible alert sounds or signals to the user such as confirming activation of the actuator system, successful or failed system access attempts, component failure attention alerts, or other useful status information.

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The communication module **209** comprises a communication port providing an input/output interface which is configured to enable two-way communications with the microcontroller and system. The communication module **163** further enables the programmable microcontroller **200** to communicate wirelessly or wired with other external electronic devices directly and/or over a wide area network (e.g. local area network, internet, etc.). Such remote devices may include for example cellular phones, wearable devices (e.g. watches wrist bands, etc.), key fobs, tablets, notebooks, computers, servers, or the like.

The display **205** may be a static or touch sensitive display in some embodiments of any suitable type for facilitating interaction with an operator. In other embodiments, the display may simply comprise status/action LEDs, lights, and/or indicators. In certain embodiments, the display **205** may be omitted and the programmable microcontroller **200** may communicate with a remote programmable user device via a wired or wireless connection using the wireless communication module **209** and use a display included with that remote unit for displaying information about the actuator system and firearm status.

Besides a battery sensor **208** and trigger sensor(s) **159**, the additional sensors noted above which are operably and communicably connected to microcontroller **200** may be used to enhance operation in some embodiments. In one example, a grip force sensor **206** may be used to wake up the microcontroller **200** (e.g. usable in Step **502** of control logic process **500** in FIG. **8**).

An intentional trigger pull to discharge the firearm may be sensed or detected in one embodiment via one or more trigger sensors **159**. At least one trigger sensor is provided. Sensor **159** is positioned proximate to rotating trigger member **104** and operable to detect movement of the trigger such as by direct engagement or proximity detection. In some embodiments, the trigger sensor **159** may be a displacement type sensor configured to sensing movement and displacement position of the trigger during its travel. Sensor **159** may alternatively be a force sensing type sensor operable to sense and measure the trigger pull force F exerted on the trigger by the user. A force sensing resistor may be used in some embodiments. Trigger sensor **159** is operably and communicably connected to the microcontroller **200** via wired and/or wireless communication links **201** (represented by the directional arrowed lines shown in FIG. **9**).

Another example of potentially desirable sensors is an accelerometer or other motion sensing device such as motion sensor **207** if the firearm is moved the user indicating potential onset of an intentional firing event. By monitoring the acceleration or motion of the firearm, the sensor **207** may be used may be used in addition to or instead of grip force sensor **206** to wake up the microcontroller **200** (e.g. usable in Step **502** of control logic process **500** in FIG. **8**).

One possible enhancement to the firearm control would be to sense the movement of the trigger using sensors **159** and actuate the firing event prior to the operator feeling the end of travel of a mechanical trigger when using the actuator in a firing mechanism release role as further described herein. This would enhance trigger follow-through and greatly reduce the operator effects of flinching as the firing event approaches. Additionally, since precise trigger event timing can be provided independent of the firing actuation event, the same firing actuator can be used with many different trigger force and displacement profiles.

One enhancement to the control system disclosed herein is the inclusion of one or more wireless communications options in some embodiments such as Bluetooth® (BLE),

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Near-Field Communication (NFC), LoRa, Wifi, etc. implemented via communications module **209** (see, e.g. FIGS. **9** and **10A**). This would allow the collection of data such as rounds fired, attempted fires, acceleration forces, performance data, maintenance data, and timing and authorization events. This data could be wirelessly shared with a cellphone or other external electronic data processing/communication device, or even directly through a WiFi hub as shown in FIG. **11**. In addition, operation of the electromagnetic actuator system including programming of the trigger pull force and displacement profile in the microcontroller **200** on the firearm may be programmed and controlled via the remote device.

Referring now to FIG. **7**, further energy conservation and repeatability enhancements can be achieved by adding a spring **125** or other resiliently flexible member to the system, and the addition of a trigger displacement sensor **159**. Spring **125** may be configured and arranged to bias the lower portion **118** (i.e. trigger **121**) upper portion **120** of the rotating trigger member **104** forward to the ready-to-fire (unactuated) position relative to the upper portion **120**. The static magnetic field generated by the permanent magnet **108** conversely holds the separately pivotable upper portion **120** of rotating trigger member **104** rearward towards the yoke **102** in the unactuated position. In various embodiments, the spring **125** may be a linear spring having a linear relationship between force and displacement, or a non-linear spring which changes spring force during trigger travel as further described herein elsewhere with respect to alternate spring **126**. The spring **125** acts as a “buffer” for the magnetically-applied force on the upper member. The spring also provides the uniform feel of the trigger pull. Spring **125** may be a linear torsion spring in one embodiment as illustrated. The force “ F ” needed to extend or compress the spring **125**, or other flexible member, by a distance “ X ” is proportional to that distance multiplied by the spring constant “ k ” (per Hooke’s Law) and provides an additional force opposed to the permanent magnet **108** static holding force. In operation, as the trigger **121** (i.e. lower portion **118**) is pulled and displaced against the biasing force of spring **125** with the separately pivotable upper portion **120** remaining stationary and engaged with permanent magnet **108**, a displacement sensor **159** determines the threshold position during trigger travel (i.e. displacement distance) for energizing the electromagnet coil **106** of the snap actuator **123**. At this point, the electromagnet coil is electrically energized to cancel out the static holding force or primary resistance created by permanent magnet **108** and creates a crisp snap-like final movement of the trigger linkage. As described elsewhere herein, permanent magnet **108** provides the primary or static magnetic field that directly constrains the movement of the trigger linkage at the beginning of the trigger travel. In this present embodiment, the final trip force is selectable by sensing the desired displacement/force point to electrically break-over the electromagnetic snap actuator **123** prior to reaching the magnetic flux open-loop break-over point of the permanent magnet.

As the trigger **121** moves rearward and is displaced against the mechanical Hooke’s law force of the spring **125**, the trigger **121** (defined by rotating trigger member **104**) can be released at any point during its travel by energizing the electromagnetic trigger mechanism **100** through the use of feedback to the microcontroller **200** provided by a trigger displacement sensor **159** operably and communicably coupled to the microcontroller. As the desired preprogrammed set-point is reached which is sensed by displacement sensor **159** and received by microcontroller **200**, the

trigger **121** is released via the microcontroller energizing the electro magnetic coil **106** in a fast snap-like action that initiates the trigger movement transfer means to activate the firing mechanism such as by releasing the striking member **130** directly engaged by the trigger mechanism **100** (see, e.g. FIG. **15**), or an intermediate sear operably linked between the trigger mechanism **100** and striking member which holds the striking member in the rearward cocked position (see, e.g. FIG. **30**).

It should be noted that spring **125** if provided affects and establishes a mechanically-based component of the force/displacement profile for the trigger **121**. Permanent magnet **108** may be considered to establish a magnetically-based component of the force/displacement profile. In one embodiment, spring **125** acts in a biasing direction counter to the holding force created by permanent magnet **108**. Spring **125** therefore acts in such an arrangement to assist the user in pulling the trigger against the static magnet holding field of the magnet **108**. Permanent magnet **108** acts to reset the rotating trigger member to the vertical unactuated position after a trigger pull event even in embodiments without a spring which may be sufficiently fast acting to support multiple trigger pulls in rapid succession. As a corollary, it bears noting that the trigger **121** of the snap actuator trigger mechanism **100** is not returned to the unactuated position by the microcontroller **200** and power source **122**. Instead, the magnet **108** and/or other mechanical means (e.g. springs) that might be provided are used to reset the trigger. This allows the actuator coil **106** to be de-energized at the end of the full trigger travel or displacement until needed during the next trigger pull event, which conserves battery power.

Additional enhancements can be combined to alter and/or improve the trigger feel. In one embodiment, a segmented trigger design shown in FIGS. **13A-B** may be used to create a non-linear trigger force displacement curve using a non-linear spring **126** or other resiliently flexible member and the electromagnetic snap actuator **123** of trigger mechanism. In this embodiment, the upper segment or portion **120** of the rotating trigger member **104** is pivotably coupled to and independently movable relative to the lower segment or portion **118**. Spring **126** has a fixed end rigidly attached to or formed integral with the lower portion **118** of trigger member **104** and a free end engaged with the upper portion **120** of the trigger member. Spring **126** engages the rear surfaces of the upper and lower portions **120**, **118** which acts to bias the trigger forward to the ready-to-fire vertical position.

In operation, as the trigger (i.e. lower portion **118**) is displaced against the biasing force of spring **126** with the separately pivotable upper portion **120** remaining stationary and engaged with permanent magnet **108**, a displacement sensor **159** determines the threshold position during trigger travel (i.e. displacement distance) for energizing the electromagnet coil **106** in the snap actuator. At this point, the electromagnet coil is electrically energized to cancel out the permanent magnet **108** generated static holding force or primary resistance and creates a crisp snap-like final movement of the trigger linkage. The final trip force is selectable by sensing the desired displacement/force point to electrically break-over the electromagnetic snap actuator prior to reaching the magnetic flux open-loop break-over point of the permanent magnet.

FIG. **12** shows a representative non-linear force-displacement curve for the proposed segmented trigger design of FIGS. **13A-B**. A non-linear means or mechanism such as a combination of springs, flexible members and linkages is used to create the trigger displacement profile shown and the

displacement sensor **159** is used to adjust the point at which the electrical trigger's break-over point is tripped. In the event of a failure of the electrical system, the default open-loop break-over point will provide a higher force trip point as a default operating point for the trigger. Many variations of the force-displacement curve could be possible using different springs, flexible members, and linkages.

In FIGS. **13A-B**, the non-linear displacement force curve characteristics are achieved using a non-linear leaf spring **126**. The first portion of the segmented trigger force-displacement curve is defined by the characteristics of the deformation of the non-linear leaf spring. When the trigger travel or displacement reaches and crosses the desired set-point, as measured using the trigger displacement trigger sensor **159** and relayed to the microcontroller **200**, an electrical signal to the actuator triggered by the microcontroller snaps the upper segment of the trigger forward to interact with a traditional trigger bar linkage, sear, or alternative firing means. Although a leaf spring **126** is disclosed herein as an example of a spring exhibiting a non-linear relationship between force and displacement, other types of non-linear springs may be used such as for example without limitation a non-linear dual pitch helical coil springs, conical/tapered springs, barrel compression springs, etc.

FIGS. **14A-B** shows another possible embodiment of the invention where the non-linear displacement force curve characteristics are achieved using a flexing member **127** combined with a secondary non-linear leaf spring **126**. In this construction, the upper segment or portion **120** of rotating trigger member **104** is hingedly connected to the lower segment or portion **118** by a structurally integral portion of the trigger member body have a reduced transverse cross section in comparison to the upper and lower portions. The cross-sectional shape may be rectilinear in one embodiment. This creates a resiliently flexible and spring-like connection between the upper and lower portions of the rotating trigger member **104**. Flexing member **127** acts as a elastically deformable living hinge. Other optional means for creating different force-displacement trigger profiles, before the magnetic break-over trip point, can be easily integrated with the magnetic snap actuation of the trigger mechanism **100** to those skilled in firearm trigger design. This could include the novel application of the magnetic snap actuation combined with mechanical trigger means used in traditional non-adjustable trigger designs. An apparent extension of the embodiment would include the application of the magnetic snap actuation combined with adjustable traditional mechanical trigger designs in a hybrid trigger design.

FIG. **15** shows the non-linear segmented trigger mechanism **100** with snap action magnetic break-over design used as a low-force sear surface and integrated into the release of a firearm striking member **130** in the form of a pivotable hammer, already described in detail above. This represents one non-limiting example of how the variable force trigger actuator could interface with existing firearm firing mechanism designs. Those skilled in firearm design can easily adapt this modular design to interface with other firing mechanisms as a direct replacement for the trigger mechanism.

The trigger member **104** in FIGS. **7** and **13-15** commonly share the design feature that the upper portion **120** of the trigger member is moveable independently of the lower portion **118** below the pivot **101** which is configured for a user's finger grip. Accordingly, in such a case, the upper portion **120** may alternatively be considered as simply a

rotating member of the electromagnetic actuator **123** which is coupled to the trigger formed by the lower portion **118**.

Referring to any of the foregoing embodiments of FIGS. **6**, **7**, and **13-15**, an overview of basic theory of operation for the trigger mechanism **100** will now be described. The permanent magnet **108** contained within a closed loop magnetic yoke arrangement provides the fixed or static holding force for resisting movement of the trigger and associated sear **131**. The holding force acts on the movable upper portion **120** of rotating trigger member **104**. The magnetic yoke cross-sectional area and soft magnetic properties are chosen to maximize the efficiency of conducting the magnetic flux lines and provide inherent immunity to external magnetic field interference. The magnetic coil **106** can be energized, in either polarity, to add to or subtract from the fixed holding force of the permanent magnet which will result in changing the release force necessary to move the trigger and release the sear formed thereon.

In the un-energized state of the actuator **123**, an operator can apply pressure to the rotating trigger member **104** until it exceeds the fixed holding force of the permanent magnet **108** at which time the trigger and its integral sear **131** will move, thereby releasing the striking member **130** (e.g. hammer or striker) to strike a chambered round and discharge the firearm. Ideally, the fixed un-energized holding force provided by the permanent magnet **108** may be chosen to product a heavy trigger pull force that would be acceptable as a manual default should battery power or a failure of the magnetic coil or control logic result in a failure to operate properly electronically. An example of this open-loop breakover trigger force profile is shown in FIG. **12**.

In normal operation, a range of trigger release forces can be chosen by applying electricity to the magnetic coil via microcontroller **200** to add to or subtract from the fixed holding force of the permanent magnet. An example of this new electrically adjusted breakover trigger force profile is also shown in FIG. **12** (dashed line curve). Because it is impractical to have the magnetic coil **106** energized at all times to extend battery life, the preprogrammed control logic executed by microcontroller **200** is used to determine the exact timing when to energize the magnetic coil, by how much (i.e. magnitude of electric voltage applied), and in what polarity (i.e. additive or subtractive).

A simple mechanical switch could be used for trigger sensor **159** in its most basic form to sense the movement of the trigger initiated by the user or shooter. Other means such as a displacement and/or force sensor can be used instead of or in combination with a mechanical switch as previously described herein to determine that an operator has taken a positive action to pull and actuate the trigger.

In its simplest form, a potentiometer **371** as shown in FIG. **33** and electrically coupled between the power source **122** and snap actuator **123** could be used as the electronic control system to mechanically adjust and select a desired amount of voltage from a battery source to be applied to the magnetic coil **106**. Potentiometer **371** provides a manually adjustable output voltage which is directed to the actuator **123** to either add to or subtract from the permanent magnetic holding force applied by permanent magnet **108**. This allows the user to select the desired static magnetic holding force and concomitantly trigger force necessary to actuate the trigger mechanism. Potentiometer includes a manually rotatable or linearly movable slider or wiper allowing the user to adjust the output voltage. Potentiometers are commercially available.

Alternatively, a simple basic electronic logic circuit or instructions implemented by microcontroller **200** and asso-

ciated circuitry could be used to control precisely the polarity, the amount of voltage, and timing of the electrical energy pulse sent to the magnetic coil **106** by the microcontroller for energizing the actuator **123** of trigger mechanism **100**. This allows the user to highly customize the trigger pull force-displacement profile. Actuation control circuit **202** (see, e.g. FIG. **9**) may be configured to include a digital potentiometer which is well known in the art. This provides adjustment of the magnitude of output voltage provided to actuator **123**, thereby concomitantly allowing the magnitude of the required peak trigger pull force to be selected in addition to the other parameters such as polarity and timing of the electric signal pulse. FIG. **8** depicts one embodiment of a core or basic control logic which may be preprogrammed into microcontroller **200** to configure operation of the microcontroller and control snap actuator **123** of trigger mechanism **100**. This control logic process may be used alone, or as the core for a more complex and detailed logic process used to control operation of the electromagnetic actuator **123** of trigger mechanism **100**.

Referring now to FIG. **8**, the control logic process **500** used to operate trigger mechanism **100** in one embodiment may start with activating and initializing the microcontroller **200** in Step **502**. This may be initiated automatically in one embodiment via a wakeup signal from the grip force sensor **206** (see, e.g. FIG. **9**) or other means. In Step **504**, user activity on the trigger is sensed and measured by the trigger sensor **159** (e.g. a trigger pull) and a corresponding real-time data signal is transmitted to microcontroller **200**. The sensor **159** may be a force or displacement type sensor in some embodiments, and the real-time data relayed to microcontroller **200** contains a respective type of information associated with the type of sensor being used (e.g. applied actual trigger pull force F or actual displacement distance of the trigger during its rearward travel). In one implementation, the displacement type sensor may be configured in its simplest form to merely measure movement of the trigger. The trigger activity real-time data may change over time during the trigger pull as the user further applies force or pressure on the trigger which is displaced by an increasingly greater distance. In Step **506**, a test is performed by the microcontroller **200** which compares the real-time trigger activity data to a force or displacement setpoint preprogrammed into the microcontroller **200** by the user. If the microcontroller determines the measured real-time actual trigger force or displacement is less than the setpoint, control passes back to Step **504** to be repeat Steps **504** and **506**. If the microcontroller determines that the measured real-time actual trigger force or displacement is greater than or equal to the preprogrammed setpoint, control passes forward to Step **508** in which the microcontroller sends an electric control pulse to actuator electromagnet coil **106**. The actuator **123** becomes energized to implement the trigger force and release profile or curve having the characteristics preset by the user in the microcontroller **200**. In Step **510**, the process circuitry is reset in anticipation of the next trigger pull event.

To achieve a crisp fast acting trigger release feel with a reliable means for varying the trigger force, one embodiment may include force or displacement type sensor **159** monitored by microcontroller **200** that determines, in real time, when the desired degree of actual trigger force or displacement is applied to the trigger by the user during a trigger pull event. At this point, a pulse of electrical energy is applied to the magnetic coil **106** by the microcontroller to quickly lower the static magnetic holding force breakover point for

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actuating the trigger mechanism **100** and releasing its integral sear **131** to discharge the firearm.

Control and adjustment of the dynamically variable force electromagnetic actuator trigger mechanism would ideally be through the use of microcontroller **200**. Such a control system could easily be configured with a wireless communication capability such as Bluetooth BLE, NFC, LoRa, WiFi or other commercial or custom communications means (see, e.g. FIG. **10A**). Additionally, wireless communications, applications using an external electronic device **372** such as smartphone, tablets, personal wearable devices, or other custom external devices could be used to control the variability of the trigger feel. Additionally, the direct sensing of the trigger means provides a rich area for the implementation of data collection on the performance and operation of the device. Shot counting, shot timing, pre-fire trigger analysis, and post firing performance analysis can be tied to internal sensing of the trigger event and electrically interfaced to the user through wired or wireless connections to the external electronic device (see, e.g. FIG. **11**).

Dual Closed Magnetic Flux Loop Path Embodiment

FIGS. **16-30** depict an electromagnetically adjustable firing system of a firearm having an alternative non-limiting embodiment of an electromagnetic trigger mechanism **300** using a second magnetic flux loop. The second magnetic flux loop or path provides additional design features that provide faster snap action at the trigger breakover point and the ability to actively pull the trigger through its full range of travel on its own under magnetic power without additional external force or displacement from the operator's finger on the trigger. This advantageously provides essentially a powered follow through motion of the trigger and elimination of the operator feeling any of the remaining resistance of movement of the sear release linkages and parts. A principle advantage of the dual loop design is that it makes the operation of the trigger less susceptible to tolerance variations in the magnetic circuits. Trying to "buck" the magnetic holding force to exactly zero in a single loop design is generally not practical.

Trigger mechanism **300** includes an electromagnetic snap actuator **350** configured to form the dual closed magnetic flux loop or paths. Actuator **350** may be a non-bistable release type electromagnetic actuator in which the actuator is not energized to change position for either initiating movement or to reset the actuator similar to trigger mechanism snap actuator **123** previously described herein. Instead, similarly to actuator **123** previously described herein, microcontroller **200** may be programmed and configured to energize the present actuator **350** of the dual flux loop design only in response to a manual trigger pull. This generates the secondary dynamic or active magnetic field which interacts with the primary fixed or static magnetic field generated by the permanent magnet **308** in either an additive or subtractive operating mode depending on the polarity of the power source **122** established via the microcontroller. The present actuator **350** is configurable by the user or shooter via the microcontroller **200** to change the trigger pull force and displacement profile in the same manner described above for single flux loop electromagnetic actuator **123**.

Referring to FIGS. **16-29**, trigger mechanism **300** generally comprises electromagnetic snap actuator **350** and a trigger member **320** which may be pivotably coupled to the actuator in one embodiment. Viewed from the perspective of being mounted in a firearm held by a user or shooter (see, e.g. FIG. **30**), actuator **350** includes a front side **310**, rear side **311**, right and left lateral sides **312**, **313**, bottom **314**, and top **315**. Actuator **350** comprises a stationary magnetic

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yoke **302**, movable central rotating member **304**, and electromagnet coil **306** which is operably connected to an electric source of power such as power source **122** onboard the firearm, as previously described herein. Yoke **302** defines mechanically robust main body or housing of the actuator, which is configured for removable mounting to a chassis or frame **22** of the firearm (see, e.g. FIG. **30**) by any suitable mechanical coupling means, such as for example without limitation fasteners, interference or press fit, mechanically interlocked surfaces, combinations thereof, or other. The yoke **302** is amenable for use in any type of small arms or light weapons using a trigger mechanism, including for example handguns (pistols and revolvers), rifles, carbines, shotguns, grenade launchers, etc.

Yoke **302** includes an outer yoke portion **305** and a central inner yoke portion **307**. The outer yoke portion **305** has a circular annular and circumferentially extending body which may be considered generally O-shaped in configuration. Outer yoke portion **305** circumscribes a central space **303**. Inner yoke portion **307** is nested inside the outer yoke **305** in the central space **603**. Outer yoke portion **305** generally comprises a common horizontal bottom section **305A**, upwardly extending rear and front vertical sections **305B**, **305C** spaced laterally apart, and a pair of inwardly-turned top sections **305D**, **305E** having a horizontal orientation. Each top section **305D**, **305E** is removably attached directly to a respective one of the vertical sections **305B** and **305C** to facilitate assembly of the actuator **350**. In one embodiment, each top section **305D**, **305E** may be attached to a vertical section by a pair of laterally spaced apart longitudinal fasteners such as cap screws **316** which extend through axial bores **318** in vertical sections **305B**, **305C** and engage corresponding threaded sockets **319** formed in the top sections. The top sections **305D**, **305E** when mounted to each of the vertical sections **305B**, **305C** are horizontally and longitudinally spaced apart to define a top gap or opening **309** therebetween which communicates with the central space **303** of the outer yoke. A working end portion **304A** of the rotating member **304** is received between the top sections **305D**, **305E** in opening **309** and movable therein when the actuator **350** is actuated, as further described herein.

The inner yoke portion **307** is generally straight and vertically elongated forming a substantially hollow structure defining an internal upper cavity **330** which movably and pivotably receives rotating member **304** therein. Inner yoke portion **307** may be formed as integral unitary structural part of the outer yoke portion **305** as shown in the figures and extends upwards from the horizontal bottom section **305A** thereof into central space **303**. Inner yoke portion **307** is cantilevered from the outer yoke portion **305** in this construction. In other embodiments, inner yoke portion **307** may be formed as a separate component attached to bottom section **305A** of outer yoke portion **305** such as via fasteners, adhesives, welding, soldering, etc. Inner yoke portion **307** is orientated parallel to the rear and front vertical sections **305B**, **305C** of the outer yoke portion **305**. The inner yoke portion **307** may be spaced approximately equidistant between the rear and front vertical sections **305B**, **305C** to facilitate winding coil **306** around the inner yoke portion in the central space **303** of actuator **350**.

Because the rotating member **304** is sheathed or shrouded by inner yoke portion **304** for a majority of its length in one embodiment as best shown in FIGS. **28** and **29**, possible physical interference between the coil **306** windings on the actuator and the rotating member is avoided. This arrangement therefore advantageously prevents impeded movement

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and response time or speed of the rotating member when actuated which might create undue pull resistance on the trigger member 320.

In one embodiment, yoke 302 comprising the outer yoke portion 305 and integral inner yoke portion 307 may be split longitudinally (i.e. lengthwise) front a right half-section 305RH and left half-section 305LH. This split casing arrangement facilitates assembly of the rotating member 304 inside the inner and outer yoke portions. The half-sections 305RH and 305LH may be mechanically coupled together by any suitable means, including for example without limitation fasteners including screws and rivets, adhesives, welding, soldering, etc. In one embodiment, threaded fasteners such as transverse cap screws 317 may be used.

Each half-section 305RH, 305LH defines a portion of the vertically elongated upper cavity 330 in inner yoke portion 307 which pivotably receives rotating member 304 partially therein. The cavity 330 communicates with a downwardly and rearwardly open internal lower cavity 331 of the actuator 350 formed in outer yoke portion 305. Lower cavity 331 pivotably receives bottom actuating section 304B of rotating member 304 therein. Lower cavity extends rearward from the central pivot region of the outer yoke portion 305 (containing pivot pin 335) to the rear side of the actuator 350 and bottom section 305A of the outer yoke portion. Upper cavity 330 extends vertically from the lower cavity 331 and penetrates the top and bottom ends of the central inner yoke portion 307.

Referring particularly to FIG. 28, upper cavity 330 in inner yoke portion 307 of yoke 302 defines a pair of opposing front and rear inner wall surfaces 307A, 307B on the front and rear of the cavity. Cavity 330 is configured to allow full pivotable actuation movement or action of the rotating member 304 about its pivot axis PAL. To achieve this functionality, the inner wall surfaces 307A-B have a non-parallel converging-diverging relationship in so far that these wall surfaces converge moving downwards in cavity 330 towards the pivot axis PA1 of the rotating member 304 and diverge moving upwards towards the top open end of the inner yoke portion 307. The front inner wall surface 307A is obliquely angled to the rear inner wall surface 307B such that upper cavity 330 of inner yoke portion 307 is wider at the top and narrower at the bottom from front to rear. In one embodiment, the front inner wall surface 307A may be obliquely angled to the vertical central axis CA of actuator 350 and rear inner wall surface 307B may be parallel to central axis CA. The foregoing arrangement permits pivotable motion of the rotating member 304 forward and rearward in the upper cavity 330.

Rotating member 304 has a vertically elongated body including a top or upper operating end section 304A, bottom or lower actuating end section 304B, and intermediate section 304C extending therebetween. Both top operating end section 304A and bottom actuating end section 304B may be enlarged and longitudinally/horizontally elongated in the front to rear direction relative to intermediate section 304C in one embodiment as shown to achieve their intended functionality. In one embodiment, intermediate section 304C may have parallel sides and be generally rectilinear in configuration and cross-sectional shape. Operating end section 304A is configured to operably interface with the both the outer yoke portion 305 of yoke 302 and the firing mechanism of the firearm as further described herein. When the electromagnetic actuator 350 is fully assembled, the operating end section 304A protrudes upwards beyond the inner yoke portion 307 of yoke 302 and is exposed to engage

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both the outer yoke portion 305 and a firing mechanism component or mechanical linkage.

The top operating end section 304A of rotating member 304 may be generally cruciform-shaped in one embodiment defining horizontally/longitudinally protruding front and rear extensions 332. This portion of operating end section 304A may be considered to generally resemble double-faced hammer in configuration and defines two opposite and outwardly facing front and rear actuation surfaces 334F, 334R (see, e.g. FIG. 28). When the actuator 350 is cycled between its two actuation positions by a user via a trigger pull, the actuation surfaces 334F, 334R are arranged to alternately engage the top sections 305D, 305E of the outer yoke portion 305. In one embodiment, rear actuation surface 334R engages permanent magnet 308 affixed to the rear top section 305D of outer yoke portion 305.

Actuator 350 may further include an engagement feature strategically located on the upper portion of central rotating member 304 and configured to interface with a component of the firearm's firing mechanism in release-type operational role. In various embodiments, the engagement feature may be an operating extension or protrusion 333 of the rotating member 304 as illustrated in FIGS. 16-29, a socket or recess formed in the rotating member (not shown), or other element of other type and/or configuration (not shown) capable of mechanically interfacing with the firing mechanism. Although the engagement feature may be described herein for convenience of description and not limitation as an operating protrusion 333, any other form of engagement feature may be provided so long as the feature is capable of mechanically interfacing with a portion of the firing mechanism.

Operating protrusion 333 extends upwards from between the front and rear extensions 332 at the top of the rotating member 304. Operating protrusion 333 may be approximately centered between actuation surfaces 334F, 334R in one embodiment; however, other positions of the operating protrusion may be used depending on the interface required with the firing mechanism component acted upon by the operating protrusion 333. The operating protrusion 333 may be configured to releasably engage a firing mechanism component or linkage in a direct release role or an indirect release role. Accordingly, operating protrusion 333 may be configured and operable to act directly on the energy storage device such as the spring-biased striking member 130 shown in FIG. 15, or indirectly by acting on a separately mounted pivotable sear 375 which in turn is releasably engaged with the striking member (see, e.g. FIGS. 16-30).

Permanent magnet 308 may be fixedly attached to rear top section 305D of outer yoke portion 305 in a position between the top section 305D and the rotating member 304. Rear top section 305D may include a flat forward facing surface 308a for mounting the permanent magnet 308. This arrangement advantageously magnetically attracts and engages rotating member 304 to create a static holding force on the rotating member. Rotating member 304 is magnetically biased rearwards towards its rearward unactuated position associated with a corresponding unactuated forward position of the trigger member 320 when not pulled by the user. Any suitable mechanical coupling means may be used to affix magnet 308 to the outer yoke portion 304, including for example without limitation adhesives, fasteners, welding, soldering, etc.

The enlarged bottom actuating end section 304B of the rotating member 304 may be completely disposed in lower cavity 331 of outer yoke portion 305 in one configuration and enclosed therein by the yoke 302. Actuating end section

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304B includes a horizontally/longitudinally elongated cantilevered rear actuating arm or extension 340 used to manually actuate the rotating member 304 via a trigger pull by the user. This may be considered to give the rotating member 304 a generally L-shaped body configuration. Actuating extension 340 extends rearward from the central pivot region of the bottom actuating end section 304B towards the rear side 311 of the actuator 350. In one embodiment, the actuating extension 340 may be formed integrally with the rotating member body as a unitary monolithic structural part thereof. Actuating extension 340 may be obliquely angled to the vertical central axis CA of actuator 350 and may extend completely to the rear side 311 of the actuator such that the free terminal rear end of the actuating extension is exposed for attachment of monitoring or sensing devices, as further described herein.

The rear actuating extension 340 includes an upwardly facing spring seating surface 341 and downwardly facing actuation surface 342. Each surface may be substantially flat or planar in one configuration. Surfaces 341 and 342 may be formed on a laterally widened paddle-shaped portion of actuating extension 340 at the terminal rear end of the extension as shown (best seen in FIGS. 20 and 21). This increases the surface area of the seating and actuation surfaces 341, 342 in contrast to portions of the actuating extension 340 extending forward from the paddle-shaped region.

Spring seating surface 341 of the rear actuating extension 340 is engaged by one end of an operating or trigger return spring 344 disposed in vertical spring socket 345 formed in yoke 302. In one embodiment, spring socket 345 may be formed in rear vertical section 305B of the outer yoke portion 305 as shown. Spring 344 may be a helical coil compression spring in one embodiment; however, other type springs may be used. Spring 344 acts to bias the rear actuating extension 340 downward, which in turn rotates the rotating member 304 about pivot pin 335 to bias the top operating end section 304A into engagement with the permanent magnet 308 when the trigger member is not pulled and actuated (e.g. ready-to-fire position).

Rotating member 304 may be pivotably mounted to yoke 302 via a pivot protuberance such as pivot pin 335 which defines a pivot axis PA1. Rotating member 304 is movable between a rearward unactuated position magnetically engaged with permanent magnet 308 (or yoke 302 in other embodiments depending on placement of the magnet), and a forward actuated position disengaged from the permanent magnet. It bears noting that the rotating member 304 may be moved between the two positions by sensing user action on the trigger member 320 which then energizes the actuator 350. Movement of the rotating member 304 then comes under the influence of the secondary electromagnetic field generated by the electromagnetic actuator 350 when energized by the microcontroller 200, which can either assist with completing the trigger pull for the user, or retard trigger travel/displacement by creating a resistance force on the trigger as previously described herein.

In one embodiment pivot axis PA1 may define a common pivot axis for mounting both the rotating member and trigger member 320 to yoke 302 of snap actuator 350 in one embodiment. Pivot pin 335 therefore defines a common center of rotation about which both the rotating member 304 and trigger member 320 each pivot or rotate independently of each other. Common pivot axis PA1 is aligned with central axis CA of the actuator 350 which passes through this pivot axis. In one embodiment, pivot pin 335 is disposed inside lower cavity 331 of the outer yoke portion 305 which serves

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as the mounting point for the rotating member and trigger member. Rotating member 304 and trigger member 320 each include laterally open pivot holes 336 and 337 respectively for inserting pivot pin 335 therethrough. Holes 336 and 337 are concentrically aligned when the trigger mechanism 300 is fully assembled.

In one construction, as shown, pivot pin 335 may comprise two right and left half-pin sections 335R, 335L each fixedly disposed on a respective right and left yoke half section 305RH, 305LH. In one embodiment, half-pin sections may be integrally formed with the right and left yoke half sections. Each half-pin section collectively forms a complete pin extending from the right to left yoke half-section when assembled together to capture both the rotating member 304 and trigger member 320 thereon and therebetween the yoke half sections. In an alternative embodiment, a single one-piece pivot pin may instead be used which extends completely through lower cavity 331 of outer yoke portion 305 from right to left. In one embodiment, pivot pin 335 is preferably circular in cross section.

Referring to the exploded views of electromagnetic actuator 350 in FIGS. 20 and 21, the foregoing split construction of yoke 302 facilitates preassembly of the rotating member 304, electromagnet coil 306, and the trigger assembly or member 320 to the yoke to form a self-supporting electromagnetic trigger unit which is configured for mounting to the firearm via any suitable mechanical manner. Because the rotating member 304 and trigger member 320 (i.e. outer trigger 321) are pivotably mounted on pin 335 inside cavity 330 of the central section or portion 307 of yoke 302, these components require mounting before the right and left half-sections 305RH, 305LH of the yoke are assembled and fastened together. A general method for assembling actuator 350 in one non-limiting scenario may therefore comprise the sequential steps of: inserting trigger spring 344 into the downwardly open spring socket 345 of the yoke 302; inserting the inner trigger 322 into the outer trigger 321; inserting the pivot pin 323 transversely through the outer and inner triggers to complete assembly of these components; inserting the bottom actuating section 304B of rotating member 304 into the U-shaped channel 361 of the outer trigger 321 (inner trigger spring 365 being pre-mounted to the underside of bottom actuating section 304B using fastener 366); pivotably mounting the rotating member 304 and trigger member 320 on pivot pins 335R or 335L on the yoke 302 inside cavity 330; assembling or joining the right and left half-sections 305RH and 305LH of yoke 302 together using fasteners 317; winding the electromagnet coil 306 around central inner yoke portion 307; and attaching and mounting each rear and front top section 305D, 305E to its respective one of the vertical sections 305B and 305C of the outer yoke portion 305 using fasteners 316 (the permanent magnet 308 being pre-mounted on the rear top section 305D). Variations of the assembly sequence are possible and not limiting of the invention. In one embodiment, the assembled electromagnetic actuator trigger unit may be dropped into an upwardly open receptacle of the firearm frame 22 (see, e.g. FIG. 30) for securing the unit to the firearm. The electromagnetic trigger unit may alternatively be mounted to the firearm frame via fasteners or other methods.

The trigger member 320 will now be described in further detail. With continuing reference to FIGS. 16-29, trigger member 320 may include an outer trigger 321 and inner safety trigger 322 movable relative to the outer trigger. Inner safety trigger 322 includes an enlarged upper mounting portion 324 and lower blade portion 326 depending down-

wards therefrom for actuation by a shooter or user. The blade portion 326 may have an open framework construction including an arcuately concave front surface configured to facilitate engagement by the shooter or user's finger. The mounting portion 324 is pivotably mounted to outer trigger 321 via a second pivot pin 323 which defines a transverse second pivot axis PA2. Pivot pin 323 extends transversely through laterally open mounting holes 329 and 328 formed in the mounting portion 324 and outer trigger 321 respectively. Safety trigger 322 is pivotable independently of both the outer trigger 321 and rotating member 304 between forward and rearward positions. Pivot axis PA2 may be parallel to transverse pivot axis PA1 about which the trigger member 320 and rotating member 304 rotate. Pivot axis PA2 may be below pivot axis PA1 and is offset rearwards from the vertical central axis CA of the actuator. A transversely oriented safety bar 325 is carried by the upper mounting portion 324 and is arranged to selectively engage or disengage an upwardly open safety notch 327 formed in the cantilevered rear actuating extension 340 of the rotating member 304. In one embodiment, actuating extension 340 runs through a an upwardly open longitudinal slot formed in the upper mounting portion 324 of safety trigger 322 and is captured beneath the safety bar 325, but movable up/down when the rotating member 304 is actuated.

The outer trigger 321 includes an upper mounting portion 362 and a lower blade portion 363 depending downwards therefrom. The blade portion includes a vertical slot 364 for movably receiving the inner safety trigger 322 therethrough when actuated by the user. Blade portion 363 may have an arcuately concave front surface configured for engagement by the user's finger. The mounting portion 362 of outer trigger 321 may have a U-shaped body in one embodiment defining a forwardly and upwardly open channel 361 which movably receives the lower actuating section 304B of rotating member 304 therein. The rear actuating extension 340 of rotating member 304 also extends through channel 361. The actuating section 304B of the rotating member is therefore nested inside the mounting portion 362 of the outer trigger 321.

Outer trigger 321 further includes a cantilevered rear operating arm or extension 360 arranged to engage the rear actuating extension 340 of the rotating member 304. In one embodiment, operating extension 360 protrudes rearwardly from the mounting portion 362 of outer trigger 321. Operating extension 360 defines a flat or planar upwardly facing operating surface 343 configured and arranged to abuttingly engage downwardly facing actuation surface 342 of rotating member 304. The interface between the operating surface 343 and actuation surface 342 is one of a flat-to-flat interface in one embodiment as shown (see, e.g. FIGS. 27-29). Operating extension 360 of outer trigger 321 is biased downward by trigger return spring 344 via rear actuating extension 340 of the rotating member (which acts on the operating extension). This in turn biases outer trigger 321 forward towards the ready-to-fire position. The spring 34 maintains continuous mutual engagement between the outer trigger 321 and the rotating member 304. Outer trigger 321 is manually movable by the shooter or user between the substantially vertical forward ready-to-fire position and pulled rearward fire position.

In one embodiment, a force/displacement sensor such as a thin film force sensing resistor 370 may be interposed at the interface between the operating surface 343 of the operating extension 360 of outer trigger 321 and actuation surface 342 of the rear actuating extension 340 of rotating member 304. Force sensing resistors measure an applied

pressure or force between two mating surfaces and are commercially available from numerous suppliers. Force sensing resistor 370 is operably and communicably coupled to microcontroller 200. Force sensing resistor 370 is configured to detect and measure a trigger force F exerted by the user on the outer trigger 321 when pulled to fire the firearm 20. When paired with trigger force setpoint preprogrammed into microcontroller 200, this serves as a basis for intermittently energizing the electromagnetic snap actuator 350 based on trigger force, as further described herein.

Inner trigger 322 is biased toward its substantially vertical forward position (see, e.g. FIGS. 27 and 28) by a spring 365. In one embodiment, spring 365 may be in the form of a spring clip having a flat thin body with an upwardly angled central arm which engages a bottom surface of the inner trigger mounting portion 324 and a pair of downwardly angled legs which engage the lower trigger within channel 361. The central arm acts on the mounting portion 324 to bias the blade portion 326 of inner trigger 322 forward. The spring clip may be mounted to the underside of rotating member 304 in one embodiment by a threaded fastener 366 received in a threaded socket in the bottom actuating section 304B of rotating member 304. The bottom of rotating member 304 may comprise a recess configured to receive the spring clip. In the forward position, the blade portion 326 of inner trigger 322 protrudes forward from the outer trigger 321 (see, e.g. FIGS. 27 and 28). In the rearward position, the blade portion protrudes rearward from the outer trigger when the inner trigger is fully depressed by the user (see, e.g. FIG. 29).

In operation, the trigger mechanism 300 will be in the ready-to-fire condition shown in FIGS. 27 and 28. Both the inner safety and outer triggers 322, 321 are in their vertical forward ready-to-fire positions via the biasing action of springs 365 and 344, respectively. In this position, the safety bar 325 on the inner trigger is engaged with the rear actuating extension 340 of the rotating member 304, thereby blocking its upward movement and preventing the firearm from being fired (best shown in FIG. 27). To discharge the firearm, the shooter or user initially applies a trigger pull force F on first the safety trigger 322 which rotates rearward to its rearward position shown in FIG. 29. The safety bar 325 seen in FIG. 27 rotates forward from the position shown and becomes vertically aligned with safety notch 327 in the rear actuating extension 340 of rotating member 304. The user's trigger finger may then fully engage and rotate the trigger member 320 (i.e. collectively outer trigger 321 with inner trigger 322) rearward to the rearward fire position. This fully actuates the trigger mechanism 300 to discharge the firearm, as further described herein. Because the safety bar 325 is aligned with safety notch 327, upward movement of rear actuating extension 340 of the rotating member 304 is no longer blocked, thereby allowing the firearm to be discharged either manually or when the snap actuator 350 is energized via normal operation.

The stationary yoke 302 and the rotating member 304 may be formed of any suitable magnetic metal capable of being magnetized, such as without limitation iron, low-carbon steel, nickel-iron, cobalt-iron, etc. Suitable fabrication methods include for example without limitation metal injection molding, casting, forging, machining, extrusion, laminated stamping, and combinations of these or other methods. The method is not limiting of the invention.

The operating theory of the electromagnetic trigger mechanism 300 with snap actuator 350 is as follows. The central rotating trigger armature or rotating member 304 is surrounded by the magnetically conductive yoke 302 con-

figured to form two possible flux loop paths. A primary fixed or static magnetic flux and associated holding force is established using the permanent magnet **308** in the right hand flux loop or path to hold the central rotating member **304** firmly to the right side of its pivotal range of motion within the yoke **302**. The primary magnetic flux path generated by the permanent magnet **308** is shown in FIG. **31** (see flux arrows representing the primary static flux **M1**). The rotating member **304** is held firmly against and abuttingly engages the permanent magnet **308** as shown in FIGS. **27** and **28**. The air gap **B** on the left side of the top of the rotating member **304** ensures that the left hand magnetic flux path is sufficiently high in magnetic reluctance that essentially all of the magnetic flux from the permanent magnet **308** is contained within the right hand loop (see, e.g. FIG. **28**). A magnetic coil **306** surrounds the rotating member and when energized, the coil will generate and provide a secondary dynamically variable magnetic flux that adds to, or subtracts from, the primary fixed or static magnetic flux generated by permanent magnet **308** depending on the polarity of the electricity provided to the coil.

Under normal operation to discharge the firearm, the operator or user pulls the outer trigger **321** which applies a trigger pull force **F** thereon that acts in an opposite direction counter to the primary fixed or static magnetic field flux and holding force generated by the permanent magnet **308**. This creates pressure on and pivotably displaces the outer trigger **321** rearwards. This applied pressure and trigger displacement provides the means for sensing physical activity with the trigger sensor **370** as input for Step **504** in the control logic process of FIG. **31**. In various embodiments, the trigger sensor(s) may be a force type sensor that measures applied force in real-time, a displacement type sensor that measures displacement distance in real-time, or a combination of force and displacement sensors may be used to provide both force and displacement information relayed to the microcontroller **200** for use in activating the snap actuator **350** in accordance with the preprogrammed trigger release profile created by the user. The force type sensor senses and provides information to the microcontroller relevant to actual trigger pull force **F** being applied on the trigger by the user. This serves as a basis for comparison to the preprogrammed breakpoint or setpoint trigger pull force used to time energizing the electromagnetic actuator **350** to alter the trigger pull force-displacement profile (see, e.g. FIG. **10B**). The displacement type sensor senses and provides information relevant to the displacement distance of the trigger which may be used as the basis by the microcontroller for energizing the actuator **350** when a displacement setpoint is preprogrammed into the control system.

In one embodiment, the sensor **370** may be a thin film force sensing resistor as previously described herein which measures the magnitude of the trigger pull force **F**. Alternative approaches such as load cells, piezo-electric force sensors, displacement sensors such as hall effect sensors, GMR sensors, and optical or mechanical switches or sensors could also be used. When the force (or displacement) reaches a preset desired trigger trip or setpoint preprogrammed into microcontroller **200** for the variable force trigger, the control system applies electrical energy to the magnetic coil **306**.

At the preset desired force or displacement trip or setpoint, the pulse of electrical energy applied to the electromagnet coil **306** by microcontroller **200** generates user-selectable and adjustable dynamic secondary dual magnetic field fluxes. The two flux loop or paths for the right-hand side and left-hand side magnetic fluxes **M2** and **M3** are

shown in FIG. **32** and represented by the flux line arrows indicated. In one implementation, as depicted, the secondary flux **M2** opposes the static magnetic flux **M1** generated by the permanent magnet **308** in the right-hand side circuit when the electric pulse from power source **122** has a first polarity as controlled by microcontroller **200**. Note that the dynamic secondary right-hand side flux **M2** generated by energizing the coil is shown to circulate in a counterclockwise direction opposite to the static clockwise flux **M1** generated by permanent magnet **308** shown in FIG. **31**. The right-hand side secondary flux **M2** created by the electromagnet coil **306** is therefore considered "subtractive" and decreases the clockwise static magnetic flux **M1** in the right-hand side of the flux circuit. The energized coil **306** also simultaneously creates the additional clockwise flux **M3** in the left-hand side of the circuit. If the current in the magnetic coil **306** is sufficiently large as in the present embodiment, then the force resulting from the magnetic flux **M3** in the left-hand circuit air gap **B** will be greater than the force in the right-hand circuit, and the central rotating member **304** will snap to the left very quickly under magnetic force without any additional pull force **F** applied to the trigger by the operator or user. As the size of the air gap **B** on the left-hand side flux loop closes, an air gap **A** opens on the opposite right-hand side flux loop between the top of the rotating member **304** and permanent magnet **308** at right (see, e.g. FIG. **29**). The magnetic reluctance of the left-hand side flux loop decreases and the magnetic reluctance of the right-hand side flux loop increases causing a rapidly increasing magnetic force of attraction pulling the central rotating member **304** to the left-most position allowed by the yoke **302** shown in FIG. **29**.

When electrical energy is removed from the magnetic coil by microcontroller **200**, the left-hand flux path collapses and the static permanent magnet **308** attractive force takes back over and pulls the rotating member **304** back to the right-hand side of the yoke **302** as shown in FIG. **28**. The trigger return spring **344** provides a preferably light biasing force ensuring the positive return of the rotating member **304** to the right-side starting or ready-to-fire position in the event the permanent magnet **308** fails to positively reset the actuator **350** or another unanticipated failure of the trigger mechanism occurs. The trigger spring, however, is not an essential component in the design in all embodiments but does provide a backup system for operating the trigger mechanism **300** completely by manual means particularly in exigent circumstances if the battery charge is lost or the microcontroller **200** malfunctions.

Under conditions when the electromagnet coil **306** is not energized, either by intentional design or failure of components or weak batteries, the operator can still cycle the firearm by applying force/displacement to the outer trigger **302** that exceeds the fixed or static holding force of the permanent magnet **308**.

An alternate embodiment and application can be envisioned where the static holding force of the permanent magnet **308** is increased by applying electrical energy to the magnetic coil **306** in an "additive" manner instead that reinforces the permanent magnet's holding force. In this instance, the microcontroller **200** is configured to apply the electric pulse to electromagnet coil **306** with an opposite second polarity. The secondary dynamic right-side flux **M2** would therefore act in the same clockwise direction as the static flux **M1** seen in FIG. **31**. This could be used to greatly increase the adjustable range of the trigger setpoint. This could also be used as a safety measure to increase the trigger holding force significantly in the event of some outside

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influence where it would be desirable to require a much higher trigger pull such as under high acceleration, drops, or shocks applications. This may be done with certain firearm configurations to ensure compliance with gun safety drop tests which is a well known test procedure in the art to confirm a firearm does not fire when accidentally dropped.

One key feature of the present variable force trigger mechanisms **100** or **300** disclosed herein is the ability to select a desired trigger pull force-based release breakpoint or breakover setpoint for the trigger that is optimal for the user's experience and shooting situation. In one embodiment, the setpoint may be preprogrammed into microcontroller **200** for use in the control logic shown in FIG. **8**. In other embodiments, the selection of the setpoint can be as simple as a manual adjustment screw or knob of the potentiometer shown in FIG. **33** that interfaces with the microcontroller **200** and its basic control logic shown in FIG. **8**. Or it can be any range of options from pre-programmed to provide preset features, or totally programmable using controls mounted on the firearm, computer, or an external electronic device such as even a cellphone application that interface with the control logic unit or microcontroller **200**. Examples of implementations that can be used include: (1) a Trigger Setpoint that is selected by manually adjusting a screw, knob, or switches of a potentiometer **371** to select either a continuous range of trigger release forces or a preset number of fixed release levels; (2) a user interface using switches, knobs, buttons, touch screen or other control interface on the firearm to set the trigger setpoint parameters and communicate them to the logic control unit or microcontroller **200** shown in FIG. **9**; and (3) a wired or wireless programming device that communicates to the firearm control logic via either a cable such as a USB cable, or wireless network connection such as Bluetooth, Wi-Fi, NFC, etc. The programming device could be a simple discrete remote control device or key fob, a computer, laptop, tablet, or cellphone running a software application which communicably interfaces with microcontroller **200** and its control logic or program instructions.

FIG. **10A** graphically shows how an external electronic device **372** such as a cellphone for example could be used to select and program microcontroller **200** located onboard the firearm **20** with a trigger release profile via wireless Bluetooth communications. The wireless communications is enabled via the communication interface or module **209** in the microcontroller **200** (see, e.g. FIG. **9**). The trigger profile parameters which may be accessed and selectively adjusted by the user in this non-limiting example may include both a trigger force breakpoint or setpoint (i.e. magnitude or value of holding or breakover trigger force F necessary to release the trigger) and timing of which point during the travel or displacement of the trigger that the trigger mechanism actuator **123** or **350** will be energized by the microcontroller **200**. An example of the breakpoint or setpoint is shown in the trigger release profile of FIG. **10B**.

The cellphone microprocessor runs a local software application or "app" comprising program instructions or control logic that allows adjustment of the trigger release profile. Two application screens which may be presented to the user on the cellphone visual touchscreen are shown in FIG. **10A** as examples. When the trigger profile setting software application is launched, a first security access screen **373** may be presented which prompts the user to enter a preselected personal identification number (PIN) in a similar manner to the security PIN required by the cellphone to change some of its core user settings. The user is then presented with a second trigger settings screen **374** contain-

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ing input fields such as active icons, adjustment sliders, or other type input fields. This the user to select/enter the desired trigger breakpoint or breakover setpoint force ("Trigger Force" icon) for energizing the actuator **350** and/or timing for energizing the actuator based instead on trigger displacement ("Displacement" icon) depending on which type sensor is used. Alternatively, both type sensors may be used in some embodiments. These input fields provide the user interface which allow adjustment of the trigger force-displacement curve (FIG. **10B**) to suit the user's preferences. In one embodiment, an active trigger release profile may be displayed in screen **374** which changes in real-time to reflect the corresponding settings for the setpoint and timing being input by the user. The external electronic device **372** then wirelessly communicates the selected changed trigger settings to the microcontroller **200** which becomes programmed with the trigger parameters entered in the cellphone trigger software application. Once the setting are complete, the user may close the trigger software application on the cellphone.

It will be appreciated that numerous variations in the configuration of the trigger profile software application are possible. The trigger profile software may also be implemented in other external electronic devices, such as a laptop, notebook, electronic pad, desktop computer, or other processor-based devices capable of communication with the onboard microcontroller **200** of the firearm.

It bears noting that particularly the electromagnetic trigger mechanism **300** is substantially immune to external magnetic field which could interfere with proper operation of the trigger mechanism electromagnetic actuator **350**. The permanent magnet **308** in the embodiment presented herein provides a fixed or static holding force for a trigger-sear release system in a closed flux loop that limits susceptibility to external magnetic fields. With the exception of the small air gap created between the rotating member **304** and stationary yoke **302**, that allows for the motion of the rotating central trigger/armature (rotating member **304**), the magnetic yoke cross sectional area, and soft magnetic material properties of the yoke and rotating member to provide a low reluctance path that captures almost all of the magnetic flux generated by energizing the magnetic coil and from the permanent magnet.

Since magnetic force within the air gap increases with magnetic cross-sectional area and decreases with the square of the air gap length or width, practical designs which are optimized for force and speed tend to minimize the length or width relative to the cross-sectional area of the yoke. A consequence of this is that variable force trigger designs based on these design principles are inherently immune to external magnetic field interference. In practice, it is virtually impossible to change the state of the variable force trigger using an external magnet (and optional soft magnetic material yoke) provided the rotating member is physically isolated from the external magnet by at least one air gap distance. This will virtually always be the case in practical firearm embodiments.

FIG. **30** shows one embodiment of a firearm **20** incorporating the electromagnetic trigger mechanism **300** with dual flux loop electromagnetic snap actuator **350** shown in FIGS. **16-29**. It bears repeating that actuator **350** does not act like a non-bistable actuator characterized by the presence of a single permanent magnet **308** in the dual flux loops. Instead, the present trigger mechanism **300** and controller in this embodiment are mutually configured and operable to use a sensed externally applied force F on the trigger member as the impetus to energize the coil of the actuator **350**. Ener-

gizing actuator **350** alters the force F required to be applied by the user to pull the trigger in accordance with the trigger release profile preprogrammed into microcontroller **200** (e.g. trigger breakpoint or breakover point previously described herein). In some configurations, the actuator **350** may actually complete the full trigger pull or travel without application of additional force by the user.

In the present firearm embodiment, electromagnetic snap actuator **350** operably interacts with and releases the energy storage device such as movable striking member **130** in an indirect manner via an intermediate firing mechanism component. The central rotating member **304** of the electromagnetic snap actuator **350** in this case operably interacts with a sear **375** operably interposed in the firing linkage between actuator **350** and striking member **130** (see also FIGS. 27-29).

In one embodiment, the firearm **20** may be a semi-automatic pistol recognizing that the trigger mechanism **300** with electromagnetic actuator **350** may be used in any type firearm having a pivotably or linearly movable striking member **130** and optionally a sear **375** or other intermediate component in some designs which operate to hold and selectively release the energy storage device (e.g. hammer or striker). Accordingly, the trigger mechanism **300** may be variously embodied in firearms including for example without limitation rifles, carbines, shotguns, revolvers, or other small arms.

Firearm **20** generally includes a frame **22**, trigger guard **23** formed as a unitary structural part of the frame or a discrete guard separately attached thereto, reciprocating slide **24**, barrel **26** mounted to the frame and/or slide **24**, and a movable energy storage device such as striking member **130**. Slide **24** is slideably mounted on frame **22** for movement in a known axially reciprocating manner between rearward open breech and forward closed breech positions under recoil after the pistol is fired. A recoil spring **29** compressed by rearward movement of the slide acts to automatically return the slide forward to reclose the breech after firing. Slide **24** may be also considered to define an axially movable receiver, in contrast to a fixed receiver mounted rigidly to the frame or chassis of a long gun such as for example a rifle, carbine, or shotgun (see, e.g. FIG. 70).

Barrel **26** is axially elongated and includes rear breech end **30**, front muzzle end **31**, and an axially extending bore **25** extending therebetween. Bore **25** defines a projectile pathway and a longitudinal axis LA of the firearm which defines an axial direction; a transverse direction being defined angularly with respect to the longitudinal axis. The breech end **30** defines a chamber **32** configured for holding an ammunition cartridge C . The slide **24** defines a vertical breech face **34** movable with the slide and arranged to abuttingly engage the rear breech end **30** of barrel **26** to form the openable/closeable breech in a well known manner. The vertically elongated rear grip portion of frame **22** comprises a downwardly open magazine well which receives a removable ammunition magazine **136** therein for uploading cartridges automatically into breech area after the firearm is discharged which are chambered into the barrel via operation of the slide **24**. All of the foregoing components and operation of semi-automatic pistols are well known in the art without requiring further elaboration.

With continuing reference to FIGS. 27-30, firearm **20** in the present embodiment includes a striking member **130** in the form of a spring-biased and linearly movable striker **40**. Striker **40** is movable in a forward linear path P for striking a chambered cartridge C . Spring **28** biases the striker **40** forwards such that when the striker is released from a

rearward cocked position, the spring drives the striker forward to strike and detonate the charge in the cartridge C . Striker **40** has a horizontally-axially elongated body including a downwardly depending catch protrusion **42** which is engageable with an upstanding sear protrusion **44** of the sear **375** to hold the striker in the rearward cocked position. Sear **375** is pivotably mounted to the firearm frame **22** about a separate transverse sear pivot axis **376**. Sear protrusion **44** may be formed on one forward end of sear **375** opposite a rear end having a transverse opening which receives a cross pin **377** that defines pivot axis **376**. In one embodiment, a rear facing vertical surface on sear protrusion **44** engages a mating front facing surface of catch protrusion **42** on striker **40** to hold the striker in the rearward cocked position. Striker **44** is movable in forward path P via a trigger pull between a rearward cocked position and a forward firing position contacting and detonating a chambered cartridge C to discharge the firearm.

Sear **375** is pivotably movable between an upward standby position in which sear protrusion **44** engages catch protrusion **42** of striker **40**, and a downward fire position in which the sear protrusion disengages the catch protrusion to release the striker for firing the firearm **20**. Sear **375** is held in the upward position by engagement with upstanding operating protrusion **333** on the central rotating member **304** of electromagnetic actuator **350** of the trigger mechanism **300** (see, e.g. FIGS. 27-28). In one embodiment, the front end of sear **375** may include a downward facing engagement surface **46** formed on a forwardly extending ledge-like protrusion of the sear which is selectively engageable with an upward facing engagement surface **48** formed on operating protrusion **333** of rotating member **304**. Mutual engagement between surfaces **46** and **48** maintains the sear **375** in the upward position. Sear **375** may be biased towards the downward fire position by a spring **45** (shown schematically in FIGS. 28 and 29).

In operation, the firing mechanism is initially in the ready-to-fire condition or state shown in FIGS. 24, 27, 28, and 30. The striker **40** is held in the rearward cocked position by sear **375** which is in the upward standby position. Engagement surface **46** of the sear is engaged with engagement surface **48** of the actuator **350** (i.e. central rotating member **304**). The trigger member **320** is not yet pulled. The microcontroller **200** is programmed with the control logic shown in FIG. 8 and may be initialized and active (Step 502), such as via the microcontroller detecting user activity on the firearm, such as the user's positive grip on the frame **22** sensed by grip force sensor **206** mounted to the frame, and/or motion of the firearm sensed by motion sensor **207** (see also FIG. 9). The rotating member is in the rearward unactuated position magnetically engaged with permanent magnet **308**.

To fire the firearm **20**, the operator or user pulls the trigger member **320** thereby applying a trigger pull force F which is sensed and measured by the trigger sensor such as thin film force sensing resistor **370**. The electromagnet coil **306** is then energized by microcontroller **200** in accordance with the control logic of FIG. 8 in the manner previously described herein. The preprogrammed trigger force and displacement profile (e.g. breakpoint or breakover setpoint) is implemented in which the microcontroller energizes the electromagnetic actuator **350** and automatically adjusts the trigger activation force according to the preprogrammed profile created by the user. The user continues to pull the trigger until the central rotating member **304** of the actuator pivots forwards to the actuated position and breaks engagement with the sear **375** as shown in FIG. 29. Sear **375** then

in turn drops and pivots downward thereby releasing the striker **40** which moves along path P to strike the chambered cartridge C and discharge the firearm **20**. After firing, actuator **350** is de-energized by the microcontroller **200** as the user completely or partially releases the trigger which resets to the ready-to-fire position for the next firing cycle. In some embodiments, the microcontroller via actuation control circuit **202** transmits merely a short momentary pulse of electric current to the coil **306** which is sufficient to change state of the electromagnetic actuator **350** for implementing the trigger release profile and alter the primary resistance force generated by the permanent magnet **308** in the flux loop. The control circuit therefore performs a quick on/off switching of the power supply to the actuator. Accordingly, no feedback control is required for the microcontroller **200** to terminate electric power to the actuator **350**.

Fire-by-Wire Dynamic Variable Force and Displacement Trigger Embodiment

Expanding on the variable force trigger concept disclosed herein, it may be ideal if both the trigger force and trigger displacement could be dynamically changed during the trigger pull and firing sequence. One way to accomplish this would be to completely separate the trigger function from the firing event. The trigger event would generate an electrical signal that would be sent by wire to a separate electromechanical actuator to fire the firearm. In this embodiment, the trigger force could be dynamically adjusted as before; but the displacement could also be dynamically adjusted. This can be accomplished by a predefined effect or with feedback using a displacement sensor **159** of a flux measurement type such as a hall-effect or alternatively a GMR (Giant Magnetoresistance Effect) sensor operably incorporated with the trigger mechanisms **100** (with single flux loop actuator **123**) or **300** (with double flux loop actuator **350**). Such a sensor could be placed near the air gap A (see, e.g. FIG. 7 or 29) to measure leakage flux at the air gap as the rotating trigger member **104/304** are moved. This measurement could be relayed to the microcontroller **200** and used to deduce the state of the electromagnetic actuator. The flux measurement displacement sensor would allow for the dynamic variation of trigger pull force based on travel or displacement and the trigger decision event could be defined as a specific displacement threshold. The possible force profiles to be defined, selected, and implemented under electrical control could be expanded to include any number of force/displacement curves with the displacement to firing being a new dynamic variable. A long easy trigger pull, verses a short heavy pull, or a long heavy pull, or even a short light hair trigger could be created by appropriately programming the microcontroller **200**. The force and displacement could conceptually be fully programmable over a plurality of all possible ranges using the control system shown in FIG. 9.

Force feedback could be combined with the dynamic adjustment of displacement and force in trigger feel to indicate the firing point. At the point of firing, the trigger force could be dynamically changed to give the operator haptic or kinesthetic feedback of the fire decision being reached. Optionally, the kinesthetic feedback could be supplied slightly after the actual firing event to minimize the possibility of the user staging or anticipating the firing event and minimizing flinching which could adversely affect point of aim.

The fire-by-wire concept has one potential weak spot in that a single fire signal could result in a single point of failure. A false positive or negative signal resulting from a short, open, or other failure could result in a failure to

function or unintended trigger event. One of several concepts that would mitigate this is to have the trigger event generate two redundant triggering signals, an armed and a fire event signal. Using the displacement sensor **159**, a minimum displacement of the trigger could be used as a signal to arm the firing system. The final fire decision could be an electrical contact or optical switch. Using two or more sensors, with different failure mechanisms, should ensure no single failure point. By adding intelligence to the relationship of the two signals, the reliability can be enhanced further. For example, it should not be possible to arm the firing sequence unless the trigger displacement has recovered to a predetermined position and the electro-mechanical switch is in an open state. The displacement sensor could be used to arm the firing signal as displacement is increased but before the mechanical switch closes. The actual closing of the mechanical switch would need to happen within a predefined time window or the arm signal would time out. This would ensure that the trigger pull event is representative of an actual firing event and would not be duplicable as a random failure of several components at the same time.

It can be envisioned that by incorporating the additional system sensors shown in FIG. 9 beyond a trigger sensor(s), a series of operating conditions could be incorporated into the control logic used to enhance operation of an electronic fire-by-wire firing mechanism. Referring to FIG. 9, some possibilities could include grip force sensors **206** to ensure a ready-to-fire secure grip of the firearm by the user preceding the firing event, to inertia or motion sensors **207** that would preclude the firearm to function under dropping or accidental movement due to a fall, trip, or other similar incident, to the incorporation of other sensors operable to confirm suitable firing conditions based on the user, location, time of day, or environment.

The fire-by-wire electronic firing system may still incorporate a modified version of either trigger mechanisms **100** or **300**. In such an application, electromagnetic actuators **123** or **350** of trigger mechanism **100** or **300** respectively would not physically engage/disengage a component of the firing mechanism as previously described herein. Instead, the actuators would simply be used to adjust the trigger release profile and breakpoint of the trigger member **104** or **320** in the manner previously described herein in accordance with the control logic of FIG. 8.

FIG. 34 shows an exemplary control logic process **400** which may be implemented by microcontroller **200** to control a fire-by-wire trigger mechanism having an electronic sear (E-sear) such as a piezo-electric actuator to detonate the cartridge. Such a system may be incorporated into any type of firearm, such as the pistol shown in FIG. 30 as one non-limiting example. FIG. 35 shows a modified control system amenable for use with such an electronic E-sear trigger mechanism. The trigger mechanism **400** may include a second mechanical trigger sensor **160** such as a mechanical switch in conjunction with a force or displacement trigger sensor **159/370** associated with the electromagnetic actuators **123/350** of firing mechanisms **100/300** depending on which firing mechanism is used with the fire-by-wire system.

Referring to FIGS. 34 and 35, the microcontroller **200** would awaken when it detects a wake-up signal generated from gripping the gun which is sensed by grip sensor **206** and communicated to microcontroller **200** (Step 402). Alternatively, this could be a motion detection wake-up signal sensed by motion sensor **207** instead of a grip sensor. On wake-up, a quick check that sufficient battery power is available and that the system is functioning is performed in

the form of a self-test (Step 404). A failure of this self-test or battery check would result in aborting the start-up sequence and informing the operator of the error/warning so that corrective action can be taken.

If however the Step 404 test is positive, the microcontroller 200 will arm the firearm and continuously monitor for a trigger event and a number of other possible state change events in Step 408; some examples of which are indicated in FIG. 34. Alternatively, these state change events could be polled periodically on a reasonable preprogrammed time schedule to ensure reliable and timely detection.

An example of one state change event that would effect authorization is the detection of loss of intent-to-fire grip that would indicate the user no longer has control of the firearm (Step 412). Another example would be the detection of an unsafe acceleration force detected by motion sensor 207 (Step 411), which is associated with falling or being bumped or jarred while holding the firearm. In the presence of a high acceleration force, the system disables the firing due to unsafe conditions. Another example of state-change events would be the detection of a system error or the detection that the battery might not have sufficient remaining power to reliably actuate the magnetic actuator (Step 416). These types of faults and warning would also drop the firearm out of the arm state and indicate a warning to the user.

An actuation event cycle also starts if a trigger event is detected by trigger sensors in Step 410, and the firearm is in an armed state and no state change event (Steps 411, 412, or 416) has occurred to disarm the firing mechanism as indicated above. Steps 422 through 430 represent a firing sequence for the firearm implemented by microcontroller 200. For added safety, two independent trigger events, "Trigger Event 1" based a signal from mechanical trigger sensor 160 and "Trigger Event 2" based on a signal from the electronic sensor 159 or 370 may be used to initiate a valid trigger event. However, a single trigger sensor and event may be used in other embodiments. After the system detects Trigger Event 1 has occurred, the system then confirms that the firearm is still under the users physical control with an intent-to-fire grip (Step 422). Next, the system detects whether an intent-to-fire Trigger Event 2 is activated. This provides the double layer of firing security. Assuming Steps 422 and 426 are positive, the electronic safety shorting clamp 251 is lifted (Step 428) to enable the firing mechanism. A high voltage electric pulse or signal from circuit 250 is sent by the microcontroller 200 via actuation control circuit 202 to the E-sear piezo actuator 252 which discharges the firearm (Step 430). The firing system is then reset for the next firing event.

During the preceding firing sequence of the fire-by-wire firing mechanism, it bears noting that the control logic of FIG. 8 is simultaneously performed and implemented by the microcontroller 200 to adjust the trigger release profile according to the preprogrammed trigger breakpoint/breakover setpoint or displacement in the manner previously described herein. The trigger release settings and electric pulse sent to actuator 123 or 350 to activate the same (depending on whether the single or double loop actuator firing mechanism is used) is represented by block 253 in FIG. 35.

Magnetically Variable Trigger Mechanisms

The following disclosure describes non-electrically operated trigger mechanisms which are magnetically variable by manually adjusting the static magnetic field of the mechanism. These trigger mechanisms function without an electric power source or electromagnet to release a spring-loaded

striking member for striking a chambered round of ammunition, but embody some of the same general magnetic operating principles of the electromagnetically operated trigger mechanisms described heretofore.

Traditional triggers for firearms provide a decisive intent-to-fire signal through mechanical motion that utilizes a displacement and force profile developed by using mechanical linkages, springs and the release of energy stored in a spring-biased hammer, striker, or sear. The trigger force and displacement curve or profile is normally fixed by these mechanical linkages and springs. A number of designs exist that provide adjustable characteristics for the force and displacement of the trigger using set screws, additional springs and other parts, or by completely changing components in order to customize the force-displacement profile of firearm triggers. Such adjustment techniques, however, modify the trigger pull force resistance in a purely mechanical manner which is limited by the physical interaction of trigger parts and associated linkages alone. To provide adjustment of the trigger pull force, these trigger mechanical linkages may therefore become quite complex, require multiple individual mechanical components, and hence are susceptible to wear and failure.

Exemplary embodiments of the present invention provide a trigger mechanism for a firing system of a firearm which is magnetically adjustable and variable, thereby providing quick and easy user-adjustment of the trigger pull force. Both closed and open magnetic flux loop designs are provided. In one implementation, the combination of a closed magnetic flux loop design and a manually translatable magnetic control device or insert configured and constructed to adjustably vary the magnetic field in the trigger mechanism produced by a permanent magnet disposed in the loop overcomes the deficiencies of purely mechanical and often complex adjustable trigger designs comprising multiple parts, springs, and linkages. The control device may comprise a "soft" magnetic material—a material preferably having a large relative magnetic permeability (i.e. the ability to support formation of a magnetic field in the material). As used in the art, "soft" magnetic materials refer to materials which are easily magnetized and demagnetized. Non-limiting examples include iron, low-carbon steel, nickel-iron, cobalt-iron, etc. The control device or insert in some embodiments is selectively and variably insertable into and retractable from a control recess or air gap (B) formed in the magnetic flux loop by varying degrees to adjust the trigger force. The control air gap B, formed by removing material from the stationary yoke, attenuates (i.e. decreases or diminishes) the maximum magnetic flux available in the loop at a working air gap (A) between the yoke and a movable trigger member which retains the trigger member magnetically to the yoke until the trigger member is pulled. Inserting the control device or insert into the control air gap B increases the magnetic flux in the closed loop at air gap A. Conversely, retracting the control device or insert from the control air gap B decreases the magnetic flux in the loop at air gap A. In some embodiments, the control device or insert may comprise the permanent magnet for the closed magnetic loop and inserting/retracting, or rotating the insert relative to the control air gap B changes the magnetic flux in the loop at air gap A. In another implementation, the combination of an open flux loop design and a manually translatable magnet configured to adjustably vary the proximity of a magnet to the trigger body provides adjustment of the trigger pull force. Each trigger mechanism design is further described herein.

In one aspect, embodiments of the magnetic trigger mechanism disclosed herein represent adjustable variable force magnetic air gap trigger designs. A permanent magnet in the closed flux loop generates a primary static magnetic field producing a fixed or static holding force for a trigger-sear release system which limits susceptibility to external magnetic fields that might affect the trigger force. By adjusting the control air gap in the closed magnetic flux loop via the magnetic control device, the fixed or static holding force can be increased or decreased to provide a variable range of trigger force breakpoints or setpoints that provide a crisp feel as the trigger pull force applied by the user to the trigger meets or crosses the fixed magnetic holding force set point during a trigger pull event. The fixed or static magnetic field generated by the permanent magnet in the closed flux loop creates a primary resistance force opposing movement of the trigger when pulled by the user. The trigger mechanism operates to release the movable sear of the firing system, which in turn releases a cocked energy storage device to discharge the firearm. The energy storage device may be a spring-biased striking member such as a pivotable hammer or linearly movable striker configured to strike and detonate a chambered ammunition cartridge; each of which is described herein.

FIGS. 36-49 depict several non-limiting example design embodiments and respective operating characteristics of closed loop non-electric magnetic only trigger mechanism having a user adjustable trigger force. Each design embodiment was evaluated using computer-aided finite element analysis (FEA) to determine the projected magnetic flux characteristics and trigger pull force profile of each design for comparison. The figures include illustrations which summarize the detailed finite element magnetic analysis of the performance of the different design embodiments and respective trigger pull force versus displacement profile graphs, thereby illustrating the characteristics and trade-offs between designs. An open magnetic loop design shown in FIGS. 50 and 51 was also computer modeled and analyzed for comparison to the closed magnetic loops designs.

The different examples of trigger mechanisms presented hereafter illustrate the relative features of the design strategies used in each design embodiment. The full analysis is not included; however, important summary performance is presented. It will be clear to those in the field that these examples are not exhaustive, but merely a sample of differing design strategies which can be implemented. It should also be clear that desirable design features of a trigger mechanism include a wide range of adjustable trigger pull force, an adjustment means that is relatively linear in response, and an adjustment means being relatively insensitive to normal mechanical tolerances.

Closed Magnetic Loop Designs

FIGS. 36 and 37 depict a first embodiment of a variable magnetically adjustable trigger mechanism 1000 configured for manually controlling the trigger force of a firearm trigger by using magnetic fields to directly constrain the movement of the trigger linkage or mechanism until a user preselected trigger release force (i.e. trigger force breakpoint or setpoint) is applied to the trigger and reached. The trigger mechanism shown in FIG. 36 is based on the electromagnetic trigger mechanism shown in FIG. 15 with non-linear leaf spring 126 and similar in construct with some revisions. Those features in common will not be discussed in detail for the sake of brevity. The electromagnetic coil 106 is notably omitted and replaced with an outwardly open control recess 1002 forming a magnetically adjustable control air gap B in yoke 102, as further described herein.

It bears noting that the magnet only trigger mechanisms described in this section of the application may also be used with any of the trigger assemblies shown in FIG. 6, 7, 13A, or 14A, and are therefore not limited in their applicability to the trigger assembly shown in FIG. 15 selected for convenience as representing represents one non-limiting embodiment.

Referring to FIGS. 36 and 37, the magnetic trigger mechanism 1000 generally includes a magnetic stationary yoke 102 and rotating trigger member 104. The yoke 102 may thus be fixedly but removably mounted to the frame 22 of the firearm, the receiver 39, or in an open receptacle of a trigger housing 1220 (see, e.g. FIG. 70) in turn attached to the frame or receiver. Any suitable mounting means may be used to fixedly mount the yoke 102 to the frame, receiver, or trigger unit housing such as for example without limitation fasteners, couplers, pins, interlocking features, etc. The mode of attachment is not limiting of the invention. Yoke 102 may be generally C-shaped in one configuration.

Rotating trigger member 104 of the trigger mechanism 1000 includes vertically elongated upper working extension or portion 120 and lower trigger portion 118 each mounted about pivot 101, as previously described herein with respect to FIG. 15. Upper working portion 120 of trigger member 104 preferably has a width commensurate with the width of the yoke 102 (i.e. yoke horizontal upper portion 110) where the working portion abuttingly but removably engages the end of the yoke at the air gap A.

The permanent magnet 108 may be disposed and arranged on or within the yoke 102 (see, e.g. FIG. 36), or alternatively on or in the upper portion 120 of the trigger member 104 at a suitable location (see, e.g. magnet 108' shown in dashed lines). In FIG. 36, the magnet 108 is embedded within the yoke 102 at a suitable location of its cross section. The magnet 108 alternatively may also be mounted on the free terminal end of the yoke 102 (e.g. horizontal upper portion 110) at the air gap A where it may engage the upper working portion 120 of trigger member 104 as one alternative non-limiting option. The permanent magnet 108 will produce the desired static magnetic field in trigger mechanism so long as the magnet is located somewhere within the closed magnetic loop formed by yoke 102 and rotating trigger member 104. Accordingly, the location of the permanent magnet 108 within the closed magnetic loop does not limit the invention.

Permanent magnet 108 preferably has dimensions and a cross-sectional area commensurate in dimensions and cross-sectional area to the cross section of the yoke 102, as shown (or alternatively the upper working portion 120 of trigger member 104 if mounted thereto as shown for example by magnet 108'). Optimal coupling of the flux lines of the magnet to the closed loop of magnetic material is achieved by such an arrangement and dimensions. If the magnet is smaller than the yoke in cross section, then flux lines will short across the gap B formed between the two yoke separated pieces in which there is no magnet, reducing the closed-loop flux in the circuit.

The yoke 102 and rotating member 104 are configured to collectively form an annular-shaped closed flux loop resistant to external magnetic fields. Yoke 102 and trigger member 104 define an enclosed open central space 1003 therebetween (see, e.g. FIG. 36). The permanent magnet 108 generates a static magnetic field or flux (see directional flux arrows) creating a fixed holding force on the rotating member 104. This creates a primary fixed or static resistance force opposing movement of the trigger mechanism when actuated by the user.

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A completely openable/closeable air gap A is formed between the yoke and rotating member. The air gap A may be vertically oriented and normally held closed by the static holding force created by the permanent magnet **108**, and opened when the trigger is pulled by the user to overcome the static holding force and discharge the firearm.

The preferably strong permanent magnet **108** arranged in the closed magnetic flux loop maintains a high static holding force threshold inhibiting the movement of the trigger portion **104** (e.g. "trigger" alternatively) around the pivot point **101**.

The magnetic control device used to alter the static magnetic field and establish a trigger force breakpoint or setpoint comprises the adjustably translatable soft magnetic material control insert **1001**. In one embodiment, the control insert **1001** may be in the form of a triangular or V-shaped wedge formed of a magnetically conductive material such as without limitation a suitable soft magnetic metal capable of being magnetized by a magnet, such as without limitation iron, low-carbon steel, nickel-iron, cobalt-iron, etc. This same material may be used for the yoke **102** and rotating trigger member **104**. The control insert **1001** is linearly translatable to project into or retract from a secondary control air gap B formed in the yoke **102** to change the reluctance. Air gap B may comprise an outwardly open and angled wedge-shaped (e.g. triangular) control recess **1002** in one embodiment as shown which may be formed in the yoke **102** by partially removing some material such that the recess does not completely sever the cross section of the yoke (see, e.g. FIGS. **36-38**). Control recess **1002** in the present wedge embodiment only partially severs the cross section of the yoke **102**. In other embodiments as shown in FIG. **39**, however, the recess **1002** may completely sever the cross section of the yoke **102**. Both the partially closed and fully open embodiments of control recess **1002** form a wedge-shaped negative space which is filled to varying degrees by the magnetically conductive wedge-shaped control insert **1001** to change and adjust the primary static magnetic field or flux. One characteristic of the partially connected design is that it would have a well defined low end holding force that is independent of the control air gap wedge insert.

To linearly translate or move the soft magnetic material control insert, a manually operable actuator **1004** may be operably coupled to the wedge-shaped control insert **1001**. The actuator **1004** may be movably mounted to the firearm frame **22**, receiver **39**, or alternatively a trigger housing **1220** (see, e.g. FIG. **70**). In either of the foregoing mounting arrangements, the actuator is ultimately supported directly or indirectly by the frame **22** to which the receiver and/or trigger housing are attached.

The actuator **1004** in one non-limiting example may be comprise an insert adjustment screw **1005** which acts on the wedge-shaped control insert **1001** as shown in FIGS. **36** and **37**. The adjustment screw **1005** converts rotary motion applied by the user to turn the screw into a linear translation of the control insert **1001** relative to control air gap B. In some possible embodiments, the control insert **1001** may be mounted directly to an end of the screw **1005** as shown. Rotating the screw in opposing directions therefore linearly projects the control insert wedge into or retracts the control insert wedge from the control air gap B created by control recess **1002** to varying degrees for adjusting the trigger pull force according to the user's preferences.

The position of the wedge-shaped control insert **1001** relative to the angled control air gap B and concomitantly the yoke **102** increases or decreases the static holding force in the closed magnetic loop of the trigger mechanism, which

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holds the upper working portion **120** of trigger member **104** against the yoke **102**. This in turn creates the user-adjustable trigger pull force which must be overcome by the user in order to pivot the trigger member about pivot **101** and open the air gap A for releasing the striking member, such as for example without limitation the spring-biased hammer **130** shown in FIG. **37**.

In sum, rotating and linearly moving actuator **1004** accordingly moves the control insert **1001** between a first position relative to the control air gap B producing a first magnetic static holding force in the closed magnetic loop, and a second position relative to the control air gap B producing a second magnetic static holding force different than the first force (e.g. more or less).

FIG. **36** shows trigger mechanism **1000** in the ready-to-fire position. Air gap A is fully closed (i.e. upper working portion **120** of trigger member **104** is abuttingly engaged with the yoke **102**). The spring-biased hammer **130** (spring not shown) is held in the rearward cocked position via engagement with a sear surface **132** formed by the trigger member working portion **120**, which defines a vertically elongated sear as described previously herein with respect to FIG. **15**. After the trigger is pulled, the trigger member working portion **120** rotates forward to break engagement between sear surface **132** and the hammer **130**, thereby releasing the hammer to strike the firing pin and discharge the firearm. Air gap A is fully open at this point as shown in FIG. **37** showing the firing position of the trigger mechanism **1000**.

FIG. **39** shows a side view of the closed-loop sliding wedge design of trigger mechanism **1000** with computer-modeled magnetic flux lines illustrated. In this case, a steel wedge (soft magnetic material) is slid in and out of similarly angled control air gap B in the magnetized stationary yoke via operation of the actuator **1004**, thereby providing a variable reluctance at air gap A based on the horizontal displacement or position of the wedge control insert **1001** relative to control air gap B. It should be noted that the analysis of FIG. **39** and FIG. **40** is performed on the alternative embodiment of FIGS. **36** and **37** in which the control recess **1002** fully severs the cross section of the yoke **102**. FIG. **40** shows the results of finite element analysis (FEA) of this design in a trigger pull force (Torque) versus displacement (D_p) profile graph. This figure shows that the torque on the trigger member **104** varies from almost 0.08 to 0.42 Nm over a trigger displacement range of about 3 mm. The variation is fairly non-linear and is more susceptible to mechanical tolerance variations than the sliding magnet or rotating magnet designs further described elsewhere herein by comparison, but nonetheless may be acceptable. Notably, the graph in FIG. **40** shows this trigger mechanism exhibits a high initial trigger pull force requirement which then relatively rapidly decreases over the remainder of the trigger displacement range to the point of discharging the firearm.

An alternate actuator **1007** for linearly translating the wedge-shaped control insert **1001** of trigger mechanism **1000** is shown in FIG. **38**. This actuator may include a gear mechanism comprising a toothed linear gear rack **1009** disposed on a linearly elongated wedge **1006** and a manually adjustable and rotatable toothed gear pinion **1010** engaged with the rack. Pinion **1010** may be mounted via a crosswise control shaft **1111** arranged transversely to the wedge and mounted in the frame, receiver, or trigger housing. The end of the control shaft **1111** may be exposed and accessible from outside the firearm frame to the user for making adjustments to the trigger pull force. The end of shaft **1111** may include a knob, or be configured with a tooling interface

(e.g. hex key interface recess, Philips or slotted screwdriver interface recess, etc.) to facilitate rotating the shaft by the user. Rotating the pinion **1010** in opposing directions similarly projects or retracts the wedge into/from control air gap B in a linear manner similar to screw actuator **1004**. The magnetic flux lines and FEA trigger pull force graph are the same as in FIGS. **39** and **40**.

By adjusting the displacement and position of a wedge control insert **1001** of magnetically conductive material relative to control air gap **1002**, the effective length of the control air gap **1002** (the distance magnetic flux lines have to travel in air) can be varied. As the effective length is shortened, the total magnetic flux in the closed loop magnetic circuit increases, and hence the flux density in the air gap A is increased resulting in greater trigger holding force (torque). An increase in the effective length of control air gap **1002** has the opposite effect. Adjusting the displacement and position of control insert **1001** therefore adjusts and changes the resulting strength of the trigger static magnetic field and holding force that creates a primary resistance force opposing movement of the trigger member when pulled by the user that must be overcome. Inserting the wedge control insert **1001** farther into control air gap B increases the static magnetic holding force to increase the required trigger pull force. Conversely, withdrawing control insert **1001** from the control air gap B decreases the static magnetic holding force to lessen the required trigger pull force.

In alternative embodiment shown in FIG. **41**, a variable control air gap B is controlled by moving a control insert **1020** in the form of a substantially planar rectangular block or plate of soft magnetic material into or out of the flux path in the trigger mechanism **1000** to varying degrees to change the reluctance and trigger pull force. Other suitable shapes may be used. The control air gap B in this embodiment completely severs the cross section of the yoke **102** at air gap B (i.e. intermediate portion **114** of the yoke). The horizontal upper and lower portions **110**, **111** and adjoining parts of the vertical intermediate portion **114** above and below the control air gap B in this case may be separately mounted to the support structure (e.g. frame, receiver, or trigger housing) via any suitable methods (e.g. fasteners, etc.). In the non-limiting illustrated embodiment, the plate-like control insert **1020** has a length and width greater than the vertical thickness of the plate. The adjustably translatable soft magnetic material control insert **1020** may similarly be formed of a magnetically conductive material such as without limitation a suitable soft magnetic metal capable of being magnetized by a magnet, such as without limitation iron, low-carbon steel, nickel-iron, cobalt-iron, etc. Any suitable manually operable actuator such as actuators **1004** and actuator **1007** previously described herein, or another type actuator may be used to adjust the position of the plate-like control insert relative to control air gap B.

The present closed-loop sliding plate design is based on a principle which allows the magnetic flux to be choked off by introducing a restriction in the magnetic loop. By contrast, it bears mention here that both the sliding magnet design and the rotating magnet design as further described below are based on varying the amount of total flux coupled from the magnet **108** into the magnetic yoke **102**.

FIG. **42** shows a side view of the closed-loop sliding plate control insert **1020** design of trigger mechanism **1000** with computer-modeled magnetic flux lines illustrated. In this case, a steel plate (soft magnetic material) is slid in and out of the control air gap B of magnetic yoke **102** providing a restriction in the magnetic loop. FIG. **43** shows the results of finite element analysis (FEA) of this design in a trigger pull

force (Torque) versus displacement (Dp) profile graph. FIG. **43** shows that the torque on the trigger member **104** varies from almost 0.08 to 0.42 Nm over a range of about 5 mm. In contrast to the sliding wedge design described herein, the graph in FIG. **43** shows the sliding plate design exhibits a low initial trigger pull force requirement which then increases over the control displacement range. The performance of the sliding plate design however is not quite as good as the sliding magnet design described elsewhere herein, but nonetheless acceptable. Contrasting FIGS. **43** (sliding plate) and **46** (sliding magnet), the range of torque is larger and the variation of displacement is more linear for the sliding magnet design. A major advantage of sliding the magnet in and out of control air gap B versus just adjusting the width of the airgap via the sliding soft magnetic material plate is that adjustment of the airgap width is a precision movement over a very small range to make a large change in torque. This will take a precision adjustment to control the small changes in width of the airgap. With the sliding magnet, the effective change in torque is distributed over a longer movement from totally open to completely centered in the yoke. It is a much less sensitive adjustment that does not require the same degree of precision adjustment tolerance. The sliding plate design relies on the principle of saturating the soft-magnetic material which is a less precise physical parameter than the physical coupling of flux lines from a permanent magnet into the yoke by varying the magnet position relative to the yoke.

FIG. **44** depicts another alternative approach and embodiment of trigger mechanism **1000** which provides a movable control insert **1031** incorporating magnet **108** in lieu of the movable soft magnetic material wedge or plate designs described above. In the moving magnet design, the permanent magnet is not mounted to the stationary yoke **102** or rotating trigger member **104** as in the moving soft magnetic material embodiments. Instead, the permanent magnet **108** may be mounted on or encapsulated in a thin wall carrier **1030** which preferably is formed a non-magnetic material such as for example without limitation nylon or other suitable polymers. Carrier **1030** may have a plate-like body in one embodiment having a width and length greater than its vertical thickness as shown. The polymeric carrier **1030** would act as both a protective cover to the magnet as well as a means and/or bearing surface for guiding the magnetic into or out of the flux path at control air gap B coupling to the trigger release surface at the interface between the yoke **102** and trigger member **104** at air gap A. The carrier **1030** with magnet **108** may be translated by a suitable actuator such as those described herein which are operably coupled to the carrier. It bears noting that control air gap B is formed by a completed severed section of the yoke **102** similarly to the sliding plate design shown in FIG. **41** and previously described herein.

FIG. **45** shows a side view of the closed-loop sliding magnet control insert **1031** design of trigger mechanism **1000** with computer-modeled magnetic flux lines illustrated. In this case, the magnet **108** mounted to the non-magnetic carrier **1030** is slid in and out of the control air gap B of magnetic yoke **102**. FIG. **46** shows the results of finite element analysis (FEA) of this design in a trigger pull force (Torque) versus displacement (Dp) profile graph. FIG. **46** shows that the torque on the trigger member **104** varies from almost 0 to 0.47 Nm over a range of about 6.5 mm. In general, this option beneficially offers wide ranges of user-adjustable holding torque with less sensitivity to mechanical displacement errors. The holding force as a function of displacement is non-linear in this closed magnetic loop

design, but it is still closer to linear which is desirable than in the open loop design case. Generally, it is desirable to have a large range of torque adjustment, and that the range of adjustment is close to linear. A uniform relationship between the amount of displacement to the change in torque over the usable range of the trigger is ideal. For example: one mm of displacement represents one unit of torque change along the whole range of possible torque settings. By contrast in FIG. 40, it is evident that torque changes much more with the same displacement change at the higher torque range than at the lower torque range. In FIG. 46, however, it can be observed that the change in torque with displacement is similar anywhere along the range except for the extreme endpoints, thereby representing a more ideal trigger setup.

Another alternative embodiment to achieve the variable coupling of the magnetic flux comprising a closed loop rotating permanent magnet control insert 1040 whose rotational position is adjustable by the user is shown in FIG. 47. The control insert 1040 may comprise the magnet 108 rotating alone (see, e.g. FIG. 47) or with support of a non-magnetic carrier 1042 (e.g. polymer) as shown in FIG. 48. When the magnet 108 is rotationally misaligned with the yoke 102 at the control air gap B with respect to its north (N) and south (S) poles, this will attenuate the flux coupling of the magnet into the closed magnetic loop. Magnet 1040 is manually and adjustably rotatable by the user about a transversely oriented rotational axis 1041 defined by the magnet itself, non-magnetic carrier 1042, or a pin/shaft coupled to the magnet. Rotary magnet 1040 may have any suitable cross-sectional shape, including as non-limiting examples rectilinear as shown (e.g. rectangular or square), polygonal (e.g. hex shaped, etc.), or non-polygonal (e.g. circular as shown in FIG. 48 or other). Control air gap B may be complementary configured to the cross-sectional shape of the magnet 1040 as shown in FIG. 48. The magnet 1040 includes opposing north (N) and south (S) poles whose orientation is changeable via rotating the magnet, thereby altering the magnetic flux field and trigger pull force. A displacement angle D_d relative to a horizontal reference line passing through the rotational axis 1041 of the magnet 1040 is therefore manually adjustable by the user to change and achieve the desired trigger pull force of the trigger mechanism 1000.

FIG. 48 shows a side view of the closed-loop rotary magnet control insert 1040 design of trigger mechanism 1000 with computer-modeled magnetic flux lines illustrated. The magnet 108 mounted to the non-magnetic carrier 1042 is rotated with respect to orientation of its north and south poles relative to the control air gap B of magnetic yoke 102. In this case, a cylindrical magnet 108 is magnetized perpendicular to its rotational axis 1041. When the magnet 108 is rotated through a displacement angle D_d , the coupled magnetic flux varies as the sine of the displacement angle with 0 being no flux coupling and 90 degrees being full flux coupling. FIG. 49 shows the results of finite element analysis (FEA) of this design in a trigger pull force (Torque) versus displacement (D_p) profile graph. FIG. 46 shows that the torque on the trigger member 104 varies from almost 0 to 0.65 Nm over an angular range of 90 degrees. Like the closed-loop sliding magnet design previously described herein, this beneficially provides a wide range of holding torques and a wide range of angular displacement with a non-linear, but well-behaved response.

It bears noting that since magnetic force within the air gap increases with magnetic cross-sectional area and decreases with the square of the air gap length, practical designs which

are optimized for force and speed tend to minimize the length relative to the cross-sectional area. A consequence of this is that actuator designs based on these design principles are advantageously inherently immune to external magnetic field interference. In practice, it is impossible to change the state of the actuator using an external magnet (and optional soft magnetic material yoke) provided the rotating trigger member 104 is physically isolated from the external magnet by at least one air gap distance. This preferably should always be the case in practical firearm embodiments utilizing the trigger mechanisms disclosed herein.

The trigger pull force in all design magnetic embodiments is adjusted by varying the magnetic flux density in the control air gap B acting on the rotating trigger bar or member 104. Ultimately the breakpoint of the trigger is determined by the magnetic flux density in the air-gap A controlled by manipulation of control air gap B via the various control inserts described herein. Even though A is very small, the holding force is determined by the flux density in this space. In general, the flux density at air gap A is varied by either changing the flux density at control air gap B, or by changing the effective coupling of flux from the magnet into the yoke. These two principles are used independently or together in each of the designs. In the case of FIGS. 36-40, the magnetic flux coupled across the gap B is varied (flux reluctance of the closed loop). In the case of FIGS. 41-49, the amount of flux injected into the closed loop is varied by either movement of the magnet into the gap B or rotating the magnet in gap B. In magnetic closed-loop designs, the flux density occupies the space between the magnetic yoke 102 and the rotating trigger member 104. In open-loop designs, the flux density is directed between the rotating trigger member 104 and the permanent magnet 108.

For open-loop designs, the flux density is dependent on the magnetic properties of the permanent magnet 108, the physical geometry of the magnet, and the displacement between the magnet and the rotating trigger member 104. For closed-loop designs, the flux density is dependent on the magnetic properties of the permanent magnet 108, the geometry of the magnet, the physical placement of the magnet within the magnetic yoke 102 and the geometry of the control air gap B. In general, the breakpoint force of the trigger mechanism is determined by the flux density at air gap A, but this flux density is varied only by (1) changing the flux using the properties of control air gap B, or (2) changing the coupled flux into the yoke by varying the position or angle of the magnet relative to the yoke at control air gap B.

In general, the magnetic flux density in closed-loop designs can be changed by a combination of changing the reluctance in the magnetic circuit and changing the described below coupling of the permanent magnet 108 into the yoke 102. In open-loop designs discussed below, the magnetic flux density is adjusted by changing the displacement of the magnet 108 relative to the rotating trigger member 104.

Open-Loop Magnetic Design

FIG. 50 shows a side view of a simple conceptual open-loop magnetic design of trigger mechanism 1100 with computer-modeled magnetic flux lines illustrated. A detailed embodiment which exemplifies this open magnetic loop design is shown in FIGS. 61-69 and further described herein. The magnet 108 is movably displaceable in position relative to the rotating trigger member 104, thereby providing a means for adjusting the control air gap B between the magnet and upper working portion 120 (e.g. sear) of the trigger member 104. Flux lines from the permanent magnet couple into the rotating trigger bar via control air gap B

formed between the upper working portion **120** of trigger member **104** and the permanent magnet **108**. These flux lines form an attractive force which results in a torque on the trigger bar or member **104** about its center of rotation defined by pivot **101**. The horizontal displacement of the magnet **108** towards or away from the trigger bar or member

FIG. **51** shows the results of finite element analysis (FEA) of this design in a trigger pull force (Torque) versus displacement (Dp) profile graph. FIG. **46** shows that the torque on the trigger member **104** varies from almost 0.18 Nm to 0.03 Nm over a displacement range of 2 mm. The trigger force profile resembles that of the foregoing sliding wedge closed magnetic loop design in so far that the pull force is also characterized by a high initial pull force which then rapidly diminishes over the remainder of the trigger displacement range. This contrasts to the other closed loop designs having the opposite trigger force profile as described above. It is important to note that in this case of the open loop and in the foregoing closed magnetic loop examples, these values are for comparative use only and not intended to indicate specific design targets for an actual trigger mechanism.

Summary of Closed and Open Loop Design Comparison Results

Based on the comparative results of the design and performance analysis for each magnetic only trigger mechanism describe above, a few summary conclusions can be offered. Each design disclosed herein is capable of achieving the design goals for a magnetically adjustable trigger mechanism, which are a wide range of adjustable trigger pull force, an adjustment means that is relatively linear, and an adjustment means that is relatively insensitive to normal mechanical tolerances.

The rotating magnet and sliding magnet have similar torque/response curves and similar holding torques. The rotating magnet and sliding magnet designs offer an optimal way of varying holding torque while being least affected by mechanical adjustment tolerances when the user manually adjusts the trigger pull force. A major advantage of the sliding magnet and rotating magnet designs in contrast to just adjusting the width of the control air gap B (via the sliding soft magnetic material plate or wedge control insert designs) is the required precision of the movement over the range necessary to change the torque. When adjusting the reluctance by opening or narrowing the control air gap B via the sliding plate or wedge, it will take a precision adjustment by the user to control the small changes in width of the air gap. Very slight precision changes in control air gap B width have a large impact on the torque. This will require a very tight manufacturing tolerance of the adjustment means to make a reliable and repeatable adjustment. Even with a fine threaded lead-screw, for example, it might only be a fraction of a turn to make a significant adjustment in the effects of the airgap. With the sliding magnet, however, the effective change in magnetic coupling is distributed over a much longer movement from totally open to completely centered in the yoke. Similarly in the rotating magnet design, the adjustment range is from 0 to 90 degrees. The sliding or rotating magnet designs are therefore offer a much less sensitive adjustment that does not require the same great degree of precision adjustment tolerance. The rotating magnet design has the added advantage of occupying less physical space, thereby advantageously allowing for a more compact trigger mechanism construction for placement in the firearm.

The open loop and closed loop sliding wedge designs both have similar torque-displacement curve shapes (i.e. high initial trigger pull holding torque requirement which diminishes over the remainder of the trigger displacement when firing the firearm). The open-loop design though has much lower holding torque due to the magnetic losses in the air which is less desirable, but nonetheless still offers an acceptable magnetic trigger mechanism design.

The analysis confirms that all the closed magnetic loop embodiments documented herein meet the magnetically adjustable trigger design goals of a wide range of adjustable trigger pull force, an adjustment means that is relatively linear, and an adjustment means that is relatively insensitive to normal mechanical tolerances. The magnetic field open loop design mentioned above provides an acceptable means for achieving a viable adjustable trigger. While not optimal in performance, the open loop design is compact and mechanically simple to construct and implement offering certain advantages.

A major feature of one non-limiting preferred closed magnetic loop design of a sliding magnet shown in FIGS. **52-60** is dependent on varying the magnetic reluctance of an air gap in the closed magnetic loop, adjusting the physical coupling of the magnetic flux from a magnet into the closed loop, or a combination of both techniques. Prior magnetic trigger mechanisms do not achieve the design goals for an adjustable trigger that include a wide range of adjustable trigger pull force, an adjustment means that is relatively linear, and an adjustment means that is relatively insensitive to normal mechanical tolerances.

Mechanically detailed preferred embodiments of closed and open magnetic loop trigger mechanism designs will now be described in further detail below, respectively.

Closed Loop Sliding Magnetic Trigger Mechanism

FIGS. **52-60** depict one non-limiting preferred embodiment of a closed magnetic loop sliding magnet type trigger mechanism **1200** which exemplifies to a certain degree the conceptual basic design of FIGS. **44-46**, but is not exactly the same in features and construction. In the present embodiment, however, the vertically extending upper working extension or portion **120** of rotating trigger member **104** defines a sear surface **132** configured to releasably engage a firing mechanism component or linkage such as rotatable sear **375** in lieu of the striking member directly such as hammer **130**. The sear **375** in turn is configured and operable to act directly on the energy storage device such as the spring-biased linearly movable striker **40** shown in FIG. **30** and previously described herein. Sear surface **132** operates hold to the striker **40** in the rear cocked position until released via a trigger pull to move forward and strike a chambered cartridge for discharging the firearm. Alternatively, the working portion **120** of trigger member **104** may instead act directly on a hammer **130** as shown in FIG. **44**. Accordingly, the trigger member **104** may be used to act directly or indirectly on and release the striking member whether it is a hammer or a striker.

The sliding magnet trigger mechanism **1200** includes a front **1230**, rear **1231**, opposing right and left lateral sides **1232** (side designations when the trigger unit is mounted in a firearm), top **1233**, and bottom **1234**. Trigger mechanism **1200** generally comprises stationary yoke **102**, rotatable trigger member **104**, sear **375**, and a movable sliding magnet control insert **1031** (a basic version of which is shown in FIG. **44** and described above). The control insert assembly is configured and constructed for varying the static magnetic field in the closed magnetic loop to provide adjustment of

the trigger pull force required to be exerted by the user via a trigger pull to release the striking member.

Yoke 102 includes horizontal upper portion 110, horizontal lower portion 111 oriented parallel to the upper portion, and vertical intermediate portion 114 extending therebetween. Control air gap B is formed in intermediate portion 114 and extends completely through the portion. The lower portion 111 may be bifurcated as shown forming a pair of laterally spaced apart arms defining a vertical through opening 1214 therebetween in which the trigger member 104 is pivotably mounted thereto by transverse trigger pivot pin 1205. Yoke 102 is fixedly mounted to the firearm frame 22, receiver 39, or a trigger housing 1220 as shown in the illustrated embodiment so as to remain stationary when the trigger is pulled.

In the embodiment shown in FIGS. 52-56, yoke 102 is fixedly mounted to a trigger housing 1220. These figures are a cutaway of the trigger housing 1220 showing only a portion of a right side plate of the housing in order to better show details of the trigger mechanism assembly. The trigger housing 1220 is mounted in turn via any suitable mechanical means (e.g. fasteners, interlocking features, etc.) to the firearm frame 22 and/or the receiver 39 depending on the type and configuration of the firearm used. Trigger housing 1220 may have any suitable shape and configuration, one example of which is shown in commonly owned U.S. Pat. No. 10,030,926 which is incorporated herein by reference. Other suitable trigger housing designs however may be used. The configuration of the trigger housing does not limit the invention. In lieu of mounting each trigger mechanism component separately in the frame or receiver, the housing makes it easier to mount, test, maintain, or repair the trigger mechanism if needed.

Rotating trigger member 104 includes upper working portion 120 and lower trigger portion 118. Trigger member 104 has a vertically elongated body. Working portion 120 may be linearly straight and have rectilinear transverse cross section (e.g. square or rectangular) in one non-limiting configuration as shown. Lower trigger portion 118 may have an arcuately curved profile by contrast.

Trigger assembly 1202 defined in part by lower trigger portion 118 of trigger member 104 may include an outer trigger 1201 and inner safety trigger 1203 movable relative to the outer trigger. Outer trigger 1201 is pivotably mounted to yoke 102 via first transverse pivot pin 1205 which defines a first pivot axis. Inner safety trigger 1203 includes an enlarged upper mounting portion 1203-1 pivotably mounted to outer trigger 1201 via a second transverse pivot pin 1206 which defines a second pivot axis parallel to the first pivot axis. The safety trigger further includes a lower blade portion 1203-2 depending downwards therefrom for actuation by a shooter or user. The blade portion 1203-2 may have a solid or an open framework construction as shown including an arcuately concave front surface configured to facilitate engagement by the shooter or user's finger. Safety trigger 1203 is pivotable independently of both the outer trigger 1201 between forward and rearward positions. A spring 1204 biases the safety trigger 1203 towards the forward position projecting forward from the vertical slot 1201-1 formed in outer trigger 1201 in which the inner safety trigger 1203 nests. The second pivot axis defined by pivot pin 1206 may be positioned below and behind the first pivot axis defined by pivot pin 1205. A vertical central axis CA and horizontal central axis HA of the trigger mechanism 1200 are defined for convenience of reference which pass through pivot pin 1205 and perpendicularly intersect each other (see, e.g. FIG. 54).

A transversely oriented split trigger safety blocking pin 1207 is fixedly coupled to the trigger housing 1220 and arranged to selectively engage or disengage a cam surface 1203-3 on top of the upper mounting portion 1203-1 of the safety trigger 1203. Safety blocking pin 1207 may have a cylindrical configuration in one embodiment; however, other shapes may be used.

The trigger member 104 may have a one-piece unitary construction such that the lower trigger portion 118 which defines the main outer trigger 1201 of the trigger member is a unitary structural part of the upper working portion 120 which engages the sear 375. Rotating the trigger 1201 about pivot pin 1205 therefore concomitantly rotates the upper working portion 120 in the same direction in unison to open air gap A and release the sear 375 to discharge the firearm. In other embodiment, the lower and upper portions 118, 120 may be separate components which are rigidly coupled together to provide the same action.

An adjustable trigger member travel stop comprises a mounting block 1213 having an internally threaded bore which rotatably receives adjustment screw 1212 there-through. Block 1213 may be fixedly mounted to the trigger housing 1220 and spaced forward from upper working portion 120 of rotatable trigger member 104 when in the upright un-pulled condition. The shaft end of adjustment screw 1212 opposite its enlarged head used to rotate the screw is variably positionable to selectively engage and bear against the upper working portion 120 of trigger member 104 when rotated forward via a trigger pull. This manually adjustable physical stop limits the travel of the rotating trigger body after release of the sear to ensure the trigger mechanism can properly reset to ready-to-fire condition. One advantageous feature of the magnetic design is that the need for the trigger return spring may be eliminated since the magnet 108 will always be drawn into the control air gap B magnetically, as previously noted. The adjustable stop may alternatively be replaced with a fixed stop in some embodiments that is not adjustable using the mounting block alone or a pin fixedly attached to the trigger housing, frame, or receiver. Based on performance and tolerances, it may be desirable to add a small trigger return spring to account for tolerances of a fixed stop. A trigger return spring may, or may not, be necessary, but if needed would still be smaller and less critical than conventional trigger return spring designs and less noticeable to the operator during trigger recovery.

The sliding magnet control insert 1031 in this embodiment shown in FIGS. 52-60 will now be further described. FIGS. 57-60 show control insert 1031 in isolation. In this embodiment, the permanent magnet 108 of control insert 1031 may be insert or over molded into, or similarly retained via adhesives or fasteners, in a polymeric carrier 1030 (or other non-magnetic material carrier). In other embodiments, the carrier may broadly be made of any suitable non-magnetic material which categorically includes polymers and non-magnetic metals such as without limitation brass, or other. Carrier 1030 preferably has a monolithic unitary body molded, cast, or otherwise formed comprising a single piece of material. In one embodiment, the non-magnetic carrier 1030 may be U-shaped comprising a vertical right and left sidewalls 1035, and rear wall 1034 extending therebetween. Rear wall 1034 includes a threaded bore 1034 which threadably engages adjustment screw 1211 for linearly translating the carrier relative to the yoke 102.

A vertically and forwardly open cavity 1036 is formed by the sidewalls 1035 and front wall 1034 of carrier 1030. Permanent magnet 108 is mounted in cavity 1036. To assist

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in retaining the magnet **108** in the cavity **1036**, a cross bar **1033** may be molded into the carrier which extends horizontally between the sidewalls **1035** at the front of the carrier body. Cross bar **1033** is insertable into control air gap B, but has no effect on the static magnetic field since the carrier is formed of a non-magnetic material.

Carrier **1030** is slideably mounted between the right and left side plates **1220-1** of trigger housing **1220** in a rearwardly open channel **1210** formed in each side plate. FIGS. **52** and **53** show only the right side plate **1220-1**, recognizing that the left side plate **1220-1** may generally be a mirror image thereof (represented schematically in FIG. **55** by dashed lines) to support the various component cross pins from each end. When mounted between the opposing pair of channels **1210** of the trigger housing **1220**, the carrier **1030** is trapped but slideably movable forward and rearward in channels **1210** to adjust the position of the carrier and magazine **108** relative to the control air gap B.

Adjustment screw **1211** is fixed in horizontal position in the trigger housing **1220** but rotatable. This can be accomplished by providing a plain unthreaded hole in a rear plate **1220-2** of the trigger housing (shown schematically in dashed lined in FIG. **54**), or other via similar approaches. The front end of the screw may abut the yoke **102** in some embodiments as shown in the cross section of FIG. **56**. When adjustment screw **1211** is rotated, the screw does not change its horizontal position.

The control insert **1031** can be slideably adjusted along the horizontal central axis HA to move the magnet **108** in carrier **1030** into and out of the control air gap B in the closed-loop magnetic trigger circuit. Rotating screw **1211** in a first direction translates the carrier **1030** forward for increasing the insertion of the permanent magnet **108** in control air gap B of yoke **102** in order to increase the magnet static holding force or torque. Rotating screw **1211** in an opposite second direction withdraws the carrier **1030** rearward for decreasing the insertion of the permanent magnet **108** in control air gap B of yoke **102** to decrease the magnet static holding force or torque. This provides a user selectable adjustment of the trigger pull force or holding torque to suit personal preferences.

It bears noting that other suitable shapes of non-magnetic carriers may be used so long as the permanent magnet **108** may be linearly translated into or out of the control air gap B of yoke **102**. Although the magnet **108** is insertable into control air gap B from the rear **1231** of the trigger mechanism **1200**, in other possible embodiment the trigger mechanism may be designed to insert the magnet from either two of the lateral sides **1232** into air gap B with equal results. This may be more convenient in some firearm designs and allows the adjustment screw **1211** to be accessible through the trigger housing **1220** from either the right or left sides of the firearm for the user.

It bears noting that the magnet **108** in the control insert **1031** will always try to pull itself into full engagement centered in the control air gap B via the magnetic attraction forces created in the closed loop, which acts like a magnetic biasing spring against the adjustment means. By turning the threaded adjustment screw **1211**, the magnet **108** can slide outward from the control air gap B, or allowed to be drawn inward into the air gap. By moving the magnet into and out off the control air gap B, the magnetic flux density in the air gap will approximately vary as a linear function. This is due to the magnetic field strength times the area being preserved across the boundaries. By changing the engagement position of the magnet **108** with yoke **102**, the magnetic static

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holding force at the air gap B between the yoke **102** and the trigger member **104** can be selectively varied by the user.

Sear **375** has already been fully described herein and will not be discussed again in depth for sake of brevity. In general, sear **375** is mounted to trigger housing **1220** via transverse cross pin **377** that defines the pivot axis **376** of the sear. Sear protrusion **44** may be formed on one forward end of sear **375** opposite a rear end having a transverse opening which receives a cross pin **377** that defines pivot axis **376**. A rear facing vertical surface on sear protrusion **44** engages a mating front facing surface of catch protrusion **42** on striker **40** to hold the striker in the rearward cocked position (see, e.g. FIG. **30**). Sear **375** shown in FIGS. **52-56** includes a rear extension **375-1** acted on by sear spring **1209** which keeps the forward sear protrusion **44** biased normally upwards into engagement with the striker's catch protrusion **42**. A mounting plate **1208** may be provided on trigger housing **1220** which acts on the end of the spring opposite the end engaging the rear extension **375-1**. Spring **1209** may be a coil compression spring in one embodiment. Other type springs may be used.

FIG. **54** shows the trigger mechanism **1200** in the ready-to-fire position. The vertically elongated upper working portion of trigger member **104** is parallel to vertical central axis CA in this position. The desired trigger pull force is previously set by the user in the manner described above.

In operation, with additional reference to FIG. **30**, as the trigger assembly **1202** of the closed magnetic loop trigger mechanism **1300** is initially pulled and displaced by the user to the right, the top trigger safety cam surface **1203-3** of the rotating inner safety trigger **1203** engages and the moves past the safety blocking pin **1207**, thereby providing the initial take-up travel of the trigger. As the user continues to pull the full trigger assembly **1202** (outer trigger **1201** and safety trigger **1203**), the final release force to rotate the trigger member **104** body and release the firing sear **375** is achieved by pulling the trigger with sufficient force to rotate upper working portion **120** of trigger member **104** forward to break the magnetic and physical engagement with the yoke **102** and open air gap A. In doing so, the static magnetic holding force created by permanent magnet **108** on the trigger member **104** is overcome. The trigger member upper working portion **120** assumes an acute angle to the vertical central axis CA. Concomitantly, contact is broken between the sear surface **132** on trigger member working portion **120**. Without support from the trigger member **104**, the front end of the sear **375** is forced and rotates downwards about its pivot axis **377-1** by the forwardly spring-biased striker **40** to disengage the sear protrusion **44** from the catch protrusion **42** on the striker. This releases the striker to move along its forward path P between the rearward cocked position and the forwarding firing position contacting and detonating a chambered cartridge C to discharge the firearm.

A method for adjusting the closed loop magnetic trigger mechanism **1200** described above will now be briefly summarized. The method comprises providing stationary yoke **102** configured for mounting in the firearm, a rotating trigger member **104** pivotably movable about a pivot axis relative to the stationary yoke, the trigger member and stationary yoke collectively configured to form a closed magnetic loop, and an openable and closeable first air gap A being formed between the trigger member and the stationary yoke. The method further includes providing a control insert **1031** comprising a non-magnetic carrier **1030** and a permanent magnet **108** operable to generate a static magnetic field in the closed magnetic loop, the static magnetic field creating a primary resistance force opposing movement of the trigger

member **104** when pulled by the user. The method includes: rotating an actuator such as screw **1211** operably coupled to the control insert in a first direction to advance the permanent magnet **108** into a second control air gap B formed in the stationary yoke **102**, the magnet creating a first static magnetic field strength in the closed magnetic loop; and rotating the actuator in an opposite second direction to withdraw the magnet from the second control air gap, the magnet creating a second static magnetic field strength in the closed magnetic loop less than the first magnetic field strength. The strength of the static magnetic field is changeable via varying position of the permanent magnet in the control insert relative to the second control air gap to adjust a trigger pull force of trigger mechanism.

Open Loop Magnetic Trigger Mechanism

FIGS. 61-69 depict one non-limiting preferred embodiment of an open magnetic loop sliding magnet type trigger mechanism **1300** which exemplifies to a certain degree the basic design concept of FIG. 50. It will be noted that design and functionality of the trigger assembly **1202** with main outer trigger **1201** and inner safety trigger **1203**, sear **375**, adjustable trigger member travel stop with travel stop **1212** and mounting block **1213**, safety blocking pin **1207**, sear **375**, and trigger housing **1220** are generally similar to that shown for the closed magnetic loop trigger mechanism **1200** shown in FIG. 52. These features will not be discussed in detail here again for brevity. Sear **375** is generally the same except for a different mounting arrangement of the sear spring **1209**, discussed below. Notably, the open magnetic loop trigger mechanism **1300** does not include a stationary yoke, thereby forming the open magnetic circuit.

With continuing reference to FIGS. 61-69, a stationary mounting block **1304** is provided for adjustably mounting a magnet holder **1302** to the trigger mechanism **1300**. FIGS. 66-69 show mounting block **1304** in isolation and greater detail. Mounting block **1304** may be fixedly mounted coupled to the trigger housing **1220**, such as without limitation to right side plate **1220-1** of the trigger housing **1220** in one embodiment by any suitable means such as fasteners, adhesives, soldering/welding, shrink fitting, or other. In one embodiment, mounting block **1304** may include a laterally extending post **1306** received in a complementary configured hole in the trigger housing **1220** for securing the block to the housing plate. Mounting block **1304** further includes an upwardly extending top post for seating sear spring **1209** thereon between the block and the underside of the sear **375**. Spring **1209** acts to bias the sear **375** upwards to a normal ready-to-fire position in which sear protrusion **44** engages catch protrusion **42** on striker **40** as previously described herein. Mounting block **1304** may have any suitable configuration.

Magnet holder mounting block **1304** includes an elongated internally threaded bore **1305** which opens forward and rearward. Bore **1305** extends horizontally parallel to horizontal central axis HA. The magnet holder **1302** may comprise an elongated threaded rod which threadably engages the bore **1305**. Holder **1302** includes a first inboard end including a forwardly open receptacle **1310** and a second outboard end which may include a tooling recess **1311** configured for engaging a tool used to turn the holder. Tooling recess **1311** may have any suitable tooling configuration, such as for example without limitation a hex shape for engaging an Allen wrench as shown, or a Philips, slotted, torx, star, square, or other shaped tooling recess for engaging a complementary configured screwdriver.

Permanent magnet **108** is insertably mounted in receptacle **1310**. Magnet **108** may be retained in the receptacle by

any suitable means, such as adhesives, fasteners, threaded caps, or other techniques. In the illustrated embodiment, magnet **108** may be cylindrical in shape and receptacle **1310** has a complementary configuration. Preferably, the front free end of the magnet **108** protrudes outwards beyond the holder **1302** and receptacle **1310** to directly engage the rear face of the upper working portion **120** of trigger member **104** as shown.

Magnet holder **1302** may be made of any suitable magnetic material or non-magnetic material. In one embodiment, the holder preferably may be made of a non-magnetic, non-ferrous metal such as brass. Non-magnetic material are essentially transparent to the magnet as long as it does not magnetically interfere into control air gap B to limit the range of motion of the magnet into the gap. Magnetic holder materials are less preferred, but may be acceptable as long as the geometry does not allow a magnetic path that would shunt magnetic flux away from the air gap B. In other possible embodiments, holder **1302** may be made of a suitably strong polymeric material.

Rotating magnet holder **1302** alternately in opposing directions advances the holder and magnet **108** towards the working portion **120**, or retracts the holder and magnet from the working portion of the trigger member. By adjusting the displacement of the magnet **108** with respect to the main rotating upper working portion **120** of the trigger member body, the static magnetic holding force of the magnet can be adjusted by increasing or decreasing the control air gap B between the magnet and the rotating trigger body.

FIG. 64 shows the trigger mechanism **1300** in the ready-to-fire position. The trigger pull and firing sequence operation for rotating the sear and releasing the striker is similar to the closed magnetic loop trigger mechanism **1200**. Those details will not be repeated here.

As the trigger assembly **1202** of the open magnetic loop trigger mechanism **1300** is initially pulled and displaced by the user to the right, the top trigger safety cam surface **1203-3** of the rotating inner safety trigger **1203** engages and the moves past the safety blocking pin **1207**, thereby providing the initial take-up travel of the trigger. As the user continues to pull the full trigger assembly **1202** (outer trigger **1201** and safety trigger **1203**), the final release force to rotate the trigger member **104** body and release the firing sear **375** is dependent on the magnetic flux density created between the magnet **108** and the rotating upper working portion **120** of the trigger body. The flux density is dependent on the magnetic properties of the permanent magnet, the physical geometry of the magnet, and the displacement between the magnet and the rotating trigger body. In general, the trigger release magnetic static holding force is adjusted by changing the displacement and position of the magnet **108** relative to the rotating trigger body at control air gap B, which in turn changes the magnetic flux contribution to the trigger release holding force.

When the trigger is reset after releasing the sear **375**, the movement of the safety trigger **1203** cams down as it resets past the safety blocking pin **1207** and applies a leveraged pressure on the rotating trigger body upper mounting portion **120** to help position the trigger body closer to the magnet. This camming action assists in driving the rotating trigger body back into the reset position where the magnetic forces are re-established and accelerates the re-establishment of the magnetic pull strength necessary to reset the sear **375**. The combination of the trigger safety camming force and the magnetic pull forces of the magnet will advantageously allow for the potential removal of the traditional trigger return spring. The elimination of the trigger return spring

allows a much crisper trigger reaction when the sear releases and more range of possible trigger pull adjustment, which is considered a significant advantage of both this open magnetic loop design and the closed magnetic loop designs.

It bears mention that the foregoing camming force of the split trigger safety and the leveraging of the magnetic attraction force at control air gap B to reset the rotating trigger arm 104 and potentially eliminate the need for a trigger return spring is a significant advantage of both the open and closed loop magnetic designs.

FIG. 70 depicts one non-limiting example of long gun 20-1 in the form of a rifle 20-1 in which the closed or open loop trigger mechanisms 1200, 1300 described above may be used. Rifle 20-1 generally includes a chassis or frame 60-1 supporting a stationary receiver 39 and an elongated barrel 23-1 coupled to the receiver. Barrel 23-1 includes a longitudinally-extending bore defining longitudinal axis LA, a rear chamber for holding the cartridge, and a forward projectile pathway through which the bullet, slug, or shot travels. Rifle 20-1 further includes buttstock 30-1 supported by the frame 60-1. Frame 60-1 includes a downwardly open magazine well 29-1 for removably receiving an ammunition magazine and optionally a grip handle 27-1. An axially movable bolt 25-1 is mounted in the receiver 39 for forming an open and closed breech. Rifle 20-1 depicts a manually operated bolt 25-1 which includes a bolt handle 25-1 for opening and closing the breech. In other embodiments, rifle 20-1 may be an automatic or semi-automatic rifle in which the bolt 25-1 reciprocates automatically upon firing to open and close the breech for ejecting a spent cartridge case and chambering a fresh cartridge. Such a firearm may have a direct or indirect gas-operated action, or be a blowback type action. Trigger mechanisms 1200 or 1300 may be mounted in a trigger unit or housing 1220 previously described herein, which is mounted to the frame 60-1. The trigger mechanisms 1200 or 1300 operate in the manner already discussed to fire the rifle 20-1.

In other possible embodiments, the closed or open loop trigger mechanisms 1200 or 1300 may instead be mounted in a handgun such as firearm 20 shown in FIG. 30 having a reciprocating slide (receiver).

It bears noting that the sear pin 377, rotatable trigger member pin 1205, safety trigger pin 1206, and the safety blocking pin 1207 are mounted in complementary configured mounting holes formed in the inner surfaces of the trigger housing 1220 right side plate 1220-1 and left side plate (not shown).

A method for adjusting the open loop magnetic trigger mechanism 1300 described above will now be briefly summarized. The method comprises providing a rotating trigger member 104 pivotably movable about a pivot axis relative to a frame 22, receiver 39, or trigger housing 1220 of a firearm 20 or 20-1, and a threaded magnet holder 1302 holding a permanent magnet 108 in proximity to the trigger member. The permanent magnet 108 is operable to generate a static magnetic field attracting the trigger member to the magnet 108, the static magnetic field creating a primary resistance force opposing movement of the trigger member 104 when pulled by the user. The method includes: rotating the magnet holder 1302 in a first direction to advance the permanent magnet 108 towards the trigger member at a control air gap B formed between the magnet and trigger member, the magnet creating a first static magnetic field strength; and rotating the magnet holder in an opposite second direction to withdraw the magnet from trigger member, the magnet now creating a second static magnetic field strength less than the first magnetic field strength. The strength of the static

magnetic field is changeable via varying position of the permanent magnet relative to the trigger member at the control air gap to adjust a trigger pull force of trigger mechanism.

The trigger mechanisms disclosed herein are all generally amenable for use in any type of small arms or light weapons using a trigger mechanism, including for example handguns (pistols and revolvers), rifles, carbines, shotguns, grenade launchers, etc.

Firing Event Tracking and Associated Event Characterization

According to another aspect of the present disclosure, the microcontroller-operated firing system with electromagnetic actuator-based trigger mechanism may be configured to provide a tracking system comprising a firing event/shot counter, and in some embodiments execute an associated post-event processing routine to characterize the type of firing event detected. One attribute of the present electromagnetic trigger system unique to microprocessor controlled firing actuation is the unique ability to electronically sense the precise moment in time that the electromagnetic actuator trigger mechanism of the firearm is directed to trip and discharge the firearm based on receiving the electric pulse or signal from the microcontroller, as previously described herein. This unique electronic trigger actuation information presents an extremely accurate timing of shots fired and can be used as a metric for firing event/shot counter that is integrated within the variable force trigger enabled firearm. This type information is especially of interest to shooters who engage in competitive shooting events. This precise timing information allows the microcontroller to track and store a running total of the cumulative number of shots fired and record an associated time/date stamp, thereby allowing the shooter to practice and improve the cadence of firing (time interval between shots). Another use of this precise firing information is the ability to use the running total of shots as an odometer to determine when maintenance of the firearm is required for parts replacements (e.g. changing barrels, etc.), routine cleaning, lubrication, or other needs.

The industry has developed versions of shot counter accessories that are standalone, attached onto the firearm, or installed within the firearm. There are multiple drawbacks with these commercial devices however which hinder their accuracy. All of these devices do not directly observe the trigger force/displacement event by the user to discharge the firearm. Instead, these shot counters generally rely on various types of sensors mounted in the firearm as the sole means for detecting a trigger pull on a "second hand" basis after the fact of an actual firing event, not simultaneously or concurrently with the occurrence of the event. These commercial shot counters typically observe the resulting effects created by the firing event (e.g. blast noise, vibrations, etc.) and must interpret those effects to determine if a shot was in fact actually fired. This presents significant difficulties in differentiating between firing events and other events that may not be related to actual firings (e.g. dropping, bumping, or manually manipulating the action of the firearm). Events such as dropping the firearm on a table, charging the firearm by chambering ammunition, extracting ammunition from the chamber, or loading or extracting an ammunition magazine could be confused with a firing event by these shot counters. Additionally, firearms that are discharged nearby such as at a shooting range during a shooting competition or the presence of other background noises may adversely affect the accuracy of sensor data, thereby making it more difficult

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to accurately predict if the event is a firing event associated with the specific firearm of interest.

The variable force electromagnetic trigger mechanism with microcontroller disclosed herein has the unique ability to precisely know electronically when the operator has intentionally pulled the trigger of the firearm without the deficiencies inherent with conventional shot sensing means and counters. This precise firing information provided by the present electromagnetic actuator trigger mechanism advantageously is unaffected by background and ambient noise, such as at shooting ranges or in other loud environments, thereby eliminating the need to differentiate which firearm has been fired and when with precision. This advantage is attributable to a shot firing event tracking system which is entirely based on the direct firing signal transmitted by the microprocessor to the electromagnetic actuator in the form of an electric pulse which activates the actuator and fires the firearm. This provides a unique advantage over existing shot counting accessories that rely on indirect and "second hand" detection of the firing event via the blast generated by firing the firearms, and which cannot reliably differentiate between blasts generated by other shooters in close proximity in some situations such as at a shooting range. In some embodiments, the microcontroller according to the present disclosure may be further configured to automatically discriminate between and classify a firing event as a "live fire" event resulting in discharge of the firearm, or a "non-fire" event which does not result in discharge (e.g. dry fire/trigger pull event or an attempted discharge event).

FIG. 71 is a control logic diagram showing one non-limiting embodiment of a firing event tracking process 520 according to the present disclosure. This figure is a modification of the existing electronic firing control logic process 500 for the electromagnetic actuator trigger mechanism already shown in FIG. 8 and discussed above, with additional functional or logic steps preprogrammed into microcontroller 200 to implement the electronic direct-sensing firing event tracking function (e.g. shot counting), and optionally in some embodiments the firing event characterization functions noted above. All steps of logic process 500 previously described herein will therefore not be repeated here for sake of brevity. It bears noting that the firing event tracking process 520 may be implemented in some embodiments without firing event characterization if the user is only interested in the total number of trigger pull and firing events including those that result in and do not result in discharge of the firearm.

Referring initially now to FIGS. 8, 9, 11, 71, and 72, the firing event tracking process 520 starts when the microcontroller 200 executes Step 508 in which the microcontroller sends an electric control pulse to electromagnet coil 106 of actuator 123 (or alternatively coil 306 of actuator 350). Any of the actuators disclosed herein may be used with the firing event tracking and characterization processes. The actuator becomes energized to implement the trigger force and release profile or curve having the characteristics preset and preprogrammed by the user into the microcontroller 200. Transmission of the electric control pulse to the actuator concurrently signals the microcontroller to record the trigger pull initiated firing event in Step 521.

In some embodiments, the microcontroller also simultaneously records/stores a time/date stamp associated with the firing event. Each time an electric control pulse is subsequently transmitted to the actuator, the microcontroller records another firing event, and so on. The microcontroller stores each of the firing events and associated time/date stamp in memory, and further maintains a running cumula-

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tive total of the number of firing events occurring. This could be a real-time date/time stamp provided by a real-time clock accessible to the microcontroller 200 in its associated circuitry. An alternate embodiment could utilize a pseudo time stamp that simply provides only a relative time stamp between firing events. This pseudo time stamp has the advantage of providing privacy to the user, and also eliminates the need to utilize a real-time clock which can result in on-firearm power savings.

In addition to recording a running total of cumulative number of rounds fired for maintenance purposes, the rate of fire which may be the timing between rounds fired or total rounds fired over a selected interval of time (may be derived by microcontroller 200 processing the foregoing recorded firing event data and its associated time/date stamps. This provides the cadence of firing or timing between firing events (shots). Timing interval scoring is used in some competitive shooting matches as a metric.

It bears noting that the trigger/firing events (e.g. number and associated time/date stamps) are recorded by the microcontroller 200 in the present embodiment based solely direct detection of the transmission of the electric control pulse or signal to the trigger mechanism actuator without reliance on any input from other secondary sensors as in known shot counters which rely the after-effects of firing (e.g. sound, vibration, motion, etc.) as an indication of a firing event. By contrast, such secondary sensor data however may be drilled down and used in the present firing event tracking process 520 as an adjunct to the direct firing event data to further characterize or classify the type of firing event which has just been detected and recorded by microcontroller 200 (e.g. live fire event or non-fire event).

The precision firing timing information recorded by the microcontroller 200 in the present firing event tracking process 520 (i.e. transmission of electric pulse to trigger mechanism actuator) may be used to help interpret the external firing-effect stimulus observed and detected by a firing event sensor 530 to differentiate between live fire events which result in discharge of the firearm, non-fire events which do not result in discharge. Since the microcontroller 200 knows precisely when the electric control signal is sent to the actuator to fire the ammunition, the microcontroller accordingly knows with precision when to poll or look for external confirmation that the actual firing event has occurred and can discriminate the beginning point of a characteristic signature of the event which should follow (e.g. acoustic, motion, etc.). Accordingly, microcontroller 200 knows exactly when the start of an acoustic, motion, or acceleration event created in reaction to tripping the trigger electronically can be expected and detected by the firing event sensor 530 due to electronic sensing of the firing event electric control pulse transmission. This greatly simplifies the complexity of parsing the detected signature or signal indicative of an after-effect observed in the firearm from an actual firing event which results in discharge of the firearm by the microcontroller 200. One of the most difficult and electrical power consuming aspects of known secondary external stimulus based shot counters previously described herein is the necessity for the microprocessor to be "always on" to continually search for and evaluate if a possible trigger actuation event has started, and then making sure it is interpreted correctly as a start of an actual discharge-related firing event and not another non-discharge event (e.g. firearm jarred/dropped, dry fire event (trigger pull), magazine inserted/ejected, etc.). This requires complex algorithms which inherently reduces reliability of known shot counters.

The foregoing processing complexity and algorithms used by convention shot counters is completely eliminated with the present firing event tracking process 520. Because the microcontroller 200 does not use the firing event sensor 530 according to the present disclosure as the primary means for detecting a trigger pull/firing event, the microcontroller need only initiate search for a signal from the firing event sensor as a secondary processing routine to characterize the event as a live fire event or non-fire event. Transmission of the electric control pulse to the trigger mechanism electromagnetic actuator provides the detection of the firing event. Accordingly, the microcontroller may include a predetermined and preprogrammed window or interval of time to actively search for confirmation of the firing event after the microcontroller senses the electric control pulse transmission to the trigger mechanism electromagnetic actuator. During this window of time, the microcontroller 200 looks for confirmation of the expected firing event characteristic/signature indicative of a live fire event detected by the firing event sensor 530. Because there is no need to guess if the detected firing event signature is the start of an actual event versus some other background or non-fire event noise, the computational analysis is greatly simplified and can result in the use of cheaper less precision sensors, lower power consumption, faster response times, and much more accurate interpretation of the data than known shot counters.

With reference to FIGS. 9 and 71A-B, once the electric control pulse is transmitted by microcontroller 200 to the electromagnetic actuator in Step 508, the firing event discrimination/characterization process begins in Step 522 with the microcontroller initializing the firing event sensor 530. The microcontroller may initiate Step 522 either in serial processing fashion after the firing event and time/date stamp is stored to memory in Step 521, or optionally in parallel processing fashion (shown in dashed lines) concurrently with Step 521. Either logic path may be used. Microcontroller 200 then starts an in-circuit electronic timer in Step 524 which initiates a signal detection time window or interval of predetermined and preprogrammed duration in which the microprocessor searches for and attempts to acquire a signal from and detected by the firing event sensor 530 (see, e.g. FIG. 72). In Step 526, the microcontroller determines whether a real-time signal has been detected by and received from firing event sensor 530 before the timer (time interval) expires. If the answer is "No," control passes to Step 534 and the microprocessor classifies the firing event as a "non-fire event" because no detection of a signal means the firearm has not detonated the chambered cartridge and been discharged. This may be attributed to a dry fire event (i.e. trigger pull and actuator activation not resulting in discharge with an empty magazine or chamber), or a failed firing attempt resulting from a miss-fire after actuation (energization) of the actuator.

If the answer is "Yes" in Step 526, control passes to Step 528. In Step 528, the microprocessor compares the detected real-time firing characteristic sensed by firing event sensor 530 to a preprogrammed firing characteristic/signature indicative of the live fire event (examples of which are shown in FIGS. 72-75 and further described below). If in Step 528 the detected firing characteristic/signature matches the preprogrammed characteristic/signature (confirmed "Yes" response), control passes to Step 534 and the firing event is classified as a "live fire" event resulting in discharge of the firearm. If the real-time detected firing characteristic/signature does not match the expected preprogrammed firing characteristic/signature ("No" response), the firing event did not result in an actual discharge of the firearm and control

passes to Step 534 which classifies the event as a "non-fire" event. Accordingly, the microcontroller determines a non-fire event if either the preprogrammed timer window or interval has lapsed, or the returned signal from the sensor 530 does not match the preprogrammed firing characteristic/signature.

As shown in FIG. 71B, control passes from either Steps 532 or 534 to Step 536 which resets the firing event sensor 530 for the next firing event. Control returns to Step 502 (FIG. 71A) to restart the firing sequence.

The firing event sensor 530 may be various types of commercially-available sensors which are capable of detecting a firing characteristic/signature indicative of a live fire event. A few non-limiting examples will now be further described.

In one embodiment, firing event sensor 530 may be a simple acoustic sensor with the range and bandwidth to differentiate the sound of a shot fired can be added to the electromagnetically variable force trigger mechanism. This can be an inexpensive piezoelectric sensor or microphone. Since the microcontroller 200 already knows the precise time when the operator pulled the trigger sufficiently to discharge the firearm and the electric control pulse was transmitted to energize the trigger mechanism actuator (FIG. 71A, Steps 506 and 508), the acoustic sensor need only be monitored at the time of the intended actuation of the firearm by the microcontroller. The initiation and duration of the monitoring function can be for the predetermined and preprogrammed window or interval of time initiated by the timer previously described herein. This timing knowledge allows a simple check for confirmation of the firing event by the microcontroller when expected, which advantageously can be accomplished with inexpensive sensors. The microcontroller 200 knows the start time of the firing event and can ignore anything that occurs outside the preprogrammed window of time such as other shooters and noises in the environment (see, e.g. FIG. 72 dotted time window box). In its simplest implementation, a simple measurement of the decibel noise level (dB) above a certain preprogrammed threshold (i.e. firing characteristic/signature) for example would be sufficient to confirm that a "live fire event" has occurred. The shooting environment can be noisy and varied, particularly at a shooting range or during competitive shooting matches. Since the blast sound of a firearm discharge has a deterministic shape that rises and falls in time fairly quickly and predictably, the microcontroller 200 can execute algorithms that enhance the discrimination of the firing event. Scale invariant filters such as Hough transformations and algorithms that look for the characteristic shape and timing of the shot fired acoustic signature (e.g. shape and magnitude/peak of the sound detected by the acoustic firing event sensor 530), can also be used beyond decibel level alone to help differentiate non-fire events in environments that have higher or lower noise thresholds. It is well within the ambit of those skilled in the art to develop such algorithms. And because the microcontroller 200 knows the exact start time of the event via the preprogrammed window or interval of time for observing a live fire event, it can precisely identify the start of the characteristic shape of the acoustic signature that results from the trigger pull event. This allows better discrimination since the peak and calibration of the shot fired sound wave can vary based on a number of conditions including variations in type and brand of ammunition, consistency of ammunition, and powder loading and bullet geometry. This also allows for interpretation of secondary events in the acoustic signature that

might normally be lost in the signal to noise ratio to be identifiable and used to help discriminate between a live-fire and non-fire events.

FIG. 72 shows a representative acoustic type firing event sensor 530 output where four rounds or shots were attempted to be fired in rapid succession. Trigger/Firing Events 1, 2, and 3 results in discharge of the firearm and produced an acoustic firing characteristic/signature indicative of a "live fire" event. When the microcontroller 200 acquires and compares those characteristics/signatures to the preprogrammed firing characteristic/signature (FIG. 71B, Step 528), a match is confirmed (e.g. dB level and/or shape of signal curve) such that Events 1, 2, and 3 would be properly classified as live fire events. Trigger/Firing Event 4, however, failed to result in discharge of the firearm, either via dry firing or a failed firing. The microcontroller 200 would readily not classify Event 4 as a live fire event since the preprogrammed firing characteristic/signature does not match the acquired characteristic/signature which is quite distinct. Event 4 would therefore be classified as a non-fire event by the microcontroller. The vertical arrows below the horizontal time axis indicates the precisely defined start of the trigger pull/firing events. The dashed observation time window/interval boxes shown in FIG. 72 represent the preprogrammed time that the microcontroller 200 scans/searches for an expected firing characteristic/signature from the firing event sensor 530. As previously described herein, the microcontroller 200 initiates each observation time window/interval only after transmission of the electric control pulse to the trigger mechanism electromagnetic actuator. Therefore, any sound or noise occurring outside of the time window/interval is not acquired by the microcontroller and advantageously need be further parsed or discriminated. This greatly simplifies signal processing by microcontroller 200, thereby eliminating the need for executing complex discrimination algorithms as previously noted.

To illustrate the above point, FIG. 73 shows the same representative acoustic sensor output (Trigger/Firing Event 1) displayed but with background noises preceding or after Event 1 associated with non-fire events. Four acoustic events are observable by sound amplitude in the graph detected by the sensor. For example, the acoustic sound of the ammunition magazine being inserted, the slide or bolt being racked back to chamber a round of ammunition, and another miscellaneous firearm sound firearm getting a jarring bump as it is dropped back onto a table after a firing event. Note that the arrow on the horizontal axis indicating the precisely defined start of the Trigger/Firing Event 1 and the preprogrammed observation window/interval of time (dashed box) allows the microcontroller 200 to readily ignore and not acquire those extraneous mechanical acoustic signatures that are not aligned with the timing of the trigger and actuator activation. Without the Trigger/Firing Event timestamp and associated observation window, it would be significantly more difficult for the microcontroller 200 to differentiate between similar acoustic events that may occur during the normal handling of a firearm.

Note that the timing of the trigger pull and trigger mechanism actuator activation event to the subsequent acoustic firing event noise pickup is very short; in the order of microseconds. Accordingly, the preprogrammed observation window/interval of time may be less than 1 second, and preferably preset and measured in fractions of a second or microseconds in some embodiments based on the typical cycle rate time for the action of the particular firearm involved. The cycle rate for the action of a firearm is generally the time required to open the breech after firing the

ammunition, extract and eject the spent cartridge case from the barrel assembly chamber via translating the bolt or slide rearward, strip a fresh cartridge from the magazine, and chamber the fresh cartridge while reclosing the breech for the next firing event. Accordingly, the preprogrammed observation window would ideally be no longer in duration than the typical action cycle rate of the particular firing system involved so that the firing event tracking system is rapidly reset and ready to track the next firing event. This ensures that each observation window, during which time the microcontroller 200 monitors and acquires a firing characteristic detected by the firing event sensor 530, does not overlap the subsequent firing event to maintain the integrity of the firing event count. As examples, a very fast shooter using a semi-automatic pistol could fire up to about 5 rounds per second. The fastest fully automatic mode machine gun can come close to 100 rounds per second. Thus the preprogrammed observation window must be preset to take into consideration the type of firearm involved and firing mode (semi-automatic or fully automatic). In one non-limiting embodiment, the observation time window may be equal to or less than approximately 1.5 times a total cycle time to cycle an action of the firearm for a semi-automatic or automatic firearm. In one non-limiting example, the preprogrammed duration of the observation window may be about 100 milliseconds maximum for a semi-automatic firearm. It bears noting that for bolt-action rifles in which the bolt is manually retracted to open the breech after each shot, the preprogrammed observation window duration would be limited to the firing event only and not include the manual racking of the bolt. Accordingly, the observation window duration would not include cycle time to retract the bolt and open breech, and closing the breech to chamber of the next round as this is a manual operation and not deterministic. For bolt-action rifles, the preprogrammed duration of the observation window of about 100 milliseconds maximum would generally also suffice for the firing event timing only for these manually operated firearms.

FIG. 74 shows the situation where several shooters may be in proximity to the shooter of interest utilizing the present firing event tracking system 520. This situation can occur at a firing range or during a shooting competition. The likelihood of another shooter firing nearby and the acoustic noise generated by another shooter discharging their firearm within the preprogrammed observation window/interval of time and being confused with the primary shooter of interest is very small. The two shots being fired close enough together to be synchronized to the same starting timestamp is very unlikely and considered a rare event that would influence the accuracy of the firing event tracking system 520. Even if this situation were to occur somehow, the firing event characterization process previously described herein would eliminate the second shooters acoustic signature since it would not match the preprogrammed acoustic signature at least in sound amplitude (dB) as shown in FIG. 74. This figure shows the acoustic signatures of shooters nearby. Four acoustic events are shown, but only the event of interest at the Trigger/Firing Event 1 timestamp within the observation window of time is acquired and classified as a valid shot count by microcontroller 200. Any sounds from nearby shooters fall outside this narrow band of time (e.g. microseconds) at the trigger event timestamp when the microcontroller is actively searching for an acoustic signature detected by firing event sensor 530.

In another embodiment, firing event sensor 530 may be a motion type sensor. The use of commercially-available

motion sensors with one, two, three or more degrees of freedom and MEMS micro-miniature single axis and multi-axis accelerometers may be used and provides the opportunity to capture a rich data signature of events during the shooting of a firearm. Motion sensors look for motion and/or acceleration of the firearm that occurs during the recoil shock of live-firing. There are a number of types of motion sensors that may be used with the present firing event tracking system **520** to discriminate between the typical slow motion changes in position or velocity of the firearm during normal handling and use, and the sudden high speed change in motion/acceleration from firing ammunition. Typically piezoelectric, piezoresistive, variable capacitance, or variable reluctance acceleration sensors (accelerometers) may be used to provide the type of high speed sensing for good motion/acceleration event discrimination in the present application. Alternatively numerous other types of motion sensors such as magnetometers, gyroscopes, inertia and position sensors may be used. Some simplistic very low cost motion sensors that simply register the movement of weighted mass or liquid can be used as the firing event sensor **530** to register the presence of the high speed motion of firing event as well. The prior knowledge of the precise timing of the firing event by the microcontroller **200** (i.e. electric control pulse transmission to trigger mechanism electromagnetic actuator) herein advantageously allows for the use of less precise in the type of sensor needed since the microcontroller is only interested in a gross measure that confirms the firing event has occurred during the observation window or interval of time as previously described. Accordingly, the term "motion sensor" for use as the firing event sensor **530** should be broadly construed to include any of the foregoing types of motions sensors and those similar.

FIG. **75** shows the use of a single axis capacitive MEMS (Micro-Electro-Mechanical Systems) Accelerometer being used as the basis for discriminating between non-fire and live-fire trigger pull events. Given the expectation of the microcontroller **200** to receive the acceleration profile shown during the preprogrammed observation time window/interval previously described herein, it is easy for the microcontroller to observe the presence or absence of the characteristic high-amplitude fired-round signatures shown thereby making it computationally simple to classify the event as respectively a "live fire" event or "non-fire" event given knowledge of the precise timing of the trigger/firing event (i.e. electric control pulse transmission to actuate the trigger mechanism actuator).

It bears noting that the firing event tracking system may be used with any of the actuators disclosed herein, including embodiments of the fire-by-wire trigger mechanism having an electronic sear (E-sear) shown in FIGS. **34-35** and previously described above. Moreover, the present firing event tracking system is broadly applicable to any firearm beyond those examples disclosed herein using a firing mechanism which relies on transmission of an electric energy pulse to detonate a chambered ammunition cartridge and discharge the firearm.

Advanced Fire Control System Interface

According to another aspect of the present disclosure, the present microcontroller-operated trigger system disclosed herein with user-adjustable electromagnetic actuator variable force trigger mechanism may be configured to cooperate and interface with an external optical-based advanced fire control targeting system also mounted onboard the firearm, such as without limitation embodiments of the Next Generation Squad Weapons Fire Control (NGSW-FC). This government initiative is intended to develop the next gen-

eration of military combat rifles which incorporates an electronic adaptive optical fire control system mountable on the rifle. The U.S. Military, through programs such as the NGSW-FC program and testing of the Israeli Smart Shooter SMASH 2000 fire control system for small arms, is evaluating the viability of integrating adaptive direct view optics with ballistic calculators, environmental/atmospheric sensors, and laser range finding devices into the next generation of firearms. The fire control system essentially assists the user with aiming the firearm and targeting for increased shot accuracy by employing a microprocessor and associated input sensors and devices. The optical-based fire control targeting system automatically compensates for user skill and a host of field variables and factors all of which affect point of aim in real-time based on the sensor data and information relayed to and processed by the on-board microcontroller.

One beneficial attribute of the present variable force electromagnetic actuator trigger system unique to microprocessor-controlled firing actuation previously described herein is the provision of an electronically interruptible trigger platform which can receive and process a shot authorization signal generated by an adaptive optics unit of the external advanced fire control targeting system which is mounted onboard the weapon. Accordingly, following a trigger pull event, the trigger unit microcontroller of the present electromagnetic trigger system in one programmed configuration may delay actual firing of the weapon until the shot authorization signal is received and detected, thereby indicating that the user has accurately acquired the target with the guidance of the advanced fire control targeting system. A critical component for integrating the intelligent adaptive optics package with a base firearm is the need for a reliable interruptible electro-mechanical trigger mechanism as disclosed herein which provides the electrical input/output control capable of operable cooperation and integration with the optics package, while doubly providing a simplistic backup manual override to default to a standard mechanical trigger means in the event of an electronics or power supply failure. Numerous operational scenarios may be preprogrammed into the interruptible electromagnetic actuator trigger mechanism disclosed herein. Embodiments of the present electromagnetic trigger mechanism disclosed herein advantageously allow the firearm to still be fired manually in exigent circumstances until such time that the electronics operating problems can be rectified.

It bears noting that the term "external" as used herein in reference to the advanced fire control targeting system merely connotes that the targeting system is separate from and hence external to the electromagnetic actuator trigger mechanism and system in that sense. Both the trigger and targeting systems may therefore still be mounted to and onboard the firearm. In other possible embodiments, it is possible that the advanced fire control targeting system may be another type of targeting system which is physically removed from the firearm and may communicate with the trigger system via any suitable wireless communication protocols.

FIG. **76** shows modification of the system block diagram of FIG. **9** depicting the programmable microprocessor-based trigger mechanism control system with microcontroller **200** previously described herein for the user-adjustable variable force electromagnetic trigger mechanism. Microcontroller **200** in this modified figure is shown operably and communicably interfaced with the electronic intelligent adaptive optics package or unit **701** of the advanced fire control targeting system **700** (e.g. NGSW-FC). Only high level

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control aspects of the advanced fire control system **700** are shown in the system block diagram.

The electronic adaptive optics unit **701** comprises a control module **711** which includes control circuitry comprising programmable targeting microcontroller **712**. Microcontroller **712** is operably coupled to the trigger unit microcontroller **200** of the electromagnetic trigger mechanism and configured for establishing two-way communications between the microcontrollers.

Control module **711** (e.g. microcontroller **712**) is operably coupled via wired and/or wireless two-way communication links to a targeting reticle display **702**, range-finding sub-module **703**, ballistics computation sub-module **704**, environmental sensing sub-module **705**, and point of aim sensing sub-module **706**. The foregoing sub-modules may each include dedicated circuitry including microprocessors, memory, application-specific integrated circuits (ASIC) chips, or other electronic or semiconductor devices configured via programming and design for performing the desired function of each sub-module. The reticle display **702** may be controlled by the control module targeting microcontroller **711**. Each of the sub-modules may be operably/communicably interlinked together to each other as shown and to the main adaptive optics unit control module **711**.

It bears noting that the term "sub-module" does not necessarily refer to discrete physical modules with individual housings separate from the optics unit control module **711**, but rather is intended to connote a grouping of electronic components (i.e. dedicated circuitry and devices as described above) organized by function which may be different parts of the main control module. Accordingly, the adaptive optics unit control module **711** may be configured and operable to perform all of the functions of the sub-modules which are further delineated below.

Range-finding sub-module **703** comprises circuitry configured to at least find the distance from the firearm to the target. Sub-module **703** therefore includes electronic devices including sensors **707** necessary to perform the intended function such as without limitation sensors including laser range finders and other devices, etc. For active range finding, methods which may be used include laser, LIDAR, radar, sonar and ultrasonic range finding. Basically, any method may be used employing the time it takes for light, radio waves, sound and ultrasonic sound waves to travel to and return from the target to measure distance to target; all with similar sensor data results.

Environmental sensing sub-module **705** comprises circuitry configured to adjust the point of aim based on factors that an expert marksman (e.g. sniper) would normally obtain and take into consideration. Sub-module **704** therefore includes electronic devices including sensors **709** necessary to perform the intended function such as without limitation sensors configured for sensing and obtaining various data and information relevant to the ambient atmospheric environment around the firearm and user which may affect point of aim. Such sensors may include for example without limitation ambient temperature, humidity, air density, wind direction and speed, altitude/elevation above sea level, etc.

Point of aim sensing sub-module **706** comprises circuitry configured to collect multi-axis positional and motion/acceleration data on the aiming of the firearm and adjust the point of aim via reticle display system **702** based on the three-dimensional (3D)/three-axis angular position of the firearm barrel in space as held by the user when aiming at the target (e.g. barrel **23-1** of firearm **20-1** in FIG. **70** or other). By combining the exact trigger pull timing information from the trigger force/displacement sensor associated with the

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trigger unit microcontroller **200** with additional high precision multi-axis position and motion sensing of sub-module **706** of the fire control targeting system, the intended direction of aim can be electronically associated with a firing decision. Sub-module **706** therefore includes electronic devices including sensors **710** necessary to perform the intended function such as without limitation sensors configured for sensing and obtaining various data and information relevant to the three-axis barrel orientation and angular position relative to the horizon, and other parameters which locate the barrel in space as the user is holding and aiming the firearm. Such information affects the point of aim and can be communicated to the ballistic computation sub-module **704** and/or the main adaptive optics unit control module **711** to alter the digital reticle display accordingly.

Examples of barrel positional and motion sensors which could be used include without limitation several MEM-based high precision fast acting accelerometers, such as the ADL001 from Analog Devices iMEMs® high-performance, high-bandwidth accelerometer line, that can give detailed acceleration and velocity data during the trigger pull event. Additionally, 3D orientation information can be acquired using off-the-shelf solutions such as a three axis orientation sensor from BOSCH (Model BMX055 or similar) that combines a MEMS (Micro-Electro-Mechanical Systems) accelerometer, magnetometer, and gyroscope on a single die with high speed ARM micro-controller and software algorithms to provide accurate real-time orientation information. By analyzing the motion and acceleration of the firearm barrel collected by the positional/motions sensors of sub-module **704** during the trigger pull event, the aimpoint of the firearm can be acquired as the user applies pressure and squeezes the trigger. This firearm barrel positional information can then be used by the trigger unit microcontroller **200** to control the timing of the actual firing event.

Ballistics computation sub-module **704** comprises circuitry configured to adjust the point of aim (or aimpoint) based on factors that an expert marksman (e.g. sniper) would normally obtain and take into consideration to improve shot accuracy. Sub-module **704** therefore includes circuitry comprising associated related electronic devices **708** configured to automatically calculate point of aim adjustments necessary to accurately sight the target through the adaptive optics unit based on receiving data obtained by the sensors associated with range-finding sub-module **703**, environmental sensing sub-module **705**, and point of aim sensing sub-module **706**. The ballistics computation sub-module **704** may also integrate characteristics of the ammunition being used (e.g. powder charge/load, projectile/bullet weight and length, etc.) to adjust the point of aim. Other sensor data used by the ballistics computation sub-module may include muzzle velocity, ballistic coefficient, and other ammunition and rifle specific inputs.

The electronic adaptive optics unit **701** comprises an optical sight including an integrated reticle display system **702**. The reticle display system may be an electronic system configured and operable to generate a digitally displayed reticle **714** of certain shape (e.g. red, green, or other color dot, cross-hairs, etc.). The adaptive optics unit **701** may be in the form of a scope or sight having a housing configured for mounting on the firearm frame, receiver, barrel, or accessory mounting rail/system in a top position normally used for mounting firearm sights via any suitable mounting interface commonly used in the art. Adaptive optics unit **701** may comprise at least one transparent sighting lens **713** usable by the user as a direct view optic for visually sighting

a target downfield therethrough. The digitally displayed reticle **714** is projected and overlaid onto the sighting lens **713**.

To improve the speed of acquiring the target and shot accuracy, the programmable targeting microcontroller **712** of the adaptive optics unit control module **711** may be configured and operable to control and adjust the position of the digitally displayed reticle **714** on sighting lens **713** based on the sensor and other data obtained and/or calculated by the sub-modules **703-706** described above. Control module **711** (e.g. targeting microcontroller **712**) is configured via programming to display a corrected reticle **714** at a position which compensates for all of the variables and parameter obtained by the sub-modules. The displayed reticle is therefore actually an electronically corrected digital reticle which accurately aligns the point of aim on the intended target.

It bears noting that all of the foregoing features of the advanced optics-based fire control targeting system **700** may all be integrated with and onboard the housing of the adaptive optics unit **701**, which is configured for mounting directly on the firearm such as for example firearm **20-1** shown in FIG. **70** or others.

Interruptible Electromagnetic Trigger Mechanism Operation

Targeting microcontroller **712** of adaptive optics unit **701** may be configured and operable to generate and then transmit a "fire" (shot authorization) signal to the trigger unit microcontroller **200** of the variable force electromagnetic actuator trigger mechanism. The trigger unit microcontroller **200** may be programmed to time electrically discharging the firearm subsequent to a trigger pull event based on the receipt and detection of the valid shot authorization signal from the advanced fire control targeting system **700**, thereby configuring the trigger mechanism to interrupt the normal firing sequence pending the shot authorization signal. Firing of the firearm is there contingent upon satisfying trigger setpoint operating parameters preprogrammed into the trigger unit microcontroller **200** and detection of the shot authorization signal from the targeting microcontroller **712**.

Numerous electronic control scenarios via programming trigger unit microcontroller **200** with appropriate control logic to operate the electromagnetic actuator trigger mechanism in an interruptible mode in conjunction with the advanced fire control targeting system **700** to discharge the firearm are possible. Several non-limiting examples of control scenarios are described below. It will be appreciated that numerous variations of the following control scenarios are possible which fall substantially within the scope of the present disclosure.

In the interruptible trigger control scenarios presented below, any of the trigger mechanisms with bistable or non-bistable design electromagnetic actuators previously described herein may be used. As one non-limiting example, electromagnetic actuator trigger mechanism **300** of the electronic trigger system shown in FIGS. **16-33** and described in detail may be used to form the present interruptible trigger mechanism for convenience of reference. The term "interruptible" as used herein refers to the operating mode and electronic control scheme (i.e. control logic or software) in which the trigger unit microcontroller **200** of the electromagnetic actuator-based trigger mechanism is configured to process and complete the firing sequence, as opposed to physical attributes of the trigger mechanism.

As previously described, trigger mechanism **300** includes electromagnetic snap actuator **350** configured to form the dual closed magnetic flux loop or paths. To briefly reiterate for convenience of reference, actuator **350** may be a non-bistable release type electromagnetic actuator in which the

actuator is not energized to change position for either initiating movement or to reset the actuator similar to trigger mechanism snap actuator **123** previously described herein. Instead, similarly to actuator **123**, microcontroller **200** may be programmed and configured to energize the present actuator **350** of the dual flux loop design via electromagnetic coil **306** in response to a manual trigger pull. This generates the secondary dynamic or active magnetic field which interacts with the primary fixed or static magnetic field generated by the permanent magnet **308** in either an additive or subtractive operating mode depending on the polarity of the power source **122** established via the microcontroller. The present electromagnetic actuator **350** is configurable by the user or shooter via programming the trigger unit microcontroller **200** to change the trigger pull force and displacement profile in the same manner described above for single flux loop electromagnetic actuator **123**. The preprogrammed trigger setpoint(s) in the interruptible trigger mechanism control scenarios below may be based on sensed trigger pull force applied by the user or measured trigger displacement via the sensors previously described herein. Trigger pull force and displacement may be characterized as trigger activity in a broad sense for brevity and includes either pull force or displacement.

FIGS. **77** and **78** illustrate an exemplary single stage interruptible trigger control logic process **730** which may be preprogrammed into and implemented by trigger unit microcontroller **200** of the electromagnetic actuator trigger mechanism control system, which is operably interfaced with the advanced fire control targeting system **700**. FIG. **77** is a high level control logic flow diagram, while FIG. **78** is a lower level flow diagram showing additional sub-steps of the basic process in FIG. **77**.

As a general high-level overview of the single stage interruptible electronic trigger system control process, the trigger unit microcontroller **200** initiates the firing or shot as the user commits to firing by increasing the pull force manually applied on the trigger (e.g. trigger member **320** in FIG. **27**) past a preprogrammed trigger activity threshold or setpoint (trigger pull force or displacement depending on whether trigger force or displacement sensors are used by the trigger system to control the firing sequence. The trigger system would then wait for the presence of a "fire" (shot authorization) signal returned by the external fire control targeting system **700** in response to receiving a "shot initiation" signal from the trigger system. Detection of both the "fire" signal and trigger activity exceeding the preprogrammed setpoint by the trigger unit microcontroller **200** initiates electrical activation of the trigger unit actuator **350** and the firing event. If the microcontroller **200** senses that the user-applied force on the trigger or its displacement falls back below the setpoint, the firing attempt is aborted and resets to await another attempted firing. If an inappropriate amount of time has passed with force still on the trigger a timeout error occurs. The system ignores the trigger event and defaults to a non-powered manual fire state at a preprogrammed maximum allowed force/displacement until the force on the trigger is removed.

Referring to FIGS. **77-78**, single stage electronic trigger control logic process **730** begins in Step **734** with trigger unit microcontroller **200** sensing user activity on the trigger (e.g. trigger pull) in the form of applied trigger pull force or trigger displacement/movement. In Step **735**, microcontroller **200** determines if the sensed activity on the trigger is above the preprogrammed trigger pull setpoint (e.g. trigger force or displacement). If a negative response (no) is returned to the inquiry, control passes back to Step **734** to

continue monitoring the sensed trigger activity. This may occur as the user contemplates firing while aiming via a partial trigger pull having an applied trigger pull force or displacement below the threshold of the preprogrammed setpoint. If a positive response (yes) is returned to the microcontroller inquiry in Step 735, control passes to Step 736.

In step 736, the trigger unit microcontroller 200 transmits/sends a "shot initiation" signal to the external adaptive optics unit control module 711 of the advanced fire control targeting system (reference system architecture schematic diagram of FIG. 76). In step 737, the adaptive optics unit control module 711 (targeting microcontroller 712) receives the "shot initiation" signal and actively adjusts the point of aim and displays a corrected reticle 714 positioned on the sighting lens 713 based on input from all of the data obtained by the sub-modules 703-706 as previously described herein. In this example, external control shot authorization by the adaptive optics unit control module 711 could be based on a corrected point of aim being satisfied by the user aligning the corrected reticle with the intended target. Given the range (distance) to the target and other ballistic information processed by ballistics computation sub-module 704, a deviation from the point of aim is calculated that includes the corresponding corrections. Targeting microcontroller 712 generates and transmits a "fire" (shot authorization) signal to trigger unit microcontroller 200 as the operator pulls the trigger smoothly and hovers aim around the intended target location when the corrected point of aim coincides with the ballistics calculation of correction indicating the user has locked onto the target.

In Step 738, trigger unit microcontroller 200 performs a test to search for both receipt of the "fire" signal from the adaptive optics unit microcontroller 712 and continued sensed activity on the trigger 320 by the user above the preprogrammed setpoint from Step 735, which would be indicative of the user's continued intent to discharge the firearm. If both signals are present and detected by trigger unit microcontroller 200 (yes), control passes to Step 739 to initiate electrical actuation of the firing event. With additional reference to FIGS. 27-29, microcontroller 200 energizes the coil 306 which trips the actuator 350 and releases sear 375 to discharge the firearm in the manner previously described herein. Control passes to Step 740 in which microcontroller 200 resets the trigger mechanism and sends a signal to adaptive optics unit module 711 (microcontroller 712) to reset for the next shot.

If in Step 738 a negative (no) response is returned, control instead passes to Step 741. The negative response indicates that the "shot initiation" signal has been sent to the adaptive optics unit control module 711, but a "fire" signal has not yet been returned to the trigger unit microcontroller 200 upon searching for receipt of this signal. The test of Step 741 performed by trigger unit microcontroller 200 confirms whether or not the trigger is still being pulled to fire the firearm. If a negative (no) response is returned, control passes to Step 740 to reset the trigger mechanism for the next trigger pull event. If the result is positive (yes) indicating that the user is still pulling the trigger, control passes to Step 742.

In Step 742, a test is performed to compare and determine if the trigger activity (i.e. trigger pull force or displacement) applied by the user exceeds a preprogrammed maximum allowed trigger pull force/displacement limit. This limit is indicative of the user's intent to apply sufficient force on the trigger to manually override the trigger system electronics and fire control targeting system 700 and discharge the

firearm. Such a situation may occur in exigent combat circumstances where immediate firing is necessary. If the test answer is negative ("no"), control passes to Step 743 to initiate a firing sequence timer of predetermined duration preprogrammed into trigger unit microcontroller 200. Any suitable duration of time may be used. In Step 743, the microcontroller 200 further performs a test to determine if the timer has expired. If not (i.e. "no" response), control passes back to Step 738 for the trigger unit microcontroller 200 to continue searching for a valid "fire" signal from the adaptive optics unit control module 711. A control loop is implemented by microcontroller 200 to repeat Steps 738, 741, 742, and 743 until the "fire" signal is detected by the trigger unit microcontroller.

If in Step 742 a positive response ("yes") results indicating that the applied actual trigger pull force/displacement imparted by the user to the trigger exceeds the preprogrammed maximum allowable trigger pull force/displacement limit, control instead bypasses the decision block in Step 743 and timer to proceed to Step 744. In Step 744, the trigger system defaults to a non-powered manual "override" condition and enters manual firing mode operably detached from electrical firing assistance normally provided by the electromagnetic actuator trigger unit. The electromagnetic actuator trigger unit acts as a conventional manually fired trigger unit for mechanically firing the firearm. Accordingly, the trigger unit microcontroller 200 electrically deactivates the electronic firing system and power to the electromagnetic actuator trigger mechanism. The increased pressure applied to the trigger by the user in excess of the preprogrammed maximum allowable trigger pull force/displacement limit indicates that the user wishes to discharge the firearm despite the absence of a valid "fire" signal from the adaptive optics unit control module 711. This condition may possibly occur: (1) if there is a malfunction with the advanced fire control targeting system electronics (e.g. dead power source onboard battery 715 (see, e.g. FIG. 76) or other component failure; or (2) if there is a malfunction with the electromagnetic trigger mechanism system electronics (e.g. dead power source onboard battery 122 (see, e.g. FIG. 76) or other component failure. The trigger system thus ignore the electronic trigger pull event and will stay in this manual firing mode until the user-applied force on the trigger is completely removed, at which time the system will reset in Step 744 and return control back to Step 734 to initiate electronic trigger mode for the next trigger pull and firing sequence. Concurrently in Step 744, the trigger unit microcontroller 200 will transmit an external reset signal to the adaptive optics unit control module 711 to terminate the present electronic firing sequence and revert to its initial state to await the beginning of a successive trigger pull event.

FIGS. 79 and 80A-B illustrate an exemplary two stage interruptible trigger control logic process 750 which may be preprogrammed into and implemented by trigger unit microcontroller 200 of the electromagnetic actuator trigger mechanism control system, which is operably interfaced with the advanced fire control targeting system 700. FIG. 79 is a high level control logic flow diagram, while FIGS. 80A and 80B are lower level flow diagrams showing additional sub-steps of the basic process in FIG. 79.

As an overview of the two stage control process 750 embodiment, the trigger pull event initiated by the user is broken into a "shot initiated" and "shot confirmed" trigger force setpoints. The "shot initiated" state of the electronic trigger system indicates that the user has initially selected an aimpoint (point of aim) for firing without yet fully commit-

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ting to discharge the firearm. The “shot confirmed” state of the electronic trigger system indicates the user has adjusted the aimpoint based on corrections made by the external fire control targeting system 700 and is now fully committing to the firing the shot at an accurately acquired target. To be clear, the “shot confirmed” signal does not indicate that the firearm has been discharged and the shot taken yet. Both the “shot initiated” and “shot confirmed” signals are control signals each associated with a respective trigger pull setpoint (Setpoint 1 and Setpoint 2) preprogrammed into trigger unit microcontroller 200. Setpoint 1 may be associated with a partial trigger pull to activate the electromagnetic trigger mechanism for aiming in preparation for firing. Setpoint 2 may be associated with the trigger activity comprising a full trigger pull for discharging the firearm via fully actuating the electromagnetic actuator trigger mechanism in the manner previously described herein after the target has been accurately acquired. Both of the “shot initiated” and “shot confirmed” signals are sent/transmitted to the external advanced fire control targeting system 700 system. Between receipt of these two control signals, the external fire control targeting system could calculate new ballistic solutions and modify an adaptive optics unit sight, confirm authorization with another entity approving authorization to take the shot, or other intelligent adjustments or control criteria such as waiting for the operator to position and align the aimpoint more precisely on target via a corrected reticle displayed on the optics unit sight before authorizing the firing event via transmitting the “fire” signal to the trigger system as a response.

Referring now to FIGS. 79 and 80A-B, control logic process 750 begins in Step 751 with trigger unit microcontroller 200 sensing user activity on the trigger (e.g. trigger pull) in the form of applied trigger pull force or trigger displacement/movement. In Step 752, microcontroller 200 compares and determines if the sensed activity on the trigger is above the preprogrammed trigger pull Setpoint 1 (e.g. trigger force or displacement depending on what type of trigger sensing design is used). If a negative response (no) is returned to the inquiry, control passes back to Step 751 to continue monitoring the sensed trigger activity until it exceeds Setpoint 1. If a positive response (yes) is returned to the microcontroller inquiry in Step 752 (i.e. trigger pull force or displacement exceeds Setpoint 1), control passes to Step 753.

In step 753, the trigger unit microcontroller 200 transmits/sends a “shot initiation” signal to the external adaptive optics unit control module 711 of the advanced fire control targeting system (reference system architecture schematic diagram of FIG. 76). In step 754, the adaptive optics unit control module 711 (targeting microcontroller 712) receives the shot initiation signal and actively adjusts the point of aim and may display a corrected reticle 714 positioned on the sighting lens 713 based on input from all of the data obtained by the sub-modules 703-706 as previously described herein. The user may adjust their aim to accurately acquire the target based on the corrected position of the reticle 714. The user may now fully pull the trigger with greater force evincing a positive intent to discharged the firearm at the acquired target.

In Step 755, the trigger unit microcontroller 200 compares the actual sensed trigger pull force or displacement (dependent on what type trigger sensing design is used) to the associated preprogrammed trigger pull Setpoint 2. Setpoint 2 will be greater than Setpoint 1 (force or displacement) since Setpoint 2 is associated with a greater threshold parameter corresponding to the user’s intent to fire the

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firearm rather than a partial trigger pull used for aiming the firearm only. If the actual sensed trigger pull force or displacement is less than Setpoint 2, control passes to Step 756 to check if there is still actual user-applied activity on the trigger (e.g. trigger force or displacement) that exceeds Setpoint 1. If not (indicating the user has backed off on or released the trigger), the shot is aborted and the system is reset for the next firing event (Step 764, FIG. 80B). If conversely the trigger activity still exceeds Setpoint 1 in Step 756, the trigger unit microcontroller initiates a timer of preprogrammed duration. If the timer has not timed out, control returns to Step 755 and a control loop is initiated for trigger unit microcontroller 200 to continue monitoring for trigger activity (force or displacement) which exceeds Setpoint 2. If the timer times out with the trigger activity still exceeding Setpoint 1 but does not reach and exceed Setpoint 2, a timeout error occurs. Control passes to Step 765 (FIG. 80B) in which the system ignores the trigger pull event and defaults to a non-powered manual fire state at maximum force/displacement until the user-applied force on the trigger is removed.

Returning to the decision block in Step 755, if the user-applied trigger activity (trigger pull force or displacement) exceeds Setpoint 2 (“yes” response to test), control passes to Step 758. The trigger unit microcontroller 200 sends a “Shot Confirmed” signal to adaptive optics unit control module 711 (microcontroller 714) indicating that the user has acquired the target with guidance from the optics unit to correct point of aim, and intends now to fire the firearm. Targeting microcontroller 712 generates and transmits a “fire” signal to trigger unit microcontroller 200 as the operator pulls the trigger smoothly and hovers aim around the intended target location when the corrected point of aim coincides with the ballistics calculation of correction indicating the user has locked onto the target.

In Step 759, the trigger unit microcontroller 200 checks for the receipt of both the “fire” signal from targeting microcontroller 712 and user trigger activity still exceeding setpoint 2 indicative of a positive intent to still discharge the firearm. If both signals are received and detected (“yes” response), the firearm is discharged (Step 760) and the electromagnetic actuator trigger unit is reset from the next firing event (Step 764).

Conversely, if both the “fire” signal and trigger activity exceeding setpoint 2 are not detected by trigger unit microcontroller 200 in Step 759, control passes to Step 761. The “fire” signal will have been generate by adaptive optics unit control module 711 and received by the trigger unit microcontroller 200; however, the user may have decided not to take the shot in the interim and backed off on the trigger. If trigger Setpoint 1 is not still exceeded by the user activity on the trigger indicating a released trigger (“no” response), the shot abort and the electromagnetic trigger mechanism is reset for the next firing event (Step 764). Conversely, if the use-applied trigger force or displacement exceeds Setpoint 1, control passes to Step 762.

In Step 762, a test is performed by trigger unit microcontroller 200 to determine if the trigger pull force/displacement applied to the trigger by the user exceeds the preprogrammed maximum allowable trigger activity limit (trigger pull force/displacement limit). If the test answer is negative, control passes to back to step 759 and a control loop and timer are initiated via Step 763 for a preprogrammed duration of time. Any suitable duration of time may be used. If the timer expires in Step 763, control passes to Step 765 described above. The electronic trigger system ignores the trigger pull event and defaults to a non-powered manual fire

state at maximum force/displacement to permit manual firing of the firearm until the user-applied force on the trigger is removed which resets the system.

If instead in Step 762 a positive response (yes) results indicating that the applied trigger pull force imparted by the user to the trigger exceeds the preprogrammed maximum allowable trigger pull force/displacement limit, control instead bypasses the decision block in Step 763 and timer, and proceeds to Step 765. The trigger system ignores initiation of the electronically-assisted trigger event and defaults now to the non-powered manual "override" condition previously described herein by entering the manual firing mode without electrical assistance of the electronic trigger unit. The increased pressure and displacement applied to the trigger by the user indicates that the user intends to discharge the firearm despite the absence of a valid "fire" signal from the adaptive optics unit control module 711. This condition may possibly occur: (1) if there is a malfunction with the advanced fire control targeting system electronics (e.g. dead power source onboard battery 715 (see, e.g. FIG. 76) or other component failure; or (2) (1) if there is a malfunction with the electromagnetic trigger mechanism system electronics (e.g. dead power source onboard battery 122 (see, e.g. FIG. 76) or other component failure. The trigger system will stay in this manual override mode until the user-applied force on the trigger is completely removed, at which time the system will reset in Step 765 and return to Step 751 and the electronic firing mode awaiting the next trigger pull. Concurrently in Step 765, the trigger unit microcontroller 200 will transmit an external reset signal to the adaptive optics unit control module 711 to terminate the present electronic firing sequence and revert to its initial state to await the beginning of a successive trigger pull event.

In summary of the foregoing two stage interruptible trigger control logic process 750, it will be appreciated that if user-applied trigger force is removed (below preprogrammed trigger Setpoint 1), after passing through Setpoint 2, the firing attempt is aborted, the system reset, and a new firing attempt can be made. If the applied trigger force is lowered from trigger Setpoint 2 back to Setpoint 1 during the trigger pull event after a firing attempt is started, the electronic electromagnetic actuator trigger system will stay in a ready-to-fire state until (1) both the "fire" signal from adaptive optics unit control module 711 and the preprogrammed trigger Setpoint 2 force or displacement profile are detected and met for the firing solution; or (2) until a preset error timeout is reached. In the event of this timeout error, the electromagnetic trigger system ignores the trigger pull event and defaults to a non-powered manual fire state at maximum force/displacement until the force on the trigger is removed. This allows the firearm to be used as regular non-electronic firearm and discharged manually without electromagnetic assistance from the electronic trigger system or the advanced fire control targeting system input. This default manual override capability is a unique and significant advantage of the present electromagnetic actuator trigger system for user safety as the firearm may still be discharged in the event of an electronics failure in exigent circumstances.

Accordingly, if during the firing sequence the trigger pull exceeds the maximum allowed adjustable trigger pull force threshold, the electromagnetic trigger system will default to the manual override mode. If the operator chooses to override the electronic/electric trigger system by increasing pressure on the trigger, before the enabling "fire" signal is received from adaptive optics unit control module 711, the

trigger system defaults to a non-powered state at maximum Force/Displacement at the point that the user-applied trigger pull force exceeds the predefined preprogrammed maximum trigger pull force. The system will stay in this manual override mode until the force on the trigger is completely removed, at which time the trigger system will reset and return to electronic trigger mode.

FIGS. 81 and 82A-B illustrate an exemplary three stage interruptible trigger control logic process 800 which may be preprogrammed into and implemented by trigger unit microcontroller 200 of the electromagnetic actuator trigger mechanism control system, which is operably interfaced with the advanced fire control targeting system 700. FIG. 81 is a high level control logic flow diagram, while FIGS. 82A and 82B are lower level flow diagrams showing additional sub-steps of the basic process in FIG. 81.

The three stage interruptible trigger control logic process 800 is essentially the same as the two stage process 750 of FIGS. 80A-B. For brevity, same steps are numbered similarly and reference is made to the description of control logic process 750 above to avoid repetition of those portions of the process implemented by trigger unit microcontroller 200 of the electromagnetic trigger system.

The three-stage trigger control logic process 800 embodiment adds a third level of trigger pull force associated with a preprogrammed third trigger activity (pull force/displacement) Setpoint 3 (Step 801, FIG. 82B) to allow the operator to continue to apply force on the trigger of a higher level, but below the default non-powered manual firing mode or state associated with the preprogrammed maximum trigger activity limit (trigger pull force/displacement) previously described herein. This allows the user to override the fire control targeting system 700 and still fire the firearm in the electronic firing mode with the electromagnetically power-assisted trigger pull despite the absence of the "fire" signal from the external adaptive optics unit control module 711. As previously described herein, the electronic trigger mechanism actuator 350 will snap over to automatically complete the firing sequence for the user when energized, by magnetically rotating the actuator rotating member 304 and releasing the firing mechanism component (e.g. sear 375 in FIG. 27 or other firing mechanism component) to discharge the firearm. The three stage trigger control process, in contrast to the default to manual trigger option in the single and two stage control logic processes 730, 750 allows for a user selectable force setpoint for the override trigger force profile.

If during the firing sequence the trigger pull force exceeds the previously described maximum allowed trigger activity (trigger pull force/displacement) limit preprogrammed into trigger unit microcontroller 200, the electronic trigger system will default to the manual override mode. If the operator chooses to override the system by increasing pressure on the trigger, before the enabling "fire" signal is received from the adaptive optics unit control module 711, the trigger system defaults to the non-powered state manual firing mode or state when the user-applied trigger force/displacement exceeds the predefined preprogrammed maximum allowable limit. The trigger system will stay in this manual override mode until the force on the trigger is completely removed at which time the system will reset and return to electronic trigger firing mode.

Accordingly, in Step 801 of the three stage trigger control logic process 800, a signal is received from the decision block of Step 759 (FIG. 82A) indicating a negative response is returned to the test whether both the "fire" signal from adaptive optics unit control module 711 and trigger Setpoint

2 is exceeded. If the “fire” signal is not detected by trigger unit microcontroller 200, Step 801 tests if the user-applied trigger pull force exceeds preprogrammed trigger Setpoint 3. Preprogrammed trigger Setpoint 3 is greater than Setpoint 2, which is greater than Setpoint 1. If trigger setpoint 3 has not been exceeded, control passes to Step 761 and the process is completed as previously described herein with respect to FIGS. 82A-B. The “no” test response here indicates that the user does not intend to manually override the firing enablement input from and control by the adaptive optics unit control module 711. If in Step 801 the trigger pull force exceeds trigger Setpoint 3, control passes to Step 760 to actuate the firing event and discharge the firearm. The “yes” test response indicates that the user intends to discharge the firearm and override control by the adaptive optics unit control module 711 of external advanced fire control targeting system 700.

It bears noting that in all of the foregoing firing control schemes of the interruptible electronic trigger system (e.g. single, two, or three stage), the trigger unit actuator is changeable between a non-powered ready-to-fire unactuated position and powered actuated firing position when the electromagnetic coil is energized, as previously described herein. User-activity sensed on the trigger unit by microcontroller 200 may awaken and activate the electronic trigger system from a sleep mode which conserves battery life. The activated trigger system is now readied to implement the foregoing firing schemes in operable cooperation with the advance fire control targeting system 700.

It further bears noting that certain steps of the foregoing control logic processes may occur rapidly within a fraction of a second, which is only made possible by bi-directional communications and cooperation between the trigger unit microcontroller 200 and targeting microcontroller 712 onboard the adaptive optics unit 701. Numerous variations of the foregoing interruptible electronic trigger system are possible within the scope of the invention. The electromagnetic actuator trigger unit retains its user-adjustable variable trigger pull force characteristics previously described herein in detail, which allows the user to customize and control at least some of the foregoing trigger Setpoints 1, 2, or 3 via programming the trigger unit microcontroller 200 to adjust the trigger.

FIG. 83 provides a user sensory feedback modification of the two stage control logic process 750 previously described herein with reference to FIGS. 79-80B. Step 805 of FIG. 83 introduces the option of providing tactile, visual, and/or audible confirmation to the user that the first stage trigger shot authorization signal has been transmitted to the targeting microcontroller 712 of the fire control targeting system 700. As the trigger pull threshold for trigger Setpoint 1 is sensed and exceeded in Step 752, the “shot initiation” signal is transmitted by trigger unit microcontroller 200 to adaptive optics unit control module 711 in Step 753 as previously described herein. In new Step 805, sending of the “shot initiation” signal triggers the trigger unit microcontroller 200 to provide confirmation to the user which may be in the form of haptic vibration through the trigger, illumination and display of a visually observable indicia (e.g. solid/blinking dot or other shaped indicia) via the digital optical display projected onto sighting lens 713 of the adaptive optics unit 701 initiated by the trigger unit microcontroller, indicator lights, and/or generation of audible sounds (e.g. clicks, etc.). Any type, form, or mode of sensory signal(s) and combinations of sensory signals perceivable by the user may be used. Accordingly, in some embodiments the trigger unit microcontroller is configured to provide a user-perceivable

sensory signal confirmation to the user that the shot authorization signal has been transmitted to the external fire control targeting system, the sensory confirmation comprising at least one of a tactile feedback, visual indicia, and audible sound. It bears noting that the sensor feedback confirmation process described above may be implemented in the same manner for the single or three stage trigger control logic processes 730 and 800 previously described herein.

FIG. 84 is a high level control diagram showing an example of the general control process which may be implemented by the adaptive optics unit control module 711 of the advanced fire control targeting system 700 and interaction with the trigger unit microcontroller 200 of the electromagnetic trigger system for the single stage trigger control mode previously described herein. The sub-modules 703-706 depicted and their function were previously described herein. The adaptive optics unit 701 and its control module 711 may be awoken from energy conserving “sleep mode” and electrically activated upon receiving the “shot initiated” signal from trigger unit microcontroller 200. The adaptive optics unit then performs its ballistic computation and overlays the digitally displayed corrected reticle generated by reticle display system 702 at a position which coincides with the proper point of aim to acquire the target, as previously described herein. Once the user aligns their aim to the corrected reticle aimpoint, the “fire” signal is generated and transmitted to the trigger unit microcontroller 200.

In the two stage trigger control mode shown in FIG. 85, the process is essentially the same as in FIG. 84 above. However, the adaptive optics unit control module 711 waits for the “shot confirmed” signal from the trigger unit microcontroller 200 before confirming that the user has aligned their point of aim with the corrected reticle aimpoint before transmitting the “fire” signal to the trigger unit microcontroller. Satisfaction of the second stage trigger Setpoint 2 by the trigger unit microcontroller is used for shot confirmation by using the corrected reticle displayed.

Safety for Electromagnetic Firing System

Utilization of an electrically-assisted fire control trigger-sear firing mechanism such as without limitation the electromagnetic firing systems disclosed herein presents unique safety and fail-safe concerns. It is critical that no single point of failure results in the possible combination of effects that could accidentally discharge the firearm. Possible failure scenarios may include an electrical error or fault attempting to electrically trip the sear without the trigger being pulled; inertia from a drop, fall, or jar causing movement of components that may cause the sear to trip; unintended movement of the trigger without intentionally placing a finger onto the trigger; or external electromagnetic field which may unintentionally interact magnetically with the electrically-actuated trigger mechanism that may cause the rotating member (e.g., trigger bar) or other movable magnetic part to change positions and release the sear.

A safety is therefore desirable which can reliably preclude the foregoing events that might cause a firearm with electrically-assisted firing mechanism to unintentionally fire.

Disclosed below are embodiments of a mechanical trigger safety for an electrically-assisted firing mechanism of a firearm which incorporates a combination of electrical and mechanical means that cooperate to provide an integrated system approach to seamlessly enable or disable the electrically-assisted fire control mechanism under the foregoing types of conditions which could unintentionally release the sear and discharge the firearm.

In the following described embodiments, a trigger safety comprising a trigger blocking mechanism is combined with

the disclosed variable force snap-acting electromagnetic trigger system previously described herein, such as the one shown in FIGS. 16-29 or others disclosed. It bears noting that the combination of the mechanical trigger blocking safety and electrically-assisted electromagnetic trigger actuation mechanism could also be represented by alternative types of electromagnetic actuators such as linear or rotating solenoids, motors, and other commercial actuators. Independent of the type of electromagnetic actuator utilized, the requirements for combining a trigger blocking safety means with such actuators would be essential to ensure robustness of design for fault tolerance and passing regulatory mandated drop safety tests required for commercial firearms. The latter comprises dropping the firearm from a prescribed height without the firearm firing.

Compliance with gun drop test safety regulations require that a firearm be dropped from a prescribed distance (e.g., 1 meter) without discharging in the absence of a trigger pull. In addition to the user and firing benefits of the dual closed magnetic flux loop path electromagnetically adjustable firing system previously shown in FIGS. 16-30 and described herein comprising electromagnetic trigger mechanism 300, it bears special noting that this design also provides a mechanical trigger safety mechanism which advantageously provides an additional layer or level of protection that prevents the electromagnetic actuator of the firing system from actuating and discharging the firearm in the absence of an intentional trigger pull. This is a design aspect that help the electromagnetic trigger mechanisms disclosed herein pass the gun drop test requirements. Such requirements are largely applicable to any type of electronically/electrically actuated firing mechanism designs.

As one example of such a mechanism trigger safety, electromagnetic trigger mechanism 300 previously described herein includes trigger member 320 which comprises outer trigger 321 and inner safety trigger 322; the latter which provides the mechanical trigger safety which physically blocks full actuation of the trigger assembly. The inner trigger in that embodiment comprises safety bar 325 which cooperates with rear actuating extension 340 of rotating member 304 to selectively block or allow movement of rotating member to discharge the firearm. Accordingly, the inner trigger 322 acts upon and physically disables actuation of the electromagnetic trigger mechanism 300 including electromagnetic snap actuator 350 to survive a drop test in the absence of an intentional trigger pull.

To briefly summarize its operation, when the inner safety trigger 322 is not pulled and actuated by the user, the transversely extending safety bar 325 mounted to the safety trigger engages the rear actuating extension 340 of the rotating member 304, thereby blocking its upward movement and preventing the firearm from being fired (best shown in FIG. 27). To discharge the firearm, the shooter or user initially applies a trigger pull force F on first the safety trigger 322 which rotates rearward to its rearward position shown in FIG. 29. The safety bar 325 rotates forward about trigger pivot pin 323 from the position shown in FIG. 27 to a new position vertically aligned with safety notch 327 in the rear actuating extension 340 of rotating member 304. Because the safety bar 325 is aligned with safety notch 327, upward movement of rear actuating extension 340 of rotating member 304 is no longer fully blocked, thereby allowing sufficient upward rotation of the outer trigger via a full trigger pull to energize and actuate the snap actuator 350 for discharging the firearm. Because the inner safety 322 does not function by electrical/electronic means, this advanta-

geously provides a mechanical safety which is resistant to the effects of external electromagnetic fields or those generated onboard the firearm.

FIGS. 86-103 show an alternative embodiment of a mechanical trigger safety based on a modified inner safety trigger 922 to enhance meeting the drop safety test requirements. The present safety trigger 922 is embodied in an alternative electromagnetic trigger unit or mechanism 900 with electromagnetic snap actuator 950. As shown, it will be apparent that electromagnetic actuator 950 is similar to electromagnetic actuator 350 previously described so as to provide a dual closed magnetic flux loop path type snap actuator with all of its trigger performance and other related benefits described above. Accordingly, those features of present electromagnetic snap actuator 950 and electromagnetic trigger mechanism 900 similar to their counterparts in trigger mechanism 300 with snap actuator 350 will not be repeated here for sake of brevity, but are assigned "900" series designations recognizing that the features function in the same manner unless noted otherwise.

Referring first generally to FIGS. 86-99, the inner safety trigger 922 of electromagnetic trigger mechanism 900 comprises an upper mounting portion 924 and lower blade portion 936 configured for grasping by the user similarly to safety trigger 322. Mounting portion 924 is pivotably mounted to outer trigger 921 via trigger pivot pin 923. The mounting portion is bifurcated similarly to mounting portion 324 of safety trigger 322 and defines an upwardly open longitudinal slot that receives rear actuating extension 940 of rotating member 904 which defines downwardly facing actuation surface 942 of the rotating member 904. Actuation surface 942 is arranged to abuttingly engage a corresponding flat or planar upwardly facing operating surface 943 formed on the rear of the outer trigger 921. The interface between the operating surface 943 and actuation surface 942 is one of a flat-to-flat interface in one embodiment as shown. The force/displacement sensor such as thin film force sensing resistor 370 previously described herein may be interposed at the interface between the operating surface 943 of the rear operating extension 960 of outer trigger 921 and actuation surface 942 of the rear actuating extension 940 of rotating member 904. This is a similar force sensor arrangement and functionality as previous electromagnetic actuator 350 and its operation to discharge the firearm in cooperation with trigger unit microcontroller 200 which detects a trigger force which meets or exceeds a pre-selected preprogrammed trigger pull force that energizes and actuates the electromagnetic actuator will not be repeated here for sake of brevity. It bears noting that ability to actuate the actuator by either energizing the coil at the preprogrammed trigger pull force threshold or via the user applying a sufficient force to manually trip the actuator and fire the firearm like electromagnetic trigger mechanism 300.

Blade portion 936 of inner safety trigger 922 is biased forward by spring 924a which acts on the upper mounting portion 924. The spring may be a helical compression spring in one embodiment; however, other suitable type springs may be used in other implementations. Spring 924a is received in longitudinal spring slot 924c formed in rotating member 904, such as in rear actuating extension 940. Spring pin 924b extends transversely through mounting portion 924 via holes 924d and slot 924c to be acted upon by one end of the spring. The other end of the spring acts on rotating member 904. Spring pin 924b provides for smooth compression and expansion of the spring within the trigger assembly when the inner safety trigger is pulled and then

released which expands the spring as it returns the inner safety trigger 922 to its forward unactuated position.

With respect to the present mechanical trigger safety, spring 924a functions to bias the upper mounting portion 924 of safety trigger 922 rearward into engagement with a blocking surface formed on the electromagnetic actuator 950 above the mounting portion. In one embodiment, the blocking surface may be defined by a stationary transverse trigger safety blocking pin 980 fixedly mounted to a stationary portion of the firing mechanism. FIGS. 88-90 and 98 show one embodiment of the blocking pin and mounting arrangement. Blocking pin 980 may be cylindrical and fixedly mounted in one arrangement to a stationary portion of electromagnetic snap actuator 950, such as for example without limitation the yoke 902 which is fixedly coupled to the frame or chassis of the firearm. Blocking pin 980 may extend completely thorough the yoke in a lateral direction from side to side as shown.

With continuing reference to FIGS. 88-90, blocking pin 980 may be transversely mounted to a rear portion of yoke 902 and extends laterally through downwardly open internal cavity 902b of the yoke. Blocking pin 980 may be cylindrical; however, other shaped pins may be used. In one embodiment, the blocking pin may include a diametrically narrowed middle section 980a which is selectively engageable with a blocking surface 980b formed on top of upper mounting portion 924 of safety trigger 922 (see also FIG. 98). The narrowed middle section of the blocking pin forms a pair of shoulders at the transition to the larger diameter outer end portions of the blocking pin 980 which helps keep the mounting portion and blocking surface thereon centered. Blocking pin 980 may be retained to yoke 902 in one embodiment via a retaining ring 981. Other means of retaining the blocking pin may be used.

It bears noting that other embodiments of a electromagnetic actuator trigger mechanism 1200 was previously described herein having an inner safety trigger 1203 and blocking pin 1207 arrangement and shown in FIGS. 52-56 and 61-65. This safety mechanism also prevents actuation of the actuator by blocking movement of the outer trigger 1201.

Returning now to present electromagnetic actuator 950, outer trigger 921 includes cantilevered rear operating arm or extension 960 arranged to engage the rear actuating extension 940 of the rotating member 904, similarly to electromagnetic actuator 350. The outer trigger 921 is pivotably coupled to the front lower portion of rotating member 904 via common pivot pin 923 received through hole 928 of the trigger and pair of spaced apart holes 928a in the rotating member (the lower portion being bifurcated in one embodiment as shown in FIG. 88). Pivot pin 923 may in turn be mounted transversely to the frame or chassis (e.g., 22 or 60-1) of the firearm which may be a handgun or long gun. Pin 923 defines the transverse pivot axis for rotating member 904. Pivot pin 923 may be transversely mounted to a middle portion of stationary yoke 902 in one embodiment such as the lower horizontal section and extends completely thorough the yoke in a lateral direction from side to side (see, e.g., FIGS. 88-90). Pin 923 may be retained to the yoke by a retaining ring 981 or other suitable retention means.

As shown in FIGS. 88-90, an additional pin such as mounting pin 982 may be provided to help secure the front portion of the yoke 902 to the firearm frame or chassis. Mounting pin 982 may be transversely mounted to the front portion of stationary yoke 902 in one embodiment and extends completely thorough the yoke in a lateral direction

from side to side. Pin 982 may similarly be retained to the yoke by a retaining ring 981 or other suitable retention means.

Outer trigger 921 is biased forward into its upright position when not actuated or pulled via trigger spring 921a. Spring 921a may be a torsion spring in one embodiment which is coiled around spring mounting protrusion 921c formed inside a downwardly open internal cavity 902b of a forward portion of stationary yoke 902 (see, e.g., FIG. 97). One free end or leg of the spring may engage a spring retention protrusion 921b formed on the front section of the outer trigger body, and the remaining free end or leg of the spring may engage a surface inside the yoke. Other types of springs and/or arrangements may be used to maintain outer trigger 921 in its upright position.

Inner safety trigger 922 is also pivotably mounted to rotating member 904 via transversely extending pivot pin 923a received through holes 923b in the safety trigger (the upper mounting portion of safety trigger being bifurcated in one embodiment as shown in FIG. 88) and hole 923c formed in rear actuating extension 940 of the rotating member. Pin 923a defines the transverse pivot axis of the inner safety trigger. In the non-limiting illustrated embodiment, the inner safety trigger 922 is movably supported entirely by the rotating member 904 independently of the outer trigger.

Additional structural features and appurtenances of present electromagnetic actuator 950 which further differentiate this actuator from previous actuator 350 will now be briefly described with initial reference to FIGS. 88-90. In the present embodiment, electromagnetic actuator 950 includes a monolithic unitary stationary yoke made of a single piece in contrast to the two-piece split yoke 302 of previous electromagnetic actuator 350 (see, e.g., FIG. 20). To enable the single piece yoke to be used, rotating member 904 is specially configured to allow insertion through a centrally-located slot 902a formed in the lower horizontal section of the yoke 902 when the actuator 950 is assembled.

With additional reference to FIGS. 100-103, the cruciform-shaped top operating end section 904A of present rotating member 904 is now a separate detachable component collectively defined by actuation assembly 976. In electromagnetic actuator 350 previously described herein, the top operating end section was integrally formed with its rotating member 304. Actuation assembly 976 similarly defines the two opposite and outwardly facing front and rear actuation surfaces 934F, 934R selectively engageable with stationary yoke 902, and the sear engagement surface 48. These function in the same manner previously described herein and will not repeated here for sake of brevity.

The actuation assembly 976 may be detachably mounted and coupled to the top end of the vertical intermediate section 904C of rotating member 904 which projects upwards from the bottom or lower actuating end section 904B. A pair of laterally spaced apart and upwardly projecting mounting prongs 972 are formed on the top of intermediate section 904C of the rotating member for coupling to actuation assembly 976.

In one embodiment, actuation assembly 976 comprises an actuation member 974 and a sear engagement insert 973. Actuation member 974 includes a front section 974a which defines front actuation surface 934F of the rotating member which is engageable with the stationary yoke 902, and a rear section 974b which defines rear actuation surface 934R engageable with permanent magnet 908 mounted on the yoke. Both insert 973 and actuation member 974 may be detachably coupled together and in turn coupled to mounting prongs 972 formed on rotating member 904 via a pair of

transversely oriented pins **973a** (see also FIGS. **88-90**). In one embodiment, split spring pins may be used; however, other suitable type pins may be used. The pins **973a** extend through complementary configured holes formed through the mounting prongs **972** and each insert **973**, **974**. As best appreciated in FIGS. **100** and **101**, sear engagement insert **973** may be at least partially received and nested within actuation member **974** via an upwardly open central passage **974a** formed therein.

When the actuation member **974** is mounted to rotating member **904**, they collectively define the cruciform-shaped top operating end section **904A** defining front and rear actuation surfaces **934F**, **934R**. Advantageously, the foregoing construction allows the one-piece unitary monolithic stationary yoke **902** to be used and slid through slot **902a** in the lower portion of the yoke. The actuation inserts are each thereafter coupled to the top end of the rotating member **904** via mounting prongs **972**. As opposed to the split yoke previously described herein which is still satisfactory, the present one-piece yoke structure advantageously offers simplified assembly and reduces fabrication costs.

With continuing general reference to FIGS. **100-103**, it bears noting that sear engagement insert **973** defines the upward facing sear engagement surface **48** of present actuator **950** which selectively engages a component of the firing mechanism such as sear **377** in a manner similarly to actuator **350** and other actuator embodiments previously described herein. In summary, when the trigger assembly (outer trigger **921** and inner safety trigger **922**) is not fully pulled, sear engagement surface **48** prevents the sear from falling to release the spring-biased striking member such as striker **40** necessary to discharge the firearm. When the trigger assembly is fully pulled and actuated, engagement surface **48** disengages the sear which drops to release the striker and fire the firearm.

The sear **37** may be formed of a ferromagnetic material such steel for strength and durability well suited for use as a component of the firing mechanism which undergoes repeated cycles of discharging the firearm. Such ferromagnetic metals however exhibit strong magnetic properties which readily conduct magnetic flux currents when exposed to the magnetic field generated by the energized electromagnetic coil **906** of the electromagnetic actuator trigger mechanism. Therefore, it is possible that electromagnetic interaction between the electromagnetic actuator might result in magnetic coupling between the sear engagement surface **48** of the actuator **950** (or other actuator previously described herein) and sear **377**. This could potentially interfere with the response time and operation of the sear for releasing the firing mechanism component (e.g., striking member such as striker **40** or a hammer) to discharge the firearm.

To counter the above potential situation, the sear engagement insert **973** of electromagnetic actuator **950** in a preferred embodiment may be made of a magnetically weak material such a titanium. Titanium has sufficient strength and durability to sustain numerous firing cycles, but is a paramagnetic material with a weak magnetic response when exposed to external magnetic fields such as those generated by electromagnetic actuator **950** when energized. This make titanium ideal for the present firing mechanism application. Other possible magnetically weak metals may be used. In other possible designs to prevent magnetic coupling between a firing mechanism component such as sear **377** and the actuator sear engagement surface **48** of the actuator, the sear instead may be formed of titanium or other magnetically weak metal while the sear engagement insert **973** may be

formed of steel. Accordingly, the sear is formed of a strongly magnetizable metallic material and the moving member such as rotating member **904** comprises a sear engagement insert formed of a weakly magnetizable metallic material by contrast which prevents magnetic coupling between sear and the sear engagement insert, or vice-versa.

Referring now to FIGS. **88-90** and **98**, stationary yoke **902** of present electromagnetic actuator **950** is generally similar to yoke **302** of actuator **350** previously described herein. Yoke **902** has a generally U-shaped body comprising common horizontal bottom section **905A**, spaced apart rear vertical section **905B** and front vertical section **905C** each extending upwardly from the bottom section, and inwardly-turned top sections **905D** and **905E** each having a horizontal orientation. In the present embodiment, however, top sections **905D**, **905E** may be integrally formed with rear and front vertical sections **905B** and **905C** respectively as a unitary structural part thereof. Yoke **902** defines downwardly open cavity **902b** formed by horizontal bottom section **905A**. Cavity **902b** extends longitudinally/horizontally from front to rear of the yoke as shown. Yoke **902** defines a central space **905F** which receives electromagnetic coil **906** therein. In one embodiment, coil **906** may be wound around non-metallic and non-magnetic spool **906a** which is insertable into central space **905F**. Spool **906a** may be formed of non-magnetizable material such as plastic in one embodiment. A coil frame **906b** may be provided which is insertable with the spool and coil into central space **905F** to couple the coil to yoke **902**. Coil frame defines a vertical central passage **906c** which receives the vertical portion of rotating member therethrough. The coil frame may be retained to the yoke **902** using the same pins **980**, **923**, and **982** and retaining rings **981** previously described herein for efficient assembly of the electromagnetic actuator **950**.

The present electromagnetic actuator **950** also includes a permanent magnet **908** which functions in the same manner and for the same purpose as magnet **308** of actuator **350** previously described herein; generating a static magnetic field in the actuator to prevent movement of rotating member **904** and release of sear **377** when the actuator is not energized via a trigger pull to discharge the firearm. The present permanent magnet **908** may also be coupled to the upper rear portion of the stationary yoke **902**. Achieving secure attachment of the flat permanent magnet to the steel yoke via adhesives may be problematic in some circumstances.

In the present embodiment with continuing reference to FIGS. **88-90** and **98**, electromagnetic actuator **950** may include a non-metallic magnet holder **975** which is configured for detachable coupling to rear vertical section **905B** of the stationary yoke **902**. Specifically, magnet holder **975** may be coupled to inwardly turned rear top section **905D** of yoke **902**. Holder **975** may be formed a semi-rigid and slightly deformable plastic in one embodiment to form a friction fit with top section **905D**. The magnet holder may have a U-shaped body in one embodiment. To provide a secure coupling and to retain magnet **908**, magnet holder **975** may include a pair of front slots **974a** which slideably engage and receive magnet **908** therein. A pair of rear slots **974b** slideably engage mating mounting rails **974c** formed on rear top section **905D** of the yoke (see, e.g., FIG. **88**). When the plastic magnet holder **975** is coupled to the yoke **902**, the plastic body of the holder slightly deforms to create the friction fit with the yoke which secures the magnet and holder in place.

In another aspect, it bears particular note that the magnet holder **975** is further configured to form an forwardly

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projecting shrouding hood **975b** extending beyond the magnet **908**. Hood defines **975b** a frontal opening **975a** which receives rear section **974b** of actuation member **974** than defines rear actuation face **934R** at least partially therein (see, e.g., FIG. **98**). Hood **975b** may be three-sided in one embodiment and extends continuously along each vertical side and across the top of the magnet holder **975** between the sides at the front of the holder **975** around the opening **975a** (best shown in FIG. **102**). Advantageously, the shrouding hood prevents stray magnetizable ferrous debris or particles inside the firearm from being drawn into the interface between permanent magnet **908** and the rear actuation face **934R** which might interfere with proper operation and pivotable action of the rotating member **904**. The hood therefore shields the interface and physically blocks the ingress of such debris.

In one embodiment, the force sensor such as a thin film force sensing resistor **370** may be supported by and mounted at least partially in a protective electronics housing **990** detachably coupled to the rear of the stationary yoke **902** of electromagnetic actuator **950**. In one embodiment, housing **990** may include a forwardly open mounting recess **992** which slideably receives a rearwardly protruding mounting protrusion **991** disposed on the rear of stationary yoke **902**. Other coupling means however may be used. The housing may preferably be made of a non-magnetizable material such as plastic.

Housing **990** may further include a second frontal electronics access opening **993** through which the force sensor such as thin film force sensing resistor **370** is accessible. The force sensing resistor **370** may be supported by housing **990** and projects outwards therefrom in a cantilevered manner into the flat-to-flat interface between upward facing operating surface **943** on the rear of outer trigger **921** and downward facing actuation surface **942** of the rotating member **904** (see, e.g., FIGS. **97-99**). The wire leads **370a** from force sensing resistor **370** may be routed through housing **990** and emerge on the bottom of the housing for operable coupling to the trigger unit microcontroller **200**. Housing **990** may support electrical components and contacts such as for the electrical leads **906b** of the electromagnetic coil **906** which may be coupled to terminals **906c** attached to the housing and operably coupled to microcontroller **200**.

Inner safety trigger **922** is pivotably movable between a first blocking position which disables the electromagnetic actuator **950**, and a second unblocking position which enables the electromagnetic actuator to discharge the firearm when the outer trigger **921** is pulled. More specifically, the inner safety trigger operates in its disabling aspect to prevent the trigger assembly from moving a sufficient amount necessary to actuate and energize electromagnetic actuator **950** when the inner safety trigger is in the first blocking position, and in its enabling aspect to allow the electromagnetic actuator to be energized via full movement of the trigger assembly to actuate the moving member (e.g., rotating member **904**) and discharge the firearm when the inner safety trigger is in the second unblocking position. Pulling the outer trigger **921** when the inner safety trigger **922** is in the second unblocking position energizes the coil via force sensor **370** in the same manner previously described herein for actuator **350** and moves the moving member (e.g., rotating member **904**) to release sear **377** and discharge the firearm. It bears noting that the inner safety trigger **922** also physically/mechanically blocks movement to the trigger assembly sufficient to inhibit manual discharge of the firearm. Accordingly, the inner trigger safety **922** and other trigger safeties disclosed herein as well all act electrically

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and physically to block and prevent actuating their respective electronic trigger mechanisms.

While the foregoing description and drawings represent exemplary (i.e. example) embodiments of the present disclosure, it will be understood that various additions, modifications and substitutions may be made therein without departing from the spirit and scope and range of equivalents of the accompanying claims. In particular, it will be clear to those skilled in the art that the present invention may be embodied in other forms, structures, arrangements, proportions, sizes, and with other elements, materials, and components, without departing from the spirit or essential characteristics thereof. In addition, numerous variations in the methods/processes described herein may be made within the scope of the present disclosure. One skilled in the art will further appreciate that the embodiments may be used with many modifications of structure, arrangement, proportions, sizes, materials, and components and otherwise, used in the practice of the disclosure, which are particularly adapted to specific environments and operative requirements without departing from the principles described herein. The presently disclosed embodiments are therefore to be considered in all respects as illustrative and not restrictive. The appended claims should be construed broadly, to include other variants and embodiments of the disclosure, which may be made by those skilled in the art without departing from the scope and range of equivalents.

What is claimed is:

1. A firearm with electronic firing system comprising:
 - a chassis or frame;
 - a trigger mechanism coupled to the chassis or frame, the trigger mechanism comprising an electromagnetic actuator including a moving member configured and operable to discharge the firearm when actuated;
 - the trigger mechanism further comprising:
 - a movable outer trigger;
 - an inner safety trigger pivotably movable with respect to the outer trigger; and
 - a coil operably coupled to an electric power source and the electromagnetic actuator, the coil when energized operable to generate a magnetic field in the electromagnetic actuator which actuates the moving member;
 - wherein the inner safety trigger is movable between a first position which prevents actuation of the electromagnetic actuator, and a second position which allows actuation of the electromagnetic actuator to discharge the firearm when the outer trigger is pulled;
 - wherein the electromagnetic actuator further comprises a permanent magnet which generates a static magnetic field creating a primary resistance force opposing movement of the outer trigger when pulled by a user.
2. The firearm according to claim 1, wherein the inner safety trigger is operable to prevent the electromagnetic actuator from being energized when the inner safety trigger is in the first position, and the inner safety trigger is operable to allow the electromagnetic actuator to be energized to actuate the moving member and discharge the firearm when the inner safety trigger is in the second position.
3. The firearm according to claim 2, wherein pulling the outer trigger when the inner safety trigger is in the second position energizes the coil and moves the moving member to discharge the firearm.
4. The firearm according to claim 3, wherein the moving member of the electromagnetic actuator is operably interfaced with a firing component of the firing mechanism of the

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firearm, the firing component being operable to discharge the firearm when actuated by the moving member.

5. The firearm according to claim 4, wherein the firing component is a rotatable sear releasably engaged with both the moving member of the electromagnetic actuator and a spring-biased striking member operable to strike a chambered cartridge in the firearm.

6. The firearm according to claim 5, wherein when the inner safety trigger is in the second position, pulling the outer trigger breaks engagement between the moving member of the electromagnetic actuator and the sear to release the striking member for discharging the firearm.

7. The firearm according to claim 6, wherein the moving member is pivotably movable between an unactuated position engaged with the sear when the coil is de-energized, and an actuated position disengaged from the sear when the coil is energized to discharge the firearm.

8. The firearm according to claim 1, wherein the inner trigger prevents the outer trigger from being pulled and moving an amount sufficient to energize the coil when the inner safety trigger is in the first position.

9. The firearm according to claim 8, further comprising a blocking surface of the trigger mechanism which is selectively engageable by the inner safety trigger when the inner safety trigger is in the first position, the blocking surface preventing the outer trigger from moving via engagement with the inner safety trigger the amount sufficient to energize the coil.

10. The firearm according to claim 9, wherein moving the inner safety trigger to the second position disengages the inner safety trigger from the blocking surface which allows the outer trigger to be pulled to energize the coil and discharge the firearm.

11. The firearm according to claim 10, wherein the blocking surface is defined by a transversely mounted pin which is selectively engageable with an upwardly protruding mounting portion of the inner safety trigger.

12. The firearm according to claim 1, wherein the electromagnetic actuator comprises:

a stationary yoke;

the moving member being disposed at least partially within the yoke, the moving member rotatably movable about a pivot axis relative to the yoke and operably coupled to a firing mechanism component operable to discharge the firearm;

the coil being disposed at least partially within the yoke; the outer trigger operably coupled to the moving member and manually movable by a user from a first unpulled position to a second pulled position for discharging the firearm.

13. The firearm according to claim 12, wherein the moving member has a vertically elongated portion comprising an upper operating section selectively engageable with the firing mechanism component and a lower actuating section, and a laterally elongated actuation extension engageable with the outer trigger for discharging the firearm.

14. The firearm according to claim 13, further comprising a force sensor disposed at an interface between the outer trigger and the moving member, the force sensor being operably coupled to the electric power source and configured to sense a trigger pull force.

15. The firearm according to claim 14, wherein the electric power source energizes the coil when the trigger pull force senses by the force sensor exceeds a predetermined threshold trigger pull force.

16. The firearm according to claim 15, further comprising a programmable controller operably coupled to the force

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sensor and the electric power source, the controller being configured to energize the coil when the predetermined threshold trigger pull force is sensed.

17. The firearm according to claim 12, wherein the outer trigger is pivotably coupled to the yoke and the inner safety trigger is pivotably coupled to the moving member.

18. The firearm according to claim 5, wherein the sear is formed of a strongly magnetizable metallic material and the moving member comprises a sear engagement insert engageable with the sear, the sear engagement insert being formed of a weakly magnetizable metallic material which prevents magnetic coupling between sear and the sear engagement insert.

19. The firearm according to claim 13, wherein the upper operating section of the moving member has a cruciform shape defining an openable and closeable first air gap between the yoke and the moving member, and an openable and closeable second air gap between the yoke and the moving member.

20. The firearm according to claim 19, wherein the permanent magnet is disposed on the yoke adjacent the second air gap, the second air gap being covered by a non-metallic shrouding hood detachably coupled to the yoke.

21. A firearm with electronic firing system comprising:
a chassis or frame;

a trigger mechanism coupled to the chassis or frame, the trigger mechanism comprising an electromagnetic actuator including a moving member configured and operable to discharge the firearm when actuated;

the trigger mechanism further comprising:

a movable outer trigger;

an inner safety trigger pivotably movable with respect to the outer trigger; and

a coil operably coupled to an electric power source and the electromagnetic actuator, the coil when energized operable to generate a magnetic field in the electromagnetic actuator which actuates the moving member; wherein the inner safety trigger is movable between a first position which prevents actuation of the electromagnetic actuator, and a second position which allows actuation of the electromagnetic actuator to discharge the firearm when the outer trigger is pulled;

wherein the inner safety trigger is operable to prevent the electromagnetic actuator from being energized when the inner safety trigger is in the first position, and the inner safety trigger is operable to allow the electromagnetic actuator to be energized to actuate the moving member and discharge the firearm when the inner safety trigger is in the second position;

wherein pulling the outer trigger when the inner safety trigger is in the second position energizes the coil and moves the moving member to discharge the firearm;

wherein the moving member of the electromagnetic actuator is operably interfaced with a firing component of the firing mechanism of the firearm, the firing component being operable to discharge the firearm when actuated by the moving member;

wherein the firing component is a rotatable sear releasably engaged with both the moving member of the electromagnetic actuator and a spring-biased striking member operable to strike a chambered cartridge in the firearm; wherein when the inner safety trigger is in the second position, pulling the outer trigger breaks engagement

between the moving member of the electromagnetic actuator and the sear to release the striking member for discharging the firearm.

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